

Project:

UQ LT LPG Export - ST & MT

Location:

Umm Qasr, Iraq

Contract:

UQ83

Engineer:

Hemanathan Gopalan

Filename:

Umm Qasr-ST

ETAP

24.0.2C

Study Case: SC_Max

Page:

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Date:

03-08-2026

SN:

SL-KENTUAE

Revision:

Base

Config.:

1_NE

Electrical Transient Analyzer Program

Short-Circuit Analysis

IEC 60909 Standard

3-Phase Fault Currents

	Swing	V-Control	Load	Total			
Number of Buses:	25	0	251	276			
	XFMR2	XFMR3	Reactor	Line/Cable/ Busway	Impedance	Tie PD	Total
Number of Branches:	29	2	0	206	0	17	254
	Synchronous Generator	Power Grid	Synchronous Motor	Induction Machines	Lumped Load	Inverter	Total
Number of Machines:	6	1	0	133	47	0	187

System Frequency:

50.00

Unit System:

Metric

Project Filename:

Umm Qasr-ST

Output Filename:

C:\Users\cpra\Desktop\UQ\BGC\SC.SIIS

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Base

1_NE

Adjustments

Tolerance	Apply Adjustments	Individual /Global	Percent
Transformer Impedance:	Yes	Individual	
Reactor Impedance:	Yes	Individual	
Overload Heater Resistance:	No		
Transmission Line Length:	No		
Cable / Busway Length:	Yes	Global	10.0
Temperature Correction	Apply Adjustments	Individual /Global	Degree C
Transmission Line Resistance:	Yes	Global	20
Cable / Busway Resistance:	Yes	Global	20

Bus Input Data

Bus					Initial Voltage	
ID	Type	Nom. kV	Base kV	Sub-sys	%Mag.	Ang.
132kV GRID	SWNG	132.000	132.000	1	100.00	0.00
200A MDB Agreko	Load	0.415	0.462	1	100.00	0.00
Area Lighting DB (A8)	Load	0.380	0.404	1	100.00	0.00
BUS-MTR1_ENCRYO	SWNG	11.000	11.000	6	100.00	0.00
BUS-MTR2_ENCRYO	SWNG	11.000	11.000	7	100.00	0.00
Bus1	Load	11.000	11.220	1	100.00	0.00
Bus2	SWNG	0.380	0.415	17	92.83	0.00
Bus4	SWNG	0.380	0.415	18	92.83	0.00
Bus7	Load	11.000	11.220	1	100.00	0.00
Bus8	Load	11.000	11.646	1	100.00	0.00
Bus9	Load	0.380	0.408	1	100.00	0.00
Bus10	Load	0.380	0.398	1	100.00	0.00
Bus11	SWNG	0.380	0.380	5	100.00	0.00
Bus12	Load	0.380	0.408	1	100.00	0.00
Bus14	Load	0.380	0.446	1	100.00	0.00
Bus15	Load	0.380	0.446	1	100.00	0.00
Bus17	Load	11.000	11.220	1	100.00	0.00
Bus18	Load	11.000	11.646	1	100.00	0.00
Bus19	Load	11.000	11.220	1	100.00	0.00
Bus20	Load	11.000	11.646	1	100.00	0.00
Bus21	Load	11.000	11.646	1	100.00	0.00
Bus22	Load	3.300	3.519	1	100.00	0.00
Bus23	Load	0.380	0.413	1	100.00	0.00
Bus24	Load	0.380	0.419	1	100.00	0.00
Bus25	Load	3.300	3.652	1	100.00	0.00
Bus27	Load	0.380	0.404	1	100.00	0.00
Bus28	Load	0.380	0.408	1	100.00	0.00
Bus29	Load	3.450	3.652	1	100.00	0.00
Bus30	Load	3.450	3.519	1	100.00	0.00
Bus31	Load	3.450	3.519	1	100.00	0.00
Bus32	Load	3.450	3.652	1	100.00	0.00
Bus33	Load	0.380	0.423	1	100.00	0.00
Bus35	SWNG	0.380	0.380	3	100.00	0.00
Bus36	Load	0.400	0.451	1	100.00	0.00
Bus37	Load	3.300	3.718	1	100.00	0.00
Bus38	Load	0.400	0.446	1	100.00	0.00
Bus40	Load	0.400	0.446	1	100.00	0.00
Bus41	SWNG	0.380	0.380	4	100.00	0.00
Bus42	Load	11.000	11.646	1	100.00	0.00
Bus43	Load	11.000	11.220	1	100.00	0.00
Bus45	Load	0.400	0.446	1	100.00	0.00

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Base

1_NE

Bus					Initial Voltage	
ID	Type	Nom. kV	Base kV	Sub-sys	%Mag.	Ang.
Bus46	Load	11.000	11.646	1	100.00	0.00
Bus47	Load	3.300	3.519	1	100.00	0.00
Bus48	Load	0.400	0.446	1	100.00	0.00
Bus49	Load	3.300	3.718	1	100.00	0.00
Bus50	Load	3.300	3.582	1	100.00	0.00
Bus51	Load	11.000	11.220	1	100.00	0.00
Bus52	Load	0.400	0.462	1	100.00	0.00
Bus53	Load	0.380	0.408	1	100.00	0.00
Bus54	Load	0.380	0.408	1	100.00	0.00
Bus55	Load	11.000	11.646	1	100.00	0.00
Bus56	Load	11.000	11.220	1	100.00	0.00
Bus57	SWNG	0.380	0.415	11	92.83	0.00
Bus58	SWNG	0.380	0.380	8	100.00	0.00
Bus59	SWNG	0.380	0.415	12	92.83	0.00
Bus60	Load	0.380	0.446	1	100.00	0.00
Bus61	SWNG	0.380	0.380	9	100.00	0.00
Bus62	Load	0.380	0.446	1	100.00	0.00
Bus63	Load	0.400	0.462	1	100.00	0.00
Bus64	Load	0.415	0.462	1	100.00	0.00
Bus65	Load	0.400	0.462	1	100.00	0.00
Bus66	Load	11.000	11.220	1	100.00	0.00
Bus67	Load	0.380	0.398	1	100.00	0.00
Bus68	SWNG	0.380	0.415	13	92.83	0.00
Bus69	Load	0.380	0.413	1	100.00	0.00
Bus70	SWNG	0.380	0.415	14	92.83	0.00
Bus71	SWNG	0.380	0.380	10	100.00	0.00
Bus72	Load	0.380	0.446	1	100.00	0.00
Bus73	Load	0.380	0.446	1	100.00	0.00
Bus74	SWNG	0.380	0.415	15	92.83	0.00
Bus75	SWNG	0.380	0.415	16	92.83	0.00
Bus76	Load	0.380	0.446	1	100.00	0.00
Bus77	Load	0.380	0.446	1	100.00	0.00
Bus78	Load	11.000	11.646	1	100.00	0.00
Bus79	Load	132.000	132.000	1	100.00	0.00
Bus80	Load	132.000	132.000	1	100.00	0.00
Bus81	Load	132.000	132.000	1	100.00	0.00
Bus82	Load	0.415	0.462	1	100.00	0.00
Bus83	Load	132.000	132.000	1	100.00	0.00
Bus84	Load	132.000	132.000	1	100.00	0.00
Bus85	Load	0.380	0.408	1	100.00	0.00
Bus87	Load	132.000	132.000	1	100.00	0.00
Bus88	Load	132.000	132.000	1	100.00	0.00
Bus89	Load	11.000	11.220	1	100.00	0.00

Bus					Initial Voltage	
ID	Type	Nom. kV	Base kV	Sub-sys	%Mag.	Ang.
Bus90	Load	11.000	11.220	1	100.00	0.00
Bus91	Load	0.380	0.406	1	100.00	0.00
Bus92	Load	11.000	11.220	1	100.00	0.00
Bus93	SWNG	0.380	0.415	19	109.21	0.00
Bus94	SWNG	0.380	0.415	20	109.21	0.00
Bus95	Load	0.400	0.462	1	100.00	0.00
Bus96	Load	0.400	0.462	1	100.00	0.00
Bus97	SWNG	0.380	0.415	21	109.21	0.00
Bus98	SWNG	0.380	0.415	22	109.21	0.00
Bus99	Load	0.400	0.462	1	100.00	0.00
Bus100	Load	0.400	0.462	1	100.00	0.00
Bus101	SWNG	0.380	0.415	23	109.21	0.00
Bus102	SWNG	0.380	0.415	24	109.21	0.00
Bus103	Load	0.400	0.462	1	100.00	0.00
Bus104	Load	0.400	0.462	1	100.00	0.00
Bus105	SWNG	0.380	0.415	25	109.21	0.00
Bus106	SWNG	0.380	0.415	26	109.21	0.00
Bus107	Load	0.400	0.462	1	100.00	0.00
Bus108	Load	0.400	0.462	1	100.00	0.00
Bus110	Load	11.000	11.220	1	100.00	0.00
Bus112	Load	11.000	11.220	1	100.00	0.00
Bus113	Load	11.000	11.646	1	100.00	0.00
Bus114	Load	11.000	11.646	1	100.00	0.00
Bus115	Load	11.000	11.646	1	100.00	0.00
Bus117	Load	11.000	11.646	1	100.00	0.00
Bus119	Load	3.300	3.519	1	100.00	0.00
Bus120	Load	3.300	3.519	1	100.00	0.00
Bus121	Load	3.300	3.519	1	100.00	0.00
Bus122	Load	3.300	3.519	1	100.00	0.00
Bus123	Load	3.300	3.519	1	100.00	0.00
Bus124	Load	3.300	3.519	1	100.00	0.00
Bus125	Load	3.300	3.519	1	100.00	0.00
Bus126	Load	3.300	3.519	1	100.00	0.00
Bus127	Load	3.300	3.519	1	100.00	0.00
Bus128	Load	3.300	3.519	1	100.00	0.00
Bus129	Load	3.300	3.519	1	100.00	0.00
Bus130	Load	3.300	3.519	1	100.00	0.00
Bus131	Load	3.300	3.519	1	100.00	0.00
Bus132	Load	3.300	3.519	1	100.00	0.00
Bus133	Load	3.300	3.652	1	100.00	0.00
Bus134	Load	3.300	3.652	1	100.00	0.00
Bus135	Load	3.300	3.652	1	100.00	0.00
Bus136	Load	3.300	3.652	1	100.00	0.00

Bus					Initial Voltage	
ID	Type	Nom. kV	Base kV	Sub-sys	%Mag.	Ang.
Bus137	Load	3.300	3.652	1	100.00	0.00
Bus138	Load	3.300	3.652	1	100.00	0.00
Bus139	Load	3.300	3.652	1	100.00	0.00
Bus140	Load	3.300	3.652	1	100.00	0.00
Bus141	Load	3.300	3.652	1	100.00	0.00
Bus142	Load	3.300	3.652	1	100.00	0.00
Bus143	Load	3.300	3.652	1	100.00	0.00
Bus144	Load	3.300	3.652	1	100.00	0.00
Bus145	Load	3.300	3.652	1	100.00	0.00
Bus146	Load	3.300	3.652	1	100.00	0.00
Bus149	Load	0.380	0.408	1	100.00	0.00
Bus151	Load	0.380	0.408	1	100.00	0.00
Bus152	Load	0.380	0.422	1	100.00	0.00
Bus154	Load	0.380	0.408	1	100.00	0.00
Bus157	Load	0.380	0.416	1	100.00	0.00
Bus158	Load	0.380	0.406	1	100.00	0.00
Bus159	Load	0.380	0.422	1	100.00	0.00
Bus160	Load	3.450	3.519	1	100.00	0.00
Bus161	Load	0.380	0.408	1	100.00	0.00
Bus162	Load	0.380	0.408	1	100.00	0.00
Bus163	Load	11.000	11.646	1	100.00	0.00
Bus164	Load	0.380	0.408	1	100.00	0.00
Bus165	Load	0.380	0.408	1	100.00	0.00
Bus166	Load	0.380	0.408	1	100.00	0.00
Bus167	Load	0.380	0.423	1	100.00	0.00
Bus168	Load	11.000	11.220	1	100.00	0.00
Bus169	Load	0.380	0.406	1	100.00	0.00
Bus170	Load	0.380	0.408	1	100.00	0.00
Bus171	Load	0.380	0.408	1	100.00	0.00
Bus172	Load	0.380	0.408	1	100.00	0.00
Bus173	Load	0.380	0.423	1	100.00	0.00
Bus174	Load	0.380	0.423	1	100.00	0.00
Bus175	Load	0.380	0.423	1	100.00	0.00
Bus176	Load	0.380	0.423	1	100.00	0.00
Bus177	Load	0.380	0.423	1	100.00	0.00
Bus178	Load	0.380	0.423	1	100.00	0.00
Bus179	Load	0.380	0.423	1	100.00	0.00
Bus180	Load	0.380	0.423	1	100.00	0.00
Bus181	Load	0.380	0.423	1	100.00	0.00
Bus182	Load	0.380	0.423	1	100.00	0.00
Bus183	Load	0.380	0.423	1	100.00	0.00
Bus184	Load	0.380	0.423	1	100.00	0.00
Bus185	Load	0.380	0.423	1	100.00	0.00

Bus					Initial Voltage	
ID	Type	Nom. kV	Base kV	Sub-sys	%Mag.	Ang.
Bus186	Load	0.380	0.423	1	100.00	0.00
Bus187	Load	0.380	0.423	1	100.00	0.00
Bus188	Load	0.380	0.423	1	100.00	0.00
Bus189	Load	0.380	0.423	1	100.00	0.00
Bus190	Load	0.380	0.423	1	100.00	0.00
Bus_LVDB-1	Load	0.415	0.462	1	100.00	0.00
Bus_LVDB-2	Load	0.415	0.446	1	100.00	0.00
Bus_MDB-1	Load	0.415	0.462	1	100.00	0.00
Bus_MDB-2	Load	0.415	0.462	1	100.00	0.00
Bus_MDB-3	Load	0.415	0.462	1	100.00	0.00
Bus_MDB-4	Load	0.415	0.462	1	100.00	0.00
Bus_MDB-5	Load	0.415	0.446	1	100.00	0.00
Bus_MDB-6	Load	0.400	0.446	1	100.00	0.00
Bus_MDB-7	Load	0.400	0.446	1	100.00	0.00
Bus_MDB-8	Load	0.400	0.446	1	100.00	0.00
BUS_S002-CMP3	Load	11.000	11.646	1	100.00	0.00
CAMP BUS	Load	11.000	11.220	1	100.00	0.00
LV SWBD_TOFS	Load	0.400	0.424	1	100.00	0.00
MV RMU-TOFS	Load	11.000	11.220	1	100.00	0.00
POWER SOCKET FEEDER	Load	0.380	0.423	1	100.00	0.00
1 B						
S002-ASPI-1 [BUS A]	Load	0.380	0.406	1	100.00	0.00
S002-ASPI-1B [BUS B]	Load	0.380	0.422	1	100.00	0.00
S002-ASPI-2A	Load	0.380	0.406	1	100.00	0.00
S002-ASPI-2B	Load	0.380	0.422	1	100.00	0.00
S002-DBASPI-2-22	Load	0.380	0.422	1	100.00	0.00
S002-MCC1-1 [BUS]	Load	0.380	0.406	1	100.00	0.00
S002-MCC1-2 [BUS]	Load	0.380	0.422	1	100.00	0.00
S002-MCC1-3 [BUS]	Load	0.380	0.406	1	100.00	0.00
S002-MCC1-4 [BUS]	Load	0.380	0.398	1	100.00	0.00
S002-MCC1-5 [BUS]	Load	0.380	0.406	1	100.00	0.00
S002-MCC1-6 [BUS]	Load	0.380	0.413	1	100.00	0.00
S002-MCC1-7 [BUS]	Load	0.380	0.406	1	100.00	0.00
S002-MCC1-8 [BUS]	Load	0.380	0.422	1	100.00	0.00
S002-MCC1-9 [BUS]	Load	0.380	0.406	1	100.00	0.00
S002-PC1-1A	Load	0.380	0.406	1	100.00	0.00
S002-PC1-1B	Load	0.380	0.422	1	100.00	0.00
S002-PC1-2A	Load	0.380	0.406	1	100.00	0.00
S002-PC1-2B	Load	0.380	0.422	1	100.00	0.00
S002-SB-1AC001 [BUS A]	Load	11.000	11.220	1	100.00	0.00
S002-SB-1AC001 [BUS B]	Load	11.000	11.646	1	100.00	0.00
S002-SB-1AD001 [BUS A]	Load	3.300	3.519	1	100.00	0.00
S002-SB-1AD001 [BUS B]	Load	3.300	3.652	1	100.00	0.00
S002-SB-1AN001 [BUS A]	Load	0.380	0.406	1	100.00	0.00

Bus					Initial Voltage	
ID	Type	Nom. kV	Base kV	Sub-sys	%Mag.	Ang.
S002-SB-1AN001 [BUS B]	Load	0.380	0.422	1	100.00	0.00
S002-SB-01E001 [BUS A]	Load	0.380	0.398	1	100.00	0.00
S002-SB-01E001 [BUS B]	Load	0.380	0.413	1	100.00	0.00
S002-SB-02N001 [BUS A]	Load	0.380	0.404	1	100.00	0.00
S002-SB-02N001 [BUS B]	Load	0.380	0.419	1	100.00	0.00
S002-SB-03E001 [BUS A]	Load	0.380	0.408	1	100.00	0.00
S002-SB-03E001 [BUS B]	Load	0.380	0.423	1	100.00	0.00
S002-SB-03N001 [BUS A]	Load	0.380	0.398	1	100.00	0.00
S002-SB-03N001 [BUS B]	Load	0.380	0.413	1	100.00	0.00
S002-SB-04E001 [BUS A]	Load	0.380	0.416	1	100.00	0.00
S002-SB-04E001 [BUS B]	Load	0.380	0.416	1	100.00	0.00
S002-SB-05N001A	Load	0.380	0.408	1	100.00	0.00
S002-SB-05N001B	Load	0.380	0.423	1	100.00	0.00
S002-SB-05N001C	Load	0.380	0.423	1	100.00	0.00
S002-SB-PSS1A01 [BUS-A]	Load	132.000	132.000	1	100.00	0.00
S002-SB-PSS01N01 [BUS A]	Load	0.380	0.398	1	100.00	0.00
S002-SB-PSS01N01 [BUS B]	Load	0.380	0.413	1	100.00	0.00
S002-TR-04DE001_PRI	Load	3.300	3.519	1	100.00	0.00
S002-TR-04DE002_PRI	Load	3.300	3.652	1	100.00	0.00
S003-DB11E001-01	Load	0.380	0.408	1	100.00	0.00
S003-DB11E001-02	Load	0.380	0.423	1	100.00	0.00
S003-DB-11N001-03 [BUS A]	Load	0.380	0.408	1	100.00	0.00
S003-DB-11N001-03 [BUS B]	Load	0.380	0.423	1	100.00	0.00
S003-DB11N001-04 [BUS A]	Load	0.380	0.408	1	100.00	0.00
S003-DB11N001-04 [BUS B]	Load	0.380	0.423	1	100.00	0.00
S003-DB11N001-11	Load	0.380	0.423	1	100.00	0.00
S003-DB11N001-12	Load	0.380	0.423	1	100.00	0.00
S003-MDP-B	Load	0.380	0.408	1	100.00	0.00
S003-SB01E1	Load	0.380	0.423	1	100.00	0.00
S003-SB-11C001 [BUS A]	Load	11.000	11.220	1	100.00	0.00
S003-SB-11C001 [BUS B]	Load	11.000	11.646	1	100.00	0.00
S003-SB-11D001 [BUS A]	Load	3.300	3.582	1	100.00	0.00
S003-SB-11D001 [BUS B]	Load	3.300	3.718	1	100.00	0.00
S003-SB-11E001 [BUS A]	Load	0.380	0.408	1	100.00	0.00
S003-SB-11E001 [BUS B]	Load	0.380	0.423	1	100.00	0.00
S003-SB-11N001 [BUS A]	Load	0.380	0.408	1	100.00	0.00
S003-SB-11N001 [BUS B]	Load	0.380	0.423	1	100.00	0.00
TR-PSS1AC01_11kV BUS	Load	11.000	11.220	1	100.00	0.00
TR-PSS1AC01_132kV BUS	Load	132.000	132.000	1	100.00	0.00
TR-PSS1AC02_11kV BUS	Load	11.000	11.646	1	100.00	0.00
TR-PSS1AC02_132kV BUS	Load	132.000	132.000	1	100.00	0.00
TYPICAL MOTOR FEEDER 1 A	Load	0.380	0.408	1	100.00	0.00

Bus					Initial Voltage	
ID	Type	Nom. kV	Base kV	Sub-sys	%Mag.	Ang.
TYPICAL MOTOR FEEDER	Load	0.380	0.423	1	100.00	0.00
1 B						
Cable40~	Load	0.446	0.446	1	100.00	0.00
Cable41~	Load	0.446	0.446	1	100.00	0.00
Cable153~	Load	0.446	0.446	1	100.00	0.00
Cable154~	Load	0.446	0.446	1	100.00	0.00
Cable163~	Load	0.446	0.446	1	100.00	0.00
Cable164~	Load	0.446	0.446	1	100.00	0.00
Cable168~	Load	0.446	0.446	1	100.00	0.00
Cable169~	Load	0.446	0.446	1	100.00	0.00
Cable188~	Load	0.462	0.462	1	100.00	0.00
Cable189~	Load	0.462	0.462	1	100.00	0.00
Cable192~	Load	0.462	0.462	1	100.00	0.00
Cable193~	Load	0.462	0.462	1	100.00	0.00
Cable196~	Load	0.462	0.462	1	100.00	0.00
Cable197~	Load	0.462	0.462	1	100.00	0.00
Cable200~	Load	0.462	0.462	1	100.00	0.00
Cable201~	Load	0.462	0.462	1	100.00	0.00
S002-TR-CMP1_VFD~2	Load	11.220	11.220	1	100.00	0.00
S002-TR-CMP1_VFD~3	Load	11.220	11.220	1	100.00	0.00
S002-TR-CMP2_VFD~2	Load	11.646	11.646	1	100.00	0.00
S002-TR-CMP2_VFD~3	Load	11.646	11.646	1	100.00	0.00

276 Buses Total

All voltages reported by ETAP are in % of bus Nominal kV.
Base kV values of buses are calculated and used internally by ETAP .

Project:	UQ LT LPG Export - ST & MT	ETAP	Page:	10
Location:	Umm Qasr, Iraq	24.0.2C	Date:	03-08-2026
Contract:	UQ83		SN:	SL-KENTUAE
Engineer:	Hemanathan Gopalan	Study Case: SC_Max	Revision:	Base
Filename:	Umm Qasr-ST		Config.:	1_NE

Line/Cable/Busway Input Data

ohms or siemens/1000 m per Conductor (Cable) or per Phase (Line/Busway)									
Line/Cable/Busway		Length							
ID	Library	Size	Adj. (m)	% Tol.	#/Phase	T (°C)	R	X	Y
Cable2	1.0MCUN3	70	345.6	-10.0	1	20	0.26822	0.07500	0.0001307
Cable3	1.0MCUN3	70	340.2	-10.0	1	20	0.26822	0.07500	0.0001307
Cable4	1.0MCUN3	70	334.8	-10.0	1	20	0.26822	0.07500	0.0001307
Cable5	1.0MCUN3	70	334.8	-10.0	1	20	0.26822	0.07500	0.0001307
Cable6	1.0MCUN3	70	329.4	-10.0	1	20	0.26822	0.07500	0.0001307
Cable7	1.0MCUN3	70	324.0	-10.0	1	20	0.26822	0.07500	0.0001307
Cable10	1.0MCUN3	70	329.4	-10.0	1	20	0.26822	0.07500	0.0001307
Cable11	1.0MCUN3	70	324.0	-10.0	1	20	0.26822	0.07500	0.0001307
Cable12	1.0MCUN3	70	318.6	-10.0	1	20	0.26822	0.07500	0.0001307
Cable13	1.0MCUN3	70	324.0	-10.0	1	20	0.26822	0.07500	0.0001307
Cable19	1.0MCUN3	70	318.6	-10.0	1	20	0.26822	0.07500	0.0001307
Cable20	1.0MCUN3	35	324.0	-10.0	1	20	0.52469	0.08000	0.0001015
Cable21	1.0MCUN3	70	313.2	-10.0	1	20	0.26822	0.07500	0.0001307
Cable22	1.0MCUN3	70	318.6	-10.0	1	20	0.26822	0.07500	0.0001307
Cable23	1.0MCUN3	16	90.0	-10.0	1	20	1.14505	0.08000	0.0000971
Cable24	1.0MCUN3	16	90.0	-10.0	1	20	1.14505	0.08000	0.0000971
Cable25	1.0MCUN3	70	313.2	-10.0	1	20	0.26822	0.07500	0.0001307
Cable26	1.0MCUN3	35	324.0	-10.0	1	20	0.52469	0.08000	0.0001015
Cable27	1.0MCUN3	70	307.8	-10.0	1	20	0.26822	0.07500	0.0001307
Cable28	1.0MCUN3	70	291.6	-10.0	1	20	0.26822	0.07500	0.0001307
Cable29	1.0MCUN3	70	297.0	-10.0	1	20	0.26822	0.07500	0.0001307
Cable30	1.0MCUN3	70	302.4	-10.0	1	20	0.26822	0.07500	0.0001307
Cable31	1.0MCUN3	70	297.0	-10.0	1	20	0.26822	0.07500	0.0001307
Cable32	1.0MCUN3	70	302.4	-10.0	1	20	0.26822	0.07500	0.0001307
Cable33	1.0MCUN3	70	307.8	-10.0	1	20	0.26822	0.07500	0.0001307
Cable34	1.0MCUN3	70	302.4	-10.0	1	20	0.26822	0.07500	0.0001307
Cable35	1.0MCUN3	70	307.8	-10.0	1	20	0.26822	0.07500	0.0001307
Cable36	1.0MCUN3	70	313.2	-10.0	1	20	0.26822	0.07500	0.0001307
Cable37	1.0MCUN3	70	307.8	-10.0	1	20	0.26822	0.07500	0.0001307
Cable38	1.0MCUN3	70	313.2	-10.0	1	20	0.26822	0.07500	0.0001307
Cable39	1.0MCUN3	70	318.6	-10.0	1	20	0.26822	0.07500	0.0001307
Cable40	1.0NCUN1	240	40.5	-10.0	2	20	0.07608	0.09200	0.0001703
Cable41	1.0NCUN1	240	45.0	-10.0	2	20	0.07608	0.09200	0.0001703
Cable42	1.0MCUN3	70	313.2	-10.0	1	20	0.26822	0.07500	0.0001307
Cable43	1.0MCUN3	70	318.6	-10.0	1	20	0.26822	0.07500	0.0001307
Cable44	1.0MCUN3	70	324.0	-10.0	1	20	0.26822	0.07500	0.0001307
Cable45	1.0MCUN3	150	248.4	-10.0	1	20	0.12470	0.07300	0.0001442

ohms or siemens/1000 m per Conductor (Cable) or per Phase (Line/Busway)

Line/Cable/Busway		Library	Size	Length		#/Phase	T (°C)	R	X	Y
				Adj. (m)	% Tol.					
Cable48		1.0MCUN3	150	117.0	-10.0	1	20	0.12470	0.07300	0.0001442
Cable49		1.0MCUN3	150	144.0	-10.0	1	20	0.12470	0.07300	0.0001442
Cable83		1.0NCUN4	16	45.0	-10.0	1	20	1.14505	0.08000	0.0000971
Cable118		1.1NCUS3	95	900.0	-10.0	1	20	0.19450	0.10200	0.0001049
Cable123		1.0NCUN2	16	49.5	-10.0	1	20	1.14505	0.08000	0.0000971
Cable128		1.0NCUN1	240	22.5	-10.0	4	20	0.07608	0.09200	0.0001703
Cable134		1.0NCUN1	240	7.2	-10.0	4	20	0.07608	0.09200	0.0001703
Cable136		1.0NCUN1	240	36.0	-10.0	7	20	0.07608	0.09200	0.0001703
Cable137		1.0NCUN1	240	7.2	-10.0	4	20	0.07608	0.09200	0.0001703
Cable144		1.0NCUN1	240	27.0	-10.0	7	20	0.07608	0.09200	0.0001703
Cable145		1.0NCUN1	240	27.0	-10.0	7	20	0.07608	0.09200	0.0001703
Cable146		1.0NCUN1	240	27.0	-10.0	7	20	0.07608	0.09200	0.0001703
Cable147		1.0NCUN1	240	7.2	-10.0	4	20	0.07608	0.09200	0.0001703
Cable148		1.0NCUN1	240	7.2	-10.0	4	20	0.07608	0.09200	0.0001703
Cable149		1.0NCUN1	240	7.2	-10.0	4	20	0.07608	0.09200	0.0001703
Cable150		1.0NCUN1	240	36.0	-10.0	7	20	0.07608	0.09200	0.0001703
Cable151		1.0NCUN1	240	22.5	-10.0	4	20	0.07608	0.09200	0.0001703
Cable152		1.0NCUN1	240	7.2	-10.0	4	20	0.07608	0.09200	0.0001703
Cable153		1.0NCUN1	240	27.0	-10.0	2	20	0.07608	0.09200	0.0001703
Cable154		1.0NCUN1	240	31.5	-10.0	2	20	0.07608	0.09200	0.0001703
Cable155		1.0NCUN3	16	31.5	-10.0	1	20	1.14505	0.08000	0.0000971
Cable156		1.0NCUN3	16	36.0	-10.0	1	20	1.14505	0.08000	0.0000971
Cable157		1.0NCUN1	240	36.0	-10.0	7	20	0.07608	0.09200	0.0001703
Cable158		1.0NCUN1	240	23.4	-10.0	4	20	0.07608	0.09200	0.0001703
Cable159		1.0NCUN1	240	21.6	-10.0	4	20	0.07608	0.09200	0.0001703
Cable160		1.0NCUN1	240	20.7	-10.0	4	20	0.07608	0.09200	0.0001703
Cable161		1.0NCUN1	240	13.5	-10.0	4	20	0.07608	0.09200	0.0001703
Cable162		1.0NCUN1	240	22.5	-10.0	4	20	0.07608	0.09200	0.0001703
Cable163		1.0NCUN1	240	10.8	-10.0	2	20	0.07608	0.09200	0.0001703
Cable164		1.0NCUN1	240	16.2	-10.0	2	20	0.07608	0.09200	0.0001703
Cable165		1.0NCUN3	16	24.3	-10.0	1	20	1.14505	0.08000	0.0000971
Cable166		1.0NCUN3	16	28.8	-10.0	1	20	1.14505	0.08000	0.0000971
Cable167		1.0NCUN1	240	22.5	-10.0	4	20	0.07608	0.09200	0.0001703
Cable168		1.0NCUN1	240	22.5	-10.0	2	20	0.07608	0.09200	0.0001703
Cable169		1.0NCUN1	240	32.4	-10.0	2	20	0.07608	0.09200	0.0001703
Cable170		1.0NCUN3	16	31.5	-10.0	1	20	1.14505	0.08000	0.0000971
Cable171		1.0NCUN3	16	41.4	-10.0	1	20	1.14505	0.08000	0.0000971
Cable176		1.0NCUN3	95	22.5	-10.0	4	20	0.19372	0.07300	0.0001433
Cable188		1.0NCUN1	240	9.0	-10.0	2	20	0.07608	0.09200	0.0001703

ohms or siemens/1000 m per Conductor (Cable) or per Phase (Line/Busway)									
Line/Cable/Busway	Library	Size	Length		#/Phase	T (°C)	R	X	Y
			Adj. (m)	% Tol.					
Cable189	1.0NCUN1	240	13.5	-10.0	2	20	0.07608	0.09200	0.0001703
Cable190	1.0NCUN3	16	22.5	-10.0	1	20	1.14505	0.08000	0.0000971
Cable191	1.0NCUN3	16	27.0	-10.0	1	20	1.14505	0.08000	0.0000971
Cable192	1.0NCUN1	240	22.5	-10.0	2	20	0.07608	0.09200	0.0001703
Cable193	1.0NCUN1	240	31.5	-10.0	2	20	0.07608	0.09200	0.0001703
Cable194	1.0NCUN3	16	31.5	-10.0	1	20	1.14505	0.08000	0.0000971
Cable195	1.0NCUN3	16	40.5	-10.0	1	20	1.14505	0.08000	0.0000971
Cable196	1.0NCUN1	240	38.7	-10.0	2	20	0.07608	0.09200	0.0001703
Cable197	1.0NCUN1	240	23.4	-10.0	2	20	0.07608	0.09200	0.0001703
Cable198	1.0NCUN3	16	45.0	-10.0	1	20	1.14505	0.08000	0.0000971
Cable199	1.0NCUN3	16	31.5	-10.0	1	20	1.14505	0.08000	0.0000971
Cable200	1.0NCUN1	240	15.3	-10.0	2	20	0.07608	0.09200	0.0001703
Cable201	1.0NCUN1	240	20.7	-10.0	2	20	0.07608	0.09200	0.0001703
Cable202	1.0NCUN3	16	22.5	-10.0	1	20	1.14505	0.08000	0.0000971
Cable203	1.0NCUN3	16	28.8	-10.0	1	20	1.14505	0.08000	0.0000971
C_688-KM-01	3.3NCUN3	95	450.0	-10.0	2	20	0.19372	0.08000	0.0001037
S002-830-1063-N-P	1.0MCUN1	630	27.0	-10.0	5	20	0.03294	0.08600	0.0002023
S002-830-1066-N-P	1.0MCUN1	630	27.0	-10.0	5	20	0.03294	0.08600	0.0002023
S002-830-1069-N-P	1.0MCUN1	630	31.5	-10.0	3	20	0.03294	0.08600	0.0002023
S002-830-1072-N-P	1.0MCUN1	630	31.5	-10.0	3	20	0.03294	0.08600	0.0002023
S002-830-1420-N-P-1	1.0MCUN1	630	22.5	-10.0	5	20	0.03294	0.08600	0.0002023
S002-830-1500-N-P-1	1.0MCUN1	630	27.0	-10.0	5	20	0.03294	0.08600	0.0002023
S002-830-9001-A-P	11MCUS1	400	342.0	-10.0	1	20	0.05019	0.10100	0.0001762
S002-830-9001-C-P	11NCUS1	1000	72.9	-10.0	8	20	0.02667	0.09300	0.0002560
S002-830-9001-D-P-1	3.3NCUN1	630	90.0	-10.0	4	20	0.03294	0.08600	0.0002023
S002-830-9004-A-P	11MCUS1	400	342.0	-10.0	1	20	0.05019	0.10100	0.0001762
S002-830-9004-D-P	3.3NCUN3	240	81.0	-10.0	1	20	0.07764	0.07300	0.0001464
S002-830-9005-D-P	3.3NCUN3	240	81.0	-10.0	1	20	0.07764	0.07300	0.0001464
S002-830-9006-D-P	3.3NCUN3	25	522.0	-10.0	1	20	0.72703	0.09600	0.0000591
S002-830-9007-D-P	3.3NCUN3	25	522.0	-10.0	1	20	0.72703	0.09600	0.0000591
S002-830-9008-C-P-1	11NCUS1	1000	94.5	-10.0	8	20	0.02667	0.09300	0.0002560
S002-830-9008-D-P	3.3NCUN3	25	360.0	-10.0	1	20	0.72703	0.09600	0.0000591
S002-830-9009-D-P	3.3NCUN3	25	360.0	-10.0	1	20	0.72703	0.09600	0.0000591
S002-830-9010-D-P	3.3NCUN3	25	621.0	-10.0	1	20	0.72703	0.09600	0.0000591
S002-830-9011-D-P	3.3NCUN3	25	621.0	-10.0	1	20	0.72703	0.09600	0.0000591
S002-830-9012-D-P	3.3NCUN3	25	783.0	-10.0	1	20	0.72703	0.09600	0.0000591
S002-830-9013-D-P	3.3NCUN3	25	783.0	-10.0	1	20	0.72703	0.09600	0.0000591
S002-830-9014-D-P	3.3NCUN3	25	495.0	-10.0	1	20	0.72703	0.09600	0.0000591
S002-830-9015-D-P	3.3NCUN3	25	504.0	-10.0	1	20	0.72703	0.09600	0.0000591

ohms or siemens/1000 m per Conductor (Cable) or per Phase (Line/Busway)									
Line/Cable/Busway	Library	Size	Length		#/Phase	T (°C)	R	X	Y
			Adj. (m)	% Tol.					
S002-830-9016-D-P	3.3NCUN3	95	396.0	-10.0	2	20	0.19372	0.08000	0.0001037
S002-830-9017-D-P	3.3NCUN3	95	405.0	-10.0	2	20	0.19372	0.08000	0.0001037
S002-830-9018-D-P	3.3NCUN3	95	423.0	-10.0	2	20	0.19372	0.08000	0.0001037
S002-830-9019-D-P	3.3NCUN3	95	144.0	-10.0	1	20	0.19372	0.08000	0.0001037
S002-830-9020-D-P	3.3NCUN3	35	243.0	-10.0	1	20	0.52469	0.09200	0.0000669
S002-830-9021-D-P-1	3.3NCUN1	630	90.0	-10.0	4	20	0.03294	0.08600	0.0002023
S002-830-9024-D-P	3.3NCUN3	240	81.0	-10.0	1	20	0.07764	0.07300	0.0001464
S002-830-9025-D-P	3.3NCUN3	240	81.0	-10.0	1	20	0.07764	0.07300	0.0001464
S002-830-9026-D-P	3.3NCUN3	25	522.0	-10.0	1	20	0.72703	0.09600	0.0000591
S002-830-9027-D-P	3.3NCUN3	25	522.0	-10.0	1	20	0.72703	0.09600	0.0000591
S002-830-9028-D-P	3.3NCUN3	25	360.0	-10.0	1	20	0.72703	0.09600	0.0000591
S002-830-9029-D-P	3.3NCUN3	25	360.0	-10.0	1	20	0.72703	0.09600	0.0000591
S002-830-9030-D-P	3.3NCUN3	25	621.0	-10.0	1	20	0.72703	0.09600	0.0000591
S002-830-9031-D-P	3.3NCUN3	25	621.0	-10.0	1	20	0.72703	0.09600	0.0000591
S002-830-9032-D-P	3.3NCUN3	25	783.0	-10.0	1	20	0.72703	0.09600	0.0000591
S002-830-9033-D-P	3.3NCUN3	25	783.0	-10.0	1	20	0.72703	0.09600	0.0000591
S002-830-9034-D-P	3.3NCUN3	25	495.0	-10.0	1	20	0.72703	0.09600	0.0000591
S002-830-9035-D-P	3.3NCUN3	25	504.0	-10.0	1	20	0.72703	0.09600	0.0000591
S002-830-9036-D-P	3.3NCUN3	95	396.0	-10.0	2	20	0.19372	0.08000	0.0001037
S002-830-9037-D-P	3.3NCUN3	95	405.0	-10.0	2	20	0.19372	0.08000	0.0001037
S002-830-9038-D-P	3.3NCUN3	95	423.0	-10.0	2	20	0.19372	0.08000	0.0001037
S002-830-9039-D-P	3.3NCUN3	25	243.0	-10.0	1	20	0.72703	0.09600	0.0000591
S002-830-9051-C-P	11NCUS3	185	108.0	-10.0	1	20	0.10039	0.09310	0.0001351
S002-830-9053-C-P	11NCUS3	185	765.0	-10.0	1	20	0.10039	0.09310	0.0001351
S002-830-9054-C-P	11NCUS3	185	864.0	-10.0	1	20	0.10039	0.09310	0.0001351
S002-830-9055-C-P	11NCUS3	185	864.0	-10.0	1	20	0.10039	0.09310	0.0001351
S002-830-9056-C-P-1	11NCUS1	630	54.0	-10.0	2	20	0.03294	0.09650	0.0002193
S002-830-9059-C-P	11NCUS3	185	427.5	-10.0	1	20	0.10039	0.09310	0.0001351
S002-830-9061-C-P-1	11NCUS3	400	297.0	-10.0	2	20	0.05019	0.08490	0.0001822
S002-830-9062-C-P	11NCUS3	185	540.0	-10.0	1	20	0.10039	0.09310	0.0001351
S002-830-9063-C-P	11NCUS3	185	531.0	-10.0	1	20	0.10039	0.09310	0.0001351
S002-830-9064-C-P	11NCUS3	185	81.0	-10.0	1	20	0.10039	0.09310	0.0001351
S002-830-9066-C-P	11NCUS3	185	765.0	-10.0	1	20	0.10039	0.09310	0.0001351
S002-830-9067-C-P	11NCUS3	185	864.0	-10.0	1	20	0.10039	0.09310	0.0001351
S002-830-9068-C-P	11NCUS3	185	864.0	-10.0	1	20	0.10039	0.09310	0.0001351
S002-830-9069-C-P-1	11NCUS1	630	54.0	-10.0	2	20	0.03294	0.09650	0.0002193
S002-830-9072-C-P	11NCUS3	185	436.5	-10.0	1	20	0.10039	0.09310	0.0001351
S002-830-9073-C-P-1	11NCUS3	400	288.0	-10.0	2	20	0.05019	0.08490	0.0001822
S002-830-9074-C-P	11NCUS3	185	432.0	-10.0	1	20	0.10039	0.09310	0.0001351

ohms or siemens/1000 m per Conductor (Cable) or per Phase (Line/Busway)									
Line/Cable/Busway	Library	Size	Length		#/Phase	T (°C)	R	X	Y
			Adj. (m)	% Tol.					
S002-830-9076-C-P	11NCUS3	185	531.0	-10.0	1	20	0.10039	0.09310	0.0001351
S002-830-9077-C-P-1	11NCUS3	150	6300.0	-10.0	2	20	0.12470	0.09620	0.0001232
S002-830-9078-C-P-1	11MCUS3	150	6300.0	-10.0	2	20	0.12470	0.09620	0.0001232
S002-830-9079-C-P	11NCUS3	95	495.0	-10.0	1	20	0.19450	0.10200	0.0001049
S002-Cable20	1.0NCUN1	630	454.5	-10.0	3	20	0.03294	0.08600	0.0002023
S002-Cable21	1.0NCUN4	185	180.0	-10.0	1	20	0.10039	0.07300	0.0001442
S002-Cable22	1.0NCUN1	630	454.5	-10.0	3	20	0.03294	0.08600	0.0002023
S002-Cable26	1.0NCUN4	95	153.0	-10.0	1	20	0.19372	0.07300	0.0001433
S002-Cable29	1.0NCUN1	800	22.5	-10.0	1	20	0.02588	0.09100	0.0002061
S002-Cable34	1.0MCUN4	185	81.0	-10.0	1	20	0.10039	0.07300	0.0001442
S002-Cable35	1.0MCUN1	630	13.5	-10.0	5	20	0.03294	0.08600	0.0002023
S002-Cable85	1.0NCUN1	800	27.0	-10.0	1	20	0.02588	0.09100	0.0002061
S002-Cable87	1.0NCUN1	800	22.5	-10.0	1	20	0.02588	0.09100	0.0002061
S002-Cable88	1.0NCUN1	800	31.5	-10.0	1	20	0.02588	0.09100	0.0002061
S002-Cable89	1.0MCUN1	630	13.5	-10.0	5	20	0.03294	0.08600	0.0002023
S002-Cable90	1.0MCUN4	185	81.0	-10.0	1	20	0.10039	0.07300	0.0001442
S002-Cable95	1.0NCUN1	800	32.4	-10.0	1	20	0.02588	0.09100	0.0002061
S002-Cable115	1.0NCUN1	630	25.2	-10.0	1	20	0.03294	0.08600	0.0002023
S002-Cable119	1.0NCUN1	630	27.0	-10.0	1	20	0.03294	0.08600	0.0002023
S002-Cable124	1.0NCUN1	800	26.1	-10.0	1	20	0.02588	0.09100	0.0002061
S002-Cable125	1.0NCUN1	800	26.1	-10.0	1	20	0.02737	0.08700	0.0000000
S002-Cable126	1.0NCUN1	800	21.6	-10.0	1	20	0.02588	0.09100	0.0002061
S002-Cable127	1.0NCUN1	800	21.6	-10.0	1	20	0.02588	0.09100	0.0002061
S002-Cable131	1.0NCUN3	95	60.3	-10.0	1	20	0.19372	0.07300	0.0001433
S002-Cable132	1.0NCUN3	95	85.5	-10.0	1	20	0.19372	0.07300	0.0001433
S002-Cable138	1.0NCUN4	16	90.0	-10.0	1	20	1.14505	0.08000	0.0000971
S002-Cable181	1.0NCUN4	35	90.0	-10.0	1	20	0.52469	0.08000	0.0001015
S002-C_TR-04DE001	3.3NCUN3	185	450.0	-10.0	1	20	0.09960	0.07500	0.0001329
S002-C_TR-04DE002	3.3NCUN3	185	450.0	-10.0	1	20	0.09960	0.07500	0.0001329
S003-830-1000-E-P	1.0MCUN1	630	45.0	-10.0	4	20	0.03294	0.08600	0.0002023
S003-830-1004-E-P	1.0MCUN1	630	45.0	-10.0	4	20	0.03294	0.08600	0.0002023
S003-830-9001-C-P	1.0MCUN3	185	67.5	-10.0	1	20	0.10039	0.07300	0.0001442
S003-830-9001-D-P	3.3NCUN1	630	47.7	-10.0	1	20	0.03294	0.08600	0.0002023
S003-830-9002-C-P	1.0MCUN3	185	45.0	-10.0	1	20	0.10039	0.07300	0.0001442
S003-830-9003-C-P	3.3NCUN3	185	45.0	-10.0	1	20	0.09960	0.07500	0.0001329
S003-830-9004-C-P	3.3NCUN3	185	40.5	-10.0	1	20	0.09960	0.07500	0.0001329
S003-830-9005-C-P	11MCUS3	185	45.0	-10.0	1	20	0.10039	0.09310	0.0001351
S003-830-9006-C-P	11MCUS3	185	60.3	-10.0	1	20	0.10039	0.09310	0.0001351
S003-830-9019-D-P	3.3NCUN1	630	47.7	-10.0	1	20	0.03294	0.08600	0.0002023

ohms or siemens/1000 m per Conductor (Cable) or per Phase (Line/Busway)									
Line/Cable/Busway		Length							
ID	Library	Size	Adj. (m)	% Tol.	#/Phase	T (°C)	R	X	Y
S003-830-9021-C-P	11NCUS3	150	544.5	-10.0	1	20	0.12470	0.09620	0.0001232
S003-830-9022-C-P	11NCUS3	150	544.5	-10.0	1	20	0.12470	0.09620	0.0001232
S003-Cable19	1.0NCUN4	70	45.0	-10.0	1	20	0.26822	0.07500	0.0001307
S003-Cable42	1.0NCUN1	630	702.0	-10.0	4	20	0.03294	0.08600	0.0002023
S003-Cable44	1.0NCUN4	70	45.0	-10.0	1	20	0.26822	0.07500	0.0001307
S003-Cable93	1.0NCUN4	70	49.5	-10.0	1	20	0.26822	0.07500	0.0001307
S003-Cable94	1.0NCUN4	70	45.0	-10.0	1	20	0.26822	0.07500	0.0001307
S003-Cable96	1.0NCUN4	70	45.0	-10.0	1	20	0.26822	0.07500	0.0001307
S003-Cable105	1.0NCUN1	630	270.0	-10.0	5	20	0.03294	0.08600	0.0002023
S003-Cable113	11NCUS3	150	765.0	-10.0	3	20	0.12470	0.09620	0.0001232
S003-Cable121	1.0NCUN4	70	45.0	-10.0	1	20	0.26822	0.07500	0.0001307
S003-Cable142	1.0MCUN4	70	144.0	-10.0	2	20	0.26822	0.07500	0.0001307
S003-Cable173	1.0MCUN4	70	144.0	-10.0	2	20	0.26822	0.07500	0.0001307

Line / Cable / Busway resistances are listed at the specified temperatures.

Project: UQ LT LPG Export - ST & MT

ETAP

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Location: Umm Qasr, Iraq

24.0.2C

Date: 03-08-2026

Contract: UQ83

SN: SL-KENTUAE

Engineer: Hemanathan Gopalan

Study Case: SC_Max

Revision: Base

Filename: Umm Qasr-ST

Config.: 1_NE

2-Winding Transformer Input Data

Transformer	Rating					Z Variation			% Tap Setting		Adjusted
ID	MVA	Prim. kV	Sec. kV	% Z	X/R	+ 5%	- 5%	% Tol.	Prim.	Sec.	% Z
FD Tranfo	0.750	3.300	0.400	5.00	3.50	0	0	0	0	0	5.0000
S002-TML1 [EXISTING]	1.000	11.000	0.416	4.42	3.50	0	0	0	5.000	0	4.4200
S002-TML1A-1 [EXISTING]	1.600	3.300	0.400	6.21	6.00	0	0	0	5.000	0	6.2100
S002-TML1A-2 [EXISTING]	1.600	3.300	0.400	6.19	6.00	0	0	0	5.000	0	6.1900
S002-TML1B-1 [EXISTING]	1.600	3.300	0.400	6.18	6.00	0	0	0	5.000	0	6.1800
S002-TML1B-2 [EXISTING]	1.600	3.300	0.400	6.16	6.00	0	0	0	5.000	0	6.1600
S002-TML2 [EXISTING]	1.000	11.000	0.416	4.27	3.50	0	0	0	5.000	0	4.2700
S002-TMM-1A [EXISTING]	12.000	11.000	3.450	8.88	20.00	0	0	0	0	0	8.8800
S002-TMM-1B [EXISTING]	12.000	11.000	3.450	8.68	20.00	0	0	0	0	0	8.6800
S002-TR-01CE001	2.000	11.000	0.400	6.00	6.00	0	0	-7.5	2.500	0	5.5500
S002-TR-01CE002	2.000	11.000	0.400	6.00	6.00	0	0	-7.5	2.500	0	5.5500
S002-TR-03CE001	1.250	11.000	0.400	4.89	5.24	0	0	0	0	0	4.8900
S002-TR-03CE002	1.250	11.000	0.400	4.89	5.25	0	0	0	0	0	4.8900
S002-TR-03CN001	2.000	11.000	0.400	6.08	6.35	0	0	0	2.500	0	6.0800
S002-TR-03CN002	2.000	11.000	0.400	6.03	6.35	0	0	0	2.500	0	6.0300
S002-TR-04DE001	0.400	3.300	0.400	4.05	1.50	0	0	-10.0	2.500	0	3.6450
S002-TR-PSS1AC01	50.000	132.000	11.500	10.30	32.50	0	0	0	2.500	0	10.3000
S002-TR-PSS1AC02	50.000	132.000	11.500	10.30	32.50	0	0	0	-1.250	0	10.3000
S003-TMM1A [EXISTING]	3.500	11.000	3.600	6.88	8.50	0	0	0	2.500	0	6.8800
S003-TMM1B [EXISTING]	3.500	11.000	3.600	6.94	8.50	0	0	0	2.500	0	6.9400
S003-TR-11CE001	1.600	11.000	0.400	6.00	3.50	0	0	-7.5	0	0	5.5500
S003-TR-11CE002	1.600	11.000	0.400	6.00	6.00	0	0	-7.5	0	0	5.5500
S003-TR-11CN001	2.500	11.000	0.400	6.00	6.00	0	0	-7.5	0	0	5.5500
S003-TR-11CN002	2.500	11.000	0.400	6.00	6.00	0	0	-7.5	0	0	5.5500
T2	1.000	11.000	0.416	5.00	3.50	0	0	0	0	0	5.0000
TR-05CE001A	2.500	11.000	0.400	9.00	6.00	0	0	-10.0	0	0	8.1000
TR-05CE001B	2.500	11.000	0.400	9.00	6.00	0	0	-10.0	0	0	8.1000
TX-A	3.150	11.000	0.415	6.50	6.00	0	0	-10.0	-5.000	0	5.8500
TX-B	3.150	11.000	0.415	6.50	6.00	0	0	-10.0	-5.000	0	5.8500

3-Winding Transformer Input Data

Transformer		Rating		Tap	Impedance				Z Variation	
ID	Winding	MVA	kV	%	% Z	X/R	MVAb	% Tol.	+ 5%	- 5%
S002-TR-CMP1_VFD	Primary:	7.500	11.000	0	Zps = 10.00	45.00	7.500	-10.0	0	0
	Secondary:	3.250	11.000	0	Zpt = 10.00	45.00	7.500	-10.0		
	Tertiary:	3.250	11.000	0	Zst = 10.00	45.00	7.500	-10.0		
S002-TR-CMP2_VFD	Primary:	7.500	11.000	0	Zps = 10.00	45.00	7.500	-10.0	0	0
	Secondary:	3.250	11.000	0	Zpt = 10.00	45.00	7.500	-10.0		
	Tertiary:	3.250	11.000	0	Zst = 10.00	45.00	7.500	-10.0		

Branch Connections

CKT/Branch		Connected Bus ID		% Impedance, Pos. Seq., 100 MVAb			
ID	Type	From Bus	To Bus	R	X	Z	Y
FD Tranfo	2W XFMR	Bus37	Bus36	139.86	489.52	509.10	
S002-TML1 [EXISTING]	2W XFMR	Bus7	Bus27	125.17	438.10	455.63	
S002-TML1A-1 [EXISTING]	2W XFMR	Bus30	S002-PC1-1A	59.53	357.16	362.09	
S002-TML1A-2 [EXISTING]	2W XFMR	Bus31	S002-PC1-2A	59.34	356.05	360.96	
S002-TML1B-1 [EXISTING]	2W XFMR	Bus29	S002-PC1-1B	54.99	329.96	334.51	
S002-TML1B-2 [EXISTING]	2W XFMR	Bus32	S002-PC1-2B	54.82	328.93	333.47	
S002-TML2 [EXISTING]	2W XFMR	Bus46	Bus24	112.33	393.16	408.90	
S002-TMM-1A [EXISTING]	2W XFMR	Bus1	Bus22	3.52	70.49	70.58	
S002-TMM-1B [EXISTING]	2W XFMR	Bus21	Bus25	3.20	64.03	64.11	
S002-TR-01CE001	2W XFMR	Bus43	Bus10	44.50	266.98	270.66	
S002-TR-01CE002	2W XFMR	Bus42	Bus23	41.30	247.80	251.22	
S002-TR-03CE001	2W XFMR	Bus17	Bus28	68.38	358.12	364.59	
S002-TR-03CE002	2W XFMR	Bus18	Bus33	63.37	332.42	338.40	
S002-TR-03CN001	2W XFMR	Bus112	Bus67	45.98	292.00	295.59	
S002-TR-03CN002	2W XFMR	Bus8	Bus69	42.34	268.87	272.18	
S002-TR-04DE001	2W XFMR	S002-TR-04DE001_PRI	S002-SB-04E001 [BUS A]	457.56	686.35	824.89	
S002-TR-PSS1AC01	2W XFMR	TR-PSS1AC01_132kV BUS	TR-PSS1AC01_11kV BUS	0.66	21.29	21.30	
S002-TR-PSS1AC02	2W XFMR	TR-PSS1AC02_132kV BUS	TR-PSS1AC02_11kV BUS	0.61	19.76	19.77	
S003-TMM1A [EXISTING]	2W XFMR	Bus168	Bus50	23.28	197.92	199.28	
S003-TMM1B [EXISTING]	2W XFMR	Bus163	Bus49	21.79	185.24	186.52	
S003-TR-11CE001	2W XFMR	Bus19	Bus166	88.54	309.88	322.28	
S003-TR-11CE002	2W XFMR	Bus20	Bus167	49.14	294.82	298.89	
S003-TR-11CN001	2W XFMR	Bus56	S003-SB-11N001 [BUS A]	33.88	203.29	206.09	
S003-TR-11CN002	2W XFMR	Bus55	S003-SB-11N001 [BUS B]	31.45	188.69	191.29	
T2	2W XFMR	Bus51	LV SWBD_TOFS	128.02	448.05	465.98	
TR-05CE001A	2W XFMR	S002-SB-1AC001 [BUS A]	S002-SB-05N001A	48.74	292.42	296.46	
TR-05CE001B	2W XFMR	S002-SB-1AC001 [BUS B]	S002-SB-05N001B	45.24	271.42	275.16	
TX-A	2W XFMR	Bus66	Bus38	25.54	153.22	155.33	
TX-B	2W XFMR	Bus78	Bus65	23.70	142.21	144.17	
S002-TR-CMP1_VFD	3W Xfmr	S002-SB-1AC001 [BUS A]	S002-TR-CMP1_VFD~2	3.93	176.63	176.68	
	3W Xfmr	S002-SB-1AC001 [BUS A]	S002-TR-CMP1_VFD~3	3.93	176.63	176.68	
	3W Xfmr	S002-TR-CMP1_VFD~2	S002-TR-CMP1_VFD~3	3.93	176.63	176.68	
	3W Xfmr	S002-SB-1AC001 [BUS B]	S002-TR-CMP2_VFD~2	3.64	163.95	163.99	
	3W Xfmr	S002-SB-1AC001 [BUS B]	S002-TR-CMP2_VFD~3	3.64	163.95	163.99	
S002-TR-CMP2_VFD	3W Xfmr	S002-TR-CMP2_VFD~2	S002-TR-CMP2_VFD~3	3.64	163.95	163.99	
Cable2	Cable	TYPICAL MOTOR FEEDER 1 A	Bus9	5569.17	1557.23	5782.78	

CKT/Branch		Connected Bus ID		% Impedance, Pos. Seq., 100 MVAb			
ID	Type	From Bus	To Bus	R	X	Z	Y
Cable3	Cable	TYPICAL MOTOR FEEDER 1 A	Bus12	5482.15	1532.90	5692.43	
Cable4	Cable	TYPICAL MOTOR FEEDER 1 A	Bus53	5395.13	1508.57	5602.07	
Cable5	Cable	TYPICAL MOTOR FEEDER 1 A	Bus54	5395.13	1508.57	5602.07	
Cable6	Cable	TYPICAL MOTOR FEEDER 1 A	Bus85	5308.11	1484.23	5511.72	
Cable7	Cable	TYPICAL MOTOR FEEDER 1 A	Bus149	5221.10	1459.90	5421.36	
Cable10	Cable	TYPICAL MOTOR FEEDER 1 A	Bus151	5308.11	1484.23	5511.72	
Cable11	Cable	TYPICAL MOTOR FEEDER 1 A	Bus154	5221.10	1459.90	5421.36	
Cable12	Cable	TYPICAL MOTOR FEEDER 1 A	Bus161	5134.08	1435.57	5331.00	
Cable13	Cable	TYPICAL MOTOR FEEDER 1 A	Bus162	5221.10	1459.90	5421.36	
Cable19	Cable	TYPICAL MOTOR FEEDER 1 A	Bus164	5134.08	1435.57	5331.00	
Cable20	Cable	S002-MCC1-8 [BUS]	Bus159	9562.17	1457.97	9672.68	
Cable21	Cable	TYPICAL MOTOR FEEDER 1 A	Bus165	5047.06	1411.24	5240.65	
Cable22	Cable	TYPICAL MOTOR FEEDER 1 A	Bus170	5134.08	1435.57	5331.00	
Cable23	Cable	S002-SB-1AN001 [BUS B]	Bus152	5796.70	404.99	5810.83	
Cable24	Cable	S002-SB-1AN001 [BUS A]	Bus158	6245.31	436.33	6260.54	
Cable25	Cable	TYPICAL MOTOR FEEDER 1 A	Bus171	5047.06	1411.24	5240.65	
Cable26	Cable	S002-MCC1-9 [BUS]	Bus169	10302.20	1570.80	10421.26	
Cable27	Cable	TYPICAL MOTOR FEEDER 1 A	Bus172	4960.04	1386.91	5150.29	
Cable28	Cable	TYPICAL MOTOR FEEDER 1 B	Bus173	4361.45	1219.53	4528.74	
Cable29	Cable	TYPICAL MOTOR FEEDER 1 B	Bus174	4442.22	1242.11	4612.60	
Cable30	Cable	TYPICAL MOTOR FEEDER 1 B	Bus175	4522.98	1264.70	4696.47	
Cable31	Cable	TYPICAL MOTOR FEEDER 1 B	Bus176	4442.22	1242.11	4612.60	
Cable32	Cable	TYPICAL MOTOR FEEDER 1 B	Bus177	4522.98	1264.70	4696.47	
Cable33	Cable	TYPICAL MOTOR FEEDER 1 B	Bus178	4603.75	1287.28	4780.34	
Cable34	Cable	TYPICAL MOTOR FEEDER 1 B	Bus179	4522.98	1264.70	4696.47	
Cable35	Cable	TYPICAL MOTOR FEEDER 1 B	Bus180	4603.75	1287.28	4780.34	
Cable36	Cable	TYPICAL MOTOR FEEDER 1 B	Bus181	4684.52	1309.87	4864.20	
Cable37	Cable	TYPICAL MOTOR FEEDER 1 B	Bus182	4603.75	1287.28	4780.34	
Cable38	Cable	TYPICAL MOTOR FEEDER 1 B	Bus183	4684.52	1309.87	4864.20	

Project: UQ LT LPG Export - ST & MT

ETAP

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Location: Umm Qasr, Iraq

24.0.2C

Date: 03-08-2026

Contract: UQ83

SN: SL-KENTUAE

Engineer: Hemanathan Gopalan

Study Case: SC_Max

Revision: Base

Filename: Umm Qasr-ST

Config.: 1_NE

CKT/Branch		Connected Bus ID		% Impedance, Pos. Seq., 100 MVAb			
ID	Type	From Bus	To Bus	R	X	Z	Y
Cable39	Cable	TYPICAL MOTOR FEEDER 1 B	Bus184	4765.29	1332.45	4948.07	
Cable40	Cable	Bus_MDB-8	Cable40~	77.60	93.84	121.77	
Cable41	Cable	Bus_MDB-8	Cable41~	86.22	104.27	135.30	
Cable42	Cable	TYPICAL MOTOR FEEDER 1 B	Bus185	4684.52	1309.87	4864.20	
Cable43	Cable	TYPICAL MOTOR FEEDER 1 B	Bus186	4765.29	1332.45	4948.07	
Cable44	Cable	TYPICAL MOTOR FEEDER 1 B	Bus187	4846.05	1355.03	5031.93	
Cable45	Cable	POWER SOCKET FEEDER 1 B	Bus188	1727.29	1011.16	2001.49	
Cable48	Cable	POWER SOCKET FEEDER 1 B	Bus189	813.58	476.27	942.73	
Cable49	Cable	POWER SOCKET FEEDER 1 B	Bus190	1001.33	586.18	1160.29	
Cable83	Cable	Bus_MDB-8	Bus14	2595.54	181.34	2601.86	
Cable118	Cable	MV RMU-TOFS	CAMP BUS	13.91	7.29	15.70	
Cable123	Cable	Bus_MDB-8	Bus15	2855.09	199.47	2862.05	
Cable128	Cable	Bus_LVDB-2	Bus_MDB-8	21.56	26.07	33.83	
Cable134	Cable	Bus52	Bus_LVDB-1	6.40	7.74	10.05	
Cable136	Cable	Bus65	Bus52	18.29	22.12	28.70	
Cable137	Cable	Bus63	Bus_LVDB-1	6.40	7.74	10.05	
Cable144	Cable	Bus38	Bus40	14.78	17.87	23.19	
Cable145	Cable	Bus38	Bus45	14.78	17.87	23.19	
Cable146	Cable	Bus38	Bus48	14.78	17.87	23.19	
Cable147	Cable	Bus40	Bus_LVDB-2	6.90	8.34	10.82	
Cable148	Cable	Bus45	Bus_LVDB-2	6.90	8.34	10.82	
Cable149	Cable	Bus48	Bus_LVDB-2	6.90	8.34	10.82	
Cable150	Cable	Bus65	Bus63	18.29	22.12	28.70	
Cable151	Cable	Bus_LVDB-2	Bus_MDB-5	21.56	26.07	33.83	
Cable152	Cable	Bus64	Bus_LVDB-1	6.40	7.74	10.05	
Cable153	Cable	Bus_MDB-5	Cable153~	51.73	62.56	81.18	
Cable154	Cable	Bus_MDB-5	Cable154~	60.36	72.99	94.71	
Cable155	Cable	Bus_MDB-5	Bus60	1816.87	126.94	1821.30	
Cable156	Cable	Bus_MDB-5	Bus62	2076.43	145.07	2081.49	
Cable157	Cable	Bus65	Bus64	18.29	22.12	28.70	
Cable158	Cable	Bus_LVDB-1	Bus_MDB-1	20.81	25.16	32.65	
Cable159	Cable	Bus_LVDB-1	Bus_MDB-2	19.21	23.23	30.14	
Cable160	Cable	Bus_LVDB-1	Bus_MDB-3	18.41	22.26	28.88	
Cable161	Cable	Bus_LVDB-1	Bus_MDB-4	12.00	14.52	18.84	
Cable162	Cable	Bus_LVDB-2	Bus_MDB-6	21.56	26.07	33.83	
Cable163	Cable	Bus_MDB-6	Cable163~	20.69	25.02	32.47	
Cable164	Cable	Bus_MDB-6	Cable164~	31.04	37.54	48.71	

Project: UQ LT LPG Export - ST & MT

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Location: Umm Qasr, Iraq

24.0.2C

Date: 03-08-2026

Contract: UQ83

SN: SL-KENTUAE

Engineer: Hemanathan Gopalan

Study Case: SC_Max

Revision: Base

Filename: Umm Qasr-ST

Config.: 1_NE

CKT/Branch		Connected Bus ID		% Impedance, Pos. Seq., 100 MVAb			
ID	Type	From Bus	To Bus	R	X	Z	Y
Cable165	Cable	Bus_MDB-6	Bus72	1401.59	97.92	1405.01	
Cable166	Cable	Bus_MDB-6	Bus73	1661.14	116.06	1665.19	
Cable167	Cable	Bus_LVDB-2	Bus_MDB-7	21.56	26.07	33.83	
Cable168	Cable	Bus_MDB-7	Cable168~	43.11	52.13	67.65	
Cable169	Cable	Bus_MDB-7	Cable169~	62.08	75.07	97.42	
Cable170	Cable	Bus_MDB-7	Bus76	1816.87	126.94	1821.30	
Cable171	Cable	Bus_MDB-7	Bus77	2387.89	166.83	2393.71	
Cable176	Cable	Bus64	Bus82	50.95	19.20	54.44	
Cable188	Cable	Bus_MDB-1	Cable188~	16.01	19.36	25.12	
Cable189	Cable	Bus_MDB-1	Cable189~	24.01	29.03	37.67	
Cable190	Cable	Bus_MDB-1	Bus95	1204.55	84.16	1207.48	
Cable191	Cable	Bus_MDB-1	Bus96	1445.46	100.99	1448.98	
Cable192	Cable	Bus_MDB-2	Cable192~	40.01	48.39	62.79	
Cable193	Cable	Bus_MDB-2	Cable193~	56.02	67.75	87.91	
Cable194	Cable	Bus_MDB-2	Bus99	1686.36	117.82	1690.48	
Cable195	Cable	Bus_MDB-2	Bus100	2168.18	151.48	2173.47	
Cable196	Cable	Bus_MDB-3	Cable196~	68.82	83.23	108.00	
Cable197	Cable	Bus_MDB-3	Cable197~	41.61	50.33	65.30	
Cable198	Cable	Bus_MDB-3	Bus103	2409.09	168.31	2414.96	
Cable199	Cable	Bus_MDB-3	Bus104	1686.36	117.82	1690.48	
Cable200	Cable	Bus_MDB-4	Cable200~	27.21	32.91	42.70	
Cable201	Cable	Bus_MDB-4	Cable201~	36.81	44.52	57.77	
Cable202	Cable	Bus_MDB-4	Bus107	1204.55	84.16	1207.48	
Cable203	Cable	Bus_MDB-4	Bus108	1541.82	107.72	1545.58	
C_688-KM-01	Cable	S002-SB-1AD001 [BUS A]	Bus160	35.20	14.54	38.08	
S002-830-1063-N-P	Cable	Bus67	S002-SB-03N001 [BUS A]	11.23	29.31	31.39	
S002-830-1066-N-P	Cable	Bus69	S002-SB-03N001 [BUS B]	10.42	27.21	29.13	
S002-830-1069-N-P	Cable	Bus28	S002-SB-03E001 [BUS A]	20.78	54.25	58.09	
S002-830-1072-N-P	Cable	Bus33	S002-SB-03E001 [BUS B]	19.29	50.35	53.92	
S002-830-1420-N-P-1	Cable	Bus10	S002-SB-01E001 [BUS A]	9.36	24.43	26.16	
S002-830-1500-N-P-1	Cable	Bus23	S002-SB-01E001 [BUS B]	10.42	27.21	29.13	
S002-830-9001-A-P	Cable	Bus81	TR-PSSIAC02_132kV BUS	0.01	0.02	0.02	
S002-830-9001-C-P	Cable	TR-PSSIAC01_11kV BUS	S002-SB-1AC001 [BUS A]	0.02	0.07	0.07	
S002-830-9001-D-P-1	Cable	Bus22	S002-SB-1AD001 [BUS A]	0.60	1.56	1.67	
S002-830-9004-A-P	Cable	Bus79	TR-PSSIAC01_132kV BUS	0.01	0.02	0.02	
S002-830-9004-D-P	Cable	S002-SB-1AD001 [BUS A]	Bus30	5.08	4.78	6.97	
S002-830-9005-D-P	Cable	S002-SB-1AD001 [BUS A]	Bus31	5.08	4.78	6.97	
S002-830-9006-D-P	Cable	S002-SB-1AD001 [BUS A]	Bus132	306.49	40.47	309.15	
S002-830-9007-D-P	Cable	S002-SB-1AD001 [BUS A]	Bus131	306.49	40.47	309.15	

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Location: Umm Qasr, Iraq

24.0.2C

Date: 03-08-2026

Contract: UQ83

SN: SL-KENTUAE

Engineer: Hemanathan Gopalan

Study Case: SC_Max

Revision: Base

Filename: Umm Qasr-ST

Config.: 1_NE

CKT/Branch		Connected Bus ID		% Impedance, Pos. Seq., 100 MVAb			
ID	Type	From Bus	To Bus	R	X	Z	Y
S002-830-9008-C-P-1	Cable	TR-PSS1AC02_11kV BUS	S002-SB-1AC001 [BUS B]	0.02	0.08	0.08	
S002-830-9008-D-P	Cable	S002-SB-1AD001 [BUS A]	Bus130	211.38	27.91	213.21	
S002-830-9009-D-P	Cable	S002-SB-1AD001 [BUS A]	Bus129	211.38	27.91	213.21	
S002-830-9010-D-P	Cable	S002-SB-1AD001 [BUS A]	Bus128	364.62	48.15	367.79	
S002-830-9011-D-P	Cable	S002-SB-1AD001 [BUS A]	Bus127	364.62	48.15	367.79	
S002-830-9012-D-P	Cable	S002-SB-1AD001 [BUS A]	Bus126	459.74	60.71	463.73	
S002-830-9013-D-P	Cable	S002-SB-1AD001 [BUS A]	Bus125	459.74	60.71	463.73	
S002-830-9014-D-P	Cable	S002-SB-1AD001 [BUS A]	Bus124	290.64	38.38	293.16	
S002-830-9015-D-P	Cable	S002-SB-1AD001 [BUS A]	Bus123	295.93	39.08	298.49	
S002-830-9016-D-P	Cable	S002-SB-1AD001 [BUS A]	Bus122	30.98	12.79	33.51	
S002-830-9017-D-P	Cable	S002-SB-1AD001 [BUS A]	Bus121	31.68	13.08	34.28	
S002-830-9018-D-P	Cable	S002-SB-1AD001 [BUS A]	Bus120	33.09	13.66	35.80	
S002-830-9019-D-P	Cable	S002-SB-1AD001 [BUS A]	Bus119	22.53	9.30	24.37	
S002-830-9020-D-P	Cable	S002-SB-1AD001 [BUS A]	Bus47	102.97	18.05	104.54	
S002-830-9021-D-P-1	Cable	Bus25	S002-SB-1AD001 [BUS B]	0.56	1.45	1.55	
S002-830-9024-D-P	Cable	S002-SB-1AD001 [BUS B]	Bus29	4.71	4.43	6.47	
S002-830-9025-D-P	Cable	S002-SB-1AD001 [BUS B]	Bus32	4.71	4.43	6.47	
S002-830-9026-D-P	Cable	S002-SB-1AD001 [BUS B]	Bus133	284.48	37.56	286.95	
S002-830-9027-D-P	Cable	S002-SB-1AD001 [BUS B]	Bus134	284.48	37.56	286.95	
S002-830-9028-D-P	Cable	S002-SB-1AD001 [BUS B]	Bus135	196.19	25.91	197.89	
S002-830-9029-D-P	Cable	S002-SB-1AD001 [BUS B]	Bus136	196.19	25.91	197.89	
S002-830-9030-D-P	Cable	S002-SB-1AD001 [BUS B]	Bus137	338.43	44.69	341.37	
S002-830-9031-D-P	Cable	S002-SB-1AD001 [BUS B]	Bus138	338.43	44.69	341.37	
S002-830-9032-D-P	Cable	S002-SB-1AD001 [BUS B]	Bus139	426.72	56.35	430.42	
S002-830-9033-D-P	Cable	S002-SB-1AD001 [BUS B]	Bus140	426.72	56.35	430.42	
S002-830-9034-D-P	Cable	S002-SB-1AD001 [BUS B]	Bus141	269.76	35.62	272.11	
S002-830-9035-D-P	Cable	S002-SB-1AD001 [BUS B]	Bus142	274.67	36.27	277.05	
S002-830-9036-D-P	Cable	S002-SB-1AD001 [BUS B]	Bus143	28.75	11.87	31.11	
S002-830-9037-D-P	Cable	S002-SB-1AD001 [BUS B]	Bus144	29.40	12.14	31.81	
S002-830-9038-D-P	Cable	S002-SB-1AD001 [BUS B]	Bus145	30.71	12.68	33.23	
S002-830-9039-D-P	Cable	S002-SB-1AD001 [BUS B]	Bus146	132.43	17.49	133.58	
S002-830-9051-C-P	Cable	S002-SB-1AC001 [BUS A]	Bus43	0.86	0.80	1.17	
S002-830-9053-C-P	Cable	S002-SB-1AC001 [BUS A]	Bus7	6.10	5.66	8.32	
S002-830-9054-C-P	Cable	S002-SB-1AC001 [BUS A]	Bus112	6.89	6.39	9.40	
S002-830-9055-C-P	Cable	S002-SB-1AC001 [BUS A]	Bus17	6.89	6.39	9.40	
S002-830-9056-C-P-1	Cable	S002-SB-1AC001 [BUS A]	Bus1	0.07	0.21	0.22	
S002-830-9059-C-P	Cable	S002-SB-1AC001 [BUS A]	Bus110	3.41	3.16	4.65	
S002-830-9061-C-P-1	Cable	S002-SB-1AC001 [BUS A]	Bus92	0.59	1.00	1.16	
S002-830-9062-C-P	Cable	S002-SB-1AC001 [BUS A]	Bus90	4.31	3.99	5.87	

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Contract:	UQ83		SN:	SL-KENTUAE
Engineer:	Hemanathan Gopalan	Study Case: SC_Max	Revision:	Base
Filename:	Umm Qasr-ST		Config.:	1_NE

CKT/Branch		Connected Bus ID		% Impedance, Pos. Seq., 100 MVAb			
ID	Type	From Bus	To Bus	R	X	Z	Y
S002-830-9063-C-P	Cable	S002-SB-1AC001 [BUS A]	Bus89	4.23	3.93	5.78	
S002-830-9064-C-P	Cable	S002-SB-1AC001 [BUS B]	Bus42	0.60	0.56	0.82	
S002-830-9066-C-P	Cable	S002-SB-1AC001 [BUS B]	Bus46	5.66	5.25	7.72	
S002-830-9067-C-P	Cable	S002-SB-1AC001 [BUS B]	Bus8	6.40	5.93	8.72	
S002-830-9068-C-P	Cable	S002-SB-1AC001 [BUS B]	Bus18	6.40	5.93	8.72	
S002-830-9069-C-P-1	Cable	S002-SB-1AC001 [BUS B]	Bus21	0.07	0.19	0.20	
S002-830-9072-C-P	Cable	S002-SB-1AC001 [BUS B]	Bus113	3.23	3.00	4.41	
S002-830-9073-C-P-1	Cable	S002-SB-1AC001 [BUS B]	Bus114	0.53	0.90	1.05	
S002-830-9074-C-P	Cable	S002-SB-1AC001 [BUS B]	Bus115	3.20	2.97	4.36	
S002-830-9076-C-P	Cable	S002-SB-1AC001 [BUS B]	Bus117	3.93	3.65	5.36	
S002-830-9077-C-P-1	Cable	S002-SB-1AC001 [BUS A]	S003-SB-11C001 [BUS A]	31.21	24.07	39.41	
S002-830-9078-C-P-1	Cable	S002-SB-1AC001 [BUS B]	S003-SB-11C001 [BUS B]	28.96	22.34	36.58	
S002-830-9079-C-P	Cable	S002-SB-1AC001 [BUS A]	MV RMU-TOFS	7.65	4.01	8.64	
S002-Cable20	Cable	S002-SB-01E001 [BUS A]	S002-SB-PSS01N01 [BUS A]	314.99	822.39	880.65	
S002-Cable21	Cable	S002-ASP1-2B	S002-DBASP1-2-22	1016.41	739.11	1256.73	
S002-Cable22	Cable	S002-SB-01E001 [BUS B]	S002-SB-PSS01N01 [BUS B]	292.37	763.31	817.39	
S002-Cable26	Cable	S002-SB-04E001 [BUS A]	Bus157	1711.66	645.01	1829.15	
S002-Cable29	Cable	S002-PC1-1B	S002-MCC1-2 [BUS]	32.76	115.17	119.74	
S002-Cable34	Cable	S002-PC1-1A	S002-SB-1AN001 [BUS A]	492.78	358.34	609.29	
S002-Cable35	Cable	Bus27	S002-SB-02N001 [BUS A]	5.45	14.22	15.23	
S002-Cable85	Cable	S002-PC1-1A	S002-MCC1-1 [BUS]	42.35	148.90	154.80	
S002-Cable87	Cable	S002-PC1-1A	S002-MCC1-3 [BUS]	35.29	124.08	129.00	
S002-Cable88	Cable	S002-PC1-2A	S002-MCC1-5 [BUS]	49.41	173.72	180.60	
S002-Cable89	Cable	Bus24	S002-SB-02N001 [BUS B]	5.06	13.20	14.13	
S002-Cable90	Cable	S002-PC1-1B	S002-SB-1AN001 [BUS B]	457.38	332.60	565.53	
S002-Cable95	Cable	S002-PC1-2A	S002-MCC1-7 [BUS]	50.82	178.68	185.76	
S002-Cable115	Cable	S002-PC1-2B	S002-MCC1-8 [BUS]	46.69	121.90	130.54	
S002-Cable119	Cable	S002-PC1-2A	S002-MCC1-9 [BUS]	53.90	140.72	150.69	
S002-Cable124	Cable	S002-PC1-1A	S002-ASP1-1 [BUS A]	40.94	143.94	149.64	
S002-Cable125	Cable	S002-PC1-1B	S002-ASP1-1B [BUS B]	40.18	127.72	133.90	
S002-Cable126	Cable	S002-PC1-2A	S002-ASP1-2A	33.88	119.12	123.84	
S002-Cable127	Cable	S002-PC1-2B	S002-ASP1-2B	31.45	110.56	114.95	
S002-Cable131	Cable	S002-SB-01E001 [BUS A]	S002-MCC1-4 [BUS]	737.31	277.85	787.93	
S002-Cable132	Cable	S002-SB-01E001 [BUS B]	S002-MCC1-6 [BUS]	970.35	365.66	1036.96	
S002-Cable138	Cable	S002-SB-02N001 [BUS A]	Area Lighting DB (A8)	6310.99	440.92	6326.38	
S002-Cable181	Cable	S002-SB-1AN001 [BUS A]	Bus91	2861.72	436.33	2894.80	
S002-C_TR-04DE001	Cable	S002-SB-1AD001 [BUS A]	S002-TR-04DE001_PRI	36.20	27.26	45.31	
S002-C_TR-04DE002	Cable	S002-SB-1AD001 [BUS B]	S002-TR-04DE002_PRI	33.60	25.30	42.06	
S003-830-1000-E-P	Cable	Bus166	S003-SB-11E001 [BUS A]	22.26	58.13	62.24	

CKT/Branch		Connected Bus ID		% Impedance, Pos. Seq., 100 MVAb			
ID	Type	From Bus	To Bus	R	X	Z	Y
S003-830-1004-E-P	Cable	Bus167	S003-SB-11E001 [BUS B]	20.66	53.95	57.77	
S003-830-9001-C-P	Cable	S003-SB-11C001 [BUS A]	Bus56	0.54	0.39	0.67	
S003-830-9001-D-P	Cable	Bus50	S003-SB-11D001 [BUS A]	1.22	3.20	3.42	
S003-830-9002-C-P	Cable	S003-SB-11C001 [BUS B]	Bus55	0.33	0.24	0.41	
S003-830-9003-C-P	Cable	S003-SB-11C001 [BUS A]	Bus19	0.36	0.27	0.45	
S003-830-9004-C-P	Cable	S003-SB-11C001 [BUS B]	Bus20	0.30	0.22	0.37	
S003-830-9005-C-P	Cable	S003-SB-11C001 [BUS A]	Bus168	0.36	0.33	0.49	
S003-830-9006-C-P	Cable	S003-SB-11C001 [BUS B]	Bus163	0.45	0.41	0.61	
S003-830-9019-D-P	Cable	Bus49	S003-SB-11D001 [BUS B]	1.14	2.97	3.18	
S003-830-9021-C-P	Cable	S003-SB-11C001 [BUS B]	Bus78	5.01	3.86	6.32	
S003-830-9022-C-P	Cable	S003-SB-11C001 [BUS A]	Bus66	5.39	4.16	6.81	
S003-Cable19	Cable	S003-SB-11N001 [BUS A]	S003-DB-11N001-03 [BUS A]	725.15	202.76	752.97	
S003-Cable42	Cable	S003-SB-11N001 [BUS B]	S003-SB01E1	322.36	841.63	901.25	
S003-Cable44	Cable	S003-SB-11N001 [BUS A]	S003-DB11N001-04 [BUS A]	725.15	202.76	752.97	
S003-Cable93	Cable	S003-SB-11E001 [BUS A]	S003-DB11E001-01	797.67	223.04	828.26	
S003-Cable94	Cable	S003-SB-11E001 [BUS B]	S003-DB11E001-02	673.06	188.20	698.88	
S003-Cable96	Cable	S003-SB-11N001 [BUS B]	S003-DB-11N001-03 [BUS B]	673.06	188.20	698.88	
S003-Cable105	Cable	S003-SB-11N001 [BUS A]	S003-MDP-B	106.86	279.00	298.77	
S003-Cable113	Cable	S003-SB-11D001 [BUS B]	Bus37	23.00	17.74	29.05	
S003-Cable121	Cable	S003-SB-11N001 [BUS B]	S003-DB11N001-04 [BUS B]	673.06	188.20	698.88	
S003-Cable142	Cable	S003-SB-11N001 [BUS B]	S003-DB11N001-11	1076.90	301.12	1118.21	
S003-Cable173	Cable	S003-SB-11N001 [BUS B]	S003-DB11N001-12	1076.90	301.12	1118.21	
=E02-Q0	Tie Breaker	132kV GRID	Bus83				
=E06-Q0	Tie Breaker	132kV GRID	Bus84				
CB1	Tie Breaker	S002-SB-05N001A	TYPICAL MOTOR FEEDER 1 A				
CB8	Tie Breaker	S002-SB-05N001B	TYPICAL MOTOR FEEDER 1 B				
CB19	Tie Breaker	S002-SB-05N001B	POWER SOCKET FEEDER 1 B				
CB23	Tie Breaker	S002-SB-05N001C	S002-SB-05N001B				
IAC001=R17	Tie Breaker	S002-SB-1AC001 [BUS B]	BUS_S002-CMP3				
QM	Tie Breaker	Bus82	200A MDB Agreko				
SB-04E001=CB3	Tie Breaker	S002-SB-04E001 [BUS B]	S002-SB-04E001 [BUS A]				
RMU TOFS	Tie Fuse	MV RMU-TOFS	Bus51				
SW3	Tie Switch	S002-SB-PSS1A01 [BUS-A]	Bus79				
SW5	Tie Switch	S002-SB-PSS1A01 [BUS-A]	Bus80				
SW7	Tie Switch	S002-SB-PSS1A01 [BUS-A]	Bus81				
SW9	Tie Switch	Bus83	S002-SB-PSS1A01 [BUS-A]				
SW11	Tie Switch	Bus84	S002-SB-PSS1A01 [BUS-A]				
SW13	Tie Switch	Bus87	S002-SB-PSS1A01 [BUS-A]				

Project:

UQ LT LPG Export - ST & MT

Location:

Umm Qasr, Iraq

Contract:

UQ83

Engineer:

Hemanathan Gopalan

Filename:

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Base

Config.:

1_NE

CKT/Branch		Connected Bus ID		% Impedance, Pos. Seq., 100 MVAb			
ID	Type	From Bus	To Bus	R	X	Z	Y
SW15	Tie Switch	Bus88	S002-SB-PSS1A01 [BUS-A]				

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Revision: Base

Config.: 1_NE

Total Connected Synchronous Generators (= 6) : 6.670 MVA

Induction Machine Input Data

Induction Machine			Connected Bus	Rating					% Impedance (Motor Base)			
ID	Type	Qty	ID	HP/kW	kVA	kV	Amp	PF	R	X"	R/X"	MW/PP
614-KM-01A	Motor	1	BUS-MTR1_ENCRYO	5400.00	6473.27	11.000	339.76	86.00	3.11	15.14	0.21	2.70
614-KM-01B	Motor	1	BUS-MTR2_ENCRYO	5400.00	6473.27	11.000	339.76	86.00	3.11	15.14	0.21	2.70
S002-40-MP-04A	Motor	1	Bus47	250.00	297.25	3.300	52.00	89.00	2.27	14.98	0.15	0.13
S002-40-MP-5C	Motor	1	Bus89	560.00	655.43	11.000	34.40	89.00	3.63	19.70	0.18	0.28
S002-40-MP-05A	Motor	1	Bus90	560.00	655.43	11.000	34.40	89.00	3.63	19.70	0.18	0.28
S002-6100-MK-1D	Motor	1	Bus92	6900.00	7991.07	11.000	419.42	90.70	4.50	21.87	0.21	3.45
S002-6100-MP-10A	Motor	1	Bus110	1050.00	1215.28	11.000	63.79	90.00	4.00	21.86	0.18	0.52
S002-6100-MP-10C	Motor	1	Bus113	1050.00	1255.98	11.000	65.92	88.00	4.00	21.86	0.18	0.52
S002-6100-MK-1C	Motor	1	Bus114	6900.00	7982.68	11.000	418.98	90.70	3.08	23.19	0.13	3.45
S002-6100-MP-10B	Motor	1	Bus115	1050.00	1255.98	11.000	65.92	88.00	4.00	21.86	0.18	0.52
S002-40-MP-05B	Motor	1	Bus117	560.00	655.43	11.000	34.40	89.00	3.63	19.70	0.18	0.28
S002-6200-MP-02A	Motor	1	Bus119	470.00	574.71	3.300	100.55	87.00	3.03	14.85	0.20	0.23
S002-6100-MP-11A	Motor	1	Bus120	950.00	1105.68	3.300	193.44	89.50	3.27	17.88	0.18	0.47
S002-6100-MP-9C	Motor	1	Bus121	950.00	1105.68	3.300	193.44	89.50	3.60	19.67	0.18	0.47
S002-6100-MP-9A	Motor	1	Bus122	950.00	1105.68	3.300	193.44	89.50	3.60	19.67	0.18	0.47
S002-6100-MP-8D	Motor	1	Bus123	200.00	248.62	3.300	43.50	86.50	3.21	13.92	0.23	0.10
S002-6100-MP-8B	Motor	1	Bus124	200.00	248.62	3.300	43.50	86.50	3.21	13.92	0.23	0.10
S002-6100-MP-7C	Motor	1	Bus125	187.00	232.46	3.300	40.67	86.50	4.44	21.77	0.20	0.09
S002-6100-MP-7A	Motor	1	Bus126	187.00	232.46	3.300	40.67	86.50	4.44	21.77	0.20	0.09
S002-6100-MP-6D	Motor	1	Bus127	187.00	232.46	3.300	40.67	86.50	4.44	21.77	0.20	0.09
S002-6100-MP-6B	Motor	1	Bus128	187.00	232.46	3.300	40.67	86.50	4.44	21.77	0.20	0.09
S002-6100-MP-4D	Motor	1	Bus129	187.00	232.46	3.300	40.67	86.50	4.44	21.77	0.20	0.09
S002-6100-MP-4B	Motor	1	Bus130	187.00	232.46	3.300	40.67	86.50	4.44	21.77	0.20	0.09
S002-6100-MP-3C	Motor	1	Bus131	187.00	232.46	3.300	40.67	86.50	4.44	21.77	0.20	0.09
S002-6100-MP-3A	Motor	1	Bus132	187.00	232.46	3.300	40.67	86.50	4.44	21.77	0.20	0.09
S002-6100-MP-3B	Motor	1	Bus133	187.00	232.46	3.300	40.67	86.50	4.44	21.77	0.20	0.09
S002-6100-MP-3D	Motor	1	Bus134	187.00	232.46	3.300	40.67	86.50	5.25	21.88	0.24	0.09
S002-6100-MP-4A	Motor	1	Bus135	187.00	232.46	3.300	40.67	86.50	5.00	21.85	0.23	0.09
S002-6100-MP-4C	Motor	1	Bus136	187.00	232.46	3.300	40.67	86.50	5.00	21.85	0.23	0.09
S002-6100-MP-6A	Motor	1	Bus137	187.00	232.46	3.300	40.67	86.50	5.42	21.91	0.25	0.09
S002-6100-MP-6C	Motor	1	Bus138	187.00	232.46	3.300	40.67	86.50	5.42	21.91	0.25	0.09
S002-6100-MP-7B	Motor	1	Bus139	187.00	232.46	3.300	40.67	86.50	4.44	21.77	0.20	0.09
S002-6100-MP-7D	Motor	1	Bus140	187.00	232.46	3.300	40.67	86.50	4.44	21.77	0.20	0.09
S002-6100-MP-8A	Motor	1	Bus141	200.00	248.62	3.300	43.50	86.50	3.21	13.92	0.23	0.10
S002-6100-MP-8C	Motor	1	Bus142	200.00	248.62	3.300	43.50	86.50	3.21	13.92	0.23	0.10
S002-6100-MP-9B	Motor	1	Bus143	950.00	1105.68	3.300	193.44	89.50	2.91	19.79	0.15	0.47
S002-6100-MP-11B	Motor	1	Bus144	950.00	1105.68	3.300	193.44	89.50	3.60	19.67	0.18	0.47
S002-6100-MP-11C	Motor	1	Bus145	950.00	1105.68	3.300	193.44	89.50	3.60	19.67	0.18	0.47
S002-40-MP-04B	Motor	1	Bus146	250.00	297.25	3.300	52.00	89.00	4.00	16.36	0.24	0.13

Induction Machine Input Data

Induction Machine			Connected Bus	Rating					% Impedance (Motor Base)			
ID	Type	Qty	ID	HP/kW	kVA	kV	Amp	PF	R	X"	R/X"	MW/PP
688-KM-01	Motor	1	Bus160	1000.00	1137.58	3.300	199.02	91.00	1.59	10.41	0.15	1.00
614-KM-11	Motor	1	BUS_S002-CMP3	3150.00	3706.10	11.000	194.52	89.00	4.02	19.63	0.20	1.58
S003-7200-MP-03A	Motor	1	S003-SB-11D001 [BUS A]	500.00	601.31	3.300	105.20	88.00	3.85	19.92	0.19	0.25
S003-7100-MK-2A	Motor	1	S003-SB-11D001 [BUS A]	210.00	251.49	3.300	44.00	90.00	4.72	19.57	0.24	0.21
S003-7100-MK-1B	Motor	1	S003-SB-11D001 [BUS A]	250.00	297.22	3.300	52.00	90.00	4.53	19.62	0.23	0.13
S003-7100-MK-1A	Motor	1	S003-SB-11D001 [BUS A]	250.00	300.66	3.300	52.60	88.00	4.58	19.63	0.23	0.25
S003-7100-MK-1C	Motor	1	S003-SB-11D001 [BUS B]	250.00	297.22	3.300	52.00	90.00	4.56	19.63	0.23	0.25
S003-7100-MK-2B	Motor	1	S003-SB-11D001 [BUS B]	210.00	251.49	3.300	44.00	90.00	4.73	19.57	0.24	0.21
S003-7100-MK-2C	Motor	1	S003-SB-11D001 [BUS B]	210.00	251.49	3.300	44.00	90.00	4.69	19.56	0.24	0.21
S003-7200-MP-02B	Motor	1	S003-SB-11D001 [BUS B]	502.00	588.77	3.300	103.01	89.00	3.59	19.77	0.18	0.50
Chiller-15	Motor	1	Bus2	295.00	357.10	0.415	496.80	86.92	4.28	13.62	0.31	0.15
Chiller-16	Motor	1	Bus4	295.00	357.10	0.415	496.80	86.92	4.28	13.62	0.31	0.15
614-FM-06AA	Motor	1	Bus9	37.00	47.06	0.380	71.50	85.00	3.08	15.07	0.20	0.02
614-FM-06AB	Motor	1	Bus12	37.00	47.06	0.380	71.50	85.00	3.08	15.07	0.20	0.02
Pump-15	Motor	1	Bus14	18.00	23.06	0.415	32.08	83.95	10.01	26.79	0.37	0.01
Pump-16	Motor	1	Bus15	18.00	23.06	0.415	32.08	83.95	10.01	26.79	0.37	0.01
614-FM-06AC	Motor	1	Bus53	37.00	47.06	0.380	71.50	85.00	3.08	15.07	0.20	0.02
614-FM-06BA	Motor	1	Bus54	37.00	47.06	0.380	71.50	85.00	3.08	15.07	0.20	0.02
Chiller-9	Motor	1	Bus57	295.00	357.10	0.415	496.80	86.92	4.28	13.62	0.31	0.15
Chiller-10	Motor	1	Bus59	295.00	357.10	0.415	496.80	86.92	4.28	13.62	0.31	0.15
Pump-9	Motor	1	Bus60	18.00	23.06	0.415	32.08	83.95	10.01	26.79	0.37	0.01
Pump-10	Motor	1	Bus62	18.00	23.06	0.415	32.08	83.95	10.01	26.79	0.37	0.01
Chiller-11	Motor	1	Bus68	295.00	357.10	0.415	496.80	86.92	4.28	13.62	0.31	0.15
Chiller-12	Motor	1	Bus70	295.00	357.10	0.415	496.80	86.92	4.28	13.62	0.31	0.15
Pump-11	Motor	1	Bus72	18.00	23.06	0.415	32.08	83.95	10.01	26.79	0.37	0.01
Pump-12	Motor	1	Bus73	18.00	23.06	0.415	32.08	83.95	10.01	26.79	0.37	0.01
Chiller-13	Motor	1	Bus74	295.00	357.10	0.415	496.80	86.92	4.28	13.62	0.31	0.15
Chiller-14	Motor	1	Bus75	295.00	357.10	0.415	496.80	86.92	4.28	13.62	0.31	0.15
Pump-13	Motor	1	Bus76	18.00	23.06	0.415	32.08	83.95	10.01	26.79	0.37	0.01
Pump-14	Motor	1	Bus77	18.00	23.06	0.415	32.08	83.95	10.01	26.79	0.37	0.01
614-FM-06BB	Motor	1	Bus85	37.00	47.06	0.380	71.50	85.00	3.08	15.07	0.20	0.02
Chiller-1	Motor	1	Bus93	295.00	357.10	0.415	496.80	86.92	4.28	13.62	0.31	0.15
Chiller-2	Motor	1	Bus94	295.00	357.10	0.415	496.80	86.92	4.28	13.62	0.31	0.15
Pump-1	Motor	1	Bus95	18.00	23.06	0.415	32.08	83.95	10.01	26.79	0.37	0.01
Pump-2	Motor	1	Bus96	18.00	23.06	0.415	32.08	83.95	10.01	26.79	0.37	0.01
Chiller-3	Motor	1	Bus97	295.00	357.10	0.415	496.80	86.92	4.28	13.62	0.31	0.15
Chiller-4	Motor	1	Bus98	295.00	357.10	0.415	496.80	86.92	4.28	13.62	0.31	0.15
Pump-3	Motor	1	Bus99	18.00	23.06	0.415	32.08	83.95	10.01	26.79	0.37	0.01
Pump-4	Motor	1	Bus100	18.00	23.06	0.415	32.08	83.95	10.01	26.79	0.37	0.01

Induction Machine Input Data

Induction Machine			Connected Bus		Rating					% Impedance (Motor Base)			
ID	Type	Qty	ID		HP/kW	kVA	kV	Amp	PF	R	X"	R/X"	MW/PP
Chiller-5	Motor	1	BusI01		295.00	357.10	0.415	496.80	86.92	4.28	13.62	0.31	0.15
Chiller-6	Motor	1	BusI02		295.00	357.10	0.415	496.80	86.92	4.28	13.62	0.31	0.15
Pump-5	Motor	1	BusI03		18.00	23.06	0.415	32.08	83.95	10.01	26.79	0.37	0.01
Pump-6	Motor	1	BusI04		18.00	23.06	0.415	32.08	83.95	10.01	26.79	0.37	0.01
Chiller-7	Motor	1	BusI05		295.00	357.10	0.415	496.80	86.92	4.28	13.62	0.31	0.15
Chiller-8	Motor	1	BusI06		295.00	357.10	0.415	496.80	86.92	4.28	13.62	0.31	0.15
Pump-7	Motor	1	BusI07		18.00	23.06	0.415	32.08	83.95	10.01	26.79	0.37	0.01
Pump-8	Motor	1	BusI08		18.00	23.06	0.415	32.08	83.95	10.01	26.79	0.37	0.01
614-FM-06BC	Motor	1	BusI49		37.00	47.06	0.380	71.50	85.00	3.08	15.07	0.20	0.02
614-FM-06CA	Motor	1	BusI51		37.00	47.06	0.380	71.50	85.00	3.08	15.07	0.20	0.02
S002-6600-MP-1B	Motor	1	BusI52		15.00	18.14	0.380	27.55	90.00	8.49	14.34	0.59	0.01
614-FM-06CB	Motor	1	BusI54		37.00	47.06	0.380	71.50	85.00	3.08	15.07	0.20	0.02
S002-6600-MP-1A	Motor	1	BusI58		15.00	18.14	0.380	27.55	90.00	8.49	14.34	0.59	0.01
S002-6600-MP-2A	Motor	1	BusI59		18.50	21.76	0.380	33.06	92.00	5.54	15.72	0.35	0.01
614-FM-06CC	Motor	1	BusI61		37.00	47.06	0.380	71.50	85.00	3.08	15.07	0.20	0.02
614-FM-06DA	Motor	1	BusI62		37.00	47.06	0.380	71.50	85.00	3.08	15.07	0.20	0.02
614-FM-06DB	Motor	1	BusI64		37.00	47.06	0.380	71.50	85.00	3.08	15.07	0.20	0.02
614-FM-06DC	Motor	1	BusI65		37.00	47.06	0.380	71.50	85.00	3.08	15.07	0.20	0.02
S002-6600-MP-2B	Motor	1	BusI69		18.50	21.76	0.380	33.06	92.00	5.54	15.72	0.35	0.01
614-FM-06EA	Motor	1	BusI70		37.00	47.06	0.380	71.50	85.00	3.08	15.07	0.20	0.02
614-FM-06EB	Motor	1	BusI71		37.00	47.06	0.380	71.50	85.00	3.08	15.07	0.20	0.02
614-FM-06EC	Motor	1	BusI72		37.00	47.06	0.380	71.50	85.00	3.08	15.07	0.20	0.02
614-FM-06KC	Motor	1	BusI73		37.00	47.06	0.380	71.50	85.00	3.08	15.07	0.20	0.02
614-FM-06KB	Motor	1	BusI74		37.00	47.06	0.380	71.50	85.00	3.08	15.07	0.20	0.02
614-FM-06KA	Motor	1	BusI75		37.00	47.06	0.380	71.50	85.00	3.08	15.07	0.20	0.02
614-FM-06JC	Motor	1	BusI76		37.00	47.06	0.380	71.50	85.00	3.08	15.07	0.20	0.02
614-FM-06JB	Motor	1	BusI77		37.00	47.06	0.380	71.50	85.00	3.08	15.07	0.20	0.02
614-FM-06JA	Motor	1	BusI78		37.00	47.06	0.380	71.50	85.00	3.08	15.07	0.20	0.02
614-FM-06HC	Motor	1	BusI79		37.00	47.06	0.380	71.50	85.00	3.08	15.07	0.20	0.02
614-FM-06HB	Motor	1	BusI80		37.00	47.06	0.380	71.50	85.00	3.08	15.07	0.20	0.02
614-FM-06HA	Motor	1	BusI81		37.00	47.06	0.380	71.50	85.00	3.08	15.07	0.20	0.02
614-FM-06GC	Motor	1	BusI82		37.00	47.06	0.380	71.50	85.00	3.08	15.07	0.20	0.02
614-FM-06GB	Motor	1	BusI83		37.00	47.06	0.380	71.50	85.00	3.08	15.07	0.20	0.02
614-FM-06GA	Motor	1	BusI84		37.00	47.06	0.380	71.50	85.00	3.08	15.07	0.20	0.02
614-FM-06FC	Motor	1	BusI85		37.00	47.06	0.380	71.50	85.00	3.08	15.07	0.20	0.02
614-FM-06FB	Motor	1	BusI86		37.00	47.06	0.380	71.50	85.00	3.08	15.07	0.20	0.02
614-FM-06FA	Motor	1	BusI87		37.00	47.06	0.380	71.50	85.00	3.08	15.07	0.20	0.02
MOV_0.93	Motor	4	S002-DBASP1-2-22		0.93	1.45	0.380	2.21	80.00	6.46	9.04	0.71	0.00
MOV_0.55	Motor	2	S002-DBASP1-2-22		0.55	0.86	0.380	1.31	80.00	6.46	9.04	0.71	0.00

Induction Machine Input Data

Induction Machine			Connected Bus		Rating				% Impedance (Motor Base)			
ID	Type	Qty	ID	HP/kW	kVA	kV	Amp	PF	R	X"	R/X"	MW/PP
MOV_1.86	Motor	6	S002-DBASP1-2-22	1.86	2.91	0.380	4.42	80.00	6.46	9.04	0.71	0.00
MOV_2.59	Motor	4	S002-DBASP1-2-22	2.59	4.05	0.380	6.15	80.00	6.46	9.04	0.71	0.00
MOV_0.4	Motor	16	S002-DBASP1-2-22	0.40	0.63	0.380	0.95	80.00	6.46	9.04	0.71	0.00
MOV_0.3	Motor	3	S002-DBASP1-2-22	0.30	0.47	0.380	0.71	80.00	6.46	9.04	0.71	0.00
MOV_2.59 [F]	Motor	3	S002-DBASP1-2-22	2.59	4.05	0.380	6.15	80.00	6.46	9.04	0.71	0.00
S002-MCC1-4_M	Motor	1	S002-MCC1-4 [BUS]	15.00	21.04	0.380	31.97	81.00	11.09	14.97	0.74	0.01
S002-6100-MP-1B	Motor	1	S002-MCC1-9 [BUS]	55.00	70.75	0.380	107.49	84.50	6.82	16.21	0.42	0.03
S002-6300-MK-1A	Motor	1	S002-SB-01E001 [BUS A]	132.00	163.29	0.380	248.09	86.00	5.80	16.91	0.34	0.07
S002-6500-MP-100B	Motor	1	S002-SB-01E001 [BUS B]	55.00	69.51	0.380	105.62	86.00	5.54	15.72	0.35	0.03
688-PM-01A	Motor	1	S002-SB-04E001 [BUS A]	22.00	28.69	0.380	43.58	84.00	8.32	14.44	0.58	0.01
688-PM-01B	Motor	1	S002-SB-04E001 [BUS A]	22.00	28.69	0.380	43.58	84.00	8.32	14.44	0.58	0.01
HVAC-001/002	Motor	1	S002-SB-04E001 [BUS B]	25.00	29.24	0.380	44.43	95.00	6.90	15.17	0.46	0.01
HVAC-002	Motor	1	S002-SB-04E001 [BUS B]	25.00	29.24	0.380	44.43	95.00	6.90	15.17	0.46	0.01
S003-7500-K-1A	Motor	1	S003-SB-11E001 [BUS B]	140.00	175.04	0.380	265.95	86.00	5.89	20.19	0.29	0.07
S003-7400-MP-4A	Motor	1	S003-SB-11N001 [BUS A]	22.00	27.16	0.380	41.27	90.00	8.11	15.27	0.53	0.01
S003-7400-P-21B	Motor	1	S003-SB-11N001 [BUS B]	90.00	108.89	0.380	165.45	87.00	6.23	16.74	0.37	0.05

Total Connected Induction Machines (= 164): 63204.6 kVA

Lumped Load Input Data

Lumped Load								Motor Loads					
Lumped Load		Connected Bus		Rating		% Load		Loading		% Impedance Machine Base			
ID	ID	kVA	kV	Amp	% PF	MTR	STAT	kW	kvar	R	X"	R/X"	MW/PP
A01	S002-SB-PSS01N01 [BUS A]	207.6	0.380	315.46	96.32	100	0	94.0	26.2	6.46	15.37	0.42	0.20
CAMP LOAD	CAMP BUS	2544.4	11.000	133.55	88.43	80	20	1800.0	950.4	1.66	16.58	0.10	1.80
Exist MOE SS	S002-PC1-2B	64.7	0.380	98.31	85.00	100	0	55.0	34.1	6.46	15.37	0.42	0.05
Lump1	S003-DB11N001-04 [BUS A]	26.3	0.380	39.88	80.00	80	20	16.8	12.6	6.46	15.37	0.42	0.02
Lump2	S003-SB-11N001 [BUS A]	366.7	0.380	557.21	83.00	80	20	219.2	147.3	6.46	15.37	0.42	0.24
Lump3	S003-DB11N001-04 [BUS B]	29.0	0.380	44.06	80.00	80	20	18.6	13.9	6.46	15.37	0.42	0.02
Lump4	S002-SB-PSS01N01 [BUS B]	111.5	0.380	169.34	85.00	100	0	94.7	58.7	6.46	15.37	0.42	0.09
Lump9	S002-MCC1-1 [BUS]	280.2	0.380	425.72	86.08	100	0	229.1	135.5	6.46	15.37	0.42	0.24
Lump10	S002-MCC1-3 [BUS]	280.2	0.380	425.72	88.81	100	0	236.4	122.3	6.46	15.37	0.42	0.25
Lump13	S002-SB-1AN001 [BUS B]	15.3	0.380	23.24	85.00	100	0	13.0	8.1	6.46	15.37	0.42	0.01
Lump14	S002-SB-1AN001 [BUS A]	99.0	0.380	150.45	53.98	20	80	10.7	16.7	6.46	15.37	0.42	0.01
Lump16	S002-MCC1-5 [BUS]	280.2	0.380	425.72	83.05	100	0	232.7	156.1	6.46	15.37	0.42	0.23
Lump17	S002-MCC1-7 [BUS]	280.2	0.380	425.72	83.00	100	0	232.6	156.3	6.46	15.37	0.42	0.23
Lump18	S002-MCC1-9 [BUS]	93.5	0.380	141.99	81.00	100	0	75.7	54.8	6.46	15.37	0.42	0.08
Lump21	S002-MCC1-8 [BUS]	135.3	0.380	205.57	81.30	100	0	110.0	78.8	6.46	15.37	0.42	0.11
Lump22	S002-SB-02N001 [BUS A]	367.8	0.380	558.86	83.00	80	20	244.2	164.1	6.46	15.37	0.42	0.24
Lump23	S002-SB-02N001 [BUS B]	205.3	0.380	311.92	83.00	87	13	148.2	99.6	6.46	15.37	0.42	0.15
Lump27	200A MDB Agreko	40.0	0.400	57.74	90.03	80	20	17.3	8.4	6.46	15.37	0.42	0.03
Lump34	S002-MCC1-6 [BUS]	63.3	0.380	96.25	81.00	100	0	51.3	37.2	6.46	15.37	0.42	0.05
Lump42	S003-SB-11E001 [BUS A]	525.9	0.380	799.00	85.00	80	20	357.6	221.6	6.46	15.37	0.42	0.36
Lump43	S003-DB11E001-01	34.3	0.380	52.13	80.00	20	80	5.4	4.0	7.75	18.44	0.42	0.01
Lump46	S003-DB11E001-02	34.1	0.380	51.87	80.40	20	80	5.5	4.1	7.75	18.44	0.42	0.01
Lump48	S003-SB-11E001 [BUS B]	399.4	0.380	606.81	82.00	80	20	262.0	182.9	6.46	15.37	0.42	0.26
Lump49	S003-DB-11N001-03 [BUS A]	25.8	0.380	39.12	80.00	100	0	20.2	15.1	6.46	15.37	0.42	0.02
Lump51	S003-SB01E1	190.0	0.380	288.68	80.00	80	20	109.4	82.1	6.46	15.37	0.42	0.12
Lump52	S003-SB-11N001 [BUS B]	417.8	0.380	634.79	82.00	80	20	274.1	191.3	6.46	15.37	0.42	0.27
Lump56	S003-DB-11N001-03 [BUS B]	32.2	0.380	48.88	80.00	80	20	20.2	15.1	6.46	15.37	0.42	0.02
OPC Fuel Depot	Bus36	800.0	0.380	1215.47	80.00	80	20	512.0	384.0	6.46	15.37	0.42	0.51
POWER FEEDER 1/2	S002-SB-05N001C	389.5	0.380	591.76	87.29	100	0	340.0	190.0	6.46	15.37	0.42	0.34
S002-688-FM-01-04	S002-SB-04E001 [BUS A]	126.3	0.380	191.94	90.00	100	0	113.7	55.1	6.46	15.37	0.42	0.11
S002-688-FM-05-07	S002-SB-04E001 [BUS A]	21.2	0.380	32.18	82.00	100	0	17.4	12.1	6.46	15.37	0.42	0.02
S002-DB03E001-20_L1	S002-SB-03E001 [BUS A]	45.1	0.380	68.56	80.00	100	0	36.1	27.1	6.46	15.37	0.42	0.04
S002-DB03N001-22_LSB	S002-SB-03E001 [BUS B]	190.0	0.380	288.68	80.00	100	0	15.2	11.4	6.46	15.37	0.42	0.15

Lumped Load Input Data

Lumped Load						Motor Loads							
Lumped Load		Connected Bus		Rating		% Load		Loading		% Impedance Machine Base			
ID	ID	kVA	kV	Amp	% PF	MTR	STAT	kW	kvar	R	X"	R/X"	MW/PP
S002-DBPC1-1-01	S002-PC1-1A	101.5	0.380	154.21	80.00	100	0	81.2	60.9	6.46	15.37	0.42	0.08
S002-MCC1-4_L	S002-MCC1-4 [BUS]	49.3	0.380	74.94	81.00	100	0	40.0	28.9	7.75	18.44	0.42	0.04
S002-MCC1-6_M_C	S002-MCC1-6 [BUS]	63.3	0.380	96.25	81.00	100	0	5.1	3.7	6.46	15.37	0.42	0.05
S002-SB-01E001 [A]_L	S002-SB-01E001 [BUS A]	574.7	0.380	873.19	87.00	10	90	50.0	28.3	6.46	15.37	0.42	0.05
S002-SB-01E001 [B]_L	S002-SB-01E001 [BUS B]	665.8	0.380	1011.51	87.60	10	90	58.3	32.1	6.46	15.37	0.42	0.06
S002-SB-03E001 [A]_C	S002-SB-03E001 [BUS A]	516.3	0.380	784.38	83.00	80	20	342.8	230.4	6.46	15.37	0.42	0.34
S002-SB-03E001 [A]_S	S002-SB-03E001 [BUS A]	221.9	0.380	337.10	80.00	100	0	17.8	13.3	6.46	15.37	0.42	0.18
S002-SB-03E001 [B]_C	S002-SB-03E001 [BUS B]	666.9	0.380	1013.21	80.00	67	33	357.4	268.1	6.46	15.37	0.42	0.36
S002-SB-03E001 [B]_I	S002-SB-03E001 [BUS B]	0.4	0.380	0.58	0.01	4	96	0.0	0.0	6.46	15.37	0.42	0.00
S002-SB-03N001 [A]_C	S002-SB-03N001 [BUS A]	1098.9	0.380	1669.59	80.77	70	30	621.3	453.5	6.46	15.37	0.42	0.62
S002-SB-03N001 [B]_C	S002-SB-03N001 [BUS B]	632.4	0.380	960.87	80.50	70	30	356.4	262.6	6.46	15.37	0.42	0.36
TOFS Load	LV SWBD_TOFS	727.0	0.400	1049.37	94.63	20	80	137.6	47.0	7.75	18.44	0.42	0.14
TYPICAL MOTOR FEEDER 2 A	S002-SB-05N001A	91.4	0.380	138.89	85.00	100	0	77.7	48.2	6.46	15.37	0.42	0.08
TYPICAL MOTOR FEEDER 2 B	S002-SB-05N001B	65.6	0.380	99.72	84.00	100	0	55.1	35.6	6.46	15.37	0.42	0.06

Total Connected Lumped Loads (= 47): 14197.5 kVA

VFD Input Data

ID	AC Rating											
	kVA	kV		FLA		Input		Output		Bypass	% Impedance	
	Output	Input	Output	Input	Output	% EFF	% PF	Hz	% I _{max}	Switch	R	X
S002-CMP1_VFD	6000.00	11.000	11.000	331.49	314.92	95.00	100.00	50.00	150.0	No	66.67	0.00
S002-CMP2_VFD	6000.00	11.000	11.000	331.49	314.92	95.00	100.00	50.00	150.0	No	66.67	0.00
VFD1	295.00	0.415	0.415	410.41	410.41	100.00	100.00	50.00	150.0	No	66.67	0.00
VFD2	295.00	0.415	0.415	410.41	410.41	100.00	100.00	50.00	150.0	No	66.67	0.00
VFD9	295.00	0.415	0.415	410.41	410.41	100.00	100.00	50.00	150.0	No	66.67	0.00
VFD10	295.00	0.415	0.415	410.41	410.41	100.00	100.00	50.00	150.0	No	66.67	0.00
VFD13	295.00	0.415	0.415	410.41	410.41	100.00	100.00	50.00	150.0	No	66.67	0.00
VFD14	295.00	0.415	0.415	410.41	410.41	100.00	100.00	50.00	150.0	No	66.67	0.00
VFD15	295.00	0.415	0.415	410.41	410.41	100.00	100.00	50.00	150.0	No	66.67	0.00
VFD16	295.00	0.415	0.415	410.41	410.41	100.00	100.00	50.00	150.0	No	66.67	0.00
VFD25	295.00	0.415	0.415	410.41	410.41	100.00	100.00	50.00	150.0	No	66.67	0.00
VFD26	295.00	0.415	0.415	410.41	410.41	100.00	100.00	50.00	150.0	No	66.67	0.00
VFD27	295.00	0.415	0.415	410.41	410.41	100.00	100.00	50.00	150.0	No	66.67	0.00
VFD28	295.00	0.415	0.415	410.41	410.41	100.00	100.00	50.00	150.0	No	66.67	0.00
VFD29	295.00	0.415	0.415	410.41	410.41	100.00	100.00	50.00	150.0	No	66.67	0.00
VFD30	295.00	0.415	0.415	410.41	410.41	100.00	100.00	50.00	150.0	No	66.67	0.00
VFD31	295.00	0.415	0.415	410.41	410.41	100.00	100.00	50.00	150.0	No	66.67	0.00
VFD32	295.00	0.415	0.415	410.41	410.41	100.00	100.00	50.00	150.0	No	66.67	0.00

SHORT-CIRCUIT REPORT

3-Phase fault at bus:

S002-TR-CMP2_VFD

Nominal kV

Voltage c Factor

Peak Value

Steady State

= 11.646

= 1.10

= 11.695

= 4.136

(Maximum If)

kA Method C

kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-TR-CMP2_VFD~2	Total	0.00	0.132	-4.312	32.6	4.314
S002-SB-1AC001 [BUS B]	S002-TR-CMP2_VFD~2	100.72	0.088	-2.875	32.6	2.876
S002-TR-CMP2_VFD~3	S002-TR-CMP2_VFD~2	47.57	0.044	-1.437	32.6	1.438

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	4.314	7.048	5.573
0.02	4.314	6.703	5.130
0.03	4.314	6.383	4.705
0.04	4.314	6.101	4.314
0.05	4.314	5.890	4.010
0.06	4.314	5.675	3.687
0.07	4.314	5.487	3.390
0.08	4.314	5.323	3.117
0.09	4.314	5.180	2.866
0.10	4.314	5.080	2.682
0.15	4.314	4.666	1.778
0.20	4.314	4.472	1.179
0.25	4.314	4.385	0.782
0.30	4.314	4.345	0.518

3-Phase fault at bus:

S002-TR-CMP2_VFD

Nominal kV = 11.646

Voltage c Factor = 1.10 (Maximum If)

Peak Value = 11.695 kA Method C

Steady State = 4.136 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-TR-CMP2_VFD~3	Total	0.00	0.132	-4.312	32.6	4.314
S002-SB-1AC001 [BUS B]	S002-TR-CMP2_VFD~3	100.72	0.088	-2.875	32.6	2.876
S002-TR-CMP2_VFD~2	S002-TR-CMP2_VFD~3	47.57	0.044	-1.437	32.6	1.438

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	4.314	7.048	5.573
0.02	4.314	6.703	5.130
0.03	4.314	6.383	4.705
0.04	4.314	6.101	4.314
0.05	4.314	5.890	4.010
0.06	4.314	5.675	3.687
0.07	4.314	5.487	3.390
0.08	4.314	5.323	3.117
0.09	4.314	5.180	2.866
0.10	4.314	5.080	2.682
0.15	4.314	4.666	1.778
0.20	4.314	4.472	1.179
0.25	4.314	4.385	0.782
0.30	4.314	4.345	0.518

3-Phase fault at bus:

S002-TR-CMP1_VFD

Nominal kV = 11.220

Voltage c Factor = 1.10 (Maximum If)

Peak Value = 11.288 kA Method C

Steady State = 3.991 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-TR-CMP1_VFD~2	Total	0.00	0.128	-4.163	32.6	4.164
S002-SB-IAC001 [BUS A]	S002-TR-CMP1_VFD~2	97.22	0.085	-2.775	32.6	2.776
S002-TR-CMP1_VFD~3	S002-TR-CMP1_VFD~2	47.66	0.043	-1.388	32.6	1.388

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	4.164	6.803	5.380
0.02	4.164	6.472	4.955
0.03	4.164	6.164	4.544
0.04	4.164	5.892	4.168
0.05	4.164	5.691	3.879
0.06	4.164	5.484	3.568
0.07	4.164	5.302	3.282
0.08	4.164	5.144	3.019
0.09	4.164	5.005	2.777
0.10	4.164	4.910	2.602
0.15	4.164	4.509	1.729
0.20	4.164	4.320	1.149
0.25	4.164	4.234	0.764
0.30	4.164	4.195	0.508

3-Phase fault at bus:

S002-TR-CMP1_VFD

Nominal kV

=

11.220

Voltage c Factor

=

1.10

(Maximum If)

Peak Value

=

11.288

kA

Method C

Steady State

=

3.991

kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-TR-CMP1_VFD~3	Total	0.00	0.128	-4.163	32.6	4.164
S002-SB-IAC001 [BUS A]	S002-TR-CMP1_VFD~3	97.22	0.085	-2.775	32.6	2.776
S002-TR-CMP1_VFD~2	S002-TR-CMP1_VFD~3	47.66	0.043	-1.388	32.6	1.388

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	4.164	6.803	5.380
0.02	4.164	6.472	4.955
0.03	4.164	6.164	4.544
0.04	4.164	5.892	4.168
0.05	4.164	5.691	3.879
0.06	4.164	5.484	3.568
0.07	4.164	5.302	3.282
0.08	4.164	5.144	3.019
0.09	4.164	5.005	2.777
0.10	4.164	4.910	2.602
0.15	4.164	4.509	1.729
0.20	4.164	4.320	1.149
0.25	4.164	4.234	0.764
0.30	4.164	4.195	0.508

Project:

Location:

Contract:

Engineer:

Filename:

UQ LT LPG Export - ST & MT

Umm Qasr, Iraq

UQ83

Hemanathan Gopalan

Umm Qasr-ST

ETAP

24.0.2C

Study Case: SC_Max

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SL-KENTUAE

Base

1_NE

3-Phase fault at bus: Cable201~

Nominal kV

Voltage c Factor

Peak Value

Steady State

= 0.462

= 1.05

= 87.082

= 46.074

(Maximum If)

kA Method C

kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Cable201~	Total	0.00	19.704	-43.914	2.2	48.132
Bus_MDB-4	Cable201~	24.82	19.704	-43.914	2.2	48.132

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	47.798	50.777	17.138
0.02	47.735	47.947	4.494
0.03	47.674	47.688	1.155
0.04	47.614	47.614	0.297
0.05	47.555	47.555	0.081
0.06	47.525	47.525	0.021
0.07	47.495	47.495	0.005
0.08	47.465	47.465	0.001
0.09	47.436	47.436	0.000
0.10	47.407	47.407	0.000
0.15	47.360	47.360	0.000
0.20	47.315	47.315	0.000
0.25	47.271	47.271	0.000
0.30	47.271	47.271	0.000

Project: UQ LT LPG Export - ST & MT
Location: Umm Qasr, Iraq
Contract: UQ83
Engineer: Hemanathan Gopalan
Filename: Umm Qasr-ST

ETAP
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Study Case: SC_Max

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Revision: Base
Config.: 1_NE

3-Phase fault at bus: **Cable200~**

Nominal kV = 0.462
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 93.051 kA Method C
Steady State = 48.545 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Cable200~	Total	0.00	20.089	-46.700	2.3	50.838
Bus_MDB-4	Cable200~	19.38	20.089	-46.700	2.3	50.838

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	50.463	53.992	19.200
0.02	50.393	50.675	5.344
0.03	50.323	50.345	1.457
0.04	50.256	50.258	0.397
0.05	50.191	50.191	0.115
0.06	50.157	50.157	0.032
0.07	50.124	50.124	0.009
0.08	50.091	50.091	0.002
0.09	50.059	50.059	0.001
0.10	50.027	50.027	0.000
0.15	49.975	49.975	0.000
0.20	49.926	49.926	0.000
0.25	49.878	49.878	0.000
0.30	49.878	49.878	0.000

Project:

Location:

Contract:

Engineer:

Filename:

UQ LT LPG Export - ST & MT

Umm Qasr, Iraq

UQ83

Hemanathan Gopalan

Umm Qasr-ST

ETAP

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SL-KENTUAE

Base

1_NE

3-Phase fault at bus: Cable153~

Nominal kV

Voltage c Factor

Peak Value

Steady State

= 0.446

= 1.05

= 74.562

= 40.591

(Maximum If)

kA Method C

kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Cable153~	Total	0.00	18.184	-37.858	2.1	41.999
Bus_MDB-5	Cable153~	28.25	18.184	-37.858	2.1	41.999

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	41.782	43.908	13.495
0.02	41.745	41.866	3.187
0.03	41.708	41.714	0.738
0.04	41.671	41.671	0.171
0.05	41.634	41.634	0.042
0.06	41.615	41.615	0.010
0.07	41.596	41.596	0.002
0.08	41.577	41.577	0.001
0.09	41.558	41.558	0.000
0.10	41.539	41.539	0.000
0.15	41.508	41.508	0.000
0.20	41.479	41.479	0.000
0.25	41.449	41.449	0.000
0.30	41.449	41.449	0.000

Project:

UQ LT LPG Export - ST & MT

Location:

Umm Qasr, Iraq

Contract:

UQ83

Engineer:

Hemanathan Gopalan

Filename:

Umm Qasr-ST

ETAP

24.0.2C

Study Case: SC_Max

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Date:

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SN:

SL-KENTUAE

Revision:

Base

Config.:

1_NE

3-Phase fault at bus: Cable192~

Nominal kV

=

0.462

Voltage c Factor

=

1.05

(Maximum If)

Peak Value

=

81.452

kA

Method C

Steady State

=

43.656

kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Cable192~	Total	0.00	19.252	-41.240	2.1	45.512
Bus_MDB-2	Cable192~	25.51	19.252	-41.240	2.1	45.512

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	45.208	47.726	15.296
0.02	45.152	45.310	3.783
0.03	45.096	45.105	0.917
0.04	45.041	45.042	0.222
0.05	44.988	44.988	0.057
0.06	44.960	44.960	0.014
0.07	44.932	44.932	0.003
0.08	44.905	44.905	0.001
0.09	44.878	44.878	0.000
0.10	44.852	44.852	0.000
0.15	44.808	44.808	0.000
0.20	44.766	44.766	0.000
0.25	44.726	44.726	0.000
0.30	44.726	44.726	0.000

Project:

UQ LT LPG Export - ST & MT

Location:

Umm Qasr, Iraq

Contract:

UQ83

Engineer:

Hemanathan Gopalan

Filename:

Umm Qasr-ST

ETAP

24.0.2C

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1_NE

3-Phase fault at bus: Cable163~

Nominal kV

=

0.446

Voltage c Factor

=

1.05

(Maximum If)

Peak Value

=

89.953

kA

Method C

Steady State

=

47.206

kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Cable163~	Total	0.00	19.378	-45.148	2.3	49.131
Bus_MDB-6	Cable163~	13.71	19.378	-45.148	2.3	49.131

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	48.829	52.241	18.571
0.02	48.777	49.050	5.171
0.03	48.725	48.746	1.411
0.04	48.674	48.676	0.385
0.05	48.625	48.625	0.112
0.06	48.599	48.599	0.031
0.07	48.573	48.573	0.009
0.08	48.547	48.547	0.002
0.09	48.522	48.522	0.001
0.10	48.497	48.497	0.000
0.15	48.457	48.457	0.000
0.20	48.419	48.419	0.000
0.25	48.381	48.381	0.000
0.30	48.381	48.381	0.000

Project: UQ LT LPG Export - ST & MT
Location: Umm Qasr, Iraq
Contract: UQ83
Engineer: Hemanathan Gopalan
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3-Phase fault at bus: **Cable197~**

Nominal kV = 0.462
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 81.043 kA Method C
Steady State = 43.480 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Cable197~	Total	0.00	19.216	-41.044	2.1	45.319
Bus_MDB-3	Cable197~	26.42	19.216	-41.044	2.1	45.319

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	45.019	47.505	15.167
0.02	44.963	45.118	3.734
0.03	44.908	44.917	0.901
0.04	44.854	44.854	0.218
0.05	44.801	44.801	0.056
0.06	44.773	44.773	0.014
0.07	44.746	44.746	0.003
0.08	44.719	44.719	0.001
0.09	44.692	44.692	0.000
0.10	44.666	44.666	0.000
0.15	44.623	44.623	0.000
0.20	44.581	44.581	0.000
0.25	44.541	44.541	0.000
0.30	44.541	44.541	0.000

Project:

Location:

Contract:

Engineer:

Filename:

UQ LT LPG Export - ST & MT

Umm Qasr, Iraq

UQ83

Hemanathan Gopalan

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Base

1_NE

3-Phase fault at bus: Cable154~

Nominal kV

Voltage c Factor

Peak Value

Steady State

= 0.446

= 1.05

= 71.177

= 39.061

(Maximum If)

kA Method C

kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Cable154~	Total	0.00	17.826	-36.211	2.0	40.361
Bus_MDB-5	Cable154~	31.67	17.826	-36.211	2.0	40.361

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	40.162	42.057	12.485
0.02	40.127	40.228	2.837
0.03	40.093	40.098	0.632
0.04	40.059	40.059	0.141
0.05	40.025	40.025	0.033
0.06	40.008	40.008	0.008
0.07	39.990	39.990	0.002
0.08	39.972	39.972	0.000
0.09	39.954	39.954	0.000
0.10	39.937	39.937	0.000
0.15	39.908	39.908	0.000
0.20	39.880	39.880	0.000
0.25	39.853	39.853	0.000
0.30	39.853	39.853	0.000

Project: UQ LT LPG Export - ST & MT
Location: Umm Qasr, Iraq
Contract: UQ83
Engineer: Hemanathan Gopalan
Filename: Umm Qasr-ST

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3-Phase fault at bus: **Cable164~**

Nominal kV = 0.446
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 84.164 kA Method C
Steady State = 44.783 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Cable164~	Total	0.00	19.009	-42.447	2.2	46.509
Bus_MDB-6	Cable164~	19.47	19.009	-42.447	2.2	46.509

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	46.240	49.119	16.570
0.02	46.193	46.397	4.346
0.03	46.147	46.161	1.117
0.04	46.102	46.103	0.287
0.05	46.057	46.057	0.078
0.06	46.034	46.034	0.020
0.07	46.011	46.011	0.005
0.08	45.988	45.988	0.001
0.09	45.965	45.965	0.000
0.10	45.942	45.942	0.000
0.15	45.906	45.906	0.000
0.20	45.870	45.870	0.000
0.25	45.836	45.836	0.000
0.30	45.836	45.836	0.000

3-Phase fault at bus: Cable168~

Nominal kV = 0.446
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 78.284 kA Method C
Steady State = 42.242 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Cable168~	Total	0.00	18.537	-39.652	2.1	43.771
Bus_MDB-7	Cable168~	25.45	18.537	-39.652	2.1	43.771

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	43.534	45.933	14.651
0.02	43.494	43.643	3.606
0.03	43.453	43.462	0.870
0.04	43.413	43.413	0.210
0.05	43.373	43.373	0.054
0.06	43.353	43.353	0.013
0.07	43.332	43.332	0.003
0.08	43.311	43.311	0.001
0.09	43.291	43.291	0.000
0.10	43.270	43.270	0.000
0.15	43.238	43.238	0.000
0.20	43.206	43.206	0.000
0.25	43.175	43.175	0.000
0.30	43.175	43.175	0.000

Project:

Location:

Contract:

Engineer:

Filename:

UQ LT LPG Export - ST & MT

Umm Qasr, Iraq

UQ83

Hemanathan Gopalan

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Base

1_NE

3-Phase fault at bus: Cable169~

Nominal kV

Voltage c Factor

Peak Value

Steady State

= 0.446

= 1.05

= 70.537

= 38.768

(Maximum If)

kA Method C

kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Cable169~	Total	0.00	17.754	-35.898	2.0	40.048
Bus_MDB-7	Cable169~	33.54	17.754	-35.898	2.0	40.048

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	39.852	41.706	12.298
0.02	39.818	39.915	2.773
0.03	39.785	39.790	0.614
0.04	39.751	39.751	0.136
0.05	39.718	39.718	0.032
0.06	39.700	39.700	0.007
0.07	39.683	39.683	0.002
0.08	39.665	39.665	0.000
0.09	39.648	39.648	0.000
0.10	39.631	39.631	0.000
0.15	39.603	39.603	0.000
0.20	39.575	39.575	0.000
0.25	39.548	39.548	0.000
0.30	39.548	39.548	0.000

Project:

UQ LT LPG Export - ST & MT

Location:

Umm Qasr, Iraq

Contract:

UQ83

Engineer:

Hemanathan Gopalan

Filename:

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3-Phase fault at bus: Cable40~

Nominal kV

=

0.446

Voltage c Factor

=

1.05

(Maximum If)

Peak Value

=

65.252

kA

Method C

Steady State

=

36.314

kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Cable40~	Total	0.00	17.108	-33.293	1.9	37.432
Bus_MDB-8	Cable40~	39.18	17.108	-33.293	1.9	37.432

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	37.262	38.798	10.809
0.02	37.233	37.303	2.289
0.03	37.204	37.207	0.476
0.04	37.174	37.174	0.099
0.05	37.145	37.145	0.022
0.06	37.130	37.130	0.005
0.07	37.114	37.114	0.001
0.08	37.099	37.099	0.000
0.09	37.084	37.084	0.000
0.10	37.068	37.068	0.000
0.15	37.043	37.043	0.000
0.20	37.019	37.019	0.000
0.25	36.994	36.994	0.000
0.30	36.994	36.994	0.000

Project: UQ LT LPG Export - ST & MT
Location: Umm Qasr, Iraq
Contract: UQ83
Engineer: Hemanathan Gopalan
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3-Phase fault at bus: **Cable41~**

Nominal kV = 0.446
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 62.644 kA Method C
Steady State = 35.077 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Cable41~	Total	0.00	16.755	-31.996	1.9	36.117
Bus_MDB-8	Cable41~	42.01	16.755	-31.996	1.9	36.117

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	35.960	37.354	10.109
0.02	35.933	35.993	2.074
0.03	35.906	35.908	0.418
0.04	35.879	35.879	0.084
0.05	35.852	35.852	0.018
0.06	35.837	35.837	0.004
0.07	35.823	35.823	0.001
0.08	35.808	35.808	0.000
0.09	35.794	35.794	0.000
0.10	35.780	35.780	0.000
0.15	35.756	35.756	0.000
0.20	35.733	35.733	0.000
0.25	35.710	35.710	0.000
0.30	35.710	35.710	0.000

Project:

Location:

Contract:

Engineer:

Filename:

UQ LT LPG Export - ST & MT

Umm Qasr, Iraq

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Base

1_NE

3-Phase fault at bus: Cable188~

Nominal kV

Voltage c Factor

Peak Value

Steady State

= 0.462

= 1.05

= 94.727

= 49.203

(Maximum If)

kA Method C

kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Cable188~	Total	0.00	20.180	-47.473	2.4	51.584
Bus_MDB-1	Cable188~	11.57	20.180	-47.473	2.4	51.584

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	51.186	54.881	19.798
0.02	51.111	51.417	5.601
0.03	51.038	51.062	1.552
0.04	50.967	50.969	0.430
0.05	50.898	50.898	0.127
0.06	50.862	50.862	0.036
0.07	50.827	50.827	0.010
0.08	50.793	50.793	0.003
0.09	50.759	50.759	0.001
0.10	50.726	50.726	0.000
0.15	50.671	50.671	0.000
0.20	50.619	50.619	0.000
0.25	50.570	50.570	0.000
0.30	50.570	50.570	0.000

Project:

Location:

Contract:

Engineer:

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UQ LT LPG Export - ST & MT

Umm Qasr, Iraq

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Base

1_NE

3-Phase fault at bus: Cable189~

Nominal kV

Voltage c Factor

Peak Value

Steady State

= 0.462

= 1.05

= 89.521

= 47.074

(Maximum If)

kA Method C

kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Cable189~	Total	0.00	19.873	-45.059	2.3	49.247
Bus_MDB-1	Cable189~	16.56	19.873	-45.059	2.3	49.247

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	48.886	52.082	17.966
0.02	48.818	49.056	4.828
0.03	48.752	48.768	1.271
0.04	48.687	48.688	0.335
0.05	48.624	48.624	0.094
0.06	48.591	48.591	0.025
0.07	48.559	48.559	0.007
0.08	48.528	48.528	0.002
0.09	48.497	48.497	0.000
0.10	48.466	48.466	0.000
0.15	48.416	48.416	0.000
0.20	48.368	48.368	0.000
0.25	48.322	48.322	0.000
0.30	48.322	48.322	0.000

Project:

Location:

Contract:

Engineer:

Filename:

UQ LT LPG Export - ST & MT

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Base

1_NE

3-Phase fault at bus: Cable193~

Nominal kV

Voltage c Factor

Peak Value

Steady State

= 0.462

= 1.05

= 74.042

= 40.381

(Maximum If)

kA Method C

kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Cable193~	Total	0.00	18.515	-37.656	2.0	41.961
Bus_MDB-2	Cable193~	32.93	18.515	-37.656	2.0	41.961

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	41.706	43.693	13.029
0.02	41.658	41.764	2.973
0.03	41.611	41.616	0.665
0.04	41.565	41.565	0.149
0.05	41.519	41.519	0.035
0.06	41.495	41.495	0.008
0.07	41.472	41.472	0.002
0.08	41.448	41.448	0.000
0.09	41.425	41.425	0.000
0.10	41.402	41.402	0.000
0.15	41.364	41.364	0.000
0.20	41.328	41.328	0.000
0.25	41.292	41.292	0.000
0.30	41.292	41.292	0.000

Project:

Location:

Contract:

Engineer:

Filename:

UQ LT LPG Export - ST & MT

Umm Qasr, Iraq

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Base

1_NE

3-Phase fault at bus: Cable196~

Nominal kV

Voltage c Factor

Peak Value

Steady State

= 0.462

= 1.05

= 69.309

= 38.222

(Maximum If)

kA Method C

kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Cable196~	Total	0.00	17.956	-35.331	2.0	39.632
Bus_MDB-3	Cable196~	38.21	17.956	-35.331	2.0	39.632

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	39.406	41.099	11.674
0.02	39.364	39.445	2.524
0.03	39.322	39.326	0.536
0.04	39.281	39.281	0.114
0.05	39.240	39.240	0.026
0.06	39.219	39.219	0.005
0.07	39.198	39.198	0.001
0.08	39.177	39.177	0.000
0.09	39.156	39.156	0.000
0.10	39.135	39.135	0.000
0.15	39.101	39.101	0.000
0.20	39.068	39.068	0.000
0.25	39.036	39.036	0.000
0.30	39.036	39.036	0.000

3-Phase fault at bus: 132kV GRID

Nominal kV = 132.000
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 45.690 kA Method C
Steady State = 18.060 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
132kV GRID	Total	0.00	2.395	-18.943	7.9	19.094
U1 [Max]	132kV GRID	110.00	2.213	-17.924	8.1	18.060
TR-PSS1AC02_132kV BUS	Bus81	0.03	0.093	-0.523	5.6	0.531
TR-PSS1AC01_132kV BUS	Bus79	0.03	0.089	-0.497	5.6	0.505
Bus79	S002-SB-PSS1A01 [BUS-A]	0.00	0.089	-0.497	5.6	0.505
S002-SB-PSS1A01 [BUS-A]	Bus80	0.00	0.000	0.000	9999.0	0.000
Bus81	S002-SB-PSS1A01 [BUS-A]	0.00	0.093	-0.523	5.6	0.531
Bus83	132kV GRID	0.00	0.091	-0.510	5.6	0.518
S002-SB-PSS1A01 [BUS-A]	Bus83	0.00	0.091	-0.510	5.6	0.518
Bus84	132kV GRID	0.00	0.091	-0.510	5.6	0.518
S002-SB-PSS1A01 [BUS-A]	Bus84	0.00	0.091	-0.510	5.6	0.518
Bus87	S002-SB-PSS1A01 [BUS-A]	0.00	0.000	0.000	9999.0	0.000
Bus88	S002-SB-PSS1A01 [BUS-A]	0.00	0.000	0.000	9999.0	0.000

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	19.026	26.339	18.214
0.02	18.989	22.639	12.327
0.03	18.947	20.697	8.328
0.04	18.899	19.718	5.627
0.05	18.838	19.221	3.818
0.06	18.807	18.983	2.582
0.07	18.776	18.857	1.746
0.08	18.746	18.783	1.180
0.09	18.716	18.733	0.798
0.10	18.687	18.695	0.542
0.15	18.623	18.624	0.077
0.20	18.562	18.562	0.011
0.25	18.503	18.503	0.002

Project: UQ LT LPG Export - ST & MT
Location: Umm Qasr, Iraq
Contract: UQ83
Engineer: Hemanathan Gopalan
Filename: Umm Qasr-ST

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3-Phase fault at bus: **132kV GRID**

Nominal kV = 132.000
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 45.690 kA Method C
Steady State = 18.060 kA rms

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

<u>TD (S)</u>	<u>Ib sym</u>	<u>Ib asym</u>	<u>Idc</u>
0.30	18.503	18.503	0.000

Project:

Location:

Contract:

Engineer:

Filename:

UQ LT LPG Export - ST & MT

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Hemanathan Gopalan

Umm Qasr-ST

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03-08-2026

SL-KENTUAE

Base

1_NE

3-Phase fault at bus: 200A MDB Agreko

Nominal kV = 0.415
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 84.661 kA Method C
Steady State = 46.822 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
200A MDB Agreko	Total	0.00	23.535	-43.246	1.8	49.236
Lump27	200A MDB Agreko	108.94	0.117	-0.278	2.4	0.302
Bus64	Bus82	23.78	23.419	-42.968	1.8	48.935
Bus82	200A MDB Agreko	0.00	23.419	-42.968	1.8	48.935

Breaking and DC Fault Current (kA)
Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	48.837	50.571	13.127
0.02	48.760	48.830	2.614
0.03	48.685	48.687	0.506
0.04	48.611	48.611	0.098
0.05	48.540	48.540	0.021
0.06	48.503	48.503	0.004
0.07	48.467	48.467	0.001
0.08	48.432	48.432	0.000
0.09	48.397	48.397	0.000
0.10	48.363	48.363	0.000
0.15	48.307	48.307	0.000
0.20	48.253	48.253	0.000
0.25	48.201	48.201	0.000
0.30	48.201	48.201	0.000

3-Phase fault at bus: **Area Lighting DB**
(A8)
Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 3.131 kA Method C
Steady State = 2.166 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Area Lighting DB (A8)	Total	0.00	2.151	-0.291	0.1	2.170
S002-SB-02N001 [BUS A]	Area Lighting DB (A8)	102.19	2.151	-0.291	0.1	2.170

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.170	2.170	0.000
0.02	2.170	2.170	0.000
0.03	2.170	2.170	0.000
0.04	2.170	2.170	0.000
0.05	2.170	2.170	0.000
0.06	2.170	2.170	0.000
0.07	2.170	2.170	0.000
0.08	2.170	2.170	0.000
0.09	2.170	2.170	0.000
0.10	2.170	2.170	0.000
0.15	2.170	2.170	0.000
0.20	2.170	2.170	0.000
0.25	2.170	2.170	0.000
0.30	2.170	2.170	0.000

Project: UQ LT LPG Export - ST & MT
Location: Umm Qasr, Iraq
Contract: UQ83
Engineer: Hemanathan Gopalan
Filename: Umm Qasr-ST

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Study Case: SC_Max

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SN: SL-KENTUAE
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Config.: 1_NE

3-Phase fault at bus: **BUS-MTR1_ENCRYO**

Nominal kV = 11.000
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 6.047 kA Method C
Steady State = 0.520 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
BUS-MTR1_ENCRYO	Total	0.00	1.007	-2.369	2.4	2.574
614-KM-01A	BUS-MTR1_ENCRYO	110.00	0.487	-2.369	4.9	2.419
S002-CMP1_VFD	BUS-MTR1_ENCRYO	110.00	0.520	0.000	0.0	0.520

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

<u>TD (S)</u>	<u>Ib sym</u>	<u>Ib asym</u>	<u>Idc</u>
0.01	2.370	2.972	1.792
0.02	2.292	2.477	0.939
0.03	2.210	2.264	0.492
0.04	2.102	2.118	0.258
0.05	1.869	1.874	0.135
0.06	1.761	1.762	0.071
0.07	1.655	1.656	0.037
0.08	1.553	1.553	0.019
0.09	1.455	1.455	0.010
0.10	1.360	1.360	0.005
0.15	1.162	1.162	0.000
0.20	0.981	0.981	0.000
0.25	0.822	0.822	0.000
0.30	0.822	0.822	0.000

3-Phase fault at bus: **BUS-MTR2_ENCRYO**

Nominal kV = 11.000
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 6.047 kA Method C
Steady State = 0.520 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
BUS-MTR2_ENCRYO	Total	0.00	1.007	-2.369	2.4	2.574
614-KM-01B	BUS-MTR2_ENCRYO	110.00	0.487	-2.369	4.9	2.419
S002-CMP2_VFD	BUS-MTR2_ENCRYO	110.00	0.520	0.000	0.0	0.520

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

<u>TD (S)</u>	<u>Ib sym</u>	<u>Ib asym</u>	<u>Idc</u>
0.01	2.370	2.972	1.792
0.02	2.292	2.477	0.939
0.03	2.210	2.264	0.492
0.04	2.102	2.118	0.258
0.05	1.869	1.874	0.135
0.06	1.761	1.762	0.071
0.07	1.655	1.656	0.037
0.08	1.553	1.553	0.019
0.09	1.455	1.455	0.010
0.10	1.360	1.360	0.005
0.15	1.162	1.162	0.000
0.20	0.981	0.981	0.000
0.25	0.822	0.822	0.000
0.30	0.822	0.822	0.000

3-Phase fault at bus: **Bus1**

Nominal kV

Voltage c Factor

Peak Value

Steady State

= 11.000

= 1.10

= 76.701

= 22.883

(Maximum If)

kA Method C

kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus1	Total	0.00	2.655	-30.130	11.3	30.246
S002-SB-1AC001 [BUS A]	Bus1	1.21	2.228	-27.819	12.5	27.908
Bus22	Bus1	34.37	0.427	-2.311	5.4	2.350

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	29.565	45.110	34.072
0.02	29.207	41.189	29.043
0.03	28.805	37.449	23.931
0.04	28.365	34.546	19.719
0.05	27.824	33.236	18.179
0.06	27.549	31.522	15.319
0.07	27.279	30.180	12.910
0.08	27.015	29.123	10.879
0.09	26.756	28.283	9.168
0.10	26.503	27.945	8.861
0.15	25.970	26.282	4.033
0.20	25.461	25.527	1.836
0.25	24.979	24.993	0.835
0.30	24.979	24.981	0.380

3-Phase fault at bus: **Bus2**

Nominal kV

Voltage c Factor

Peak Value

Steady State

= 0.380

= 1.05

= 7.485

= 0.592

(Maximum If)

kA Method C

kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus2	Total	0.00	1.596	-3.192	2.0	3.569
Chiller-15	Bus2	96.14	1.004	-3.192	3.2	3.346
VFD15	Bus2	96.14	0.592	0.000	0.0	0.592

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.933	3.421	1.761
0.02	2.610	2.691	0.655
0.03	2.299	2.312	0.244
0.04	2.005	2.007	0.091
0.05	1.730	1.731	0.034
0.06	1.599	1.599	0.013
0.07	1.473	1.473	0.005
0.08	1.353	1.353	0.002
0.09	1.238	1.238	0.001
0.10	1.130	1.130	0.000
0.15	0.936	0.936	0.000
0.20	0.773	0.773	0.000
0.25	0.650	0.650	0.000
0.30	0.650	0.650	0.000

3-Phase fault at bus: **Bus4**

Nominal kV

Voltage c Factor

Peak Value

Steady State

= 0.380

= 1.05

= 7.485

= 0.592

(Maximum If)

kA Method C

kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus4	Total	0.00	1.596	-3.192	2.0	3.569
Chiller-16	Bus4	96.14	1.004	-3.192	3.2	3.346
VFD16	Bus4	96.14	0.592	0.000	0.0	0.592

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.933	3.421	1.761
0.02	2.610	2.691	0.655
0.03	2.299	2.312	0.244
0.04	2.005	2.007	0.091
0.05	1.730	1.731	0.034
0.06	1.599	1.599	0.013
0.07	1.473	1.473	0.005
0.08	1.353	1.353	0.002
0.09	1.238	1.238	0.001
0.10	1.130	1.130	0.000
0.15	0.936	0.936	0.000
0.20	0.773	0.773	0.000
0.25	0.650	0.650	0.000
0.30	0.650	0.650	0.000

3-Phase fault at bus: **Bus7**

Nominal kV

Voltage c Factor

Peak Value

Steady State

= 11.000

= 1.10

= 44.357

= 18.184

(Maximum If)

kA Method C

kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus7	Total	0.00	6.811	-21.221	3.1	22.287
S002-SB-1AC001 [BUS A]	Bus7	36.59	6.772	-21.127	3.1	22.186
Bus27	Bus7	9.54	0.038	-0.094	2.4	0.101

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	21.996	25.184	12.263
0.02	21.834	22.460	5.266
0.03	21.647	21.753	2.153
0.04	21.435	21.453	0.880
0.05	21.165	21.169	0.428
0.06	21.026	21.026	0.181
0.07	20.888	20.888	0.077
0.08	20.751	20.751	0.032
0.09	20.616	20.616	0.014
0.10	20.484	20.484	0.007
0.15	20.195	20.195	0.000
0.20	19.915	19.915	0.000
0.25	19.645	19.645	0.000
0.30	19.645	19.645	0.000

3-Phase fault at bus: **Bus8**

Nominal kV

Voltage c Factor

Peak Value

Steady State

= 11.000

= 1.10

= 41.748

= 17.536

(Maximum If)

kA Method C

kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus8	Total	0.00	6.850	-20.170	2.9	21.301
S002-SB-1AC001 [BUS B]	Bus8	39.40	6.795	-20.032	2.9	21.153
Bus69	Bus8	8.84	0.055	-0.138	2.5	0.148

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	21.043	23.756	11.024
0.02	20.893	21.356	4.423
0.03	20.722	20.791	1.695
0.04	20.516	20.526	0.650
0.05	20.258	20.260	0.293
0.06	20.125	20.125	0.116
0.07	19.993	19.993	0.046
0.08	19.862	19.862	0.018
0.09	19.733	19.733	0.007
0.10	19.605	19.605	0.004
0.15	19.325	19.325	0.000
0.20	19.052	19.052	0.000
0.25	18.788	18.788	0.000
0.30	18.788	18.788	0.000

3-Phase fault at bus: **Bus9**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 4.021 kA Method C
Steady State = 2.337 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus9	Total	0.00	2.325	-1.201	0.5	2.617
TYPICAL MOTOR FEEDER 1 A	Bus9	102.76	2.228	-0.723	0.3	2.342
614-FM-06AA	Bus9	105.00	0.098	-0.478	4.9	0.488

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.493	2.540	0.484
0.02	2.468	2.476	0.191
0.03	2.445	2.445	0.043
0.04	2.423	2.423	0.010
0.05	2.403	2.403	0.005
0.06	2.393	2.393	0.001
0.07	2.383	2.383	0.000
0.08	2.374	2.374	0.000
0.09	2.366	2.366	0.000
0.10	2.358	2.358	0.000
0.15	2.345	2.345	0.000
0.20	2.333	2.333	0.000
0.25	2.322	2.322	0.000
0.30	2.322	2.322	0.000

Project: UQ LT LPG Export - ST & MT
Location: Umm Qasr, Iraq
Contract: UQ83
Engineer: Hemanathan Gopalan
Filename: Umm Qasr-ST

ETAP
24.0.2C

Study Case: SC_Max

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3-Phase fault at bus: **Bus10**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 123.006 kA Method C
Steady State = 49.213 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus10	Total	0.00	9.736	-53.498	5.5	54.377
S002-SB-01E001 [BUS A]	Bus10	0.81	1.618	-3.983	2.5	4.299
Bus43	Bus10	95.50	8.118	-49.515	6.1	50.176

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	53.451	69.787	44.869
0.02	53.089	59.503	26.873
0.03	52.738	55.079	15.886
0.04	52.400	53.235	9.391
0.05	52.079	52.393	5.728
0.06	51.922	52.034	3.407
0.07	51.769	51.809	2.027
0.08	51.620	51.634	1.206
0.09	51.475	51.480	0.717
0.10	51.334	51.336	0.441
0.15	51.073	51.073	0.033
0.20	50.826	50.826	0.003
0.25	50.594	50.594	0.000
0.30	50.594	50.594	0.000

3-Phase fault at bus: **Bus11**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 32.860 kA Method C
Steady State = 2.287 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus11	Total	0.00	0.742	-14.106	19.0	14.126
DG-1 (TOFS)	Bus11	105.00	0.742	-14.106	19.0	14.126

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	12.620	21.118	16.933
0.02	12.045	18.737	14.352
0.03	11.456	16.710	12.165
0.04	10.857	14.973	10.311
0.05	10.250	13.470	8.740
0.06	10.003	12.447	7.408
0.07	9.755	11.601	6.279
0.08	9.506	10.894	5.322
0.09	9.256	10.296	4.511
0.10	9.005	9.783	3.823
0.15	8.691	8.851	1.673
0.20	8.380	8.412	0.732
0.25	8.071	8.077	0.320
0.30	8.071	8.072	0.140

3-Phase fault at bus: **Bus12**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 4.071 kA Method C
Steady State = 2.373 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus12	Total	0.00	2.360	-1.214	0.5	2.654
TYPICAL MOTOR FEEDER 1 A	Bus12	102.73	2.262	-0.736	0.3	2.379
614-FM-06AB	Bus12	105.00	0.098	-0.478	4.9	0.488

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.530	2.575	0.482
0.02	2.505	2.512	0.189
0.03	2.481	2.481	0.042
0.04	2.459	2.459	0.010
0.05	2.439	2.439	0.005
0.06	2.429	2.429	0.001
0.07	2.420	2.420	0.000
0.08	2.411	2.411	0.000
0.09	2.402	2.402	0.000
0.10	2.394	2.394	0.000
0.15	2.381	2.381	0.000
0.20	2.369	2.369	0.000
0.25	2.358	2.358	0.000
0.30	2.358	2.358	0.000

3-Phase fault at bus: **Bus14**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 6.241 kA Method C
Steady State = 4.266 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus14	Total	0.00	4.261	-0.748	0.2	4.327
Bus_MDB-8	Bus14	95.57	4.224	-0.647	0.2	4.273
Pump-15	Bus14	96.14	0.038	-0.101	2.7	0.108

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	4.301	4.301	0.000
0.02	4.296	4.296	0.000
0.03	4.292	4.292	0.000
0.04	4.288	4.288	0.000
0.05	4.284	4.284	0.000
0.06	4.282	4.282	0.000
0.07	4.279	4.279	0.000
0.08	4.277	4.277	0.000
0.09	4.275	4.275	0.000
0.10	4.273	4.273	0.000
0.15	4.270	4.270	0.000
0.20	4.267	4.267	0.000
0.25	4.264	4.264	0.000
0.30	4.264	4.264	0.000

3-Phase fault at bus: **Bus15**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 5.703 kA Method C
Steady State = 3.894 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus15	Total	0.00	3.897	-0.664	0.2	3.954
Bus_MDB-8	Bus15	95.96	3.860	-0.563	0.1	3.900
Pump-16	Bus15	96.14	0.038	-0.101	2.7	0.108

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	3.928	3.928	0.000
0.02	3.924	3.924	0.000
0.03	3.919	3.919	0.000
0.04	3.915	3.915	0.000
0.05	3.911	3.911	0.000
0.06	3.909	3.909	0.000
0.07	3.907	3.907	0.000
0.08	3.905	3.905	0.000
0.09	3.903	3.903	0.000
0.10	3.901	3.901	0.000
0.15	3.898	3.898	0.000
0.20	3.894	3.894	0.000
0.25	3.892	3.892	0.000
0.30	3.892	3.892	0.000

3-Phase fault at bus: **Bus17**

Nominal kV = 11.000
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 42.146 kA Method C
Steady State = 17.662 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus17	Total	0.00	6.975	-20.388	2.9	21.548
S002-SB-1AC001 [BUS A]	Bus17	39.73	6.896	-20.184	2.9	21.330
Bus28	Bus17	16.61	0.078	-0.204	2.6	0.218

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	21.277	23.986	11.074
0.02	21.125	21.586	4.438
0.03	20.948	21.016	1.694
0.04	20.749	20.759	0.646
0.05	20.496	20.498	0.294
0.06	20.365	20.365	0.116
0.07	20.235	20.235	0.046
0.08	20.106	20.106	0.018
0.09	19.980	19.980	0.007
0.10	19.855	19.855	0.004
0.15	19.582	19.582	0.000
0.20	19.317	19.317	0.000
0.25	19.062	19.062	0.000
0.30	19.062	19.062	0.000

3-Phase fault at bus: **Bus18**

Nominal kV = 11.000

Voltage c Factor = 1.10 (Maximum If)

Peak Value = 41.828 kA Method C

Steady State = 17.536 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus18	Total	0.00	6.848	-20.205	3.0	21.334
S002-SB-1AC001 [BUS B]	Bus18	39.35	6.773	-20.013	3.0	21.128
Bus33	Bus18	15.72	0.074	-0.193	2.6	0.207

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	21.071	23.793	11.052
0.02	20.919	21.384	4.434
0.03	20.746	20.815	1.700
0.04	20.537	20.547	0.652
0.05	20.277	20.279	0.294
0.06	20.143	20.143	0.116
0.07	20.010	20.010	0.046
0.08	19.878	19.878	0.018
0.09	19.748	19.748	0.007
0.10	19.619	19.619	0.004
0.15	19.337	19.337	0.000
0.20	19.062	19.062	0.000
0.25	18.796	18.796	0.000
0.30	18.796	18.796	0.000

3-Phase fault at bus: **Bus19**

Nominal kV = 11.000
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 17.669 kA Method C
Steady State = 9.518 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus19	Total	0.00	6.463	-8.801	1.4	10.919
S003-SB-11C001 [BUS A]	Bus19	0.95	6.408	-8.665	1.4	10.777
Bus166	Bus19	9.89	0.056	-0.136	2.5	0.147

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	10.816	10.994	1.972
0.02	10.758	10.763	0.344
0.03	10.699	10.699	0.051
0.04	10.637	10.637	0.008
0.05	10.577	10.577	0.002
0.06	10.547	10.547	0.000
0.07	10.518	10.518	0.000
0.08	10.489	10.489	0.000
0.09	10.461	10.461	0.000
0.10	10.433	10.433	0.000
0.15	10.379	10.379	0.000
0.20	10.326	10.326	0.000
0.25	10.274	10.274	0.000
0.30	10.274	10.274	0.000

3-Phase fault at bus: **Bus20**

Nominal kV

Voltage c Factor

Peak Value

Steady State

= 11.000

= 1.10

= 18.093

= 9.491

(Maximum If)

kA Method C

kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus20	Total	0.00	6.467	-9.024	1.4	11.102
S003-SB-11C001 [BUS B]	Bus20	0.87	6.413	-8.875	1.4	10.950
Bus167	Bus20	10.65	0.054	-0.149	2.8	0.159

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	10.983	11.191	2.148
0.02	10.915	10.923	0.400
0.03	10.843	10.844	0.064
0.04	10.773	10.773	0.010
0.05	10.706	10.706	0.002
0.06	10.673	10.673	0.000
0.07	10.641	10.641	0.000
0.08	10.609	10.609	0.000
0.09	10.578	10.578	0.000
0.10	10.548	10.548	0.000
0.15	10.489	10.489	0.000
0.20	10.432	10.432	0.000
0.25	10.378	10.378	0.000
0.30	10.378	10.378	0.000

3-Phase fault at bus: **Bus21**

Nominal kV = 11.000

Voltage c Factor = 1.10 (Maximum If)

Peak Value = 75.642 kA Method C

Steady State = 22.670 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus21	Total	0.00	2.596	-29.689	11.4	29.803
S002-SB-1AC001 [BUS B]	Bus21	1.22	2.279	-28.128	12.3	28.220
Bus25	Bus21	22.80	0.317	-1.561	4.9	1.593

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	29.132	44.483	33.616
0.02	28.776	40.550	28.570
0.03	28.381	36.861	23.522
0.04	27.922	33.980	19.366
0.05	27.371	32.602	17.713
0.06	27.092	30.916	14.893
0.07	26.818	29.598	12.523
0.08	26.549	28.561	10.529
0.09	26.285	27.736	8.853
0.10	26.027	27.376	8.485
0.15	25.481	25.764	3.807
0.20	24.958	25.017	1.708
0.25	24.461	24.473	0.766
0.30	24.461	24.464	0.344

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3-Phase fault at bus: **Bus22**

Nominal kV

Voltage c Factor

Peak Value

Steady State

= 3.300

= 1.10

= 70.335

= 17.848

(Maximum If)

kA Method C

kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus22	Total	0.00	3.465	-28.560	8.2	28.769
S002-SB-1AD001 [BUS A]	Bus22	1.11	2.408	-9.878	4.1	10.168
Bus1	Bus22	82.09	1.057	-18.681	17.7	18.711

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	27.418	40.606	29.952
0.02	26.659	36.378	24.752
0.03	25.778	32.206	19.306
0.04	24.884	29.085	15.058
0.05	24.032	27.964	14.298
0.06	23.623	26.318	11.600
0.07	23.224	25.058	9.411
0.08	22.836	24.078	7.635
0.09	22.458	23.297	6.194
0.10	22.092	22.993	6.373
0.15	21.374	21.523	2.522
0.20	20.696	20.720	0.998
0.25	20.058	20.062	0.395
0.30	20.058	20.059	0.156

3-Phase fault at bus: **Bus23**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 121.337 kA Method C
Steady State = 49.213 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus23	Total	0.00	9.432	-52.654	5.6	53.492
S002-SB-01E001 [BUS B]	Bus23	0.77	1.350	-3.135	2.3	3.414
Bus42	Bus23	95.49	8.082	-49.518	6.1	50.174

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	52.559	68.836	44.452
0.02	52.264	58.705	26.734
0.03	51.979	54.354	15.893
0.04	51.707	52.563	9.448
0.05	51.448	51.771	5.773
0.06	51.322	51.438	3.451
0.07	51.199	51.240	2.063
0.08	51.079	51.094	1.233
0.09	50.963	50.969	0.737
0.10	50.851	50.853	0.453
0.15	50.650	50.650	0.035
0.20	50.460	50.460	0.003
0.25	50.283	50.283	0.000
0.30	50.283	50.283	0.000

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3-Phase fault at bus: **Bus24**

Nominal kV

Voltage c Factor

Peak Value

Steady State

= 0.380

= 1.05

= 64.579

= 29.935

(Maximum If)

kA Method C

kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus24	Total	0.00	9.028	-30.682	3.4	31.983
S002-SB-02N001 [BUS B]	Bus24	0.19	0.661	-1.573	2.4	1.707
Bus46	Bus24	95.25	8.367	-29.109	3.5	30.287

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	31.663	36.483	18.123
0.02	31.497	32.339	7.331
0.03	31.337	31.476	2.952
0.04	31.184	31.206	1.188
0.05	31.037	31.041	0.485
0.06	30.967	30.968	0.196
0.07	30.898	30.898	0.079
0.08	30.831	30.831	0.032
0.09	30.766	30.766	0.013
0.10	30.703	30.703	0.005
0.15	30.584	30.584	0.000
0.20	30.471	30.471	0.000
0.25	30.366	30.366	0.000
0.30	30.366	30.366	0.000

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3-Phase fault at bus: **Bus25**

Nominal kV

Voltage c Factor

Peak Value

Steady State

= 3.300

= 1.10

= 62.606

= 18.093

(Maximum If)

kA Method C

kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus25	Total	0.00	2.502	-24.915	10.0	25.040
S002-SB-1AD001 [BUS B]	Bus25	0.66	1.419	-5.869	4.1	6.038
Bus21	Bus25	81.90	1.083	-19.046	17.6	19.076

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	24.151	36.487	27.350
0.02	23.652	32.950	22.941
0.03	23.107	29.579	18.465
0.04	22.581	27.033	14.863
0.05	22.077	25.965	13.667
0.06	21.833	24.583	11.298
0.07	21.595	23.528	9.339
0.08	21.362	22.714	7.720
0.09	21.135	22.077	6.381
0.10	20.914	21.806	6.171
0.15	20.481	20.642	2.576
0.20	20.070	20.099	1.075
0.25	19.682	19.687	0.449
0.30	19.682	19.683	0.187

3-Phase fault at bus: **Bus27**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 64.701 kA Method C
Steady State = 29.020 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus27	Total	0.00	9.198	-30.797	3.3	32.142
S002-SB-02N001 [BUS A]	Bus27	0.32	1.088	-2.589	2.4	2.808
Bus7	Bus27	95.47	8.110	-28.208	3.5	29.351

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	31.770	36.530	18.032
0.02	31.493	32.315	7.242
0.03	31.224	31.357	2.891
0.04	30.965	30.987	1.154
0.05	30.719	30.723	0.468
0.06	30.601	30.602	0.188
0.07	30.486	30.486	0.075
0.08	30.373	30.373	0.030
0.09	30.264	30.264	0.012
0.10	30.158	30.158	0.005
0.15	29.952	29.952	0.000
0.20	29.757	29.757	0.000
0.25	29.574	29.574	0.000
0.30	29.574	29.574	0.000

3-Phase fault at bus: **Bus28**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 89.771 kA Method C
Steady State = 34.953 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus28	Total	0.00	9.424	-40.603	4.3	41.682
S002-SB-03E001 [BUS A]	Bus28	2.79	2.449	-5.843	2.4	6.336
Bus17	Bus28	93.16	6.974	-34.760	5.0	35.453

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	40.861	50.575	29.803
0.02	40.259	43.188	15.632
0.03	39.675	40.484	8.050
0.04	39.113	39.332	4.146
0.05	38.577	38.641	2.223
0.06	38.319	38.337	1.154
0.07	38.068	38.072	0.599
0.08	37.823	37.824	0.311
0.09	37.585	37.585	0.162
0.10	37.354	37.354	0.087
0.15	36.902	36.902	0.003
0.20	36.476	36.476	0.000
0.25	36.076	36.076	0.000
0.30	36.076	36.076	0.000

3-Phase fault at bus: **Bus29**

Nominal kV = 3.450
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 55.331 kA Method C
Steady State = 17.642 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus29	Total	0.00	3.975	-23.644	5.9	23.976
S002-SB-1AD001 [BUS B]	Bus29	10.85	3.956	-23.610	6.0	23.939
S002-PC1-1B	Bus29	0.91	0.019	-0.034	1.8	0.039

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	23.184	31.463	21.270
0.02	22.738	27.034	14.623
0.03	22.251	24.234	9.603
0.04	21.779	22.674	6.307
0.05	21.327	21.864	4.814
0.06	21.108	21.358	3.258
0.07	20.894	21.010	2.205
0.08	20.685	20.739	1.492
0.09	20.481	20.506	1.010
0.10	20.283	20.299	0.817
0.15	19.893	19.893	0.127
0.20	19.522	19.522	0.020
0.25	19.172	19.172	0.003
0.30	19.172	19.172	0.000

3-Phase fault at bus: **Bus30**

Nominal kV = 3.450
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 61.728 kA Method C
Steady State = 17.420 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus30	Total	0.00	5.222	-26.941	5.2	27.442
S002-SB-1AD001 [BUS A]	Bus30	12.12	4.968	-26.277	5.3	26.742
S002-PC1-1A	Bus30	16.78	0.255	-0.664	2.6	0.711

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	26.239	34.763	22.804
0.02	25.560	29.735	15.195
0.03	24.771	26.533	9.508
0.04	23.971	24.698	5.949
0.05	23.209	23.668	4.642
0.06	22.842	23.043	3.036
0.07	22.485	22.572	1.985
0.08	22.137	22.175	1.298
0.09	21.798	21.815	0.849
0.10	21.470	21.482	0.728
0.15	20.825	20.825	0.100
0.20	20.215	20.215	0.014
0.25	19.643	19.643	0.002
0.30	19.643	19.643	0.000

3-Phase fault at bus: **Bus31**

Nominal kV = 3.450
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 61.747 kA Method C
Steady State = 17.420 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus31	Total	0.00	5.220	-26.947	5.2	27.448
S002-SB-1AD001 [BUS A]	Bus31	12.10	4.957	-26.252	5.3	26.716
S002-PC1-2A	Bus31	17.47	0.263	-0.695	2.6	0.743

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	26.242	34.772	22.813
0.02	25.563	29.741	15.200
0.03	24.774	26.537	9.512
0.04	23.974	24.702	5.952
0.05	23.211	23.671	4.643
0.06	22.844	23.045	3.037
0.07	22.487	22.574	1.986
0.08	22.138	22.176	1.299
0.09	21.800	21.816	0.849
0.10	21.471	21.484	0.728
0.15	20.826	20.826	0.100
0.20	20.216	20.216	0.014
0.25	19.643	19.643	0.002
0.30	19.643	19.643	0.000

3-Phase fault at bus: **Bus32**

Nominal kV = 3.450
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 55.479 kA Method C
Steady State = 17.642 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus32	Total	0.00	3.965	-23.699	6.0	24.028
S002-SB-1AD001 [BUS B]	Bus32	10.74	3.814	-23.388	6.1	23.697
S002-PC1-2B	Bus32	8.07	0.151	-0.310	2.1	0.345

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	23.215	31.529	21.333
0.02	22.764	27.076	14.661
0.03	22.271	24.264	9.630
0.04	21.794	22.694	6.326
0.05	21.338	21.876	4.824
0.06	21.117	21.368	3.265
0.07	20.901	21.017	2.209
0.08	20.689	20.743	1.495
0.09	20.484	20.509	1.012
0.10	20.283	20.300	0.818
0.15	19.890	19.890	0.127
0.20	19.516	19.516	0.020
0.25	19.164	19.164	0.003
0.30	19.164	19.164	0.000

3-Phase fault at bus: **Bus33**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 89.031 kA Method C
Steady State = 34.933 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus33	Total	0.00	9.254	-40.221	4.3	41.272
S002-SB-03E001 [BUS B]	Bus33	2.62	2.298	-5.481	2.4	5.944
Bus18	Bus33	93.10	6.956	-34.740	5.0	35.430

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	40.595	50.264	29.639
0.02	40.027	42.953	15.584
0.03	39.474	40.287	8.053
0.04	38.943	39.164	4.161
0.05	38.436	38.500	2.234
0.06	38.192	38.210	1.163
0.07	37.954	37.959	0.606
0.08	37.723	37.724	0.315
0.09	37.497	37.498	0.164
0.10	37.279	37.279	0.089
0.15	36.849	36.849	0.003
0.20	36.442	36.442	0.000
0.25	36.061	36.061	0.000
0.30	36.061	36.061	0.000

3-Phase fault at bus: **Bus35**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 32.588 kA Method C
Steady State = 4.384 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus35	Total	0.00	0.736	-13.990	19.0	14.009
S002-EDG-03E001	Bus35	105.00	0.736	-13.990	19.0	14.009

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	12.918	21.187	16.793
0.02	12.504	18.946	14.233
0.03	12.057	17.056	12.064
0.04	11.580	15.448	10.226
0.05	11.077	14.065	8.667
0.06	10.888	13.134	7.346
0.07	10.695	12.376	6.227
0.08	10.499	11.751	5.278
0.09	10.299	11.229	4.473
0.10	10.097	10.785	3.792
0.15	9.780	9.920	1.659
0.20	9.453	9.481	0.726
0.25	9.119	9.124	0.317
0.30	9.119	9.120	0.139

3-Phase fault at bus: **Bus36**

Nominal kV = 0.400
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 44.983 kA Method C
Steady State = 15.470 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus36	Total	0.00	7.157	-21.717	3.0	22.866
Bus37	Bus36	73.65	4.659	-15.772	3.4	16.446
OPC Fuel Depot	Bus36	110.53	2.498	-5.945	2.4	6.448

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	22.337	25.269	11.814
0.02	21.849	22.290	4.413
0.03	21.205	21.268	1.630
0.04	20.586	20.595	0.602
0.05	19.996	19.998	0.228
0.06	19.715	19.715	0.085
0.07	19.441	19.441	0.031
0.08	19.174	19.174	0.012
0.09	18.915	18.915	0.004
0.10	18.664	18.664	0.002
0.15	18.158	18.158	0.000
0.20	17.680	17.680	0.000
0.25	17.232	17.232	0.000
0.30	17.232	17.232	0.000

3-Phase fault at bus: **Bus37**

Nominal kV = 3.300
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 15.040 kA Method C
Steady State = 5.774 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus37	Total	0.00	2.355	-7.261	3.1	7.633
S003-SB-11D001 [BUS B]	Bus37	14.74	2.121	-6.662	3.1	6.991
Bus36	Bus37	23.76	0.234	-0.599	2.6	0.643

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	7.491	8.468	3.948
0.02	7.366	7.511	1.466
0.03	7.218	7.238	0.540
0.04	7.074	7.077	0.199
0.05	6.935	6.935	0.075
0.06	6.868	6.868	0.028
0.07	6.802	6.802	0.010
0.08	6.737	6.737	0.004
0.09	6.674	6.674	0.001
0.10	6.612	6.612	0.001
0.15	6.487	6.487	0.000
0.20	6.366	6.366	0.000
0.25	6.253	6.253	0.000
0.30	6.253	6.253	0.000

Project: UQ LT LPG Export - ST & MT
Location: Umm Qasr, Iraq
Contract: UQ83
Engineer: Hemanathan Gopalan
Filename: Umm Qasr-ST

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3-Phase fault at bus: **Bus38**

Nominal kV = 0.400
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 119.736 kA Method C
Steady State = 56.822 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus38	Total	0.00	17.612	-57.162	3.2	59.814
Bus40	Bus38	0.06	0.110	-0.279	2.5	0.300
Bus45	Bus38	0.06	0.110	-0.279	2.5	0.300
Bus48	Bus38	0.06	0.110	-0.279	2.5	0.300
Bus66	Bus38	72.04	17.282	-56.325	3.3	58.917

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	59.389	68.019	33.158
0.02	59.316	60.850	13.577
0.03	59.243	59.493	5.439
0.04	59.172	59.212	2.179
0.05	59.104	59.111	0.931
0.06	59.068	59.069	0.378
0.07	59.033	59.033	0.153
0.08	58.998	58.998	0.062
0.09	58.964	58.964	0.025
0.10	58.930	58.930	0.011
0.15	58.876	58.876	0.000
0.20	58.825	58.825	0.000
0.25	58.775	58.775	0.000
0.30	58.775	58.775	0.000

Project: UQ LT LPG Export - ST & MT
Location: Umm Qasr, Iraq
Contract: UQ83
Engineer: Hemanathan Gopalan
Filename: Umm Qasr-ST

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Study Case: SC_Max

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3-Phase fault at bus: **Bus40**

Nominal kV = 0.400
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 111.106 kA Method C
Steady State = 53.880 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus40	Total	0.00	17.861	-53.729	3.0	56.620
Bus38	Bus40	6.10	9.633	-29.034	3.0	30.590
Bus_LVDB-2	Bus40	2.33	8.229	-24.694	3.0	26.029

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	56.213	63.283	29.066
0.02	56.143	57.214	11.017
0.03	56.073	56.222	4.087
0.04	56.005	56.026	1.516
0.05	55.939	55.942	0.600
0.06	55.905	55.905	0.225
0.07	55.871	55.871	0.085
0.08	55.837	55.837	0.032
0.09	55.804	55.804	0.012
0.10	55.772	55.772	0.005
0.15	55.720	55.720	0.000
0.20	55.670	55.670	0.000
0.25	55.622	55.622	0.000
0.30	55.622	55.622	0.000

3-Phase fault at bus: **Bus41**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 60.258 kA Method C
Steady State = 5.846 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus41	Total	0.00	1.361	-25.868	19.0	25.904
S002-EDG-01E001	Bus41	105.00	1.361	-25.868	19.0	25.904

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	23.441	38.905	31.051
0.02	22.493	34.621	26.319
0.03	21.502	30.983	22.308
0.04	20.474	27.869	18.908
0.05	19.416	25.176	16.026
0.06	18.997	23.354	13.584
0.07	18.573	21.852	11.514
0.08	18.146	20.604	9.759
0.09	17.714	19.550	8.272
0.10	17.279	18.648	7.011
0.15	16.675	16.955	3.067
0.20	16.068	16.124	1.342
0.25	15.460	15.472	0.587
0.30	15.460	15.463	0.257

3-Phase fault at bus: **Bus42**

Nominal kV

Voltage c Factor

Peak Value

Steady State

= 11.000

= 1.10

= 71.229

= 22.284

(Maximum If)

kA Method C

kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus42	Total	0.00	3.349	-28.883	8.6	29.077
S002-SB-1AC001 [BUS B]	Bus42	5.06	3.302	-28.770	8.7	28.959
Bus23	Bus42	6.76	0.047	-0.113	2.4	0.123

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	28.453	41.399	30.073
0.02	28.120	36.686	23.561
0.03	27.751	32.987	17.834
0.04	27.319	30.473	13.499
0.05	26.800	29.151	11.470
0.06	26.537	27.985	8.885
0.07	26.278	27.164	6.882
0.08	26.023	26.564	5.331
0.09	25.774	26.103	4.130
0.10	25.530	25.796	3.698
0.15	25.011	25.035	1.109
0.20	24.514	24.516	0.332
0.25	24.041	24.041	0.100
0.30	24.041	24.041	0.030

3-Phase fault at bus: **Bus43**

Nominal kV = 11.000

Voltage c Factor = 1.10 (Maximum If)

Peak Value = 70.431 kA Method C

Steady State = 22.296 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus43	Total	0.00	3.677	-28.908	7.9	29.141
S002-SB-1AC001 [BUS A]	Bus43	6.75	3.622	-28.766	7.9	28.993
Bus10	Bus43	8.38	0.055	-0.142	2.6	0.152

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	28.527	40.786	29.150
0.02	28.203	35.913	22.233
0.03	27.837	32.274	16.331
0.04	27.437	29.944	11.995
0.05	26.940	28.737	10.002
0.06	26.688	27.731	7.535
0.07	26.440	27.043	5.677
0.08	26.197	26.543	4.277
0.09	25.958	26.157	3.222
0.10	25.725	25.881	2.838
0.15	25.232	25.243	0.745
0.20	24.759	24.760	0.195
0.25	24.311	24.311	0.051
0.30	24.311	24.311	0.013

3-Phase fault at bus: **Bus45**

Nominal kV = 0.400
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 111.106 kA Method C
Steady State = 53.880 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus45	Total	0.00	17.861	-53.729	3.0	56.620
Bus38	Bus45	6.10	9.633	-29.034	3.0	30.590
Bus_LVDB-2	Bus45	2.33	8.229	-24.694	3.0	26.029

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	56.213	63.283	29.066
0.02	56.143	57.214	11.017
0.03	56.073	56.222	4.087
0.04	56.005	56.026	1.516
0.05	55.939	55.942	0.600
0.06	55.905	55.905	0.225
0.07	55.871	55.871	0.085
0.08	55.837	55.837	0.032
0.09	55.804	55.804	0.012
0.10	55.772	55.772	0.005
0.15	55.720	55.720	0.000
0.20	55.670	55.670	0.000
0.25	55.622	55.622	0.000
0.30	55.622	55.622	0.000

3-Phase fault at bus: **Bus46**

Nominal kV = 11.000
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 43.972 kA Method C
Steady State = 18.050 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus46	Total	0.00	6.683	-21.006	3.1	22.044
S002-SB-1AC001 [BUS B]	Bus46	36.25	6.658	-20.947	3.1	21.980
Bus24	Bus46	5.80	0.024	-0.059	2.4	0.064

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	21.759	24.955	12.220
0.02	21.599	22.229	5.252
0.03	21.417	21.525	2.156
0.04	21.196	21.214	0.885
0.05	20.920	20.925	0.428
0.06	20.778	20.779	0.181
0.07	20.637	20.638	0.077
0.08	20.498	20.498	0.033
0.09	20.360	20.360	0.014
0.10	20.224	20.224	0.007
0.15	19.927	19.927	0.000
0.20	19.637	19.637	0.000
0.25	19.358	19.358	0.000
0.30	19.358	19.358	0.000

3-Phase fault at bus: **Bus47**

Nominal kV = 3.300
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 18.999 kA Method C
Steady State = 10.753 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus47	Total	0.00	10.313	-7.586	0.7	12.802
S002-SB-1AD001 [BUS A]	Bus47	85.19	10.256	-7.213	0.7	12.539
S002-40-MP-04A	Bus47	110.00	0.057	-0.373	6.6	0.378

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	12.655	12.676	0.728
0.02	12.567	12.567	0.113
0.03	12.461	12.461	0.009
0.04	12.350	12.350	0.001
0.05	12.243	12.243	0.000
0.06	12.190	12.190	0.000
0.07	12.138	12.138	0.000
0.08	12.087	12.087	0.000
0.09	12.037	12.037	0.000
0.10	11.987	11.987	0.000
0.15	11.885	11.885	0.000
0.20	11.787	11.787	0.000
0.25	11.693	11.693	0.000
0.30	11.693	11.693	0.000

3-Phase fault at bus: **Bus48**

Nominal kV = 0.400
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 111.106 kA Method C
Steady State = 53.880 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus48	Total	0.00	17.861	-53.729	3.0	56.620
Bus38	Bus48	6.10	9.633	-29.034	3.0	30.590
Bus_LVDB-2	Bus48	2.33	8.229	-24.694	3.0	26.029

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	56.213	63.283	29.066
0.02	56.143	57.214	11.017
0.03	56.073	56.222	4.087
0.04	56.005	56.026	1.516
0.05	55.939	55.942	0.600
0.06	55.905	55.905	0.225
0.07	55.871	55.871	0.085
0.08	55.837	55.837	0.032
0.09	55.804	55.804	0.012
0.10	55.772	55.772	0.005
0.15	55.720	55.720	0.000
0.20	55.670	55.670	0.000
0.25	55.622	55.622	0.000
0.30	55.622	55.622	0.000

3-Phase fault at bus: **Bus49**

Nominal kV = 3.300
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 18.211 kA Method C
Steady State = 6.414 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus49	Total	0.00	1.970	-8.308	4.2	8.538
S003-SB-11D001 [BUS B]	Bus49	0.45	0.511	-1.882	3.7	1.951
Bus163	Bus49	83.79	1.459	-6.425	4.4	6.589

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	8.363	10.219	5.873
0.02	8.214	8.719	2.922
0.03	8.042	8.169	1.438
0.04	7.874	7.906	0.707
0.05	7.713	7.721	0.358
0.06	7.635	7.637	0.177
0.07	7.559	7.560	0.088
0.08	7.485	7.485	0.043
0.09	7.413	7.413	0.021
0.10	7.342	7.342	0.011
0.15	7.199	7.199	0.000
0.20	7.063	7.063	0.000
0.25	6.935	6.935	0.000
0.30	6.935	6.935	0.000

3-Phase fault at bus: **Bus50**

Nominal kV = 3.300
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 17.197 kA Method C
Steady State = 6.465 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus50	Total	0.00	1.778	-7.814	4.4	8.014
S003-SB-11D001 [BUS A]	Bus50	0.32	0.293	-1.346	4.6	1.378
Bus168	Bus50	83.70	1.485	-6.468	4.4	6.636

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	7.821	9.612	5.588
0.02	7.690	8.177	2.781
0.03	7.561	7.686	1.378
0.04	7.437	7.468	0.683
0.05	7.318	7.326	0.343
0.06	7.260	7.262	0.171
0.07	7.204	7.205	0.085
0.08	7.149	7.149	0.042
0.09	7.095	7.095	0.021
0.10	7.043	7.043	0.011
0.15	6.940	6.940	0.000
0.20	6.842	6.842	0.000
0.25	6.750	6.750	0.000
0.30	6.750	6.750	0.000

3-Phase fault at bus: **Bus51**

Nominal kV = 11.000
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 44.311 kA Method C
Steady State = 18.915 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus51	Total	0.00	8.703	-21.818	2.5	23.490
LV SWBD_TOFS	Bus51	4.20	0.017	-0.040	2.4	0.044
CAMP BUS	MV RMU-TOFS	2.17	0.081	-0.693	8.6	0.697
S002-SB-1AC001 [BUS A]	MV RMU-TOFS	38.98	8.606	-21.085	2.5	22.774
MV RMU-TOFS	Bus51	0.00	8.687	-21.778	2.5	23.446

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Ide
0.01	23.147	25.460	10.604
0.02	22.963	23.321	4.073
0.03	22.751	22.796	1.426
0.04	22.501	22.507	0.499
0.05	22.186	22.188	0.244
0.06	22.026	22.026	0.091
0.07	21.867	21.867	0.034
0.08	21.711	21.711	0.013
0.09	21.557	21.557	0.005
0.10	21.406	21.406	0.003
0.15	21.080	21.080	0.000
0.20	20.766	20.766	0.000
0.25	20.465	20.465	0.000
0.30	20.465	20.465	0.000

3-Phase fault at bus: **Bus52**

Nominal kV = 0.400
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 110.376 kA Method C
Steady State = 53.170 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus52	Total	0.00	17.869	-53.408	3.0	56.318
Bus_LVDB-1	Bus52	2.53	8.961	-26.710	3.0	28.173
Bus65	Bus52	7.48	8.908	-26.698	3.0	28.145

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	55.813	62.799	28.786
0.02	55.718	56.768	10.871
0.03	55.624	55.769	4.016
0.04	55.534	55.554	1.484
0.05	55.446	55.450	0.585
0.06	55.402	55.402	0.219
0.07	55.358	55.358	0.082
0.08	55.314	55.314	0.031
0.09	55.272	55.272	0.011
0.10	55.230	55.230	0.005
0.15	55.162	55.162	0.000
0.20	55.097	55.097	0.000
0.25	55.035	55.035	0.000
0.30	55.035	55.035	0.000

3-Phase fault at bus: **Bus53**

Nominal kV

Voltage c Factor

Peak Value

Steady State

= 0.380

= 1.05

= 4.122

= 2.410

(Maximum If)

kA Method C

kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus53	Total	0.00	2.395	-1.227	0.5	2.691
TYPICAL MOTOR FEEDER 1 A	Bus53	102.69	2.297	-0.749	0.3	2.416
614-FM-06AC	Bus53	105.00	0.098	-0.478	4.9	0.488

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.567	2.612	0.480
0.02	2.542	2.549	0.187
0.03	2.518	2.519	0.041
0.04	2.496	2.497	0.009
0.05	2.476	2.476	0.005
0.06	2.467	2.467	0.001
0.07	2.457	2.457	0.000
0.08	2.448	2.448	0.000
0.09	2.440	2.440	0.000
0.10	2.432	2.432	0.000
0.15	2.418	2.418	0.000
0.20	2.406	2.406	0.000
0.25	2.395	2.395	0.000
0.30	2.395	2.395	0.000

3-Phase fault at bus: **Bus54**

Nominal kV

Voltage c Factor

Peak Value

Steady State

= 0.380

= 1.05

= 4.122

= 2.410

(Maximum If)

kA Method C

kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus54	Total	0.00	2.395	-1.227	0.5	2.691
TYPICAL MOTOR FEEDER 1 A	Bus54	102.69	2.297	-0.749	0.3	2.416
614-FM-06BA	Bus54	105.00	0.098	-0.478	4.9	0.488

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.567	2.612	0.480
0.02	2.542	2.549	0.187
0.03	2.518	2.519	0.041
0.04	2.496	2.497	0.009
0.05	2.476	2.476	0.005
0.06	2.467	2.467	0.001
0.07	2.457	2.457	0.000
0.08	2.448	2.448	0.000
0.09	2.440	2.440	0.000
0.10	2.432	2.432	0.000
0.15	2.418	2.418	0.000
0.20	2.406	2.406	0.000
0.25	2.395	2.395	0.000
0.30	2.395	2.395	0.000

3-Phase fault at bus: **Bus55**

Nominal kV

Voltage c Factor

Peak Value

Steady State

= 11.000

= 1.10

= 18.078

= 9.485

(Maximum If)

kA Method C

kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus55	Total	0.00	6.465	-9.016	1.4	11.095
S003-SB-11C001 [BUS B]	Bus55	0.96	6.385	-8.812	1.4	10.882
S003-SB-11N001 [BUS B]	Bus55	9.45	0.080	-0.205	2.6	0.220

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	10.975	11.183	2.143
0.02	10.908	10.915	0.399
0.03	10.836	10.836	0.064
0.04	10.766	10.766	0.010
0.05	10.699	10.699	0.002
0.06	10.666	10.666	0.000
0.07	10.634	10.634	0.000
0.08	10.602	10.602	0.000
0.09	10.571	10.571	0.000
0.10	10.541	10.541	0.000
0.15	10.481	10.481	0.000
0.20	10.424	10.424	0.000
0.25	10.370	10.370	0.000
0.30	10.370	10.370	0.000

Project: UQ LT LPG Export - ST & MT
Location: Umm Qasr, Iraq
Contract: UQ83
Engineer: Hemanathan Gopalan
Filename: Umm Qasr-ST

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3-Phase fault at bus: **Bus56**

Nominal kV = 11.000
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 17.586 kA Method C
Steady State = 9.485 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus56	Total	0.00	6.450	-8.757	1.4	10.876
S003-SB-11C001 [BUS A]	Bus56	1.42	6.398	-8.633	1.3	10.746
S003-SB-11N001 [BUS A]	Bus56	5.75	0.051	-0.123	2.4	0.134

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	10.773	10.948	1.950
0.02	10.715	10.721	0.338
0.03	10.657	10.657	0.050
0.04	10.596	10.596	0.007
0.05	10.535	10.535	0.002
0.06	10.506	10.506	0.000
0.07	10.477	10.477	0.000
0.08	10.448	10.448	0.000
0.09	10.420	10.420	0.000
0.10	10.393	10.393	0.000
0.15	10.341	10.341	0.000
0.20	10.289	10.289	0.000
0.25	10.235	10.235	0.000
0.30	10.235	10.235	0.000

Project: UQ LT LPG Export - ST & MT
Location: Umm Qasr, Iraq
Contract: UQ83
Engineer: Hemanathan Gopalan
Filename: Umm Qasr-ST

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3-Phase fault at bus: **Bus57**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 7.485 kA Method C
Steady State = 0.592 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus57	Total	0.00	1.596	-3.192	2.0	3.569
Chiller-9	Bus57	96.14	1.004	-3.192	3.2	3.346
VFD1	Bus57	96.14	0.592	0.000	0.0	0.592

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.933	3.421	1.761
0.02	2.610	2.691	0.655
0.03	2.299	2.312	0.244
0.04	2.005	2.007	0.091
0.05	1.730	1.731	0.034
0.06	1.599	1.599	0.013
0.07	1.473	1.473	0.005
0.08	1.353	1.353	0.002
0.09	1.238	1.238	0.001
0.10	1.130	1.130	0.000
0.15	0.936	0.936	0.000
0.20	0.773	0.773	0.000
0.25	0.650	0.650	0.000
0.30	0.650	0.650	0.000

3-Phase fault at bus: **Bus58**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 11.972 kA Method C
Steady State = 0.935 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus58	Total	0.00	0.270	-5.139	19.0	5.146
S003-A-77301A	Bus58	105.00	0.270	-5.139	19.0	5.146

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	4.613	7.703	6.169
0.02	4.408	6.839	5.229
0.03	4.198	6.104	4.432
0.04	3.983	5.475	3.757
0.05	3.764	4.930	3.184
0.06	3.675	4.560	2.699
0.07	3.586	4.254	2.287
0.08	3.497	3.998	1.939
0.09	3.406	3.782	1.643
0.10	3.316	3.597	1.393
0.15	3.200	3.257	0.609
0.20	3.084	3.095	0.267
0.25	2.969	2.971	0.117
0.30	2.969	2.969	0.051

Project: UQ LT LPG Export - ST & MT
Location: Umm Qasr, Iraq
Contract: UQ83
Engineer: Hemanathan Gopalan
Filename: Umm Qasr-ST

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3-Phase fault at bus: **Bus59**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 7.485 kA Method C
Steady State = 0.592 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus59	Total	0.00	1.596	-3.192	2.0	3.569
Chiller-10	Bus59	96.14	1.004	-3.192	3.2	3.346
VFD2	Bus59	96.14	0.592	0.000	0.0	0.592

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.933	3.421	1.761
0.02	2.610	2.691	0.655
0.03	2.299	2.312	0.244
0.04	2.005	2.007	0.091
0.05	1.730	1.731	0.034
0.06	1.599	1.599	0.013
0.07	1.473	1.473	0.005
0.08	1.353	1.353	0.002
0.09	1.238	1.238	0.001
0.10	1.130	1.130	0.000
0.15	0.936	0.936	0.000
0.20	0.773	0.773	0.000
0.25	0.650	0.650	0.000
0.30	0.650	0.650	0.000

Project: UQ LT LPG Export - ST & MT
Location: Umm Qasr, Iraq
Contract: UQ83
Engineer: Hemanathan Gopalan
Filename: Umm Qasr-ST

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3-Phase fault at bus: **Bus60**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 8.715 kA Method C
Steady State = 5.970 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus60	Total	0.00	5.921	-1.203	0.2	6.042
Bus_MDB-5	Bus60	90.32	5.883	-1.102	0.2	5.985
Pump-9	Bus60	96.14	0.038	-0.101	2.7	0.108

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	6.015	6.015	0.000
0.02	6.010	6.010	0.000
0.03	6.005	6.005	0.000
0.04	6.001	6.001	0.000
0.05	5.996	5.996	0.000
0.06	5.994	5.994	0.000
0.07	5.992	5.992	0.000
0.08	5.990	5.990	0.000
0.09	5.988	5.988	0.000
0.10	5.985	5.985	0.000
0.15	5.982	5.982	0.000
0.20	5.979	5.979	0.000
0.25	5.976	5.976	0.000
0.30	5.976	5.976	0.000

3-Phase fault at bus: **Bus61**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 29.723 kA Method C
Steady State = 2.287 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus61	Total	0.00	0.672	-12.760	19.0	12.777
S003-DG-1 [EXISTING]	Bus61	105.00	0.672	-12.760	19.0	12.777

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	11.455	19.126	15.316
0.02	10.949	16.983	12.982
0.03	10.427	15.159	11.003
0.04	9.893	13.596	9.326
0.05	9.350	12.244	7.905
0.06	9.131	11.325	6.700
0.07	8.910	10.566	5.679
0.08	8.688	9.932	4.814
0.09	8.465	9.397	4.080
0.10	8.240	8.937	3.458
0.15	7.951	8.093	1.513
0.20	7.662	7.691	0.662
0.25	7.376	7.382	0.290
0.30	7.376	7.378	0.127

3-Phase fault at bus: **Bus62**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 7.697 kA Method C
Steady State = 5.269 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus62	Total	0.00	5.241	-1.002	0.2	5.336
Bus_MDB-5	Bus62	91.07	5.203	-0.901	0.2	5.281
Pump-10	Bus62	96.14	0.038	-0.101	2.7	0.108

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	5.309	5.309	0.000
0.02	5.305	5.305	0.000
0.03	5.300	5.300	0.000
0.04	5.296	5.296	0.000
0.05	5.292	5.292	0.000
0.06	5.289	5.289	0.000
0.07	5.287	5.287	0.000
0.08	5.285	5.285	0.000
0.09	5.283	5.283	0.000
0.10	5.281	5.281	0.000
0.15	5.278	5.278	0.000
0.20	5.274	5.274	0.000
0.25	5.271	5.271	0.000
0.30	5.271	5.271	0.000

3-Phase fault at bus: **Bus63**

Nominal kV = 0.400
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 110.376 kA Method C
Steady State = 53.170 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus63	Total	0.00	17.869	-53.408	3.0	56.318
Bus_LVDB-1	Bus63	2.53	8.961	-26.710	3.0	28.173
Bus65	Bus63	7.48	8.908	-26.698	3.0	28.145

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	55.813	62.799	28.786
0.02	55.718	56.768	10.871
0.03	55.624	55.769	4.016
0.04	55.534	55.554	1.484
0.05	55.446	55.450	0.585
0.06	55.402	55.402	0.219
0.07	55.358	55.358	0.082
0.08	55.314	55.314	0.031
0.09	55.272	55.272	0.011
0.10	55.230	55.230	0.005
0.15	55.162	55.162	0.000
0.20	55.097	55.097	0.000
0.25	55.035	55.035	0.000
0.30	55.035	55.035	0.000

3-Phase fault at bus: **Bus64**

Nominal kV = 0.415
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 114.565 kA Method C
Steady State = 55.164 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus64	Total	0.00	18.540	-55.433	3.0	58.451
Bus_LVDB-1	Bus64	2.60	9.211	-27.519	3.0	29.020
Bus65	Bus64	7.74	9.212	-27.635	3.0	29.130
Bus82	Bus64	0.15	0.117	-0.278	2.4	0.302

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	57.917	65.172	29.885
0.02	57.816	58.907	11.287
0.03	57.717	57.867	4.171
0.04	57.621	57.642	1.541
0.05	57.529	57.532	0.607
0.06	57.482	57.482	0.227
0.07	57.436	57.436	0.085
0.08	57.390	57.390	0.032
0.09	57.346	57.346	0.012
0.10	57.302	57.302	0.005
0.15	57.231	57.231	0.000
0.20	57.164	57.164	0.000
0.25	57.100	57.100	0.000
0.30	57.100	57.100	0.000

3-Phase fault at bus: **Bus65**

Nominal kV = 0.400
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 121.036 kA Method C
Steady State = 56.763 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus65	Total	0.00	17.554	-57.637	3.3	60.251
Bus52	Bus65	0.10	0.137	-0.346	2.5	0.372
Bus63	Bus65	0.10	0.137	-0.346	2.5	0.372
Bus64	Bus65	0.11	0.167	-0.415	2.5	0.447
Bus78	Bus65	72.22	17.113	-56.531	3.3	59.064

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Ide
0.01	59.719	68.653	33.865
0.02	59.619	61.257	14.070
0.03	59.521	59.795	5.718
0.04	59.426	59.471	2.323
0.05	59.334	59.342	1.007
0.06	59.287	59.288	0.415
0.07	59.241	59.241	0.171
0.08	59.195	59.195	0.070
0.09	59.151	59.151	0.029
0.10	59.107	59.107	0.013
0.15	59.037	59.037	0.000
0.20	58.969	58.969	0.000
0.25	58.905	58.905	0.000
0.30	58.905	58.905	0.000

Project: UQ LT LPG Export - ST & MT
Location: Umm Qasr, Iraq
Contract: UQ83
Engineer: Hemanathan Gopalan
Filename: Umm Qasr-ST

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3-Phase fault at bus: **Bus66**

Nominal kV = 11.000
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 15.536 kA Method C
Steady State = 8.620 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus66	Total	0.00	6.006	-7.677	1.3	9.747
S003-SB-11C001 [BUS A]	Bus66	13.11	5.991	-7.639	1.3	9.709
Bus38	Bus66	1.36	0.015	-0.038	2.6	0.040

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	9.669	9.785	1.503
0.02	9.624	9.627	0.222
0.03	9.579	9.579	0.028
0.04	9.534	9.534	0.004
0.05	9.490	9.490	0.001
0.06	9.469	9.469	0.000
0.07	9.448	9.448	0.000
0.08	9.428	9.428	0.000
0.09	9.408	9.408	0.000
0.10	9.388	9.388	0.000
0.15	9.349	9.349	0.000
0.20	9.312	9.312	0.000
0.25	9.277	9.277	0.000
0.30	9.277	9.277	0.000

3-Phase fault at bus: **Bus67**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 116.165 kA Method C
Steady State = 44.482 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus67	Total	0.00	10.467	-51.289	4.9	52.346
S002-SB-03N001 [BUS A]	Bus67	1.64	2.804	-6.683	2.4	7.248
Bus112	Bus67	94.07	7.663	-44.606	5.8	45.259

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	51.798	66.145	41.136
0.02	51.420	56.674	23.831
0.03	50.721	52.492	13.521
0.04	50.046	50.630	7.671
0.05	49.400	49.610	4.557
0.06	49.092	49.162	2.609
0.07	48.792	48.815	1.494
0.08	48.499	48.506	0.856
0.09	48.214	48.216	0.490
0.10	47.936	47.937	0.293
0.15	47.369	47.369	0.018
0.20	46.833	46.833	0.001
0.25	46.328	46.328	0.000
0.30	46.328	46.328	0.000

3-Phase fault at bus: **Bus68**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 7.485 kA Method C
Steady State = 0.592 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus68	Total	0.00	1.596	-3.192	2.0	3.569
Chiller-11	Bus68	96.14	1.004	-3.192	3.2	3.346
VFD9	Bus68	96.14	0.592	0.000	0.0	0.592

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.933	3.421	1.761
0.02	2.610	2.691	0.655
0.03	2.299	2.312	0.244
0.04	2.005	2.007	0.091
0.05	1.730	1.731	0.034
0.06	1.599	1.599	0.013
0.07	1.473	1.473	0.005
0.08	1.353	1.353	0.002
0.09	1.238	1.238	0.001
0.10	1.130	1.130	0.000
0.15	0.936	0.936	0.000
0.20	0.773	0.773	0.000
0.25	0.650	0.650	0.000
0.30	0.650	0.650	0.000

Project: UQ LT LPG Export - ST & MT
Location: Umm Qasr, Iraq
Contract: UQ83
Engineer: Hemanathan Gopalan
Filename: Umm Qasr-ST

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3-Phase fault at bus: **Bus69**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 111.237 kA Method C
Steady State = 44.773 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus69	Total	0.00	9.342	-48.776	5.2	49.662
S002-SB-03N001 [BUS B]	Bus69	0.95	1.625	-3.872	2.4	4.199
Bus8	Bus69	93.95	7.716	-44.904	5.8	45.562

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	49.290	63.384	39.850
0.02	48.880	54.120	23.234
0.03	48.481	50.289	13.363
0.04	48.097	48.707	7.686
0.05	47.731	47.948	4.560
0.06	47.556	47.629	2.639
0.07	47.384	47.409	1.527
0.08	47.218	47.226	0.884
0.09	47.056	47.058	0.512
0.10	46.898	46.899	0.306
0.15	46.585	46.585	0.020
0.20	46.289	46.289	0.001
0.25	46.011	46.011	0.000
0.30	46.011	46.011	0.000

3-Phase fault at bus: **Bus70**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 7.485 kA Method C
Steady State = 0.592 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus70	Total	0.00	1.596	-3.192	2.0	3.569
Chiller-12	Bus70	96.14	1.004	-3.192	3.2	3.346
VFD10	Bus70	96.14	0.592	0.000	0.0	0.592

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.933	3.421	1.761
0.02	2.610	2.691	0.655
0.03	2.299	2.312	0.244
0.04	2.005	2.007	0.091
0.05	1.730	1.731	0.034
0.06	1.599	1.599	0.013
0.07	1.473	1.473	0.005
0.08	1.353	1.353	0.002
0.09	1.238	1.238	0.001
0.10	1.130	1.130	0.000
0.15	0.936	0.936	0.000
0.20	0.773	0.773	0.000
0.25	0.650	0.650	0.000
0.30	0.650	0.650	0.000

3-Phase fault at bus: **Bus71**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 28.447 kA Method C
Steady State = 3.653 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus71	Total	0.00	0.643	-12.212	19.0	12.229
S003-EDG-11E001	Bus71	105.00	0.643	-12.212	19.0	12.229

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	11.244	18.474	14.659
0.02	10.868	16.507	12.425
0.03	10.465	14.846	10.531
0.04	10.036	13.431	8.926
0.05	9.586	12.212	7.566
0.06	9.416	11.392	6.413
0.07	9.242	10.722	5.435
0.08	9.065	10.169	4.607
0.09	8.886	9.706	3.905
0.10	8.704	9.312	3.310
0.15	8.424	8.547	1.448
0.20	8.136	8.161	0.633
0.25	7.843	7.848	0.277
0.30	7.843	7.844	0.121

Project: UQ LT LPG Export - ST & MT
Location: Umm Qasr, Iraq
Contract: UQ83
Engineer: Hemanathan Gopalan
Filename: Umm Qasr-ST

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3-Phase fault at bus: **Bus72**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 11.055 kA Method C
Steady State = 7.578 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus72	Total	0.00	7.464	-1.735	0.2	7.663
Bus_MDB-6	Bus72	91.84	7.427	-1.634	0.2	7.604
Pump-11	Bus72	96.14	0.038	-0.101	2.7	0.108

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	7.635	7.635	0.000
0.02	7.630	7.630	0.000
0.03	7.625	7.625	0.000
0.04	7.621	7.621	0.000
0.05	7.616	7.616	0.000
0.06	7.614	7.614	0.000
0.07	7.611	7.611	0.000
0.08	7.609	7.609	0.000
0.09	7.607	7.607	0.000
0.10	7.604	7.604	0.000
0.15	7.601	7.601	0.000
0.20	7.597	7.597	0.000
0.25	7.594	7.594	0.000
0.30	7.594	7.594	0.000

3-Phase fault at bus: **Bus73**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 9.467 kA Method C
Steady State = 6.486 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus73	Total	0.00	6.419	-1.363	0.2	6.563
Bus_MDB-6	Bus73	93.12	6.382	-1.262	0.2	6.505
Pump-12	Bus73	96.14	0.038	-0.101	2.7	0.108

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	6.535	6.535	0.000
0.02	6.530	6.530	0.000
0.03	6.526	6.526	0.000
0.04	6.521	6.521	0.000
0.05	6.517	6.517	0.000
0.06	6.514	6.514	0.000
0.07	6.512	6.512	0.000
0.08	6.510	6.510	0.000
0.09	6.508	6.508	0.000
0.10	6.505	6.505	0.000
0.15	6.502	6.502	0.000
0.20	6.499	6.499	0.000
0.25	6.495	6.495	0.000
0.30	6.495	6.495	0.000

Project: UQ LT LPG Export - ST & MT
Location: Umm Qasr, Iraq
Contract: UQ83
Engineer: Hemanathan Gopalan
Filename: Umm Qasr-ST

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3-Phase fault at bus: **Bus74**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 7.485 kA Method C
Steady State = 0.592 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus74	Total	0.00	1.596	-3.192	2.0	3.569
Chiller-13	Bus74	96.14	1.004	-3.192	3.2	3.346
VFD13	Bus74	96.14	0.592	0.000	0.0	0.592

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.933	3.421	1.761
0.02	2.610	2.691	0.655
0.03	2.299	2.312	0.244
0.04	2.005	2.007	0.091
0.05	1.730	1.731	0.034
0.06	1.599	1.599	0.013
0.07	1.473	1.473	0.005
0.08	1.353	1.353	0.002
0.09	1.238	1.238	0.001
0.10	1.130	1.130	0.000
0.15	0.936	0.936	0.000
0.20	0.773	0.773	0.000
0.25	0.650	0.650	0.000
0.30	0.650	0.650	0.000

3-Phase fault at bus: **Bus75**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 7.485 kA Method C
Steady State = 0.592 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus75	Total	0.00	1.596	-3.192	2.0	3.569
Chiller-14	Bus75	96.14	1.004	-3.192	3.2	3.346
VFD14	Bus75	96.14	0.592	0.000	0.0	0.592

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.933	3.421	1.761
0.02	2.610	2.691	0.655
0.03	2.299	2.312	0.244
0.04	2.005	2.007	0.091
0.05	1.730	1.731	0.034
0.06	1.599	1.599	0.013
0.07	1.473	1.473	0.005
0.08	1.353	1.353	0.002
0.09	1.238	1.238	0.001
0.10	1.130	1.130	0.000
0.15	0.936	0.936	0.000
0.20	0.773	0.773	0.000
0.25	0.650	0.650	0.000
0.30	0.650	0.650	0.000

3-Phase fault at bus: **Bus76**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 8.715 kA Method C
Steady State = 5.970 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus76	Total	0.00	5.921	-1.203	0.2	6.042
Bus_MDB-7	Bus76	93.71	5.883	-1.102	0.2	5.985
Pump-13	Bus76	96.14	0.038	-0.101	2.7	0.108

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	6.015	6.015	0.000
0.02	6.010	6.010	0.000
0.03	6.005	6.005	0.000
0.04	6.001	6.001	0.000
0.05	5.996	5.996	0.000
0.06	5.994	5.994	0.000
0.07	5.992	5.992	0.000
0.08	5.990	5.990	0.000
0.09	5.988	5.988	0.000
0.10	5.985	5.985	0.000
0.15	5.982	5.982	0.000
0.20	5.979	5.979	0.000
0.25	5.976	5.976	0.000
0.30	5.976	5.976	0.000

3-Phase fault at bus: **Bus77**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 6.752 kA Method C
Steady State = 4.617 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus77	Total	0.00	4.606	-0.833	0.2	4.681
Bus_MDB-7	Bus77	95.20	4.568	-0.732	0.2	4.626
Pump-14	Bus77	96.14	0.038	-0.101	2.7	0.108

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	4.654	4.654	0.000
0.02	4.650	4.650	0.000
0.03	4.646	4.646	0.000
0.04	4.641	4.641	0.000
0.05	4.637	4.637	0.000
0.06	4.635	4.635	0.000
0.07	4.633	4.633	0.000
0.08	4.631	4.631	0.000
0.09	4.629	4.629	0.000
0.10	4.627	4.627	0.000
0.15	4.623	4.623	0.000
0.20	4.620	4.620	0.000
0.25	4.617	4.617	0.000
0.30	4.617	4.617	0.000

3-Phase fault at bus: **Bus78**

Nominal kV

Voltage c Factor

Peak Value

Steady State

= 11.000

= 1.10

= 15.854

= 8.593

(Maximum If)

kA Method C

kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus78	Total	0.00	6.018	-7.850	1.3	9.891
S003-SB-11C001 [BUS B]	Bus78	13.29	5.998	-7.800	1.3	9.840
Bus65	Bus78	1.79	0.020	-0.050	2.5	0.053

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	9.799	9.932	1.623
0.02	9.748	9.751	0.255
0.03	9.693	9.693	0.034
0.04	9.640	9.640	0.005
0.05	9.588	9.588	0.001
0.06	9.563	9.563	0.000
0.07	9.538	9.538	0.000
0.08	9.514	9.514	0.000
0.09	9.490	9.490	0.000
0.10	9.467	9.467	0.000
0.15	9.421	9.421	0.000
0.20	9.377	9.377	0.000
0.25	9.334	9.334	0.000
0.30	9.334	9.334	0.000

3-Phase fault at bus: **Bus79**

Nominal kV = 132.000
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 45.690 kA Method C
Steady State = 18.060 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus79	Total	0.00	2.395	-18.943	7.9	19.094
TR-PSS1AC01_132kV BUS	Bus79	0.03	0.089	-0.497	5.6	0.505
U1 [Max]	132kV GRID	110.00	2.213	-17.924	8.1	18.060
TR-PSS1AC02_132kV BUS	Bus81	0.03	0.093	-0.523	5.6	0.531
S002-SB-PSS1A01 [BUS-A]	Bus79	0.00	2.306	-18.446	8.0	18.590
S002-SB-PSS1A01 [BUS-A]	Bus80	0.00	0.000	0.000	9999.0	0.000
Bus81	S002-SB-PSS1A01 [BUS-A]	0.00	0.093	-0.523	5.6	0.531
132kV GRID	Bus83	0.00	1.106	-8.962	8.1	9.030
Bus83	S002-SB-PSS1A01 [BUS-A]	0.00	1.106	-8.962	8.1	9.030
132kV GRID	Bus84	0.00	1.106	-8.962	8.1	9.030
Bus84	S002-SB-PSS1A01 [BUS-A]	0.00	1.106	-8.962	8.1	9.030
Bus87	S002-SB-PSS1A01 [BUS-A]	0.00	0.000	0.000	9999.0	0.000
Bus88	S002-SB-PSS1A01 [BUS-A]	0.00	0.000	0.000	9999.0	0.000

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	19.026	26.339	18.214
0.02	18.989	22.639	12.327
0.03	18.947	20.697	8.328
0.04	18.899	19.718	5.627
0.05	18.838	19.221	3.818
0.06	18.807	18.983	2.582
0.07	18.776	18.857	1.746
0.08	18.746	18.783	1.180
0.09	18.716	18.733	0.798
0.10	18.687	18.695	0.542
0.15	18.623	18.624	0.077
0.20	18.562	18.562	0.011
0.25	18.503	18.503	0.002

Project: UQ LT LPG Export - ST & MT
Location: Umm Qasr, Iraq
Contract: UQ83
Engineer: Hemanathan Gopalan
Filename: Umm Qasr-ST

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3-Phase fault at bus: **Bus79**

Nominal kV	=	132.000	
Voltage c Factor	=	1.10	(Maximum If)
Peak Value	=	45.690	kA Method C
Steady State	=	18.060	kA rms

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

<u>TD (S)</u>	<u>Ib sym</u>	<u>Ib asym</u>	<u>Idc</u>
0.30	18.503	18.503	0.000

Project: UQ LT LPG Export - ST & MT
Location: Umm Qasr, Iraq
Contract: UQ83
Engineer: Hemanathan Gopalan
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3-Phase fault at bus: **Bus80**

Nominal kV = 132.000
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 45.690 kA Method C
Steady State = 18.060 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus80	Total	0.00	2.395	-18.943	7.9	19.094
U1 [Max]	132kV GRID	110.00	2.213	-17.924	8.1	18.060
TR-PSS1AC02_132kV BUS	Bus81	0.03	0.093	-0.523	5.6	0.531
TR-PSS1AC01_132kV BUS	Bus79	0.03	0.089	-0.497	5.6	0.505
Bus79	S002-SB-PSS1A01 [BUS-A]	0.00	0.089	-0.497	5.6	0.505
S002-SB-PSS1A01 [BUS-A]	Bus80	0.00	2.395	-18.943	7.9	19.094
Bus81	S002-SB-PSS1A01 [BUS-A]	0.00	0.093	-0.523	5.6	0.531
132kV GRID	Bus83	0.00	1.106	-8.962	8.1	9.030
Bus83	S002-SB-PSS1A01 [BUS-A]	0.00	1.106	-8.962	8.1	9.030
132kV GRID	Bus84	0.00	1.106	-8.962	8.1	9.030
Bus84	S002-SB-PSS1A01 [BUS-A]	0.00	1.106	-8.962	8.1	9.030
Bus87	S002-SB-PSS1A01 [BUS-A]	0.00	0.000	0.000	9999.0	0.000
Bus88	S002-SB-PSS1A01 [BUS-A]	0.00	0.000	0.000	9999.0	0.000

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	19.026	26.339	18.214
0.02	18.989	22.639	12.327
0.03	18.947	20.697	8.328
0.04	18.899	19.718	5.627
0.05	18.838	19.221	3.818
0.06	18.807	18.983	2.582
0.07	18.776	18.857	1.746
0.08	18.746	18.783	1.180
0.09	18.716	18.733	0.798
0.10	18.687	18.695	0.542
0.15	18.623	18.624	0.077
0.20	18.562	18.562	0.011
0.25	18.503	18.503	0.002

Project: UQ LT LPG Export - ST & MT
Location: Umm Qasr, Iraq
Contract: UQ83
Engineer: Hemanathan Gopalan
Filename: Umm Qasr-ST

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3-Phase fault at bus: **Bus80**

Nominal kV	=	132.000	
Voltage c Factor	=	1.10	(Maximum If)
Peak Value	=	45.690	kA Method C
Steady State	=	18.060	kA rms

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

<u>TD (S)</u>	<u>Ib sym</u>	<u>Ib asym</u>	<u>Idc</u>
0.30	18.503	18.503	0.000

3-Phase fault at bus: **Bus81**

Nominal kV = 132.000
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 45.690 kA Method C
Steady State = 18.060 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus81	Total	0.00	2.395	-18.943	7.9	19.094
TR-PSS1AC02_132kV BUS	Bus81	0.03	0.093	-0.523	5.6	0.531
U1 [Max]	132kV GRID	110.00	2.213	-17.924	8.1	18.060
TR-PSS1AC01_132kV BUS	Bus79	0.03	0.089	-0.497	5.6	0.505
Bus79	S002-SB-PSS1A01 [BUS-A]	0.00	0.089	-0.497	5.6	0.505
S002-SB-PSS1A01 [BUS-A]	Bus80	0.00	0.000	0.000	9999.0	0.000
S002-SB-PSS1A01 [BUS-A]	Bus81	0.00	2.302	-18.421	8.0	18.564
132kV GRID	Bus83	0.00	1.106	-8.962	8.1	9.030
Bus83	S002-SB-PSS1A01 [BUS-A]	0.00	1.106	-8.962	8.1	9.030
132kV GRID	Bus84	0.00	1.106	-8.962	8.1	9.030
Bus84	S002-SB-PSS1A01 [BUS-A]	0.00	1.106	-8.962	8.1	9.030
Bus87	S002-SB-PSS1A01 [BUS-A]	0.00	0.000	0.000	9999.0	0.000
Bus88	S002-SB-PSS1A01 [BUS-A]	0.00	0.000	0.000	9999.0	0.000

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	19.026	26.339	18.214
0.02	18.989	22.639	12.327
0.03	18.947	20.697	8.328
0.04	18.899	19.718	5.627
0.05	18.838	19.221	3.818
0.06	18.807	18.983	2.582
0.07	18.776	18.857	1.746
0.08	18.746	18.783	1.180
0.09	18.716	18.733	0.798
0.10	18.687	18.695	0.542
0.15	18.623	18.624	0.077
0.20	18.562	18.562	0.011
0.25	18.503	18.503	0.002

Project: UQ LT LPG Export - ST & MT
Location: Umm Qasr, Iraq
Contract: UQ83
Engineer: Hemanathan Gopalan
Filename: Umm Qasr-ST

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3-Phase fault at bus: **Bus81**

Nominal kV	=	132.000	
Voltage c Factor	=	1.10	(Maximum If)
Peak Value	=	45.690	kA Method C
Steady State	=	18.060	kA rms

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

<u>TD (S)</u>	<u>Ib sym</u>	<u>Ib asym</u>	<u>Idc</u>
0.30	18.503	18.503	0.000

3-Phase fault at bus: **Bus82**

Nominal kV = 0.415
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 84.661 kA Method C
Steady State = 46.822 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus82	Total	0.00	23.535	-43.246	1.8	49.236
Bus64	Bus82	23.78	23.419	-42.968	1.8	48.935
Lump27	200A MDB Agreko	108.94	0.117	-0.278	2.4	0.302
200A MDB Agreko	Bus82	0.00	0.117	-0.278	2.4	0.302

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	48.837	50.571	13.127
0.02	48.760	48.830	2.614
0.03	48.685	48.687	0.506
0.04	48.611	48.611	0.098
0.05	48.540	48.540	0.021
0.06	48.503	48.503	0.004
0.07	48.467	48.467	0.001
0.08	48.432	48.432	0.000
0.09	48.397	48.397	0.000
0.10	48.363	48.363	0.000
0.15	48.307	48.307	0.000
0.20	48.253	48.253	0.000
0.25	48.201	48.201	0.000
0.30	48.201	48.201	0.000

3-Phase fault at bus: **Bus83**

Nominal kV = 132.000
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 45.690 kA Method C
Steady State = 18.060 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus83	Total	0.00	2.395	-18.943	7.9	19.094
TR-PSS1AC02_132kV BUS	Bus81	0.03	0.093	-0.523	5.6	0.531
TR-PSS1AC01_132kV BUS	Bus79	0.03	0.089	-0.497	5.6	0.505
U1 [Max]	132kV GRID	110.00	2.213	-17.924	8.1	18.060
Bus79	S002-SB-PSS1A01 [BUS-A]	0.00	0.089	-0.497	5.6	0.505
S002-SB-PSS1A01 [BUS-A]	Bus80	0.00	0.000	0.000	9999.0	0.000
Bus81	S002-SB-PSS1A01 [BUS-A]	0.00	0.093	-0.523	5.6	0.531
132kV GRID	Bus83	0.00	2.304	-18.434	8.0	18.577
S002-SB-PSS1A01 [BUS-A]	Bus83	0.00	0.091	-0.510	5.6	0.518
Bus84	132kV GRID	0.00	0.091	-0.510	5.6	0.518
S002-SB-PSS1A01 [BUS-A]	Bus84	0.00	0.091	-0.510	5.6	0.518
Bus87	S002-SB-PSS1A01 [BUS-A]	0.00	0.000	0.000	9999.0	0.000
Bus88	S002-SB-PSS1A01 [BUS-A]	0.00	0.000	0.000	9999.0	0.000

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	19.026	26.339	18.214
0.02	18.989	22.639	12.327
0.03	18.947	20.697	8.328
0.04	18.899	19.718	5.627
0.05	18.838	19.221	3.818
0.06	18.807	18.983	2.582
0.07	18.776	18.857	1.746
0.08	18.746	18.783	1.180
0.09	18.716	18.733	0.798
0.10	18.687	18.695	0.542
0.15	18.623	18.624	0.077
0.20	18.562	18.562	0.011
0.25	18.503	18.503	0.002

Project: UQ LT LPG Export - ST & MT
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3-Phase fault at bus: **Bus83**

Nominal kV	=	132.000	
Voltage c Factor	=	1.10	(Maximum If)
Peak Value	=	45.690	kA Method C
Steady State	=	18.060	kA rms

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

<u>TD (S)</u>	<u>Ib sym</u>	<u>Ib asym</u>	<u>Idc</u>
0.30	18.503	18.503	0.000

3-Phase fault at bus: **Bus84**

Nominal kV = 132.000
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 45.690 kA Method C
Steady State = 18.060 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus84	Total	0.00	2.395	-18.943	7.9	19.094
TR-PSS1AC02_132kV BUS	Bus81	0.03	0.093	-0.523	5.6	0.531
TR-PSS1AC01_132kV BUS	Bus79	0.03	0.089	-0.497	5.6	0.505
U1 [Max]	132kV GRID	110.00	2.213	-17.924	8.1	18.060
Bus79	S002-SB-PSS1A01 [BUS-A]	0.00	0.089	-0.497	5.6	0.505
S002-SB-PSS1A01 [BUS-A]	Bus80	0.00	0.000	0.000	9999.0	0.000
Bus81	S002-SB-PSS1A01 [BUS-A]	0.00	0.093	-0.523	5.6	0.531
Bus83	132kV GRID	0.00	0.091	-0.510	5.6	0.518
S002-SB-PSS1A01 [BUS-A]	Bus83	0.00	0.091	-0.510	5.6	0.518
132kV GRID	Bus84	0.00	2.304	-18.434	8.0	18.577
S002-SB-PSS1A01 [BUS-A]	Bus84	0.00	0.091	-0.510	5.6	0.518
Bus87	S002-SB-PSS1A01 [BUS-A]	0.00	0.000	0.000	9999.0	0.000
Bus88	S002-SB-PSS1A01 [BUS-A]	0.00	0.000	0.000	9999.0	0.000

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	19.026	26.339	18.214
0.02	18.989	22.639	12.327
0.03	18.947	20.697	8.328
0.04	18.899	19.718	5.627
0.05	18.838	19.221	3.818
0.06	18.807	18.983	2.582
0.07	18.776	18.857	1.746
0.08	18.746	18.783	1.180
0.09	18.716	18.733	0.798
0.10	18.687	18.695	0.542
0.15	18.623	18.624	0.077
0.20	18.562	18.562	0.011
0.25	18.503	18.503	0.002

Project: UQ LT LPG Export - ST & MT
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Contract: UQ83
Engineer: Hemanathan Gopalan
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3-Phase fault at bus: **Bus84**

Nominal kV	=	132.000	
Voltage c Factor	=	1.10	(Maximum If)
Peak Value	=	45.690	kA Method C
Steady State	=	18.060	kA rms

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

<u>TD (S)</u>	<u>Ib sym</u>	<u>Ib asym</u>	<u>Idc</u>
0.30	18.503	18.503	0.000

3-Phase fault at bus: **Bus85**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 4.175 kA Method C
Steady State = 2.449 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus85	Total	0.00	2.431	-1.241	0.5	2.729
TYPICAL MOTOR FEEDER 1 A	Bus85	102.65	2.333	-0.762	0.3	2.455
614-FM-06BB	Bus85	105.00	0.098	-0.478	4.9	0.488

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.606	2.649	0.478
0.02	2.581	2.587	0.185
0.03	2.557	2.557	0.041
0.04	2.535	2.535	0.009
0.05	2.515	2.515	0.005
0.06	2.505	2.505	0.001
0.07	2.496	2.496	0.000
0.08	2.487	2.487	0.000
0.09	2.479	2.479	0.000
0.10	2.470	2.470	0.000
0.15	2.457	2.457	0.000
0.20	2.445	2.445	0.000
0.25	2.434	2.434	0.000
0.30	2.434	2.434	0.000

3-Phase fault at bus: **Bus87**

Nominal kV = 132.000
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 45.690 kA Method C
Steady State = 18.060 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus87	Total	0.00	2.395	-18.943	7.9	19.094
U1 [Max]	132kV GRID	110.00	2.213	-17.924	8.1	18.060
TR-PSS1AC02_132kV BUS	Bus81	0.03	0.093	-0.523	5.6	0.531
TR-PSS1AC01_132kV BUS	Bus79	0.03	0.089	-0.497	5.6	0.505
Bus79	S002-SB-PSS1A01 [BUS-A]	0.00	0.089	-0.497	5.6	0.505
S002-SB-PSS1A01 [BUS-A]	Bus80	0.00	0.000	0.000	9999.0	0.000
Bus81	S002-SB-PSS1A01 [BUS-A]	0.00	0.093	-0.523	5.6	0.531
132kV GRID	Bus83	0.00	1.106	-8.962	8.1	9.030
Bus83	S002-SB-PSS1A01 [BUS-A]	0.00	1.106	-8.962	8.1	9.030
132kV GRID	Bus84	0.00	1.106	-8.962	8.1	9.030
Bus84	S002-SB-PSS1A01 [BUS-A]	0.00	1.106	-8.962	8.1	9.030
S002-SB-PSS1A01 [BUS-A]	Bus87	0.00	2.395	-18.943	7.9	19.094
Bus88	S002-SB-PSS1A01 [BUS-A]	0.00	0.000	0.000	9999.0	0.000

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	19.026	26.339	18.214
0.02	18.989	22.639	12.327
0.03	18.947	20.697	8.328
0.04	18.899	19.718	5.627
0.05	18.838	19.221	3.818
0.06	18.807	18.983	2.582
0.07	18.776	18.857	1.746
0.08	18.746	18.783	1.180
0.09	18.716	18.733	0.798
0.10	18.687	18.695	0.542
0.15	18.623	18.624	0.077
0.20	18.562	18.562	0.011
0.25	18.503	18.503	0.002

Project: UQ LT LPG Export - ST & MT
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Contract: UQ83
Engineer: Hemanathan Gopalan
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3-Phase fault at bus: **Bus87**

Nominal kV	=	132.000	
Voltage c Factor	=	1.10	(Maximum If)
Peak Value	=	45.690	kA Method C
Steady State	=	18.060	kA rms

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

<u>TD (S)</u>	<u>Ib sym</u>	<u>Ib asym</u>	<u>Idc</u>
0.30	18.503	18.503	0.000

3-Phase fault at bus: **Bus88**

Nominal kV = 132.000
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 45.690 kA Method C
Steady State = 18.060 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus88	Total	0.00	2.395	-18.943	7.9	19.094
U1 [Max]	132kV GRID	110.00	2.213	-17.924	8.1	18.060
TR-PSS1AC02_132kV BUS	Bus81	0.03	0.093	-0.523	5.6	0.531
TR-PSS1AC01_132kV BUS	Bus79	0.03	0.089	-0.497	5.6	0.505
Bus79	S002-SB-PSS1A01 [BUS-A]	0.00	0.089	-0.497	5.6	0.505
S002-SB-PSS1A01 [BUS-A]	Bus80	0.00	0.000	0.000	9999.0	0.000
Bus81	S002-SB-PSS1A01 [BUS-A]	0.00	0.093	-0.523	5.6	0.531
132kV GRID	Bus83	0.00	1.106	-8.962	8.1	9.030
Bus83	S002-SB-PSS1A01 [BUS-A]	0.00	1.106	-8.962	8.1	9.030
132kV GRID	Bus84	0.00	1.106	-8.962	8.1	9.030
Bus84	S002-SB-PSS1A01 [BUS-A]	0.00	1.106	-8.962	8.1	9.030
Bus87	S002-SB-PSS1A01 [BUS-A]	0.00	0.000	0.000	9999.0	0.000
S002-SB-PSS1A01 [BUS-A]	Bus88	0.00	2.395	-18.943	7.9	19.094

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	19.026	26.339	18.214
0.02	18.989	22.639	12.327
0.03	18.947	20.697	8.328
0.04	18.899	19.718	5.627
0.05	18.838	19.221	3.818
0.06	18.807	18.983	2.582
0.07	18.776	18.857	1.746
0.08	18.746	18.783	1.180
0.09	18.716	18.733	0.798
0.10	18.687	18.695	0.542
0.15	18.623	18.624	0.077
0.20	18.562	18.562	0.011
0.25	18.503	18.503	0.002

Project:

Location:

Contract:

Engineer:

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UQ LT LPG Export - ST & MT

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3-Phase fault at bus: **Bus88**

Nominal kV

Voltage c Factor

Peak Value

Steady State

=

=

=

=

132.000

1.10

45.690

18.060

(Maximum If)

kA Method C

kA rms

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

<u>TD (S)</u>	<u>Ib sym</u>	<u>Ib asym</u>	<u>Idc</u>
0.30	18.503	18.503	0.000

3-Phase fault at bus: **Bus89**

Nominal kV = 11.000
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 51.297 kA Method C
Steady State = 19.516 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus89	Total	0.00	6.160	-23.649	3.8	24.438
S002-SB-1AC001 [BUS A]	Bus89	27.76	6.126	-23.463	3.8	24.250
S002-40-MP-5C	Bus89	110.00	0.034	-0.186	5.4	0.189

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	24.058	29.027	16.242
0.02	23.849	25.276	8.373
0.03	23.610	23.966	4.121
0.04	23.343	23.431	2.028
0.05	23.007	23.037	1.170
0.06	22.835	22.842	0.595
0.07	22.665	22.667	0.302
0.08	22.497	22.497	0.153
0.09	22.332	22.332	0.078
0.10	22.169	22.169	0.048
0.15	21.820	21.820	0.002
0.20	21.482	21.482	0.000
0.25	21.159	21.159	0.000
0.30	21.159	21.159	0.000

Project: UQ LT LPG Export - ST & MT
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3-Phase fault at bus: **Bus90**

Nominal kV = 11.000
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 50.999 kA Method C
Steady State = 19.462 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus90	Total	0.00	6.191	-23.550	3.8	24.351
S002-SB-1AC001 [BUS A]	Bus90	28.13	6.157	-23.365	3.8	24.162
S002-40-MP-05A	Bus90	110.00	0.034	-0.186	5.4	0.189

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	23.974	28.858	16.063
0.02	23.767	25.149	8.221
0.03	23.530	23.870	4.017
0.04	23.266	23.348	1.963
0.05	22.932	22.960	1.125
0.06	22.761	22.768	0.567
0.07	22.592	22.594	0.286
0.08	22.426	22.426	0.144
0.09	22.262	22.262	0.073
0.10	22.101	22.101	0.045
0.15	21.754	21.754	0.002
0.20	21.418	21.418	0.000
0.25	21.097	21.097	0.000
0.30	21.097	21.097	0.000

3-Phase fault at bus: **Bus91**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 5.601 kA Method C
Steady State = 3.829 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus91	Total	0.00	3.684	-1.225	0.3	3.882
S002-SB-1AN001 [BUS A]	Bus91	84.52	3.684	-1.225	0.3	3.882

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	3.882	3.882	0.001
0.02	3.882	3.882	0.000
0.03	3.882	3.882	0.000
0.04	3.882	3.882	0.000
0.05	3.882	3.882	0.000
0.06	3.882	3.882	0.000
0.07	3.882	3.882	0.000
0.08	3.882	3.882	0.000
0.09	3.882	3.882	0.000
0.10	3.882	3.882	0.000
0.15	3.882	3.882	0.000
0.20	3.882	3.882	0.000
0.25	3.882	3.882	0.000
0.30	3.882	3.882	0.000

3-Phase fault at bus: **Bus92**

Nominal kV = 11.000

Voltage c Factor = 1.10 (Maximum If)

Peak Value = 71.816 kA Method C

Steady State = 22.133 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus92	Total	0.00	3.163	-28.927	9.1	29.099
S002-SB-1AC001 [BUS A]	Bus92	6.24	2.747	-26.903	9.8	27.043
S002-6100-MK-1D	Bus92	110.00	0.416	-2.024	4.9	2.066

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	28.473	41.825	30.636
0.02	28.145	37.233	24.374
0.03	27.776	33.517	18.759
0.04	27.370	30.945	14.437
0.05	26.859	29.591	12.419
0.06	26.598	28.336	9.773
0.07	26.341	27.441	7.691
0.08	26.090	26.782	6.052
0.09	25.843	26.278	4.763
0.10	25.602	25.960	4.300
0.15	25.091	25.130	1.390
0.20	24.603	24.607	0.449
0.25	24.139	24.139	0.145
0.30	24.139	24.139	0.047

3-Phase fault at bus: **Bus93**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 7.485 kA Method C
Steady State = 0.592 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus93	Total	0.00	1.596	-3.192	2.0	3.569
Chiller-1	Bus93	96.14	1.004	-3.192	3.2	3.346
VFD25	Bus93	96.14	0.592	0.000	0.0	0.592

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.933	3.421	1.761
0.02	2.610	2.691	0.655
0.03	2.299	2.312	0.244
0.04	2.005	2.007	0.091
0.05	1.730	1.731	0.034
0.06	1.599	1.599	0.013
0.07	1.473	1.473	0.005
0.08	1.353	1.353	0.002
0.09	1.238	1.238	0.001
0.10	1.130	1.130	0.000
0.15	0.936	0.936	0.000
0.20	0.773	0.773	0.000
0.25	0.650	0.650	0.000
0.30	0.650	0.650	0.000

3-Phase fault at bus: **Bus94**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 7.485 kA Method C
Steady State = 0.592 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus94	Total	0.00	1.596	-3.192	2.0	3.569
Chiller-2	Bus94	96.14	1.004	-3.192	3.2	3.346
VFD26	Bus94	96.14	0.592	0.000	0.0	0.592

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.933	3.421	1.761
0.02	2.610	2.691	0.655
0.03	2.299	2.312	0.244
0.04	2.005	2.007	0.091
0.05	1.730	1.731	0.034
0.06	1.599	1.599	0.013
0.07	1.473	1.473	0.005
0.08	1.353	1.353	0.002
0.09	1.238	1.238	0.001
0.10	1.130	1.130	0.000
0.15	0.936	0.936	0.000
0.20	0.773	0.773	0.000
0.25	0.650	0.650	0.000
0.30	0.650	0.650	0.000

Project: UQ LT LPG Export - ST & MT
Location: Umm Qasr, Iraq
Contract: UQ83
Engineer: Hemanathan Gopalan
Filename: Umm Qasr-ST

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3-Phase fault at bus: **Bus95**

Nominal kV = 0.400
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 12.458 kA Method C
Steady State = 8.531 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus95	Total	0.00	8.391	-2.046	0.2	8.636
Bus_MDB-1	Bus95	92.41	8.351	-1.939	0.2	8.573
Pump-1	Bus95	101.20	0.040	-0.106	2.7	0.114

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	8.606	8.606	0.000
0.02	8.601	8.601	0.000
0.03	8.595	8.595	0.000
0.04	8.590	8.590	0.000
0.05	8.586	8.586	0.000
0.06	8.583	8.583	0.000
0.07	8.581	8.581	0.000
0.08	8.578	8.578	0.000
0.09	8.576	8.576	0.000
0.10	8.573	8.573	0.000
0.15	8.569	8.569	0.000
0.20	8.566	8.566	0.000
0.25	8.562	8.562	0.000
0.30	8.562	8.562	0.000

Project: UQ LT LPG Export - ST & MT
Location: Umm Qasr, Iraq
Contract: UQ83
Engineer: Hemanathan Gopalan
Filename: Umm Qasr-ST

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3-Phase fault at bus: **Bus96**

Nominal kV = 0.400
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 10.562 kA Method C
Steady State = 7.232 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus96	Total	0.00	7.151	-1.576	0.2	7.322
Bus_MDB-1	Bus96	93.92	7.111	-1.469	0.2	7.261
Pump-2	Bus96	101.20	0.040	-0.106	2.7	0.114

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	7.293	7.293	0.000
0.02	7.288	7.288	0.000
0.03	7.283	7.283	0.000
0.04	7.278	7.278	0.000
0.05	7.273	7.273	0.000
0.06	7.271	7.271	0.000
0.07	7.268	7.268	0.000
0.08	7.266	7.266	0.000
0.09	7.264	7.264	0.000
0.10	7.261	7.261	0.000
0.15	7.258	7.258	0.000
0.20	7.254	7.254	0.000
0.25	7.251	7.251	0.000
0.30	7.251	7.251	0.000

3-Phase fault at bus: **Bus97**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 7.485 kA Method C
Steady State = 0.592 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus97	Total	0.00	1.596	-3.192	2.0	3.569
Chiller-3	Bus97	96.14	1.004	-3.192	3.2	3.346
VFD27	Bus97	96.14	0.592	0.000	0.0	0.592

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.933	3.421	1.761
0.02	2.610	2.691	0.655
0.03	2.299	2.312	0.244
0.04	2.005	2.007	0.091
0.05	1.730	1.731	0.034
0.06	1.599	1.599	0.013
0.07	1.473	1.473	0.005
0.08	1.353	1.353	0.002
0.09	1.238	1.238	0.001
0.10	1.130	1.130	0.000
0.15	0.936	0.936	0.000
0.20	0.773	0.773	0.000
0.25	0.650	0.650	0.000
0.30	0.650	0.650	0.000

Project: UQ LT LPG Export - ST & MT
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Engineer: Hemanathan Gopalan
Filename: Umm Qasr-ST

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3-Phase fault at bus: **Bus98**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 7.485 kA Method C
Steady State = 0.592 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus98	Total	0.00	1.596	-3.192	2.0	3.569
Chiller-4	Bus98	96.14	1.004	-3.192	3.2	3.346
VFD28	Bus98	96.14	0.592	0.000	0.0	0.592

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.933	3.421	1.761
0.02	2.610	2.691	0.655
0.03	2.299	2.312	0.244
0.04	2.005	2.007	0.091
0.05	1.730	1.731	0.034
0.06	1.599	1.599	0.013
0.07	1.473	1.473	0.005
0.08	1.353	1.353	0.002
0.09	1.238	1.238	0.001
0.10	1.130	1.130	0.000
0.15	0.936	0.936	0.000
0.20	0.773	0.773	0.000
0.25	0.650	0.650	0.000
0.30	0.650	0.650	0.000

3-Phase fault at bus: **Bus99**

Nominal kV = 0.400
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 9.176 kA Method C
Steady State = 6.281 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus99	Total	0.00	6.234	-1.265	0.2	6.361
Bus_MDB-2	Bus99	95.10	6.194	-1.159	0.2	6.302
Pump-3	Bus99	101.20	0.040	-0.106	2.7	0.114

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	6.332	6.332	0.000
0.02	6.328	6.328	0.000
0.03	6.323	6.323	0.000
0.04	6.318	6.318	0.000
0.05	6.313	6.313	0.000
0.06	6.311	6.311	0.000
0.07	6.309	6.309	0.000
0.08	6.306	6.306	0.000
0.09	6.304	6.304	0.000
0.10	6.302	6.302	0.000
0.15	6.298	6.298	0.000
0.20	6.295	6.295	0.000
0.25	6.292	6.292	0.000
0.30	6.292	6.292	0.000

3-Phase fault at bus: **Bus100**

Nominal kV = 0.400
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 7.257 kA Method C
Steady State = 4.961 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus100	Total	0.00	4.949	-0.902	0.2	5.031
Bus_MDB-2	Bus100	96.50	4.909	-0.795	0.2	4.973
Pump-4	Bus100	101.20	0.040	-0.106	2.7	0.114

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	5.003	5.003	0.000
0.02	4.998	4.998	0.000
0.03	4.994	4.994	0.000
0.04	4.989	4.989	0.000
0.05	4.985	4.985	0.000
0.06	4.982	4.982	0.000
0.07	4.980	4.980	0.000
0.08	4.978	4.978	0.000
0.09	4.976	4.976	0.000
0.10	4.974	4.974	0.000
0.15	4.970	4.970	0.000
0.20	4.967	4.967	0.000
0.25	4.964	4.964	0.000
0.30	4.964	4.964	0.000

Project: UQ LT LPG Export - ST & MT
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Engineer: Hemanathan Gopalan
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3-Phase fault at bus: **Bus101**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 7.485 kA Method C
Steady State = 0.592 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus101	Total	0.00	1.596	-3.192	2.0	3.569
Chiller-5	Bus101	96.14	1.004	-3.192	3.2	3.346
VFD29	Bus101	96.14	0.592	0.000	0.0	0.592

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.933	3.421	1.761
0.02	2.610	2.691	0.655
0.03	2.299	2.312	0.244
0.04	2.005	2.007	0.091
0.05	1.730	1.731	0.034
0.06	1.599	1.599	0.013
0.07	1.473	1.473	0.005
0.08	1.353	1.353	0.002
0.09	1.238	1.238	0.001
0.10	1.130	1.130	0.000
0.15	0.936	0.936	0.000
0.20	0.773	0.773	0.000
0.25	0.650	0.650	0.000
0.30	0.650	0.650	0.000

3-Phase fault at bus: **Bus102**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 7.485 kA Method C
Steady State = 0.592 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus102	Total	0.00	1.596	-3.192	2.0	3.569
Chiller-6	Bus102	96.14	1.004	-3.192	3.2	3.346
VFD30	Bus102	96.14	0.592	0.000	0.0	0.592

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.933	3.421	1.761
0.02	2.610	2.691	0.655
0.03	2.299	2.312	0.244
0.04	2.005	2.007	0.091
0.05	1.730	1.731	0.034
0.06	1.599	1.599	0.013
0.07	1.473	1.473	0.005
0.08	1.353	1.353	0.002
0.09	1.238	1.238	0.001
0.10	1.130	1.130	0.000
0.15	0.936	0.936	0.000
0.20	0.773	0.773	0.000
0.25	0.650	0.650	0.000
0.30	0.650	0.650	0.000

3-Phase fault at bus: **Bus103**

Nominal kV = 0.400
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 6.573 kA Method C
Steady State = 4.490 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus103	Total	0.00	4.488	-0.786	0.2	4.557
Bus_MDB-3	Bus103	97.02	4.449	-0.680	0.2	4.500
Pump-5	Bus103	101.20	0.040	-0.106	2.7	0.114

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	4.529	4.529	0.000
0.02	4.525	4.525	0.000
0.03	4.520	4.520	0.000
0.04	4.516	4.516	0.000
0.05	4.511	4.511	0.000
0.06	4.509	4.509	0.000
0.07	4.507	4.507	0.000
0.08	4.505	4.505	0.000
0.09	4.503	4.503	0.000
0.10	4.501	4.501	0.000
0.15	4.497	4.497	0.000
0.20	4.494	4.494	0.000
0.25	4.491	4.491	0.000
0.30	4.491	4.491	0.000

3-Phase fault at bus: **Bus104**

Nominal kV = 0.400
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 9.181 kA Method C
Steady State = 6.284 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus104	Total	0.00	6.238	-1.263	0.2	6.364
Bus_MDB-3	Bus104	95.15	6.198	-1.157	0.2	6.305
Pump-6	Bus104	101.20	0.040	-0.106	2.7	0.114

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	6.336	6.336	0.000
0.02	6.331	6.331	0.000
0.03	6.326	6.326	0.000
0.04	6.321	6.321	0.000
0.05	6.317	6.317	0.000
0.06	6.314	6.314	0.000
0.07	6.312	6.312	0.000
0.08	6.310	6.310	0.000
0.09	6.308	6.308	0.000
0.10	6.305	6.305	0.000
0.15	6.302	6.302	0.000
0.20	6.298	6.298	0.000
0.25	6.295	6.295	0.000
0.30	6.295	6.295	0.000

Project: UQ LT LPG Export - ST & MT
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Engineer: Hemanathan Gopalan
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3-Phase fault at bus: **Bus105**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 7.485 kA Method C
Steady State = 0.592 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus105	Total	0.00	1.596	-3.192	2.0	3.569
Chiller-7	Bus105	96.14	1.004	-3.192	3.2	3.346
VFD31	Bus105	96.14	0.592	0.000	0.0	0.592

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.933	3.421	1.761
0.02	2.610	2.691	0.655
0.03	2.299	2.312	0.244
0.04	2.005	2.007	0.091
0.05	1.730	1.731	0.034
0.06	1.599	1.599	0.013
0.07	1.473	1.473	0.005
0.08	1.353	1.353	0.002
0.09	1.238	1.238	0.001
0.10	1.130	1.130	0.000
0.15	0.936	0.936	0.000
0.20	0.773	0.773	0.000
0.25	0.650	0.650	0.000
0.30	0.650	0.650	0.000

Project: UQ LT LPG Export - ST & MT
Location: Umm Qasr, Iraq
Contract: UQ83
Engineer: Hemanathan Gopalan
Filename: Umm Qasr-ST

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3-Phase fault at bus: **Bus106**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 7.485 kA Method C
Steady State = 0.592 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus106	Total	0.00	1.596	-3.192	2.0	3.569
Chiller-8	Bus106	96.14	1.004	-3.192	3.2	3.346
VFD32	Bus106	96.14	0.592	0.000	0.0	0.592

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.933	3.421	1.761
0.02	2.610	2.691	0.655
0.03	2.299	2.312	0.244
0.04	2.005	2.007	0.091
0.05	1.730	1.731	0.034
0.06	1.599	1.599	0.013
0.07	1.473	1.473	0.005
0.08	1.353	1.353	0.002
0.09	1.238	1.238	0.001
0.10	1.130	1.130	0.000
0.15	0.936	0.936	0.000
0.20	0.773	0.773	0.000
0.25	0.650	0.650	0.000
0.30	0.650	0.650	0.000

Project: UQ LT LPG Export - ST & MT
Location: Umm Qasr, Iraq
Contract: UQ83
Engineer: Hemanathan Gopalan
Filename: Umm Qasr-ST

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3-Phase fault at bus: **Bus107**

Nominal kV = 0.400
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 12.560 kA Method C
Steady State = 8.602 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus107	Total	0.00	8.472	-2.008	0.2	8.707
Bus_MDB-4	Bus107	93.18	8.433	-1.902	0.2	8.644
Pump-7	Bus107	101.20	0.040	-0.106	2.7	0.114

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	8.677	8.677	0.000
0.02	8.672	8.672	0.000
0.03	8.667	8.667	0.000
0.04	8.662	8.662	0.000
0.05	8.657	8.657	0.000
0.06	8.654	8.654	0.000
0.07	8.652	8.652	0.000
0.08	8.649	8.649	0.000
0.09	8.647	8.647	0.000
0.10	8.645	8.645	0.000
0.15	8.641	8.641	0.000
0.20	8.637	8.637	0.000
0.25	8.634	8.634	0.000
0.30	8.634	8.634	0.000

3-Phase fault at bus: **Bus108**

Nominal kV = 0.400
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 10.019 kA Method C
Steady State = 6.860 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus108	Total	0.00	6.800	-1.412	0.2	6.945
Bus_MDB-4	Bus108	95.00	6.761	-1.305	0.2	6.886
Pump-8	Bus108	101.20	0.040	-0.106	2.7	0.114

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	6.917	6.917	0.000
0.02	6.912	6.912	0.000
0.03	6.907	6.907	0.000
0.04	6.902	6.902	0.000
0.05	6.897	6.897	0.000
0.06	6.895	6.895	0.000
0.07	6.893	6.893	0.000
0.08	6.890	6.890	0.000
0.09	6.888	6.888	0.000
0.10	6.886	6.886	0.000
0.15	6.882	6.882	0.000
0.20	6.879	6.879	0.000
0.25	6.876	6.876	0.000
0.30	6.876	6.876	0.000

3-Phase fault at bus: **Bus110**

Nominal kV = 11.000
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 55.131 kA Method C
Steady State = 20.151 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus110	Total	0.00	5.728	-24.868	4.3	25.519
S002-SB-1AC001 [BUS A]	Bus110	23.23	5.671	-24.557	4.3	25.204
S002-6100-MP-10A	Bus110	110.00	0.057	-0.311	5.5	0.316

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	25.091	31.234	18.601
0.02	24.858	26.970	10.464
0.03	24.588	25.225	5.635
0.04	24.290	24.478	3.034
0.05	23.916	23.991	1.897
0.06	23.725	23.748	1.052
0.07	23.537	23.544	0.584
0.08	23.351	23.353	0.324
0.09	23.169	23.170	0.180
0.10	22.990	22.990	0.120
0.15	22.606	22.606	0.007
0.20	22.236	22.236	0.000
0.25	21.883	21.883	0.000
0.30	21.883	21.883	0.000

Project: UQ LT LPG Export - ST & MT
Location: Umm Qasr, Iraq
Contract: UQ83
Engineer: Hemanathan Gopalan
Filename: Umm Qasr-ST

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Study Case: SC_Max

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3-Phase fault at bus: **Bus112**

Nominal kV = 11.000
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 42.178 kA Method C
Steady State = 17.662 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus112	Total	0.00	6.974	-20.402	2.9	21.561
S002-SB-1AC001 [BUS A]	Bus112	39.71	6.887	-20.176	2.9	21.320
Bus67	Bus112	14.55	0.087	-0.226	2.6	0.242

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	21.295	24.007	11.084
0.02	21.146	21.608	4.442
0.03	20.967	21.036	1.695
0.04	20.767	20.777	0.647
0.05	20.511	20.513	0.294
0.06	20.379	20.379	0.116
0.07	20.248	20.248	0.046
0.08	20.119	20.119	0.018
0.09	19.992	19.992	0.007
0.10	19.866	19.866	0.004
0.15	19.591	19.591	0.000
0.20	19.324	19.324	0.000
0.25	19.067	19.067	0.000
0.30	19.067	19.067	0.000

3-Phase fault at bus: **Bus113**

Nominal kV = 11.000
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 54.292 kA Method C
Steady State = 19.930 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus113	Total	0.00	5.642	-24.493	4.3	25.134
S002-SB-1AC001 [BUS B]	Bus113	23.35	5.583	-24.172	4.3	24.808
S002-6100-MP-10C	Bus113	110.00	0.059	-0.321	5.5	0.326

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	24.717	30.747	18.288
0.02	24.486	26.528	10.207
0.03	24.222	24.832	5.469
0.04	23.911	24.090	2.931
0.05	23.532	23.601	1.805
0.06	23.338	23.359	0.995
0.07	23.147	23.153	0.548
0.08	22.958	22.960	0.302
0.09	22.772	22.773	0.166
0.10	22.590	22.590	0.109
0.15	22.196	22.196	0.006
0.20	21.816	21.816	0.000
0.25	21.451	21.451	0.000
0.30	21.451	21.451	0.000

3-Phase fault at bus: **Bus114**

Nominal kV = 11.000
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 71.120 kA Method C
Steady State = 21.961 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus114	Total	0.00	3.062	-28.569	9.3	28.732
S002-SB-1AC001 [BUS B]	Bus114	5.99	2.803	-26.616	9.5	26.763
S002-6100-MK-1C	Bus114	110.00	0.259	-1.953	7.5	1.970

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	28.116	41.457	30.466
0.02	27.789	36.928	24.320
0.03	27.425	33.258	18.815
0.04	27.000	30.674	14.556
0.05	26.479	29.264	12.458
0.06	26.214	27.998	9.835
0.07	25.953	27.090	7.764
0.08	25.697	26.418	6.129
0.09	25.446	25.902	4.838
0.10	25.200	25.567	4.320
0.15	24.676	24.716	1.409
0.20	24.174	24.178	0.459
0.25	23.696	23.697	0.150
0.30	23.696	23.696	0.049

3-Phase fault at bus: **Bus115**

Nominal kV = 11.000
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 54.458 kA Method C
Steady State = 19.958 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus115	Total	0.00	5.622	-24.544	4.4	25.179
S002-SB-1AC001 [BUS B]	Bus115	23.15	5.563	-24.223	4.4	24.853
S002-6100-MP-10B	Bus115	110.00	0.059	-0.321	5.5	0.326

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	24.760	30.844	18.393
0.02	24.528	26.604	10.304
0.03	24.263	24.888	5.543
0.04	23.951	24.136	2.982
0.05	23.570	23.642	1.843
0.06	23.375	23.397	1.019
0.07	23.183	23.190	0.564
0.08	22.994	22.996	0.312
0.09	22.807	22.808	0.172
0.10	22.624	22.624	0.114
0.15	22.229	22.229	0.006
0.20	21.847	21.847	0.000
0.25	21.482	21.482	0.000
0.30	21.482	21.482	0.000

3-Phase fault at bus: **Bus117**

Nominal kV = 11.000
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 50.847 kA Method C
Steady State = 19.361 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus117	Total	0.00	6.031	-23.401	3.9	24.166
S002-SB-1AC001 [BUS B]	Bus117	27.45	5.997	-23.215	3.9	23.977
S002-40-MP-05B	Bus117	110.00	0.034	-0.186	5.4	0.189

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	23.792	28.765	16.167
0.02	23.584	25.013	8.333
0.03	23.348	23.708	4.115
0.04	23.068	23.157	2.032
0.05	22.724	22.754	1.164
0.06	22.547	22.555	0.592
0.07	22.373	22.375	0.301
0.08	22.201	22.202	0.153
0.09	22.032	22.032	0.078
0.10	21.865	21.865	0.048
0.15	21.503	21.503	0.002
0.20	21.154	21.154	0.000
0.25	20.818	20.818	0.000
0.30	20.818	20.818	0.000

3-Phase fault at bus: **Bus119**

Nominal kV = 3.300
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 42.865 kA Method C
Steady State = 15.504 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus119	Total	0.00	8.968	-21.217	2.4	23.034
S002-SB-1AD001 [BUS A]	Bus119	35.35	8.822	-20.501	2.3	22.319
S002-6200-MP-02A	Bus119	110.00	0.146	-0.715	4.9	0.730

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	22.245	24.329	9.852
0.02	21.782	22.082	3.632
0.03	21.240	21.275	1.213
0.04	20.688	20.692	0.405
0.05	20.160	20.161	0.187
0.06	19.905	19.906	0.067
0.07	19.656	19.656	0.024
0.08	19.413	19.413	0.008
0.09	19.176	19.176	0.003
0.10	18.946	18.946	0.002
0.15	18.488	18.488	0.000
0.20	18.054	18.054	0.000
0.25	17.645	17.645	0.000
0.30	17.645	17.645	0.000

3-Phase fault at bus: **Bus120**

Nominal kV = 3.300
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 37.092 kA Method C
Steady State = 14.538 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus120	Total	0.00	9.663	-18.610	1.9	20.969
S002-SB-1AD001 [BUS A]	Bus120	46.19	9.453	-17.459	1.8	19.853
S002-6100-MP-11A	Bus120	110.00	0.211	-1.151	5.5	1.170

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	20.366	21.582	7.142
0.02	19.988	20.118	2.284
0.03	19.537	19.547	0.634
0.04	19.077	19.078	0.176
0.05	18.635	18.635	0.074
0.06	18.421	18.421	0.022
0.07	18.212	18.212	0.007
0.08	18.007	18.007	0.002
0.09	17.807	17.807	0.001
0.10	17.612	17.612	0.000
0.15	17.221	17.221	0.000
0.20	16.848	16.848	0.000
0.25	16.497	16.497	0.000
0.30	16.497	16.497	0.000

3-Phase fault at bus: **Bus121**

Nominal kV = 3.300
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 37.654 kA Method C
Steady State = 14.665 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus121	Total	0.00	9.619	-18.890	2.0	21.198
S002-SB-1AD001 [BUS A]	Bus121	44.95	9.427	-17.843	1.9	20.181
S002-6100-MP-9C	Bus121	110.00	0.192	-1.047	5.5	1.064

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	20.582	21.855	7.351
0.02	20.199	20.337	2.363
0.03	19.743	19.754	0.663
0.04	19.276	19.277	0.186
0.05	18.827	18.828	0.078
0.06	18.611	18.611	0.024
0.07	18.398	18.398	0.007
0.08	18.191	18.191	0.002
0.09	17.988	17.988	0.001
0.10	17.790	17.790	0.000
0.15	17.393	17.393	0.000
0.20	17.016	17.016	0.000
0.25	16.659	16.659	0.000
0.30	16.659	16.659	0.000

3-Phase fault at bus: **Bus122**

Nominal kV = 3.300
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 38.039 kA Method C
Steady State = 14.728 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus122	Total	0.00	9.580	-19.070	2.0	21.341
S002-SB-1AD001 [BUS A]	Bus122	44.26	9.388	-18.023	1.9	20.322
S002-6100-MP-9A	Bus122	110.00	0.192	-1.047	5.5	1.064

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	20.714	22.039	7.528
0.02	20.325	20.472	2.448
0.03	19.861	19.874	0.697
0.04	19.388	19.389	0.199
0.05	18.933	18.933	0.084
0.06	18.713	18.713	0.026
0.07	18.497	18.497	0.008
0.08	18.287	18.287	0.002
0.09	18.081	18.081	0.001
0.10	17.881	17.881	0.000
0.15	17.478	17.478	0.000
0.20	17.096	17.096	0.000
0.25	16.735	16.735	0.000
0.30	16.735	16.735	0.000

3-Phase fault at bus: **Bus123**

Nominal kV = 3.300
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 7.961 kA Method C
Steady State = 5.127 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus123	Total	0.00	5.123	-1.967	0.4	5.487
S002-SB-1AD001 [BUS A]	Bus123	102.96	5.047	-1.640	0.3	5.307
S002-6100-MP-8D	Bus123	110.00	0.075	-0.326	4.3	0.335

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	5.442	5.444	0.116
0.02	5.424	5.424	0.013
0.03	5.407	5.407	0.001
0.04	5.391	5.391	0.000
0.05	5.376	5.376	0.000
0.06	5.369	5.369	0.000
0.07	5.362	5.362	0.000
0.08	5.355	5.355	0.000
0.09	5.348	5.348	0.000
0.10	5.342	5.342	0.000
0.15	5.331	5.331	0.000
0.20	5.320	5.320	0.000
0.25	5.310	5.310	0.000
0.30	5.310	5.310	0.000

3-Phase fault at bus: **Bus124**

Nominal kV = 3.300
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 8.091 kA Method C
Steady State = 5.207 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus124	Total	0.00	5.202	-2.010	0.4	5.577
S002-SB-1AD001 [BUS A]	Bus124	102.81	5.127	-1.684	0.3	5.396
S002-6100-MP-8B	Bus124	110.00	0.075	-0.326	4.3	0.335

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	5.532	5.533	0.116
0.02	5.514	5.514	0.013
0.03	5.496	5.496	0.001
0.04	5.480	5.480	0.000
0.05	5.465	5.465	0.000
0.06	5.458	5.458	0.000
0.07	5.451	5.451	0.000
0.08	5.444	5.444	0.000
0.09	5.437	5.437	0.000
0.10	5.431	5.431	0.000
0.15	5.420	5.420	0.000
0.20	5.409	5.409	0.000
0.25	5.399	5.399	0.000
0.30	5.399	5.399	0.000

3-Phase fault at bus: **Bus125**

Nominal kV = 3.300
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 5.211 kA Method C
Steady State = 3.449 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus125	Total	0.00	3.440	-1.070	0.3	3.602
S002-SB-1AD001 [BUS A]	Bus125	105.77	3.399	-0.873	0.3	3.510
S002-6100-MP-7C	Bus125	110.00	0.040	-0.197	4.9	0.201

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	3.580	3.581	0.055
0.02	3.571	3.571	0.008
0.03	3.563	3.563	0.000
0.04	3.555	3.555	0.000
0.05	3.547	3.547	0.000
0.06	3.543	3.543	0.000
0.07	3.540	3.540	0.000
0.08	3.536	3.536	0.000
0.09	3.532	3.532	0.000
0.10	3.529	3.529	0.000
0.15	3.523	3.523	0.000
0.20	3.517	3.517	0.000
0.25	3.511	3.511	0.000
0.30	3.511	3.511	0.000

Project: UQ LT LPG Export - ST & MT
Location: Umm Qasr, Iraq
Contract: UQ83
Engineer: Hemanathan Gopalan
Filename: Umm Qasr-ST

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3-Phase fault at bus: **Bus126**

Nominal kV = 3.300
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 5.211 kA Method C
Steady State = 3.449 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus126	Total	0.00	3.440	-1.070	0.3	3.602
S002-SB-1AD001 [BUS A]	Bus126	105.77	3.399	-0.873	0.3	3.510
S002-6100-MP-7A	Bus126	110.00	0.040	-0.197	4.9	0.201

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	3.580	3.581	0.055
0.02	3.571	3.571	0.008
0.03	3.563	3.563	0.000
0.04	3.555	3.555	0.000
0.05	3.547	3.547	0.000
0.06	3.543	3.543	0.000
0.07	3.540	3.540	0.000
0.08	3.536	3.536	0.000
0.09	3.532	3.532	0.000
0.10	3.529	3.529	0.000
0.15	3.523	3.523	0.000
0.20	3.517	3.517	0.000
0.25	3.511	3.511	0.000
0.30	3.511	3.511	0.000

3-Phase fault at bus: **Bus127**

Nominal kV = 3.300
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 6.461 kA Method C
Steady State = 4.263 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus127	Total	0.00	4.240	-1.409	0.3	4.468
S002-SB-1AD001 [BUS A]	Bus127	104.49	4.200	-1.212	0.3	4.371
S002-6100-MP-6D	Bus127	110.00	0.040	-0.197	4.9	0.201

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	4.446	4.446	0.048
0.02	4.436	4.436	0.005
0.03	4.428	4.428	0.000
0.04	4.419	4.419	0.000
0.05	4.411	4.411	0.000
0.06	4.407	4.407	0.000
0.07	4.403	4.403	0.000
0.08	4.399	4.399	0.000
0.09	4.396	4.396	0.000
0.10	4.392	4.392	0.000
0.15	4.385	4.385	0.000
0.20	4.379	4.379	0.000
0.25	4.373	4.373	0.000
0.30	4.373	4.373	0.000

3-Phase fault at bus: **Bus128**

Nominal kV = 3.300
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 6.461 kA Method C
Steady State = 4.263 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus128	Total	0.00	4.240	-1.409	0.3	4.468
S002-SB-1AD001 [BUS A]	Bus128	104.49	4.200	-1.212	0.3	4.371
S002-6100-MP-6B	Bus128	110.00	0.040	-0.197	4.9	0.201

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	4.446	4.446	0.048
0.02	4.436	4.436	0.005
0.03	4.428	4.428	0.000
0.04	4.419	4.419	0.000
0.05	4.411	4.411	0.000
0.06	4.407	4.407	0.000
0.07	4.403	4.403	0.000
0.08	4.399	4.399	0.000
0.09	4.396	4.396	0.000
0.10	4.392	4.392	0.000
0.15	4.385	4.385	0.000
0.20	4.379	4.379	0.000
0.25	4.373	4.373	0.000
0.30	4.373	4.373	0.000

3-Phase fault at bus: **Bus129**

Nominal kV = 3.300
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 10.565 kA Method C
Steady State = 6.779 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus129	Total	0.00	6.717	-2.857	0.4	7.300
S002-SB-1AD001 [BUS A]	Bus129	99.59	6.677	-2.660	0.4	7.187
S002-6100-MP-4D	Bus129	110.00	0.040	-0.197	4.9	0.201

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	7.273	7.273	0.067
0.02	7.262	7.262	0.004
0.03	7.249	7.249	0.000
0.04	7.235	7.235	0.000
0.05	7.222	7.222	0.000
0.06	7.215	7.215	0.000
0.07	7.208	7.208	0.000
0.08	7.202	7.202	0.000
0.09	7.195	7.195	0.000
0.10	7.189	7.189	0.000
0.15	7.176	7.176	0.000
0.20	7.164	7.164	0.000
0.25	7.152	7.152	0.000
0.30	7.152	7.152	0.000

3-Phase fault at bus: **Bus130**

Nominal kV = 3.300
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 10.565 kA Method C
Steady State = 6.779 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus130	Total	0.00	6.717	-2.857	0.4	7.300
S002-SB-1AD001 [BUS A]	Bus130	99.59	6.677	-2.660	0.4	7.187
S002-6100-MP-4B	Bus130	110.00	0.040	-0.197	4.9	0.201

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	7.273	7.273	0.067
0.02	7.262	7.262	0.004
0.03	7.249	7.249	0.000
0.04	7.235	7.235	0.000
0.05	7.222	7.222	0.000
0.06	7.215	7.215	0.000
0.07	7.208	7.208	0.000
0.08	7.202	7.202	0.000
0.09	7.195	7.195	0.000
0.10	7.189	7.189	0.000
0.15	7.176	7.176	0.000
0.20	7.164	7.164	0.000
0.25	7.152	7.152	0.000
0.30	7.152	7.152	0.000

3-Phase fault at bus: **Bus131**

Nominal kV = 3.300
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 7.577 kA Method C
Steady State = 4.973 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus131	Total	0.00	4.938	-1.753	0.4	5.240
S002-SB-1AD001 [BUS A]	Bus131	103.26	4.898	-1.556	0.3	5.139
S002-6100-MP-3C	Bus131	110.00	0.040	-0.197	4.9	0.201

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	5.217	5.217	0.048
0.02	5.207	5.207	0.004
0.03	5.198	5.198	0.000
0.04	5.189	5.189	0.000
0.05	5.181	5.181	0.000
0.06	5.177	5.177	0.000
0.07	5.172	5.172	0.000
0.08	5.169	5.169	0.000
0.09	5.165	5.165	0.000
0.10	5.161	5.161	0.000
0.15	5.154	5.154	0.000
0.20	5.147	5.147	0.000
0.25	5.141	5.141	0.000
0.30	5.141	5.141	0.000

3-Phase fault at bus: **Bus132**

Nominal kV = 3.300
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 7.577 kA Method C
Steady State = 4.973 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus132	Total	0.00	4.938	-1.753	0.4	5.240
S002-SB-1AD001 [BUS A]	Bus132	103.26	4.898	-1.556	0.3	5.139
S002-6100-MP-3A	Bus132	110.00	0.040	-0.197	4.9	0.201

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	5.217	5.217	0.048
0.02	5.207	5.207	0.004
0.03	5.198	5.198	0.000
0.04	5.189	5.189	0.000
0.05	5.181	5.181	0.000
0.06	5.177	5.177	0.000
0.07	5.172	5.172	0.000
0.08	5.169	5.169	0.000
0.09	5.165	5.165	0.000
0.10	5.161	5.161	0.000
0.15	5.154	5.154	0.000
0.20	5.147	5.147	0.000
0.25	5.141	5.141	0.000
0.30	5.141	5.141	0.000

3-Phase fault at bus: **Bus133**

Nominal kV = 3.300
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 7.529 kA Method C
Steady State = 4.981 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus133	Total	0.00	4.855	-1.871	0.4	5.203
S002-SB-1AD001 [BUS B]	Bus133	102.41	4.814	-1.674	0.3	5.097
S002-6100-MP-3B	Bus133	110.00	0.040	-0.197	4.9	0.201

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	5.178	5.179	0.059
0.02	5.168	5.168	0.005
0.03	5.159	5.159	0.000
0.04	5.149	5.149	0.000
0.05	5.141	5.141	0.000
0.06	5.136	5.136	0.000
0.07	5.132	5.132	0.000
0.08	5.128	5.128	0.000
0.09	5.124	5.124	0.000
0.10	5.120	5.120	0.000
0.15	5.112	5.112	0.000
0.20	5.105	5.105	0.000
0.25	5.099	5.099	0.000
0.30	5.099	5.099	0.000

3-Phase fault at bus: **Bus134**

Nominal kV = 3.300
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 7.532 kA Method C
Steady State = 4.981 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus134	Total	0.00	4.861	-1.867	0.4	5.207
S002-SB-1AD001 [BUS B]	Bus134	102.42	4.815	-1.674	0.3	5.097
S002-6100-MP-3D	Bus134	110.00	0.046	-0.193	4.2	0.199

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	5.181	5.181	0.043
0.02	5.170	5.170	0.002
0.03	5.160	5.160	0.000
0.04	5.150	5.150	0.000
0.05	5.140	5.140	0.000
0.06	5.136	5.136	0.000
0.07	5.131	5.131	0.000
0.08	5.127	5.127	0.000
0.09	5.122	5.122	0.000
0.10	5.118	5.118	0.000
0.15	5.110	5.110	0.000
0.20	5.102	5.102	0.000
0.25	5.095	5.095	0.000
0.30	5.095	5.095	0.000

3-Phase fault at bus: **Bus135**

Nominal kV = 3.300
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 10.444 kA Method C
Steady State = 6.798 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus135	Total	0.00	6.526	-3.057	0.5	7.207
S002-SB-1AD001 [BUS B]	Bus135	98.18	6.482	-2.862	0.4	7.086
S002-6100-MP-4A	Bus135	110.00	0.045	-0.195	4.4	0.200

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	7.176	7.176	0.077
0.02	7.163	7.163	0.003
0.03	7.151	7.151	0.000
0.04	7.139	7.139	0.000
0.05	7.127	7.127	0.000
0.06	7.121	7.121	0.000
0.07	7.116	7.116	0.000
0.08	7.110	7.110	0.000
0.09	7.105	7.105	0.000
0.10	7.099	7.099	0.000
0.15	7.089	7.089	0.000
0.20	7.080	7.080	0.000
0.25	7.070	7.070	0.000
0.30	7.070	7.070	0.000

Project: UQ LT LPG Export - ST & MT
Location: Umm Qasr, Iraq
Contract: UQ83
Engineer: Hemanathan Gopalan
Filename: Umm Qasr-ST

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3-Phase fault at bus: **Bus136**

Nominal kV = 3.300
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 10.444 kA Method C
Steady State = 6.798 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus136	Total	0.00	6.526	-3.057	0.5	7.207
S002-SB-1AD001 [BUS B]	Bus136	98.18	6.482	-2.862	0.4	7.086
S002-6100-MP-4C	Bus136	110.00	0.045	-0.195	4.4	0.200

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	7.176	7.176	0.077
0.02	7.163	7.163	0.003
0.03	7.151	7.151	0.000
0.04	7.139	7.139	0.000
0.05	7.127	7.127	0.000
0.06	7.121	7.121	0.000
0.07	7.116	7.116	0.000
0.08	7.110	7.110	0.000
0.09	7.105	7.105	0.000
0.10	7.099	7.099	0.000
0.15	7.089	7.089	0.000
0.20	7.080	7.080	0.000
0.25	7.070	7.070	0.000
0.30	7.070	7.070	0.000

Project: UQ LT LPG Export - ST & MT
Location: Umm Qasr, Iraq
Contract: UQ83
Engineer: Hemanathan Gopalan
Filename: Umm Qasr-ST

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3-Phase fault at bus: **Bus137**

Nominal kV = 3.300
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 6.435 kA Method C
Steady State = 4.269 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus137	Total	0.00	4.192	-1.494	0.4	4.450
S002-SB-1AD001 [BUS B]	Bus137	103.83	4.144	-1.301	0.3	4.344
S002-6100-MP-6A	Bus137	110.00	0.048	-0.192	4.0	0.198

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	4.424	4.425	0.037
0.02	4.414	4.414	0.002
0.03	4.404	4.404	0.000
0.04	4.394	4.394	0.000
0.05	4.385	4.385	0.000
0.06	4.380	4.380	0.000
0.07	4.376	4.376	0.000
0.08	4.371	4.371	0.000
0.09	4.367	4.367	0.000
0.10	4.363	4.363	0.000
0.15	4.355	4.355	0.000
0.20	4.347	4.347	0.000
0.25	4.340	4.340	0.000
0.30	4.340	4.340	0.000

3-Phase fault at bus: **Bus138**

Nominal kV = 3.300
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 6.435 kA Method C
Steady State = 4.269 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus138	Total	0.00	4.192	-1.494	0.4	4.450
S002-SB-1AD001 [BUS B]	Bus138	103.83	4.144	-1.301	0.3	4.344
S002-6100-MP-6C	Bus138	110.00	0.048	-0.192	4.0	0.198

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	4.424	4.425	0.037
0.02	4.414	4.414	0.002
0.03	4.404	4.404	0.000
0.04	4.394	4.394	0.000
0.05	4.385	4.385	0.000
0.06	4.380	4.380	0.000
0.07	4.376	4.376	0.000
0.08	4.371	4.371	0.000
0.09	4.367	4.367	0.000
0.10	4.363	4.363	0.000
0.15	4.355	4.355	0.000
0.20	4.347	4.347	0.000
0.25	4.340	4.340	0.000
0.30	4.340	4.340	0.000

3-Phase fault at bus: **Bus139**

Nominal kV = 3.300
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 5.196 kA Method C
Steady State = 3.452 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus139	Total	0.00	3.407	-1.130	0.3	3.590
S002-SB-1AD001 [BUS B]	Bus139	105.30	3.367	-0.932	0.3	3.494
S002-6100-MP-7B	Bus139	110.00	0.040	-0.197	4.9	0.201

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	3.567	3.568	0.061
0.02	3.558	3.558	0.009
0.03	3.549	3.549	0.000
0.04	3.541	3.541	0.000
0.05	3.533	3.533	0.000
0.06	3.529	3.529	0.000
0.07	3.525	3.525	0.000
0.08	3.521	3.521	0.000
0.09	3.518	3.518	0.000
0.10	3.514	3.514	0.000
0.15	3.507	3.507	0.000
0.20	3.501	3.501	0.000
0.25	3.495	3.495	0.000
0.30	3.495	3.495	0.000

3-Phase fault at bus: **Bus140**

Nominal kV = 3.300
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 5.196 kA Method C
Steady State = 3.452 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus140	Total	0.00	3.407	-1.130	0.3	3.590
S002-SB-1AD001 [BUS B]	Bus140	105.30	3.367	-0.932	0.3	3.494
S002-6100-MP-7D	Bus140	110.00	0.040	-0.197	4.9	0.201

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	3.567	3.568	0.061
0.02	3.558	3.558	0.009
0.03	3.549	3.549	0.000
0.04	3.541	3.541	0.000
0.05	3.533	3.533	0.000
0.06	3.529	3.529	0.000
0.07	3.525	3.525	0.000
0.08	3.521	3.521	0.000
0.09	3.518	3.518	0.000
0.10	3.514	3.514	0.000
0.15	3.507	3.507	0.000
0.20	3.501	3.501	0.000
0.25	3.495	3.495	0.000
0.30	3.495	3.495	0.000

3-Phase fault at bus: **Bus141**

Nominal kV = 3.300
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 8.045 kA Method C
Steady State = 5.216 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus141	Total	0.00	5.106	-2.140	0.4	5.536
S002-SB-1AD001 [BUS B]	Bus141	101.89	5.031	-1.813	0.4	5.348
S002-6100-MP-8A	Bus141	110.00	0.075	-0.326	4.3	0.335

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	5.490	5.491	0.133
0.02	5.471	5.471	0.015
0.03	5.453	5.453	0.001
0.04	5.436	5.436	0.000
0.05	5.420	5.420	0.000
0.06	5.412	5.412	0.000
0.07	5.405	5.405	0.000
0.08	5.398	5.398	0.000
0.09	5.391	5.391	0.000
0.10	5.384	5.384	0.000
0.15	5.373	5.373	0.000
0.20	5.361	5.361	0.000
0.25	5.351	5.351	0.000
0.30	5.351	5.351	0.000

3-Phase fault at bus: **Bus142**

Nominal kV = 3.300
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 7.918 kA Method C
Steady State = 5.135 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus142	Total	0.00	5.031	-2.093	0.4	5.449
S002-SB-1AD001 [BUS B]	Bus142	102.06	4.956	-1.767	0.4	5.261
S002-6100-MP-8C	Bus142	110.00	0.075	-0.326	4.3	0.335

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	5.402	5.404	0.133
0.02	5.384	5.384	0.015
0.03	5.366	5.366	0.001
0.04	5.349	5.349	0.000
0.05	5.333	5.333	0.000
0.06	5.325	5.325	0.000
0.07	5.318	5.318	0.000
0.08	5.311	5.311	0.000
0.09	5.304	5.304	0.000
0.10	5.298	5.298	0.000
0.15	5.286	5.286	0.000
0.20	5.275	5.275	0.000
0.25	5.265	5.265	0.000
0.30	5.265	5.265	0.000

3-Phase fault at bus: **Bus143**

Nominal kV = 3.300
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 35.797 kA Method C
Steady State = 14.888 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus143	Total	0.00	7.779	-17.757	2.3	19.386
S002-SB-1AD001 [BUS B]	Bus143	39.99	7.624	-16.705	2.2	18.362
S002-6100-MP-9B	Bus143	110.00	0.155	-1.053	6.8	1.064

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	18.921	20.574	8.078
0.02	18.629	18.868	2.992
0.03	18.301	18.327	0.989
0.04	17.981	17.983	0.327
0.05	17.671	17.672	0.153
0.06	17.521	17.521	0.054
0.07	17.373	17.373	0.019
0.08	17.228	17.228	0.007
0.09	17.086	17.086	0.002
0.10	16.948	16.948	0.001
0.15	16.669	16.669	0.000
0.20	16.403	16.403	0.000
0.25	16.150	16.150	0.000
0.30	16.150	16.150	0.000

3-Phase fault at bus: **Bus144**

Nominal kV = 3.300
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 35.400 kA Method C
Steady State = 14.823 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus144	Total	0.00	7.849	-17.624	2.2	19.293
S002-SB-1AD001 [BUS B]	Bus144	40.68	7.657	-16.578	2.2	18.261
S002-6100-MP-11B	Bus144	110.00	0.192	-1.047	5.5	1.064

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	18.834	20.356	7.723
0.02	18.544	18.730	2.634
0.03	18.217	18.235	0.818
0.04	17.899	17.901	0.254
0.05	17.591	17.592	0.104
0.06	17.442	17.442	0.034
0.07	17.295	17.295	0.011
0.08	17.151	17.151	0.004
0.09	17.010	17.010	0.001
0.10	16.872	16.872	0.001
0.15	16.595	16.595	0.000
0.20	16.330	16.330	0.000
0.25	16.078	16.078	0.000
0.30	16.078	16.078	0.000

3-Phase fault at bus: **Bus145**

Nominal kV = 3.300
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 34.749 kA Method C
Steady State = 14.693 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus145	Total	0.00	7.937	-17.342	2.2	19.072
S002-SB-1AD001 [BUS B]	Bus145	41.98	7.746	-16.295	2.1	18.043
S002-6100-MP-11C	Bus145	110.00	0.192	-1.047	5.5	1.064

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	18.627	20.041	7.396
0.02	18.344	18.509	2.466
0.03	18.025	18.040	0.746
0.04	17.713	17.715	0.225
0.05	17.413	17.413	0.090
0.06	17.267	17.267	0.029
0.07	17.123	17.123	0.009
0.08	16.982	16.982	0.003
0.09	16.844	16.844	0.001
0.10	16.708	16.708	0.000
0.15	16.437	16.437	0.000
0.20	16.177	16.177	0.000
0.25	15.930	15.930	0.000
0.30	15.930	15.930	0.000

3-Phase fault at bus: **Bus146**

Nominal kV = 3.300
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 14.626 kA Method C
Steady State = 9.071 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus146	Total	0.00	8.490	-5.260	0.6	9.988
S002-SB-1AD001 [BUS B]	Bus146	91.18	8.410	-4.930	0.6	9.748
S002-40-MP-04B	Bus146	110.00	0.081	-0.330	4.1	0.340

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	9.922	9.926	0.286
0.02	9.880	9.880	0.016
0.03	9.833	9.833	0.001
0.04	9.787	9.787	0.000
0.05	9.742	9.742	0.000
0.06	9.720	9.720	0.000
0.07	9.697	9.697	0.000
0.08	9.676	9.676	0.000
0.09	9.654	9.654	0.000
0.10	9.633	9.633	0.000
0.15	9.589	9.589	0.000
0.20	9.547	9.547	0.000
0.25	9.506	9.506	0.000
0.30	9.506	9.506	0.000

Project: UQ LT LPG Export - ST & MT
Location: Umm Qasr, Iraq
Contract: UQ83
Engineer: Hemanathan Gopalan
Filename: Umm Qasr-ST

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Config.: 1_NE

3-Phase fault at bus: **Bus149**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 4.230 kA Method C
Steady State = 2.488 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus149	Total	0.00	2.468	-1.255	0.5	2.769
TYPICAL MOTOR FEEDER 1 A	Bus149	102.61	2.371	-0.777	0.3	2.495
614-FM-06BC	Bus149	105.00	0.098	-0.478	4.9	0.488

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.646	2.688	0.475
0.02	2.621	2.627	0.183
0.03	2.597	2.597	0.040
0.04	2.575	2.575	0.009
0.05	2.555	2.555	0.005
0.06	2.545	2.545	0.001
0.07	2.536	2.536	0.000
0.08	2.527	2.527	0.000
0.09	2.518	2.518	0.000
0.10	2.510	2.510	0.000
0.15	2.497	2.497	0.000
0.20	2.485	2.485	0.000
0.25	2.474	2.474	0.000
0.30	2.474	2.474	0.000

3-Phase fault at bus: **Bus151**

Nominal kV = 0.380

Voltage c Factor = 1.05 (Maximum If)

Peak Value = 4.175 kA Method C

Steady State = 2.449 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus151	Total	0.00	2.431	-1.241	0.5	2.729
TYPICAL MOTOR FEEDER 1 A	Bus151	102.65	2.333	-0.762	0.3	2.455
614-FM-06CA	Bus151	105.00	0.098	-0.478	4.9	0.488

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.606	2.649	0.478
0.02	2.581	2.587	0.185
0.03	2.557	2.557	0.041
0.04	2.535	2.535	0.009
0.05	2.515	2.515	0.005
0.06	2.505	2.505	0.001
0.07	2.496	2.496	0.000
0.08	2.487	2.487	0.000
0.09	2.479	2.479	0.000
0.10	2.470	2.470	0.000
0.15	2.457	2.457	0.000
0.20	2.445	2.445	0.000
0.25	2.434	2.434	0.000
0.30	2.434	2.434	0.000

3-Phase fault at bus: **Bus152**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 3.084 kA Method C
Steady State = 2.017 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus152	Total	0.00	2.077	-0.505	0.2	2.138
S002-SB-1AN001 [BUS B]	Bus152	95.13	1.989	-0.356	0.2	2.020
S002-6600-MP-1B	Bus152	105.00	0.088	-0.149	1.7	0.174

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.076	2.076	0.001
0.02	2.066	2.066	0.000
0.03	2.056	2.056	0.000
0.04	2.047	2.047	0.000
0.05	2.038	2.038	0.000
0.06	2.034	2.034	0.000
0.07	2.030	2.030	0.000
0.08	2.026	2.026	0.000
0.09	2.022	2.022	0.000
0.10	2.019	2.019	0.000
0.15	2.013	2.013	0.000
0.20	2.008	2.008	0.000
0.25	2.004	2.004	0.000
0.30	2.004	2.004	0.000

3-Phase fault at bus: **Bus154**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 4.230 kA Method C
Steady State = 2.488 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus154	Total	0.00	2.468	-1.255	0.5	2.769
TYPICAL MOTOR FEEDER 1 A	Bus154	102.61	2.371	-0.777	0.3	2.495
614-FM-06CB	Bus154	105.00	0.098	-0.478	4.9	0.488

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.646	2.688	0.475
0.02	2.621	2.627	0.183
0.03	2.597	2.597	0.040
0.04	2.575	2.575	0.009
0.05	2.555	2.555	0.005
0.06	2.545	2.545	0.001
0.07	2.536	2.536	0.000
0.08	2.527	2.527	0.000
0.09	2.518	2.518	0.000
0.10	2.510	2.510	0.000
0.15	2.497	2.497	0.000
0.20	2.485	2.485	0.000
0.25	2.474	2.474	0.000
0.30	2.474	2.474	0.000

3-Phase fault at bus: **Bus157**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 7.743 kA Method C
Steady State = 5.027 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus157	Total	0.00	4.531	-2.800	0.6	5.326
S002-SB-04E001 [BUS A]	Bus157	76.89	4.531	-2.800	0.6	5.326

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	5.326	5.326	0.049
0.02	5.326	5.326	0.000
0.03	5.326	5.326	0.000
0.04	5.326	5.326	0.000
0.05	5.326	5.326	0.000
0.06	5.326	5.326	0.000
0.07	5.326	5.326	0.000
0.08	5.326	5.326	0.000
0.09	5.326	5.326	0.000
0.10	5.326	5.326	0.000
0.15	5.326	5.326	0.000
0.20	5.326	5.326	0.000
0.25	5.326	5.326	0.000
0.30	5.326	5.326	0.000

3-Phase fault at bus: **Bus158**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 3.087 kA Method C
Steady State = 2.017 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus158	Total	0.00	2.084	-0.484	0.2	2.140
S002-SB-1AN001 [BUS A]	Bus158	95.30	1.996	-0.335	0.2	2.024
S002-6600-MP-1A	Bus158	105.00	0.088	-0.149	1.7	0.174

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.079	2.079	0.000
0.02	2.069	2.069	0.000
0.03	2.059	2.059	0.000
0.04	2.050	2.050	0.000
0.05	2.042	2.042	0.000
0.06	2.038	2.038	0.000
0.07	2.034	2.034	0.000
0.08	2.030	2.030	0.000
0.09	2.026	2.026	0.000
0.10	2.023	2.023	0.000
0.15	2.017	2.017	0.000
0.20	2.012	2.012	0.000
0.25	2.007	2.007	0.000
0.30	2.007	2.007	0.000

Project: UQ LT LPG Export - ST & MT
Location: Umm Qasr, Iraq
Contract: UQ83
Engineer: Hemanathan Gopalan
Filename: Umm Qasr-ST

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3-Phase fault at bus: **Bus159**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 2.084 kA Method C
Steady State = 1.312 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus159	Total	0.00	1.357	-0.454	0.3	1.431
S002-MCC1-8 [BUS]	Bus159	102.92	1.288	-0.257	0.2	1.313
S002-6600-MP-2A	Bus159	105.00	0.069	-0.196	2.8	0.208

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	1.370	1.370	0.040
0.02	1.359	1.359	0.003
0.03	1.350	1.350	0.000
0.04	1.341	1.341	0.000
0.05	1.333	1.333	0.000
0.06	1.328	1.328	0.000
0.07	1.325	1.325	0.000
0.08	1.321	1.321	0.000
0.09	1.317	1.317	0.000
0.10	1.314	1.314	0.000
0.15	1.308	1.308	0.000
0.20	1.303	1.303	0.000
0.25	1.299	1.299	0.000
0.30	1.299	1.299	0.000

3-Phase fault at bus: **Bus160**

Nominal kV = 3.450
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 39.475 kA Method C
Steady State = 15.002 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus160	Total	0.00	9.923	-19.593	2.0	21.963
S002-SB-1AD001 [BUS A]	Bus160	49.28	9.596	-17.444	1.8	19.909
688-KM-01	Bus160	115.00	0.327	-2.150	6.6	2.174

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	21.268	22.885	8.449
0.02	20.886	21.151	3.336
0.03	20.392	20.421	1.093
0.04	19.865	19.868	0.358
0.05	19.360	19.361	0.194
0.06	19.118	19.118	0.070
0.07	18.880	18.880	0.026
0.08	18.649	18.649	0.009
0.09	18.423	18.423	0.003
0.10	18.204	18.204	0.002
0.15	17.767	17.767	0.000
0.20	17.352	17.352	0.000
0.25	16.960	16.960	0.000
0.30	16.960	16.960	0.000

3-Phase fault at bus: **Bus161**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 4.287 kA Method C
Steady State = 2.529 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus161	Total	0.00	2.507	-1.269	0.5	2.810
TYPICAL MOTOR FEEDER 1 A	Bus161	102.57	2.409	-0.791	0.3	2.536
614-FM-06CC	Bus161	105.00	0.098	-0.478	4.9	0.488

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.687	2.728	0.473
0.02	2.662	2.668	0.181
0.03	2.638	2.639	0.039
0.04	2.616	2.617	0.008
0.05	2.596	2.596	0.004
0.06	2.587	2.587	0.001
0.07	2.577	2.577	0.000
0.08	2.568	2.568	0.000
0.09	2.560	2.560	0.000
0.10	2.552	2.552	0.000
0.15	2.538	2.538	0.000
0.20	2.526	2.526	0.000
0.25	2.515	2.515	0.000
0.30	2.515	2.515	0.000

3-Phase fault at bus: **Bus162**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 4.230 kA Method C
Steady State = 2.488 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus162	Total	0.00	2.468	-1.255	0.5	2.769
TYPICAL MOTOR FEEDER 1 A	Bus162	102.61	2.371	-0.777	0.3	2.495
614-FM-06DA	Bus162	105.00	0.098	-0.478	4.9	0.488

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.646	2.688	0.475
0.02	2.621	2.627	0.183
0.03	2.597	2.597	0.040
0.04	2.575	2.575	0.009
0.05	2.555	2.555	0.005
0.06	2.545	2.545	0.001
0.07	2.536	2.536	0.000
0.08	2.527	2.527	0.000
0.09	2.518	2.518	0.000
0.10	2.510	2.510	0.000
0.15	2.497	2.497	0.000
0.20	2.485	2.485	0.000
0.25	2.474	2.474	0.000
0.30	2.474	2.474	0.000

3-Phase fault at bus: **Bus163**

Nominal kV = 11.000
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 18.022 kA Method C
Steady State = 9.449 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus163	Total	0.00	6.436	-8.988	1.4	11.054
S003-SB-11C001 [BUS B]	Bus163	1.37	6.310	-8.467	1.3	10.560
Bus49	Bus163	22.70	0.126	-0.521	4.1	0.535

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	10.936	11.145	2.150
0.02	10.868	10.876	0.405
0.03	10.796	10.796	0.065
0.04	10.726	10.726	0.010
0.05	10.659	10.659	0.003
0.06	10.626	10.626	0.000
0.07	10.594	10.594	0.000
0.08	10.562	10.562	0.000
0.09	10.532	10.532	0.000
0.10	10.501	10.501	0.000
0.15	10.442	10.442	0.000
0.20	10.385	10.385	0.000
0.25	10.331	10.331	0.000
0.30	10.331	10.331	0.000

3-Phase fault at bus: **Bus164**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 4.287 kA Method C
Steady State = 2.529 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus164	Total	0.00	2.507	-1.269	0.5	2.810
TYPICAL MOTOR FEEDER 1 A	Bus164	102.57	2.409	-0.791	0.3	2.536
614-FM-06DB	Bus164	105.00	0.098	-0.478	4.9	0.488

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.687	2.728	0.473
0.02	2.662	2.668	0.181
0.03	2.638	2.639	0.039
0.04	2.616	2.617	0.008
0.05	2.596	2.596	0.004
0.06	2.587	2.587	0.001
0.07	2.577	2.577	0.000
0.08	2.568	2.568	0.000
0.09	2.560	2.560	0.000
0.10	2.552	2.552	0.000
0.15	2.538	2.538	0.000
0.20	2.526	2.526	0.000
0.25	2.515	2.515	0.000
0.30	2.515	2.515	0.000

3-Phase fault at bus: **Bus165**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 4.345 kA Method C
Steady State = 2.572 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus165	Total	0.00	2.547	-1.284	0.5	2.853
TYPICAL MOTOR FEEDER 1 A	Bus165	102.53	2.449	-0.806	0.3	2.579
614-FM-06DC	Bus165	105.00	0.098	-0.478	4.9	0.488

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.730	2.770	0.471
0.02	2.705	2.710	0.179
0.03	2.681	2.681	0.038
0.04	2.659	2.659	0.008
0.05	2.639	2.639	0.004
0.06	2.629	2.629	0.001
0.07	2.620	2.620	0.000
0.08	2.611	2.611	0.000
0.09	2.602	2.602	0.000
0.10	2.594	2.594	0.000
0.15	2.581	2.581	0.000
0.20	2.569	2.569	0.000
0.25	2.558	2.558	0.000
0.30	2.558	2.558	0.000

3-Phase fault at bus: **Bus166**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 80.053 kA Method C
Steady State = 36.597 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus166	Total	0.00	13.535	-39.028	2.9	41.308
S003-SB-11E001 [BUS A]	Bus166	1.89	1.551	-3.696	2.4	4.008
Bus19	Bus166	86.66	11.985	-35.332	2.9	37.309

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	40.929	45.588	20.079
0.02	40.534	41.152	7.103
0.03	40.150	40.227	2.477
0.04	39.781	39.790	0.864
0.05	39.428	39.429	0.314
0.06	39.259	39.259	0.110
0.07	39.093	39.094	0.039
0.08	38.933	38.933	0.014
0.09	38.776	38.776	0.005
0.10	38.624	38.624	0.002
0.15	38.322	38.322	0.000
0.20	38.036	38.036	0.000
0.25	37.767	37.767	0.000
0.30	37.767	37.767	0.000

3-Phase fault at bus: **Bus167**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 88.807 kA Method C
Steady State = 36.803 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus167	Total	0.00	9.985	-40.700	4.1	41.907
S003-SB-11E001 [BUS B]	Bus167	2.05	1.547	-4.063	2.6	4.348
Bus20	Bus167	87.26	8.439	-36.637	4.3	37.596

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	41.202	50.001	28.328
0.02	40.798	43.136	14.008
0.03	40.405	40.975	6.811
0.04	40.026	40.163	3.311
0.05	39.665	39.700	1.684
0.06	39.489	39.498	0.826
0.07	39.318	39.320	0.405
0.08	39.151	39.152	0.199
0.09	38.989	38.989	0.098
0.10	38.831	38.831	0.050
0.15	38.527	38.527	0.001
0.20	38.239	38.239	0.000
0.25	37.969	37.969	0.000
0.30	37.969	37.969	0.000

3-Phase fault at bus: **Bus168**

Nominal kV = 11.000
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 17.668 kA Method C
Steady State = 9.509 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus168	Total	0.00	6.450	-8.801	1.4	10.912
S003-SB-11C001 [BUS A]	Bus168	1.02	6.370	-8.408	1.3	10.549
Bus50	Bus168	16.86	0.080	-0.393	4.9	0.401

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	10.808	10.989	1.985
0.02	10.750	10.755	0.351
0.03	10.691	10.691	0.053
0.04	10.629	10.629	0.008
0.05	10.568	10.568	0.002
0.06	10.538	10.538	0.000
0.07	10.509	10.509	0.000
0.08	10.480	10.480	0.000
0.09	10.452	10.452	0.000
0.10	10.424	10.424	0.000
0.15	10.370	10.370	0.000
0.20	10.317	10.317	0.000
0.25	10.265	10.265	0.000
0.30	10.265	10.265	0.000

3-Phase fault at bus: **Bus169**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 2.083 kA Method C
Steady State = 1.311 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus169	Total	0.00	1.359	-0.449	0.3	1.431
S002-MCC1-9 [BUS]	Bus169	102.98	1.289	-0.253	0.2	1.314
S002-6600-MP-2B	Bus169	105.00	0.069	-0.196	2.8	0.208

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	1.370	1.371	0.040
0.02	1.360	1.360	0.003
0.03	1.350	1.350	0.000
0.04	1.341	1.341	0.000
0.05	1.333	1.333	0.000
0.06	1.329	1.329	0.000
0.07	1.325	1.325	0.000
0.08	1.322	1.322	0.000
0.09	1.318	1.318	0.000
0.10	1.315	1.315	0.000
0.15	1.309	1.309	0.000
0.20	1.304	1.304	0.000
0.25	1.300	1.300	0.000
0.30	1.300	1.300	0.000

3-Phase fault at bus: **Bus170**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 4.287 kA Method C
Steady State = 2.529 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus170	Total	0.00	2.507	-1.269	0.5	2.810
TYPICAL MOTOR FEEDER 1 A	Bus170	102.57	2.409	-0.791	0.3	2.536
614-FM-06EA	Bus170	105.00	0.098	-0.478	4.9	0.488

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.687	2.728	0.473
0.02	2.662	2.668	0.181
0.03	2.638	2.639	0.039
0.04	2.616	2.617	0.008
0.05	2.596	2.596	0.004
0.06	2.587	2.587	0.001
0.07	2.577	2.577	0.000
0.08	2.568	2.568	0.000
0.09	2.560	2.560	0.000
0.10	2.552	2.552	0.000
0.15	2.538	2.538	0.000
0.20	2.526	2.526	0.000
0.25	2.515	2.515	0.000
0.30	2.515	2.515	0.000

3-Phase fault at bus: **Bus171**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 4.345 kA Method C
Steady State = 2.572 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus171	Total	0.00	2.547	-1.284	0.5	2.853
TYPICAL MOTOR FEEDER 1 A	Bus171	102.53	2.449	-0.806	0.3	2.579
614-FM-06EB	Bus171	105.00	0.098	-0.478	4.9	0.488

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.730	2.770	0.471
0.02	2.705	2.710	0.179
0.03	2.681	2.681	0.038
0.04	2.659	2.659	0.008
0.05	2.639	2.639	0.004
0.06	2.629	2.629	0.001
0.07	2.620	2.620	0.000
0.08	2.611	2.611	0.000
0.09	2.602	2.602	0.000
0.10	2.594	2.594	0.000
0.15	2.581	2.581	0.000
0.20	2.569	2.569	0.000
0.25	2.558	2.558	0.000
0.30	2.558	2.558	0.000

3-Phase fault at bus: **Bus172**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 4.406 kA Method C
Steady State = 2.616 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus172	Total	0.00	2.588	-1.300	0.5	2.896
TYPICAL MOTOR FEEDER 1 A	Bus172	102.48	2.491	-0.822	0.3	2.623
614-FM-06EC	Bus172	105.00	0.098	-0.478	4.9	0.488

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.774	2.813	0.468
0.02	2.749	2.754	0.177
0.03	2.725	2.725	0.037
0.04	2.703	2.703	0.008
0.05	2.683	2.683	0.004
0.06	2.673	2.673	0.001
0.07	2.664	2.664	0.000
0.08	2.655	2.655	0.000
0.09	2.647	2.647	0.000
0.10	2.638	2.638	0.000
0.15	2.625	2.625	0.000
0.20	2.613	2.613	0.000
0.25	2.602	2.602	0.000
0.30	2.602	2.602	0.000

3-Phase fault at bus: **Bus173**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 4.603 kA Method C
Steady State = 2.756 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus173	Total	0.00	2.727	-1.343	0.5	3.040
TYPICAL MOTOR FEEDER 1 B	Bus173	102.45	2.629	-0.865	0.3	2.768
614-FM-06KC	Bus173	105.00	0.098	-0.478	4.9	0.488

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.918	2.954	0.458
0.02	2.893	2.898	0.169
0.03	2.870	2.870	0.034
0.04	2.848	2.848	0.007
0.05	2.828	2.828	0.003
0.06	2.818	2.818	0.001
0.07	2.809	2.809	0.000
0.08	2.800	2.800	0.000
0.09	2.791	2.791	0.000
0.10	2.783	2.783	0.000
0.15	2.770	2.770	0.000
0.20	2.758	2.758	0.000
0.25	2.747	2.747	0.000
0.30	2.747	2.747	0.000

Project:

Location:

Contract:

Engineer:

Filename:

UQ LT LPG Export - ST & MT

Umm Qasr, Iraq

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Hemanathan Gopalan

Umm Qasr-ST

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Study Case: SC_Max

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Base

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3-Phase fault at bus: **Bus174**

Nominal kV

Voltage c Factor

Peak Value

Steady State

= 0.380

= 1.05

= 4.535

= 2.708

(Maximum If)

kA Method C

kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus174	Total	0.00	2.681	-1.326	0.5	2.991
TYPICAL MOTOR FEEDER 1 B	Bus174	102.50	2.583	-0.848	0.3	2.719
614-FM-06KB	Bus174	105.00	0.098	-0.478	4.9	0.488

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.869	2.906	0.461
0.02	2.844	2.849	0.172
0.03	2.821	2.821	0.035
0.04	2.799	2.799	0.007
0.05	2.779	2.779	0.004
0.06	2.769	2.769	0.001
0.07	2.760	2.760	0.000
0.08	2.751	2.751	0.000
0.09	2.742	2.742	0.000
0.10	2.734	2.734	0.000
0.15	2.721	2.721	0.000
0.20	2.709	2.709	0.000
0.25	2.698	2.698	0.000
0.30	2.698	2.698	0.000

3-Phase fault at bus: **Bus175**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 4.470 kA Method C
Steady State = 2.661 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus175	Total	0.00	2.636	-1.309	0.5	2.943
TYPICAL MOTOR FEEDER 1 B	Bus175	102.55	2.539	-0.831	0.3	2.671
614-FM-06KA	Bus175	105.00	0.098	-0.478	4.9	0.488

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.822	2.860	0.464
0.02	2.797	2.802	0.174
0.03	2.773	2.773	0.036
0.04	2.751	2.751	0.007
0.05	2.731	2.731	0.004
0.06	2.722	2.722	0.001
0.07	2.712	2.712	0.000
0.08	2.703	2.703	0.000
0.09	2.695	2.695	0.000
0.10	2.687	2.687	0.000
0.15	2.674	2.674	0.000
0.20	2.661	2.661	0.000
0.25	2.650	2.650	0.000
0.30	2.650	2.650	0.000

3-Phase fault at bus: **Bus176**

Nominal kV = 0.380

Voltage c Factor = 1.05 (Maximum If)

Peak Value = 4.535 kA Method C

Steady State = 2.708 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus176	Total	0.00	2.681	-1.326	0.5	2.991
TYPICAL MOTOR FEEDER 1 B	Bus176	102.50	2.583	-0.848	0.3	2.719
614-FM-06JC	Bus176	105.00	0.098	-0.478	4.9	0.488

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.869	2.906	0.461
0.02	2.844	2.849	0.172
0.03	2.821	2.821	0.035
0.04	2.799	2.799	0.007
0.05	2.779	2.779	0.004
0.06	2.769	2.769	0.001
0.07	2.760	2.760	0.000
0.08	2.751	2.751	0.000
0.09	2.742	2.742	0.000
0.10	2.734	2.734	0.000
0.15	2.721	2.721	0.000
0.20	2.709	2.709	0.000
0.25	2.698	2.698	0.000
0.30	2.698	2.698	0.000

3-Phase fault at bus: **Bus177**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 4.470 kA Method C
Steady State = 2.661 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus177	Total	0.00	2.636	-1.309	0.5	2.943
TYPICAL MOTOR FEEDER 1 B	Bus177	102.55	2.539	-0.831	0.3	2.671
614-FM-06JB	Bus177	105.00	0.098	-0.478	4.9	0.488

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.822	2.860	0.464
0.02	2.797	2.802	0.174
0.03	2.773	2.773	0.036
0.04	2.751	2.751	0.007
0.05	2.731	2.731	0.004
0.06	2.722	2.722	0.001
0.07	2.712	2.712	0.000
0.08	2.703	2.703	0.000
0.09	2.695	2.695	0.000
0.10	2.687	2.687	0.000
0.15	2.674	2.674	0.000
0.20	2.661	2.661	0.000
0.25	2.650	2.650	0.000
0.30	2.650	2.650	0.000

3-Phase fault at bus: **Bus178**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 4.407 kA Method C
Steady State = 2.615 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus178	Total	0.00	2.593	-1.293	0.5	2.898
TYPICAL MOTOR FEEDER 1 B	Bus178	102.59	2.496	-0.815	0.3	2.626
614-FM-06JA	Bus178	105.00	0.098	-0.478	4.9	0.488

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.776	2.815	0.467
0.02	2.751	2.756	0.176
0.03	2.727	2.728	0.037
0.04	2.706	2.706	0.008
0.05	2.686	2.686	0.004
0.06	2.676	2.676	0.001
0.07	2.667	2.667	0.000
0.08	2.658	2.658	0.000
0.09	2.649	2.649	0.000
0.10	2.641	2.641	0.000
0.15	2.628	2.628	0.000
0.20	2.616	2.616	0.000
0.25	2.605	2.605	0.000
0.30	2.605	2.605	0.000

3-Phase fault at bus: **Bus179**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 4.470 kA Method C
Steady State = 2.661 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus179	Total	0.00	2.636	-1.309	0.5	2.943
TYPICAL MOTOR FEEDER 1 B	Bus179	102.55	2.539	-0.831	0.3	2.671
614-FM-06HC	Bus179	105.00	0.098	-0.478	4.9	0.488

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.822	2.860	0.464
0.02	2.797	2.802	0.174
0.03	2.773	2.773	0.036
0.04	2.751	2.751	0.007
0.05	2.731	2.731	0.004
0.06	2.722	2.722	0.001
0.07	2.712	2.712	0.000
0.08	2.703	2.703	0.000
0.09	2.695	2.695	0.000
0.10	2.687	2.687	0.000
0.15	2.674	2.674	0.000
0.20	2.661	2.661	0.000
0.25	2.650	2.650	0.000
0.30	2.650	2.650	0.000

3-Phase fault at bus: **Bus180**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 4.407 kA Method C
Steady State = 2.615 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus180	Total	0.00	2.593	-1.293	0.5	2.898
TYPICAL MOTOR FEEDER 1 B	Bus180	102.59	2.496	-0.815	0.3	2.626
614-FM-06HB	Bus180	105.00	0.098	-0.478	4.9	0.488

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.776	2.815	0.467
0.02	2.751	2.756	0.176
0.03	2.727	2.728	0.037
0.04	2.706	2.706	0.008
0.05	2.686	2.686	0.004
0.06	2.676	2.676	0.001
0.07	2.667	2.667	0.000
0.08	2.658	2.658	0.000
0.09	2.649	2.649	0.000
0.10	2.641	2.641	0.000
0.15	2.628	2.628	0.000
0.20	2.616	2.616	0.000
0.25	2.605	2.605	0.000
0.30	2.605	2.605	0.000

3-Phase fault at bus: **Bus181**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 4.346 kA Method C
Steady State = 2.572 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus181	Total	0.00	2.552	-1.278	0.5	2.854
TYPICAL MOTOR FEEDER 1 B	Bus181	102.63	2.454	-0.800	0.3	2.581
614-FM-06HA	Bus181	105.00	0.098	-0.478	4.9	0.488

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.732	2.772	0.469
0.02	2.707	2.712	0.178
0.03	2.683	2.683	0.038
0.04	2.661	2.661	0.008
0.05	2.641	2.641	0.004
0.06	2.632	2.632	0.001
0.07	2.622	2.622	0.000
0.08	2.613	2.613	0.000
0.09	2.605	2.605	0.000
0.10	2.597	2.597	0.000
0.15	2.584	2.584	0.000
0.20	2.572	2.572	0.000
0.25	2.560	2.560	0.000
0.30	2.560	2.560	0.000

3-Phase fault at bus: **Bus182**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 4.407 kA Method C
Steady State = 2.615 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus182	Total	0.00	2.593	-1.293	0.5	2.898
TYPICAL MOTOR FEEDER 1 B	Bus182	102.59	2.496	-0.815	0.3	2.626
614-FM-06GC	Bus182	105.00	0.098	-0.478	4.9	0.488

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.776	2.815	0.467
0.02	2.751	2.756	0.176
0.03	2.727	2.728	0.037
0.04	2.706	2.706	0.008
0.05	2.686	2.686	0.004
0.06	2.676	2.676	0.001
0.07	2.667	2.667	0.000
0.08	2.658	2.658	0.000
0.09	2.649	2.649	0.000
0.10	2.641	2.641	0.000
0.15	2.628	2.628	0.000
0.20	2.616	2.616	0.000
0.25	2.605	2.605	0.000
0.30	2.605	2.605	0.000

3-Phase fault at bus: **Bus183**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 4.346 kA Method C
Steady State = 2.572 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus183	Total	0.00	2.552	-1.278	0.5	2.854
TYPICAL MOTOR FEEDER 1 B	Bus183	102.63	2.454	-0.800	0.3	2.581
614-FM-06GB	Bus183	105.00	0.098	-0.478	4.9	0.488

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.732	2.772	0.469
0.02	2.707	2.712	0.178
0.03	2.683	2.683	0.038
0.04	2.661	2.661	0.008
0.05	2.641	2.641	0.004
0.06	2.632	2.632	0.001
0.07	2.622	2.622	0.000
0.08	2.613	2.613	0.000
0.09	2.605	2.605	0.000
0.10	2.597	2.597	0.000
0.15	2.584	2.584	0.000
0.20	2.572	2.572	0.000
0.25	2.560	2.560	0.000
0.30	2.560	2.560	0.000

3-Phase fault at bus: **Bus184**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 4.287 kA Method C
Steady State = 2.529 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus184	Total	0.00	2.512	-1.263	0.5	2.811
TYPICAL MOTOR FEEDER 1 B	Bus184	102.67	2.414	-0.785	0.3	2.539
614-FM-06GA	Bus184	105.00	0.098	-0.478	4.9	0.488

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.689	2.730	0.472
0.02	2.664	2.670	0.181
0.03	2.640	2.641	0.038
0.04	2.619	2.619	0.008
0.05	2.599	2.599	0.004
0.06	2.589	2.589	0.001
0.07	2.580	2.580	0.000
0.08	2.571	2.571	0.000
0.09	2.562	2.562	0.000
0.10	2.554	2.554	0.000
0.15	2.541	2.541	0.000
0.20	2.529	2.529	0.000
0.25	2.518	2.518	0.000
0.30	2.518	2.518	0.000

3-Phase fault at bus: **Bus185**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 4.346 kA Method C
Steady State = 2.572 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus185	Total	0.00	2.552	-1.278	0.5	2.854
TYPICAL MOTOR FEEDER 1 B	Bus185	102.63	2.454	-0.800	0.3	2.581
614-FM-06FC	Bus185	105.00	0.098	-0.478	4.9	0.488

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.732	2.772	0.469
0.02	2.707	2.712	0.178
0.03	2.683	2.683	0.038
0.04	2.661	2.661	0.008
0.05	2.641	2.641	0.004
0.06	2.632	2.632	0.001
0.07	2.622	2.622	0.000
0.08	2.613	2.613	0.000
0.09	2.605	2.605	0.000
0.10	2.597	2.597	0.000
0.15	2.584	2.584	0.000
0.20	2.572	2.572	0.000
0.25	2.560	2.560	0.000
0.30	2.560	2.560	0.000

3-Phase fault at bus: **Bus186**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 4.287 kA Method C
Steady State = 2.529 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus186	Total	0.00	2.512	-1.263	0.5	2.811
TYPICAL MOTOR FEEDER 1 B	Bus186	102.67	2.414	-0.785	0.3	2.539
614-FM-06FB	Bus186	105.00	0.098	-0.478	4.9	0.488

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.689	2.730	0.472
0.02	2.664	2.670	0.181
0.03	2.640	2.641	0.038
0.04	2.619	2.619	0.008
0.05	2.599	2.599	0.004
0.06	2.589	2.589	0.001
0.07	2.580	2.580	0.000
0.08	2.571	2.571	0.000
0.09	2.562	2.562	0.000
0.10	2.554	2.554	0.000
0.15	2.541	2.541	0.000
0.20	2.529	2.529	0.000
0.25	2.518	2.518	0.000
0.30	2.518	2.518	0.000

3-Phase fault at bus: **Bus187**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 4.230 kA Method C
Steady State = 2.488 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus187	Total	0.00	2.473	-1.248	0.5	2.770
TYPICAL MOTOR FEEDER 1 B	Bus187	102.71	2.375	-0.770	0.3	2.497
614-FM-06FA	Bus187	105.00	0.098	-0.478	4.9	0.488

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	2.648	2.690	0.474
0.02	2.623	2.629	0.183
0.03	2.599	2.599	0.039
0.04	2.577	2.577	0.009
0.05	2.557	2.557	0.004
0.06	2.548	2.548	0.001
0.07	2.538	2.538	0.000
0.08	2.529	2.529	0.000
0.09	2.521	2.521	0.000
0.10	2.513	2.513	0.000
0.15	2.500	2.500	0.000
0.20	2.487	2.487	0.000
0.25	2.476	2.476	0.000
0.30	2.476	2.476	0.000

3-Phase fault at bus: **Bus188**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 8.674 kA Method C
Steady State = 5.834 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus188	Total	0.00	4.862	-3.400	0.7	5.933
POWER SOCKET FEEDER 1 B	Bus188	97.06	4.862	-3.400	0.7	5.933

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	5.933	5.934	0.098
0.02	5.933	5.933	0.001
0.03	5.933	5.933	0.000
0.04	5.933	5.933	0.000
0.05	5.933	5.933	0.000
0.06	5.933	5.933	0.000
0.07	5.933	5.933	0.000
0.08	5.933	5.933	0.000
0.09	5.933	5.933	0.000
0.10	5.933	5.933	0.000
0.15	5.933	5.933	0.000
0.20	5.933	5.933	0.000
0.25	5.933	5.933	0.000
0.30	5.933	5.933	0.000

3-Phase fault at bus: **Bus189**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 17.075 kA Method C
Steady State = 11.128 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus189	Total	0.00	8.922	-7.319	0.8	11.540
POWER SOCKET FEEDER 1 B	Bus189	88.93	8.922	-7.319	0.8	11.540

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	11.540	11.546	0.377
0.02	11.540	11.540	0.009
0.03	11.540	11.540	0.000
0.04	11.540	11.540	0.000
0.05	11.540	11.540	0.000
0.06	11.540	11.540	0.000
0.07	11.540	11.540	0.000
0.08	11.540	11.540	0.000
0.09	11.540	11.540	0.000
0.10	11.540	11.540	0.000
0.15	11.540	11.540	0.000
0.20	11.540	11.540	0.000
0.25	11.540	11.540	0.000
0.30	11.540	11.540	0.000

3-Phase fault at bus: **Bus190**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 14.242 kA Method C
Steady State = 9.389 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus190	Total	0.00	7.631	-5.939	0.8	9.670
POWER SOCKET FEEDER 1 B	Bus190	91.71	7.631	-5.939	0.8	9.670

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	9.670	9.673	0.255
0.02	9.670	9.670	0.005
0.03	9.670	9.670	0.000
0.04	9.670	9.670	0.000
0.05	9.670	9.670	0.000
0.06	9.670	9.670	0.000
0.07	9.670	9.670	0.000
0.08	9.670	9.670	0.000
0.09	9.670	9.670	0.000
0.10	9.670	9.670	0.000
0.15	9.670	9.670	0.000
0.20	9.670	9.670	0.000
0.25	9.670	9.670	0.000
0.30	9.670	9.670	0.000

3-Phase fault at bus: **Bus_LVDB-1**

Nominal kV = 0.415
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 115.763 kA Method C
Steady State = 55.556 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus_LVDB-1	Total	0.00	18.517	-55.920	3.0	58.906
Bus52	Bus_LVDB-1	1.79	6.030	-18.282	3.0	19.251
Bus63	Bus_LVDB-1	1.79	6.030	-18.282	3.0	19.251
Bus64	Bus_LVDB-1	1.75	6.117	-18.484	3.0	19.470
Bus_MDB-1	Bus_LVDB-1	0.07	0.085	-0.218	2.6	0.234
Bus_MDB-2	Bus_LVDB-1	0.06	0.086	-0.217	2.5	0.234
Bus_MDB-3	Bus_LVDB-1	0.06	0.086	-0.217	2.5	0.233
Bus_MDB-4	Bus_LVDB-1	0.04	0.085	-0.218	2.6	0.234

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

<u>TD (S)</u>	<u>Ib sym</u>	<u>Ib asym</u>	<u>Idc</u>
0.01	58.354	65.813	30.433
0.02	58.250	59.397	11.615
0.03	58.149	58.310	4.337
0.04	58.050	58.073	1.619
0.05	57.956	57.959	0.645
0.06	57.907	57.908	0.244
0.07	57.860	57.860	0.092
0.08	57.813	57.813	0.035
0.09	57.768	57.768	0.013
0.10	57.723	57.723	0.005
0.15	57.651	57.651	0.000
0.20	57.582	57.582	0.000
0.25	57.516	57.516	0.000
0.30	57.516	57.516	0.000

Project:

UQ LT LPG Export - ST & MT

Location:

Umm Qasr, Iraq

Contract:

UQ83

Engineer:

Hemanathan Gopalan

Filename:

Umm Qasr-ST

ETAP

24.0.2C

Study Case: SC_Max

Page:

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Date:

03-08-2026

SN:

SL-KENTUAE

Revision:

Base

Config.:

1_NE

3-Phase fault at bus: **Bus_LVDB-2**

Nominal kV

=

0.415

Voltage c Factor

=

1.05

(Maximum If)

Peak Value

=

116.230

kA

Method C

Steady State

=

56.203

kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus_LVDB-2	Total	0.00	18.513	-56.133	3.0	59.107
Bus_MDB-8	Bus_LVDB-2	0.07	0.087	-0.216	2.5	0.233
Bus40	Bus_LVDB-2	1.80	6.057	-18.421	3.0	19.392
Bus45	Bus_LVDB-2	1.80	6.057	-18.421	3.0	19.392
Bus48	Bus_LVDB-2	1.80	6.057	-18.421	3.0	19.392
Bus_MDB-5	Bus_LVDB-2	0.07	0.085	-0.218	2.5	0.234
Bus_MDB-6	Bus_LVDB-2	0.07	0.085	-0.218	2.6	0.234
Bus_MDB-7	Bus_LVDB-2	0.07	0.086	-0.217	2.5	0.234

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

<u>TD (S)</u>	<u>Ib sym</u>	<u>Ib asym</u>	<u>Idc</u>
0.01	58.665	66.162	30.592
0.02	58.588	59.743	11.690
0.03	58.512	58.675	4.372
0.04	58.438	58.461	1.635
0.05	58.367	58.370	0.652
0.06	58.330	58.330	0.247
0.07	58.293	58.294	0.094
0.08	58.258	58.258	0.035
0.09	58.222	58.222	0.013
0.10	58.188	58.188	0.005
0.15	58.132	58.132	0.000
0.20	58.079	58.079	0.000
0.25	58.029	58.029	0.000
0.30	58.029	58.029	0.000

3-Phase fault at bus: **Bus_MDB-1**

Nominal kV = 0.415
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 96.178 kA Method C
Steady State = 48.514 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus_MDB-1	Total	0.00	18.521	-47.634	2.6	51.108
Bus_LVDB-1	Bus_MDB-1	14.83	18.437	-47.416	2.6	50.874
Cable188~	Bus_MDB-1	0.00	0.000	0.000	999.9	0.000
Cable189~	Bus_MDB-1	0.00	0.000	0.000	999.9	0.000
Bus95	Bus_MDB-1	1.31	0.042	-0.109	2.6	0.117
Bus96	Bus_MDB-1	1.57	0.042	-0.109	2.6	0.117

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	50.671	55.242	22.003
0.02	50.589	51.070	6.990
0.03	50.509	50.556	2.174
0.04	50.431	50.435	0.676
0.05	50.355	50.355	0.224
0.06	50.316	50.316	0.071
0.07	50.277	50.277	0.022
0.08	50.239	50.239	0.007
0.09	50.202	50.202	0.002
0.10	50.165	50.165	0.001
0.15	50.106	50.106	0.000
0.20	50.049	50.049	0.000
0.25	49.994	49.994	0.000
0.30	49.994	49.994	0.000

3-Phase fault at bus: **Bus_MDB-2**

Nominal kV = 0.415
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 97.446 kA Method C
Steady State = 48.995 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus_MDB-2	Total	0.00	18.548	-48.192	2.6	51.638
Bus_LVDB-1	Bus_MDB-2	13.83	18.462	-47.975	2.6	51.404
Cable192~	Bus_MDB-2	0.00	0.000	0.000	999.9	0.000
Cable193~	Bus_MDB-2	0.00	0.000	0.000	999.9	0.000
Bus99	Bus_MDB-2	1.83	0.043	-0.109	2.6	0.117
Bus100	Bus_MDB-2	2.35	0.043	-0.109	2.5	0.117

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	51.194	55.924	22.510
0.02	51.111	51.621	7.242
0.03	51.029	51.080	2.280
0.04	50.949	50.954	0.718
0.05	50.872	50.873	0.241
0.06	50.833	50.833	0.077
0.07	50.794	50.794	0.025
0.08	50.755	50.755	0.008
0.09	50.717	50.717	0.002
0.10	50.680	50.680	0.001
0.15	50.620	50.620	0.000
0.20	50.562	50.562	0.000
0.25	50.506	50.506	0.000
0.30	50.506	50.506	0.000

3-Phase fault at bus: **Bus_MDB-3**

Nominal kV = 0.415
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 98.093 kA Method C
Steady State = 49.239 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus_MDB-3	Total	0.00	18.560	-48.475	2.6	51.907
Bus_LVDB-1	Bus_MDB-3	13.32	18.474	-48.258	2.6	51.673
Cable196~	Bus_MDB-3	0.00	0.000	0.000	999.9	0.000
Cable197~	Bus_MDB-3	0.00	0.000	0.000	999.9	0.000
Bus103	Bus_MDB-3	2.61	0.043	-0.108	2.5	0.117
Bus104	Bus_MDB-3	1.83	0.043	-0.109	2.6	0.117

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	51.459	56.272	22.771
0.02	51.375	51.901	7.373
0.03	51.293	51.346	2.336
0.04	51.212	51.218	0.740
0.05	51.135	51.135	0.250
0.06	51.095	51.095	0.080
0.07	51.055	51.055	0.026
0.08	51.017	51.017	0.008
0.09	50.978	50.978	0.003
0.10	50.941	50.941	0.001
0.15	50.880	50.880	0.000
0.20	50.822	50.822	0.000
0.25	50.766	50.766	0.000
0.30	50.766	50.766	0.000

3-Phase fault at bus: **Bus_MDB-4**

Nominal kV = 0.415
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 103.596 kA Method C
Steady State = 51.278 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus_MDB-4	Total	0.00	18.621	-50.855	2.7	54.156
Bus_LVDB-1	Bus_MDB-4	9.07	18.536	-50.636	2.7	53.922
Cable200~	Bus_MDB-4	0.00	0.000	0.000	999.9	0.000
Cable201~	Bus_MDB-4	0.00	0.000	0.000	999.9	0.000
Bus107	Bus_MDB-4	1.31	0.042	-0.109	2.6	0.117
Bus108	Bus_MDB-4	1.68	0.042	-0.109	2.6	0.117

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	53.676	59.233	25.049
0.02	53.586	54.265	8.554
0.03	53.498	53.574	2.859
0.04	53.412	53.421	0.955
0.05	53.329	53.330	0.340
0.06	53.287	53.287	0.115
0.07	53.245	53.245	0.039
0.08	53.204	53.204	0.013
0.09	53.163	53.163	0.004
0.10	53.123	53.123	0.002
0.15	53.059	53.059	0.000
0.20	52.997	52.997	0.000
0.25	52.939	52.939	0.000
0.30	52.939	52.939	0.000

3-Phase fault at bus: **Bus_MDB-5**

Nominal kV = 0.415
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 97.134 kA Method C
Steady State = 49.259 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus_MDB-5	Total	0.00	18.548	-48.073	2.6	51.527
Bus_LVDB-2	Bus_MDB-5	14.38	18.463	-47.856	2.6	51.293
Cable153~	Bus_MDB-5	0.00	0.000	0.000	999.9	0.000
Cable154~	Bus_MDB-5	0.00	0.000	0.000	999.9	0.000
Bus60	Bus_MDB-5	1.93	0.043	-0.109	2.6	0.117
Bus62	Bus_MDB-5	2.20	0.043	-0.109	2.5	0.117

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	51.168	55.833	22.343
0.02	51.106	51.603	7.146
0.03	51.044	51.093	2.238
0.04	50.984	50.989	0.701
0.05	50.925	50.925	0.234
0.06	50.894	50.894	0.074
0.07	50.864	50.864	0.024
0.08	50.834	50.834	0.007
0.09	50.805	50.805	0.002
0.10	50.776	50.776	0.001
0.15	50.729	50.729	0.000
0.20	50.684	50.684	0.000
0.25	50.641	50.641	0.000
0.30	50.641	50.641	0.000

3-Phase fault at bus: **Bus_MDB-6**

Nominal kV = 0.400
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 93.624 kA Method C
Steady State = 47.478 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus_MDB-6	Total	0.00	17.877	-46.336	2.6	49.665
Bus_LVDB-2	Bus_MDB-6	13.86	17.796	-46.125	2.6	49.439
Cable163~	Bus_MDB-6	0.00	0.000	0.000	999.9	0.000
Cable164~	Bus_MDB-6	0.00	0.000	0.000	999.9	0.000
Bus72	Bus_MDB-6	1.44	0.041	-0.105	2.6	0.113
Bus73	Bus_MDB-6	1.70	0.041	-0.105	2.6	0.113

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	49.319	53.816	21.536
0.02	49.260	49.739	6.888
0.03	49.200	49.248	2.157
0.04	49.142	49.147	0.676
0.05	49.085	49.086	0.225
0.06	49.056	49.056	0.071
0.07	49.026	49.026	0.023
0.08	48.997	48.997	0.007
0.09	48.969	48.969	0.002
0.10	48.940	48.940	0.001
0.15	48.895	48.895	0.000
0.20	48.851	48.851	0.000
0.25	48.808	48.808	0.000
0.30	48.808	48.808	0.000

3-Phase fault at bus: **Bus_MDB-7**

Nominal kV = 0.400
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 93.623 kA Method C
Steady State = 47.478 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus_MDB-7	Total	0.00	17.878	-46.336	2.6	49.665
Bus_LVDB-2	Bus_MDB-7	13.86	17.795	-46.126	2.6	49.440
Cable168~	Bus_MDB-7	0.00	0.000	0.000	999.9	0.000
Cable169~	Bus_MDB-7	0.00	0.000	0.000	999.9	0.000
Bus76	Bus_MDB-7	1.86	0.041	-0.105	2.6	0.113
Bus77	Bus_MDB-7	2.44	0.041	-0.105	2.5	0.112

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	49.319	53.816	21.535
0.02	49.260	49.739	6.888
0.03	49.200	49.248	2.157
0.04	49.142	49.147	0.675
0.05	49.085	49.086	0.225
0.06	49.056	49.056	0.071
0.07	49.026	49.026	0.023
0.08	48.997	48.997	0.007
0.09	48.969	48.969	0.002
0.10	48.940	48.940	0.001
0.15	48.895	48.895	0.000
0.20	48.851	48.851	0.000
0.25	48.808	48.808	0.000
0.30	48.808	48.808	0.000

3-Phase fault at bus: **Bus_MDB-8**

Nominal kV = 0.400
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 93.623 kA Method C
Steady State = 47.478 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
Bus_MDB-8	Total	0.00	17.878	-46.335	2.6	49.665
Cable40~	Bus_MDB-8	0.00	0.000	0.000	999.9	0.000
Cable41~	Bus_MDB-8	0.00	0.000	0.000	999.9	0.000
Bus14	Bus_MDB-8	2.65	0.042	-0.104	2.5	0.112
Bus15	Bus_MDB-8	2.91	0.042	-0.104	2.5	0.112
Bus_LVDB-2	Bus_MDB-8	13.86	17.795	-46.127	2.6	49.440

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	49.319	53.815	21.534
0.02	49.260	49.739	6.887
0.03	49.201	49.248	2.157
0.04	49.142	49.147	0.675
0.05	49.085	49.086	0.225
0.06	49.056	49.056	0.071
0.07	49.027	49.027	0.023
0.08	48.997	48.997	0.007
0.09	48.969	48.969	0.002
0.10	48.940	48.940	0.001
0.15	48.895	48.895	0.000
0.20	48.851	48.851	0.000
0.25	48.808	48.808	0.000
0.30	48.808	48.808	0.000

3-Phase fault at bus: **BUS_S002-CMP3**

Nominal kV = 11.000
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 76.691 kA Method C
Steady State = 22.866 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
BUS_S002-CMP3	Total	0.00	2.542	-30.001	11.8	30.108
614-KM-11	BUS_S002-CMP3	110.00	0.214	-1.046	4.9	1.068
TR-PSS1AC02_11kV BUS	S002-SB-1AC001 [BUS B]	0.41	0.983	-22.919	23.3	22.940
Bus42	S002-SB-1AC001 [BUS B]	0.02	0.047	-0.113	2.4	0.123
Bus46	S002-SB-1AC001 [BUS B]	0.11	0.024	-0.059	2.4	0.064
Bus8	S002-SB-1AC001 [BUS B]	0.28	0.055	-0.137	2.5	0.148
Bus18	S002-SB-1AC001 [BUS B]	0.38	0.075	-0.192	2.6	0.206
Bus21	S002-SB-1AC001 [BUS B]	0.07	0.317	-1.560	4.9	1.592
Bus113	S002-SB-1AC001 [BUS B]	0.31	0.059	-0.320	5.4	0.326
Bus114	S002-SB-1AC001 [BUS B]	0.44	0.261	-1.945	7.4	1.963
Bus115	S002-SB-1AC001 [BUS B]	0.30	0.059	-0.320	5.4	0.326
Bus117	S002-SB-1AC001 [BUS B]	0.22	0.034	-0.185	5.4	0.189
S003-SB-11C001 [BUS B]	S002-SB-1AC001 [BUS B]	7.13	0.297	-0.864	2.9	0.913
S002-SB-05N001B	S002-SB-1AC001 [BUS B]	22.21	0.116	-0.340	2.9	0.359
S002-TR-CMP2_VFD~2	S002-SB-1AC001 [BUS B]	0.00	0.000	0.000	999.9	0.000
S002-TR-CMP2_VFD~3	S002-SB-1AC001 [BUS B]	0.00	0.000	0.000	999.9	0.000
S002-SB-1AC001 [BUS B]	BUS_S002-CMP3	0.00	2.328	-28.955	12.4	29.048

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	29.424	45.159	34.257
0.02	29.061	41.322	29.376
0.03	28.658	37.639	24.400
0.04	28.191	34.720	20.267
0.05	27.630	33.366	18.706
0.06	27.346	31.616	15.868
0.07	27.067	30.229	13.461
0.08	26.793	29.125	11.419
0.09	26.525	28.238	9.687

Project:

UQ LT LPG Export - ST & MT

Location:

Umm Qasr, Iraq

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3-Phase fault at bus: **BUS_S002-CMP3**

Nominal kV

=

11.000

Voltage c Factor

=

1.10

(Maximum If)

Peak Value

=

76.691

kA Method C

Steady State

=

22.866

kA rms

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

<u>TD (S)</u>	<u>Ib sym</u>	<u>Ib asym</u>	<u>Idc</u>
0.10	26.262	27.883	9.369
0.15	25.707	26.080	4.395
0.20	25.176	25.260	2.061
0.25	24.671	24.690	0.967
0.30	24.671	24.676	0.454

3-Phase fault at bus: **CAMP BUS**

Nominal kV = 11.000
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 24.999 kA Method C
Steady State = 13.240 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
CAMP BUS	Total	0.00	9.083	-12.334	1.4	15.318
MV RMU-TOFS	CAMP BUS	45.80	9.013	-11.633	1.3	14.716
CAMP LOAD	CAMP BUS	110.00	0.070	-0.702	10.0	0.705

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	15.212	15.572	3.327
0.02	15.159	15.200	1.124
0.03	15.095	15.097	0.256
0.04	14.998	14.998	0.058
0.05	14.870	14.870	0.052
0.06	14.804	14.804	0.015
0.07	14.740	14.740	0.005
0.08	14.676	14.676	0.001
0.09	14.612	14.612	0.000
0.10	14.550	14.550	0.001
0.15	14.411	14.411	0.000
0.20	14.276	14.276	0.000
0.25	14.144	14.144	0.000
0.30	14.144	14.144	0.000

3-Phase fault at bus: LV SWBD_TOFS

Nominal kV = 0.400
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 57.679 kA Method C
Steady State = 27.193 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
LV SWBD_TOFS	Total	0.00	8.115	-27.434	3.4	28.609
Bus51	LV SWBD_TOFS	96.10	7.688	-26.418	3.4	27.514
TOFS Load	LV SWBD_TOFS	105.00	0.427	-1.016	2.4	1.102

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	28.403	32.659	16.120
0.02	28.300	29.036	6.496
0.03	28.199	28.319	2.603
0.04	28.101	28.120	1.043
0.05	28.007	28.010	0.426
0.06	27.961	27.962	0.171
0.07	27.916	27.916	0.069
0.08	27.872	27.872	0.028
0.09	27.830	27.830	0.011
0.10	27.788	27.788	0.005
0.15	27.708	27.708	0.000
0.20	27.632	27.632	0.000
0.25	27.560	27.560	0.000
0.30	27.560	27.560	0.000

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3-Phase fault at bus: MV RMU-TOFS

Nominal kV

=

11.000

Voltage c Factor

=

1.10

(Maximum If)

Peak Value

=

44.311

kA

Method C

Steady State

=

18.915

kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
MV RMU-TOFS	Total	0.00	8.703	-21.818	2.5	23.490
CAMP BUS	MV RMU-TOFS	2.17	0.081	-0.693	8.6	0.697
S002-SB-1AC001 [BUS A]	MV RMU-TOFS	38.98	8.606	-21.085	2.5	22.774
LV SWBD_TOFS	Bus51	4.20	0.017	-0.040	2.4	0.044
Bus51	MV RMU-TOFS	0.00	0.017	-0.040	2.4	0.044

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Ide
0.01	23.147	25.460	10.604
0.02	22.963	23.321	4.073
0.03	22.751	22.796	1.426
0.04	22.501	22.507	0.499
0.05	22.186	22.188	0.244
0.06	22.026	22.026	0.091
0.07	21.867	21.867	0.034
0.08	21.711	21.711	0.013
0.09	21.557	21.557	0.005
0.10	21.406	21.406	0.003
0.15	21.080	21.080	0.000
0.20	20.766	20.766	0.000
0.25	20.465	20.465	0.000
0.30	20.465	20.465	0.000

3-Phase fault at bus: **POWER SOCKET**
FEEDER 1 B
Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 121.703 kA Method C
Steady State = 43.174 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
POWER SOCKET FEEDER 1 B	Total	0.00	10.984	-53.665	4.9	54.777
Bus188	POWER SOCKET FEEDER 1 B	0.00	0.000	0.000	999.9	0.000
Bus189	POWER SOCKET FEEDER 1 B	0.00	0.000	0.000	999.9	0.000
Bus190	POWER SOCKET FEEDER 1 B	0.00	0.000	0.000	999.9	0.000
S002-SB-1AC001 [BUS B]	S002-SB-05N001B	93.86	7.008	-43.367	6.2	43.930
TYPICAL MOTOR FEEDER 2 B	S002-SB-05N001B	105.00	0.243	-0.579	2.4	0.628
POWER FEEDER 1/2	S002-SB-05N001C	105.00	1.444	-3.437	2.4	3.728
Bus173	TYPICAL MOTOR FEEDER 1 B	16.58	0.150	-0.422	2.8	0.448
Bus174	TYPICAL MOTOR FEEDER 1 B	16.86	0.151	-0.421	2.8	0.447
Bus175	TYPICAL MOTOR FEEDER 1 B	17.14	0.152	-0.420	2.8	0.446
Bus176	TYPICAL MOTOR FEEDER 1 B	16.86	0.151	-0.421	2.8	0.447
Bus177	TYPICAL MOTOR FEEDER 1 B	17.14	0.152	-0.420	2.8	0.446
Bus178	TYPICAL MOTOR FEEDER 1 B	17.42	0.153	-0.419	2.7	0.446
Bus179	TYPICAL MOTOR FEEDER 1 B	17.14	0.152	-0.420	2.8	0.446
Bus180	TYPICAL MOTOR FEEDER 1 B	17.42	0.153	-0.419	2.7	0.446
Bus181	TYPICAL MOTOR FEEDER 1 B	17.69	0.153	-0.418	2.7	0.445
Bus182	TYPICAL MOTOR FEEDER 1 B	17.42	0.153	-0.419	2.7	0.446
Bus183	TYPICAL MOTOR FEEDER 1 B	17.69	0.153	-0.418	2.7	0.445
Bus184	TYPICAL MOTOR FEEDER 1 B	17.96	0.154	-0.417	2.7	0.444
Bus185	TYPICAL MOTOR FEEDER 1 B	17.69	0.153	-0.418	2.7	0.445
Bus186	TYPICAL MOTOR FEEDER 1 B	17.96	0.154	-0.417	2.7	0.444
Bus187	TYPICAL MOTOR FEEDER 1 B	18.24	0.155	-0.416	2.7	0.443
S002-SB-05N001B	POWER SOCKET FEEDER 1 B	0.00	10.984	-53.665	4.9	54.777
TYPICAL MOTOR FEEDER 1 B	S002-SB-05N001B	0.00	2.289	-6.281	2.7	6.685

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Engineer:	Hemanathan Gopalan	Study Case: SC_Max	Revision:	Base
Filename:	Umm Qasr-ST		Config.:	1_NE

(Cont.)

3-Phase fault at bus: **POWER SOCKET**
FEEDER 1 B
Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 121.703 kA Method C
Steady State = 43.174 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-SB-05N001C	S002-SB-05N001B	0.00	1.444	-3.437	2.4	3.728

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	51.801	67.590	43.417
0.02	50.852	56.994	25.737
0.03	49.939	52.096	14.835
0.04	49.068	49.807	8.551
0.05	48.245	48.530	5.250
0.06	47.843	47.941	3.064
0.07	47.452	47.486	1.789
0.08	47.074	47.085	1.044
0.09	46.707	46.711	0.609
0.10	46.352	46.354	0.378
0.15	45.718	45.718	0.026
0.20	45.123	45.123	0.002
0.25	44.568	44.568	0.000
0.30	44.568	44.568	0.000

Project:

UQ LT LPG Export - ST & MT

Location:

Umm Qasr, Iraq

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3-Phase fault at bus: S002-ASP1-1 [BUS A]

Nominal kV

=

0.380

Voltage c Factor

=

1.05

(Maximum If)

Peak Value

=

59.348

kA

Method C

Steady State

=

22.802

kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-ASP1-1 [BUS A]	Total	0.00	6.211	-26.804	4.3	27.514
S002-PC1-1A	S002-ASP1-1 [BUS A]	30.97	6.211	-26.804	4.3	27.514

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	27.049	33.542	19.834
0.02	26.764	28.799	10.632
0.03	26.483	27.060	5.557
0.04	26.207	26.368	2.905
0.05	25.940	25.991	1.624
0.06	25.809	25.824	0.860
0.07	25.680	25.684	0.456
0.08	25.554	25.555	0.241
0.09	25.429	25.429	0.128
0.10	25.307	25.307	0.073
0.15	25.065	25.065	0.003
0.20	24.832	24.832	0.000
0.25	24.611	24.611	0.000
0.30	24.611	24.611	0.000

3-Phase fault at bus: **S002-ASP1-1B [BUS B]**
Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 53.465 kA Method C
Steady State = 23.132 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-ASP1-1B [BUS B]	Total	0.00	4.898	-23.836	4.9	24.334
S002-PC1-1B	S002-ASP1-1B [BUS B]	26.40	4.898	-23.836	4.9	24.334

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	24.261	30.393	18.306
0.02	24.248	26.193	9.904
0.03	24.236	24.811	5.313
0.04	24.224	24.391	2.850
0.05	24.212	24.263	1.572
0.06	24.206	24.221	0.848
0.07	24.200	24.205	0.458
0.08	24.195	24.196	0.247
0.09	24.189	24.189	0.133
0.10	24.184	24.184	0.074
0.15	24.175	24.175	0.003
0.20	24.167	24.167	0.000
0.25	24.159	24.159	0.000
0.30	24.159	24.159	0.000

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3-Phase fault at bus: S002-ASP1-2A

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 63.168 kA Method C
Steady State = 23.846 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-ASP1-2A	Total	0.00	6.542	-28.474	4.4	29.216
S002-PC1-2A	S002-ASP1-2A	27.21	6.542	-28.474	4.4	29.216

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	28.625	35.647	21.244
0.02	28.297	30.554	11.525
0.03	27.975	28.629	6.087
0.04	27.660	27.846	3.215
0.05	27.355	27.416	1.823
0.06	27.206	27.223	0.977
0.07	27.059	27.064	0.523
0.08	26.915	26.916	0.280
0.09	26.773	26.774	0.150
0.10	26.635	26.635	0.087
0.15	26.364	26.364	0.004
0.20	26.104	26.104	0.000
0.25	25.857	25.857	0.000
0.30	25.857	25.857	0.000

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3-Phase fault at bus: S002-ASP1-2B

Nominal kV

=

0.380

Voltage c Factor

=

1.05

Peak Value

=

59.249

Steady State

=

23.967

(Maximum If)

Method C

kA

kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-ASP1-2B	Total	0.00	5.779	-26.412	4.6	27.037
S002-DBASP1-2-22	S002-ASP1-2B	8.71	0.515	-0.683	1.3	0.856
S002-PC1-2B	S002-ASP1-2B	24.46	5.264	-25.729	4.9	26.262

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	26.192	33.210	20.418
0.02	26.031	28.386	11.320
0.03	25.876	26.599	6.160
0.04	25.728	25.945	3.352
0.05	25.589	25.660	1.911
0.06	25.519	25.541	1.049
0.07	25.452	25.458	0.576
0.08	25.386	25.388	0.317
0.09	25.323	25.323	0.174
0.10	25.261	25.262	0.100
0.15	25.160	25.160	0.005
0.20	25.064	25.064	0.000
0.25	24.974	24.974	0.000
0.30	24.974	24.974	0.000

3-Phase fault at bus: S002-DBASP1-2-22

Nominal kV

=

0.380

Voltage c Factor

=

1.05

(Maximum If)

Peak Value

=

13.490

kA

Method C

Steady State

=

7.672

kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-DBASP1-2-22	Total	0.00	5.817	-6.541	1.1	8.753
S002-ASP1-2B	S002-DBASP1-2-22	79.74	5.277	-5.785	1.1	7.830
MOV_0.93	S002-DBASP1-2-22	105.00	0.049	-0.068	1.4	0.083
MOV_0.55	S002-DBASP1-2-22	105.00	0.014	-0.020	1.4	0.025
MOV_1.86	S002-DBASP1-2-22	105.00	0.146	-0.204	1.4	0.250
MOV_2.59	S002-DBASP1-2-22	105.00	0.135	-0.189	1.4	0.232
MOV_0.4	S002-DBASP1-2-22	105.00	0.083	-0.117	1.4	0.144
MOV_0.3	S002-DBASP1-2-22	105.00	0.012	-0.016	1.4	0.020
MOV_2.59 [F]	S002-DBASP1-2-22	105.00	0.101	-0.142	1.4	0.174

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	8.034	8.072	0.787
0.02	7.962	7.962	0.051
0.03	7.895	7.895	0.003
0.04	7.834	7.834	0.000
0.05	7.779	7.779	0.000
0.06	7.751	7.751	0.000
0.07	7.724	7.724	0.000
0.08	7.699	7.699	0.000
0.09	7.675	7.675	0.000
0.10	7.652	7.652	0.000
0.15	7.626	7.626	0.000
0.20	7.603	7.603	0.000
0.25	7.581	7.581	0.000
0.30	7.581	7.581	0.000

3-Phase fault at bus: S002-MCC1-1 [BUS]

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 61.390 kA Method C
Steady State = 22.612 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-MCC1-1 [BUS]	Total	0.00	6.649	-27.830	4.2	28.613
S002-PC1-1A	S002-MCC1-1 [BUS]	30.24	5.610	-25.357	4.5	25.971
Lump9	S002-MCC1-1 [BUS]	105.00	1.039	-2.473	2.4	2.682

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	27.923	34.546	20.340
0.02	27.485	29.537	10.818
0.03	27.057	27.629	5.594
0.04	26.644	26.800	2.892
0.05	26.247	26.296	1.609
0.06	26.055	26.069	0.844
0.07	25.867	25.871	0.443
0.08	25.683	25.684	0.232
0.09	25.504	25.504	0.122
0.10	25.329	25.329	0.069
0.15	24.989	24.989	0.003
0.20	24.665	24.665	0.000
0.25	24.360	24.360	0.000
0.30	24.360	24.360	0.000

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3-Phase fault at bus: S002-MCC1-2 [BUS]

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 55.432 kA Method C
Steady State = 23.712 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-MCC1-2 [BUS]	Total	0.00	4.804	-24.514	5.1	24.980
S002-PC1-1B	S002-MCC1-2 [BUS]	24.24	4.804	-24.514	5.1	24.980

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	24.903	31.546	19.364
0.02	24.889	27.130	10.796
0.03	24.876	25.582	5.968
0.04	24.863	25.081	3.299
0.05	24.851	24.921	1.875
0.06	24.844	24.866	1.043
0.07	24.838	24.845	0.580
0.08	24.832	24.835	0.322
0.09	24.827	24.827	0.179
0.10	24.821	24.821	0.103
0.15	24.812	24.812	0.006
0.20	24.804	24.804	0.000
0.25	24.796	24.796	0.000
0.30	24.796	24.796	0.000

3-Phase fault at bus: S002-MCC1-3 [BUS]

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 64.429 kA Method C
Steady State = 23.593 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-MCC1-3 [BUS]	Total	0.00	6.848	-29.117	4.3	29.912
S002-PC1-1A	S002-MCC1-3 [BUS]	26.46	5.809	-26.645	4.6	27.271
Lump10	S002-MCC1-3 [BUS]	105.00	1.039	-2.473	2.4	2.682

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	29.187	36.275	21.541
0.02	28.730	30.992	11.621
0.03	28.285	28.933	6.091
0.04	27.854	28.036	3.192
0.05	27.442	27.501	1.803
0.06	27.242	27.259	0.959
0.07	27.046	27.051	0.510
0.08	26.855	26.856	0.272
0.09	26.669	26.669	0.145
0.10	26.488	26.488	0.083
0.15	26.135	26.135	0.004
0.20	25.799	25.799	0.000
0.25	25.483	25.483	0.000
0.30	25.483	25.483	0.000

3-Phase fault at bus: S002-MCC1-4 [BUS]

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 22.717 kA Method C
Steady State = 14.674 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-MCC1-4 [BUS]	Total	0.00	12.429	-9.162	0.7	15.441
S002-SB-01E001 [BUS A]	S002-MCC1-4 [BUS]	84.97	12.170	-8.655	0.7	14.933
S002-MCC1-4_M	S002-MCC1-4 [BUS]	105.00	0.107	-0.145	1.3	0.180
S002-MCC1-4_L	S002-MCC1-4 [BUS]	105.00	0.152	-0.363	2.4	0.393

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	15.249	15.255	0.411
0.02	15.205	15.205	0.009
0.03	15.162	15.162	0.000
0.04	15.121	15.121	0.000
0.05	15.082	15.082	0.000
0.06	15.063	15.063	0.000
0.07	15.044	15.044	0.000
0.08	15.025	15.025	0.000
0.09	15.007	15.007	0.000
0.10	14.990	14.990	0.000
0.15	14.961	14.961	0.000
0.20	14.933	14.933	0.000
0.25	14.907	14.907	0.000
0.30	14.907	14.907	0.000

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3-Phase fault at bus: S002-MCC1-5 [BUS]

Nominal kV

=

0.380

Voltage c Factor

=

1.05

(Maximum If)

Peak Value

=

59.007

kA Method C

Steady State

=

21.747

kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-MCC1-5 [BUS]	Total	0.00	6.526	-26.843	4.1	27.625
S002-PC1-2A	S002-MCC1-5 [BUS]	33.93	5.487	-24.370	4.4	24.980
Lump16	S002-MCC1-5 [BUS]	105.00	1.039	-2.473	2.4	2.682

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	26.907	33.146	19.356
0.02	26.479	28.356	10.146
0.03	26.060	26.568	5.171
0.04	25.655	25.790	2.635
0.05	25.267	25.308	1.444
0.06	25.077	25.089	0.747
0.07	24.892	24.895	0.386
0.08	24.711	24.712	0.200
0.09	24.535	24.535	0.103
0.10	24.363	24.363	0.058
0.15	24.028	24.028	0.002
0.20	23.709	23.709	0.000
0.25	23.409	23.409	0.000
0.30	23.409	23.409	0.000

3-Phase fault at bus: S002-MCC1-6 [BUS]

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 18.018 kA Method C
Steady State = 11.062 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-MCC1-6 [BUS]	Total	0.00	9.914	-7.056	0.7	12.168
S002-SB-01E001 [BUS B]	S002-MCC1-6 [BUS]	90.00	9.444	-5.938	0.6	11.155
Lump34	S002-MCC1-6 [BUS]	105.00	0.235	-0.559	2.4	0.606
S002-MCC1-6_M_C	S002-MCC1-6 [BUS]	105.00	0.235	-0.559	2.4	0.606

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	11.854	11.865	0.504
0.02	11.758	11.758	0.021
0.03	11.666	11.666	0.001
0.04	11.579	11.579	0.000
0.05	11.497	11.497	0.000
0.06	11.456	11.456	0.000
0.07	11.417	11.417	0.000
0.08	11.380	11.380	0.000
0.09	11.343	11.343	0.000
0.10	11.308	11.308	0.000
0.15	11.245	11.245	0.000
0.20	11.186	11.186	0.000
0.25	11.132	11.132	0.000
0.30	11.132	11.132	0.000

3-Phase fault at bus: S002-MCC1-7 [BUS]

Nominal kV

Voltage c Factor

Peak Value

Steady State

= 0.380

= 1.05

= 58.482

= 21.574

(Maximum If)

kA Method C

kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-MCC1-7 [BUS]	Total	0.00	6.489	-26.618	4.1	27.398
S002-PC1-2A	S002-MCC1-7 [BUS]	34.58	5.450	-24.146	4.4	24.753
Lump17	S002-MCC1-7 [BUS]	105.00	1.039	-2.473	2.4	2.682

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	26.687	32.849	19.154
0.02	26.262	28.107	10.015
0.03	25.847	26.343	5.092
0.04	25.444	25.576	2.589
0.05	25.058	25.098	1.415
0.06	24.870	24.881	0.730
0.07	24.687	24.689	0.377
0.08	24.507	24.508	0.194
0.09	24.331	24.332	0.100
0.10	24.160	24.160	0.056
0.15	23.828	23.828	0.002
0.20	23.511	23.511	0.000
0.25	23.212	23.212	0.000
0.30	23.212	23.212	0.000

3-Phase fault at bus: S002-MCC1-8 [BUS]

Nominal kV

Voltage c Factor

Peak Value

Steady State

= 0.380

= 1.05

= 56.856

= 23.361

(Maximum If)

kA Method C

kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-MCC1-8 [BUS]	Total	0.00	6.252	-25.866	4.1	26.611
Bus159	S002-MCC1-8 [BUS]	15.09	0.086	-0.172	2.0	0.193
S002-PC1-2B	S002-MCC1-8 [BUS]	26.60	5.664	-24.500	4.3	25.146
Lump21	S002-MCC1-8 [BUS]	105.00	0.502	-1.194	2.4	1.295

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	25.857	31.810	18.528
0.02	25.659	27.346	9.457
0.03	25.469	25.906	4.741
0.04	25.287	25.398	2.377
0.05	25.115	25.146	1.248
0.06	25.030	25.038	0.631
0.07	24.948	24.950	0.319
0.08	24.868	24.868	0.162
0.09	24.790	24.791	0.082
0.10	24.715	24.716	0.044
0.15	24.585	24.585	0.001
0.20	24.463	24.463	0.000
0.25	24.349	24.349	0.000
0.30	24.349	24.349	0.000

3-Phase fault at bus: S002-MCC1-9 [BUS]

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 59.944 kA Method C
Steady State = 22.869 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-MCC1-9 [BUS]	Total	0.00	7.267	-27.668	3.8	28.606
Bus169	S002-MCC1-9 [BUS]	15.09	0.086	-0.172	2.0	0.193
S002-PC1-2A	S002-MCC1-9 [BUS]	30.48	6.585	-26.079	4.0	26.898
S002-6100-MP-1B	S002-MCC1-9 [BUS]	105.00	0.249	-0.592	2.4	0.642
Lump18	S002-MCC1-9 [BUS]	105.00	0.347	-0.825	2.4	0.895

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Ide
0.01	27.778	33.544	18.805
0.02	27.396	28.913	9.243
0.03	27.023	27.381	4.418
0.04	26.661	26.744	2.112
0.05	26.313	26.335	1.087
0.06	26.143	26.148	0.527
0.07	25.976	25.977	0.256
0.08	25.812	25.813	0.124
0.09	25.653	25.653	0.060
0.10	25.497	25.497	0.032
0.15	25.198	25.198	0.001
0.20	24.913	24.913	0.000
0.25	24.643	24.643	0.000
0.30	24.643	24.643	0.000

3-Phase fault at bus: S002-PC1-1A

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 86.320 kA Method C
Steady State = 30.119 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-PC1-1A	Total	0.00	8.017	-38.161	4.8	38.994
S002-SB-1AN001 [BUS A]	S002-PC1-1A	1.60	0.166	-0.307	1.8	0.349
S002-MCC1-1 [BUS]	S002-PC1-1A	3.03	1.001	-2.405	2.4	2.605
S002-MCC1-3 [BUS]	S002-PC1-1A	2.54	1.007	-2.416	2.4	2.618
S002-ASP1-1 [BUS A]	S002-PC1-1A	0.00	0.000	0.000	999.9	0.000
Bus30	S002-PC1-1A	84.71	5.467	-32.137	5.9	32.599
S002-DBPC1-1-01	S002-PC1-1A	105.00	0.376	-0.896	2.4	0.972

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	37.934	48.767	30.647
0.02	37.320	41.517	18.189
0.03	36.725	38.182	10.446
0.04	36.153	36.648	6.000
0.05	35.610	35.807	3.755
0.06	35.347	35.415	2.194
0.07	35.092	35.115	1.282
0.08	34.843	34.851	0.749
0.09	34.601	34.604	0.438
0.10	34.367	34.368	0.281
0.15	33.917	33.917	0.020
0.20	33.492	33.492	0.001
0.25	33.094	33.094	0.000
0.30	33.094	33.094	0.000

3-Phase fault at bus: S002-PC1-1B

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 74.055 kA Method C
Steady State = 30.313 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-PC1-1B	Total	0.00	5.439	-31.973	5.9	32.433
S002-MCC1-2 [BUS]	S002-PC1-1B	0.00	0.000	0.000	999.9	0.000
S002-SB-1AN001 [BUS B]	S002-PC1-1B	1.41	0.150	-0.268	1.8	0.307
S002-ASP1-1B [BUS B]	S002-PC1-1B	0.00	0.000	0.000	999.9	0.000
Bus29	S002-PC1-1B	83.13	5.289	-31.705	6.0	32.143

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	32.299	42.328	27.357
0.02	32.275	36.318	16.653
0.03	32.252	33.777	10.035
0.04	32.230	32.792	6.047
0.05	32.209	32.429	3.767
0.06	32.199	32.280	2.285
0.07	32.189	32.219	1.386
0.08	32.180	32.191	0.841
0.09	32.170	32.174	0.510
0.10	32.161	32.163	0.322
0.15	32.147	32.147	0.027
0.20	32.134	32.134	0.002
0.25	32.122	32.122	0.000
0.30	32.122	32.122	0.000

3-Phase fault at bus: S002-PC1-2A

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 87.155 kA Method C
Steady State = 30.192 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-PC1-2A	Total	0.00	8.136	-38.564	4.7	39.413
S002-MCC1-5 [BUS]	S002-PC1-2A	3.52	0.995	-2.394	2.4	2.593
S002-MCC1-7 [BUS]	S002-PC1-2A	3.62	0.993	-2.392	2.4	2.590
S002-MCC1-9 [BUS]	S002-PC1-2A	1.92	0.668	-1.560	2.3	1.697
S002-ASP1-2A	S002-PC1-2A	0.00	0.000	0.000	999.9	0.000
Bus31	S002-PC1-2A	84.65	5.480	-32.217	5.9	32.680

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	38.225	49.149	30.894
0.02	37.593	41.818	18.316
0.03	36.981	38.442	10.500
0.04	36.392	36.887	6.019
0.05	35.833	36.030	3.768
0.06	35.562	35.630	2.199
0.07	35.298	35.321	1.283
0.08	35.041	35.049	0.748
0.09	34.792	34.794	0.437
0.10	34.550	34.551	0.281
0.15	34.088	34.088	0.020
0.20	33.653	33.653	0.001
0.25	33.244	33.244	0.000
0.30	33.244	33.244	0.000

Project:

UQ LT LPG Export - ST & MT

Location:

Umm Qasr, Iraq

Contract:

UQ83

Engineer:

Hemanathan Gopalan

Filename:

Umm Qasr-ST

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3-Phase fault at bus: S002-PC1-2B

Nominal kV

=

0.380

Voltage c Factor

=

1.05

Peak Value

=

78.774

Steady State

=

30.387

(Maximum If)

Method C

kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-PC1-2B	Total	0.00	6.598	-34.333	5.2	34.961
S002-MCC1-8 [BUS]	S002-PC1-2B	1.55	0.579	-1.346	2.3	1.465
S002-ASP1-2B	S002-PC1-2B	0.79	0.510	-0.679	1.3	0.850
Bus32	S002-PC1-2B	82.94	5.270	-31.736	6.0	32.171
Exist MOE SS	S002-PC1-2B	105.00	0.240	-0.571	2.4	0.619

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Ide
0.01	33.895	44.424	28.716
0.02	33.653	37.879	17.388
0.03	33.420	34.975	10.311
0.04	33.200	33.759	6.115
0.05	32.994	33.215	3.823
0.06	32.892	32.972	2.292
0.07	32.793	32.822	1.373
0.08	32.698	32.708	0.823
0.09	32.606	32.610	0.493
0.10	32.518	32.519	0.313
0.15	32.368	32.368	0.025
0.20	32.228	32.228	0.002
0.25	32.097	32.097	0.000
0.30	32.097	32.097	0.000

3-Phase fault at bus: **S002-SB-1AC001**
[BUS A]
Nominal kV = 11.000
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 77.725 kA Method C
Steady State = 23.082 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-SB-1AC001 [BUS A]	Total	0.00	2.602	-30.433	11.7	30.544
TR-PSS1AC01_11kV BUS	S002-SB-1AC001 [BUS A]	0.32	0.971	-23.135	23.8	23.156
Bus43	S002-SB-1AC001 [BUS A]	0.04	0.055	-0.142	2.6	0.152
Bus7	S002-SB-1AC001 [BUS A]	0.17	0.038	-0.094	2.4	0.101
Bus112	S002-SB-1AC001 [BUS A]	0.45	0.087	-0.225	2.6	0.241
Bus17	S002-SB-1AC001 [BUS A]	0.41	0.078	-0.203	2.6	0.218
Bus1	S002-SB-1AC001 [BUS A]	0.10	0.427	-2.309	5.4	2.348
Bus110	S002-SB-1AC001 [BUS A]	0.29	0.057	-0.310	5.4	0.315
Bus92	S002-SB-1AC001 [BUS A]	0.47	0.417	-2.015	4.8	2.058
Bus90	S002-SB-1AC001 [BUS A]	0.22	0.034	-0.185	5.4	0.189
Bus89	S002-SB-1AC001 [BUS A]	0.22	0.034	-0.185	5.4	0.189
S003-SB-11C001 [BUS A]	S002-SB-1AC001 [BUS A]	5.39	0.212	-0.657	3.1	0.690
MV RMU-TOFS	S002-SB-1AC001 [BUS A]	1.26	0.104	-0.727	7.0	0.735
S002-SB-05N001A	S002-SB-1AC001 [BUS A]	16.11	0.086	-0.246	2.9	0.260
S002-TR-CMP1_VFD~2	S002-SB-1AC001 [BUS A]	0.00	0.000	0.000	999.9	0.000
S002-TR-CMP1_VFD~3	S002-SB-1AC001 [BUS A]	0.00	0.000	0.000	999.9	0.000

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

<u>TD (S)</u>	<u>Ib sym</u>	<u>Ib asym</u>	<u>Idc</u>
0.01	29.849	45.773	34.701
0.02	29.485	41.952	29.844
0.03	29.076	38.220	24.806
0.04	28.629	35.281	20.619
0.05	28.079	34.008	19.186
0.06	27.800	32.232	16.312
0.07	27.525	30.822	13.868
0.08	27.257	29.697	11.790
0.09	26.994	28.795	10.024
0.10	26.737	28.470	9.781

Project:

Location:

Contract:

Engineer:

Filename:

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Umm Qasr, Iraq

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3-Phase fault at bus: S002-SB-1AC001

[BUS A]

Nominal kV

=

11.000

Voltage c Factor

=

1.10

(Maximum If)

Peak Value

=

77.725

kA Method C

Steady State

=

23.082

kA rms

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.15	26.196	26.607	4.654
0.20	25.680	25.776	2.215
0.25	25.191	25.213	1.054
0.30	25.191	25.196	0.502

3-Phase fault at bus: **S002-SB-1AC001**
[BUS B]
Nominal kV = 11.000
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 76.691 kA Method C
Steady State = 22.866 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-SB-1AC001 [BUS B]	Total	0.00	2.542	-30.001	11.8	30.108
TR-PSS1AC02_11kV BUS	S002-SB-1AC001 [BUS B]	0.41	0.983	-22.919	23.3	22.940
Bus42	S002-SB-1AC001 [BUS B]	0.02	0.047	-0.113	2.4	0.123
Bus46	S002-SB-1AC001 [BUS B]	0.11	0.024	-0.059	2.4	0.064
Bus8	S002-SB-1AC001 [BUS B]	0.28	0.055	-0.137	2.5	0.148
Bus18	S002-SB-1AC001 [BUS B]	0.38	0.075	-0.192	2.6	0.206
Bus21	S002-SB-1AC001 [BUS B]	0.07	0.317	-1.560	4.9	1.592
Bus113	S002-SB-1AC001 [BUS B]	0.31	0.059	-0.320	5.4	0.326
Bus114	S002-SB-1AC001 [BUS B]	0.44	0.261	-1.945	7.4	1.963
Bus115	S002-SB-1AC001 [BUS B]	0.30	0.059	-0.320	5.4	0.326
Bus117	S002-SB-1AC001 [BUS B]	0.22	0.034	-0.185	5.4	0.189
S003-SB-11C001 [BUS B]	S002-SB-1AC001 [BUS B]	7.13	0.297	-0.864	2.9	0.913
S002-SB-05N001B	S002-SB-1AC001 [BUS B]	22.21	0.116	-0.340	2.9	0.359
S002-TR-CMP2_VFD~2	S002-SB-1AC001 [BUS B]	0.00	0.000	0.000	999.9	0.000
S002-TR-CMP2_VFD~3	S002-SB-1AC001 [BUS B]	0.00	0.000	0.000	999.9	0.000
614-KM-11	BUS_S002-CMP3	110.00	0.214	-1.046	4.9	1.068
BUS_S002-CMP3	S002-SB-1AC001 [BUS B]	0.00	0.214	-1.046	4.9	1.068

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	29.424	45.159	34.257
0.02	29.061	41.322	29.376
0.03	28.658	37.639	24.400
0.04	28.191	34.720	20.267
0.05	27.630	33.366	18.706
0.06	27.346	31.616	15.868
0.07	27.067	30.229	13.461
0.08	26.793	29.125	11.419
0.09	26.525	28.238	9.687

Project: UQ LT LPG Export - ST & MT
Location: Umm Qasr, Iraq
Contract: UQ83
Engineer: Hemanathan Gopalan
Filename: Umm Qasr-ST

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3-Phase fault at bus: **S002-SB-1AC001**
[BUS B]
Nominal kV = 11.000
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 76.691 kA Method C
Steady State = 22.866 kA rms

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

<u>TD (S)</u>	<u>Ib sym</u>	<u>Ib asym</u>	<u>Idc</u>
0.10	26.262	27.883	9.369
0.15	25.707	26.080	4.395
0.20	25.176	25.260	2.061
0.25	24.671	24.690	0.967
0.30	24.671	24.676	0.454

3-Phase fault at bus: **S002-SB-1AD001**
[BUS A]
Nominal kV = 3.300
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 69.410 kA Method C
Steady State = 17.554 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-SB-1AD001 [BUS A]	Total	0.00	3.560	-28.333	8.0	28.555
Bus160	S002-SB-1AD001 [BUS A]	4.80	0.384	-1.993	5.2	2.029
Bus22	S002-SB-1AD001 [BUS A]	2.00	1.141	-18.351	16.1	18.387
Bus30	S002-SB-1AD001 [BUS A]	0.29	0.244	-0.633	2.6	0.679
Bus31	S002-SB-1AD001 [BUS A]	0.31	0.252	-0.663	2.6	0.709
Bus132	S002-SB-1AD001 [BUS A]	3.99	0.046	-0.193	4.2	0.199
Bus131	S002-SB-1AD001 [BUS A]	3.99	0.046	-0.193	4.2	0.199
Bus130	S002-SB-1AD001 [BUS A]	2.77	0.045	-0.195	4.4	0.200
Bus129	S002-SB-1AD001 [BUS A]	2.77	0.045	-0.195	4.4	0.200
Bus128	S002-SB-1AD001 [BUS A]	4.74	0.048	-0.193	4.1	0.198
Bus127	S002-SB-1AD001 [BUS A]	4.74	0.048	-0.193	4.1	0.198
Bus126	S002-SB-1AD001 [BUS A]	5.95	0.049	-0.191	3.9	0.197
Bus125	S002-SB-1AD001 [BUS A]	5.95	0.049	-0.191	3.9	0.197
Bus124	S002-SB-1AD001 [BUS A]	6.25	0.090	-0.315	3.5	0.328
Bus123	S002-SB-1AD001 [BUS A]	6.36	0.091	-0.315	3.5	0.328
Bus122	S002-SB-1AD001 [BUS A]	2.29	0.207	-1.031	5.0	1.052
Bus121	S002-SB-1AD001 [BUS A]	2.34	0.208	-1.031	5.0	1.051
Bus120	S002-SB-1AD001 [BUS A]	2.69	0.231	-1.131	4.9	1.155
Bus119	S002-SB-1AD001 [BUS A]	1.15	0.151	-0.710	4.7	0.726
Bus47	S002-SB-1AD001 [BUS A]	2.55	0.064	-0.369	5.7	0.375
S002-TR-04DE001_PRI	S002-SB-1AD001 [BUS A]	0.81	0.121	-0.247	2.0	0.276

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	27.170	39.928	29.258
0.02	26.393	35.469	23.695
0.03	25.492	31.293	18.150
0.04	24.579	28.238	13.903
0.05	23.709	26.981	12.879

Project: UQ LT LPG Export - ST & MT
Location: Umm Qasr, Iraq
Contract: UQ83
Engineer: Hemanathan Gopalan
Filename: Umm Qasr-ST

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3-Phase fault at bus: **S002-SB-1AD001**
[BUS A]
Nominal kV = 3.300
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 69.410 kA Method C
Steady State = 17.554 kA rms

Breaking and DC Fault Current (kA)
Based on Total Bus Fault Current

<u>TD (S)</u>	<u>Ib sym</u>	<u>Ib asym</u>	<u>Idc</u>
0.06	23.291	25.445	10.247
0.07	22.883	24.293	8.154
0.08	22.487	23.404	6.488
0.09	22.102	22.696	5.162
0.10	21.728	22.335	5.172
0.15	20.996	21.078	1.851
0.20	20.305	20.316	0.662
0.25	19.655	19.657	0.237
0.30	19.655	19.656	0.085

3-Phase fault at bus: **S002-SB-1AD001**
[BUS B]
Nominal kV = 3.300
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 61.464 kA Method C
Steady State = 17.790 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-SB-1AD001 [BUS B]	Total	0.00	2.592	-24.608	9.5	24.744
Bus25	S002-SB-1AD001 [BUS B]	2.04	1.169	-18.703	16.0	18.739
Bus29	S002-SB-1AD001 [BUS B]	0.02	0.018	-0.032	1.8	0.037
Bus32	S002-SB-1AD001 [BUS B]	0.14	0.144	-0.296	2.1	0.330
Bus133	S002-SB-1AD001 [BUS B]	3.99	0.046	-0.193	4.2	0.199
Bus134	S002-SB-1AD001 [BUS B]	3.94	0.052	-0.189	3.6	0.196
Bus135	S002-SB-1AD001 [BUS B]	2.74	0.049	-0.192	3.9	0.198
Bus136	S002-SB-1AD001 [BUS B]	2.74	0.049	-0.192	3.9	0.198
Bus137	S002-SB-1AD001 [BUS B]	4.66	0.054	-0.187	3.5	0.195
Bus138	S002-SB-1AD001 [BUS B]	4.66	0.054	-0.187	3.5	0.195
Bus139	S002-SB-1AD001 [BUS B]	5.95	0.049	-0.191	3.9	0.197
Bus140	S002-SB-1AD001 [BUS B]	5.95	0.049	-0.191	3.9	0.197
Bus141	S002-SB-1AD001 [BUS B]	6.25	0.090	-0.315	3.5	0.328
Bus142	S002-SB-1AD001 [BUS B]	6.36	0.091	-0.315	3.5	0.328
Bus143	S002-SB-1AD001 [BUS B]	2.29	0.172	-1.038	6.0	1.052
Bus144	S002-SB-1AD001 [BUS B]	2.34	0.208	-1.031	5.0	1.051
Bus145	S002-SB-1AD001 [BUS B]	2.45	0.208	-1.030	4.9	1.051
Bus146	S002-SB-1AD001 [BUS B]	3.14	0.088	-0.324	3.7	0.336
S002-TR-04DE002_PRI	S002-SB-1AD001 [BUS B]	0.00	0.000	0.000	999.9	0.000

Breaking and DC Fault Current (kA)
Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	23.842	35.701	26.573
0.02	23.335	31.976	21.862
0.03	22.783	28.595	17.280
0.04	22.249	26.107	13.658
0.05	21.739	24.971	12.288
0.06	21.491	23.690	9.967
0.07	21.250	22.736	8.085

Project: UQ LT LPG Export - ST & MT
Location: Umm Qasr, Iraq
Contract: UQ83
Engineer: Hemanathan Gopalan
Filename: Umm Qasr-ST

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3-Phase fault at bus: **S002-SB-1AD001**

[BUS B]
Nominal kV = 3.300
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 61.464 kA Method C
Steady State = 17.790 kA rms

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

<u>TD (S)</u>	<u>Ib sym</u>	<u>Ib asym</u>	<u>Idc</u>
0.08	21.014	22.014	6.558
0.09	20.784	21.454	5.319
0.10	20.561	21.166	5.027
0.15	20.122	20.212	1.905
0.20	19.706	19.720	0.722
0.25	19.314	19.316	0.274
0.30	19.314	19.315	0.104

3-Phase fault at bus: **S002-SB-1AN001**
[BUS A]
Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 24.793 kA Method C
Steady State = 14.079 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-SB-1AN001 [BUS A]	Total	0.00	9.719	-12.325	1.3	15.696
Bus158	S002-SB-1AN001 [BUS A]	7.81	0.093	-0.137	1.5	0.166
S002-PC1-1A	S002-SB-1AN001 [BUS A]	70.34	9.553	-12.013	1.3	15.348
Bus91	S002-SB-1AN001 [BUS A]	0.00	0.000	0.000	999.9	0.000
Lump14	S002-SB-1AN001 [BUS A]	105.00	0.073	-0.175	2.4	0.190

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Ide
0.01	15.453	15.585	2.030
0.02	15.362	15.364	0.201
0.03	15.272	15.272	0.019
0.04	15.183	15.183	0.002
0.05	15.095	15.095	0.000
0.06	15.051	15.051	0.000
0.07	15.007	15.007	0.000
0.08	14.964	14.964	0.000
0.09	14.920	14.920	0.000
0.10	14.878	14.878	0.000
0.15	14.794	14.794	0.000
0.20	14.713	14.713	0.000
0.25	14.633	14.633	0.000
0.30	14.633	14.633	0.000

3-Phase fault at bus: **S002-SB-1AN001**
[BUS B]
Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 23.677 kA Method C
Steady State = 14.116 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-SB-1AN001 [BUS B]	Total	0.00	8.554	-11.990	1.4	14.728
Bus152	S002-SB-1AN001 [BUS B]	7.81	0.093	-0.137	1.5	0.166
S002-PC1-1B	S002-SB-1AN001 [BUS B]	66.08	8.404	-11.718	1.4	14.420
Lump13	S002-SB-1AN001 [BUS B]	105.00	0.057	-0.135	2.4	0.146

Breaking and DC Fault Current (kA)
Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	14.581	14.756	2.264
0.02	14.554	14.557	0.253
0.03	14.529	14.529	0.028
0.04	14.505	14.505	0.003
0.05	14.483	14.483	0.000
0.06	14.471	14.471	0.000
0.07	14.461	14.461	0.000
0.08	14.450	14.450	0.000
0.09	14.440	14.440	0.000
0.10	14.430	14.430	0.000
0.15	14.415	14.415	0.000
0.20	14.400	14.400	0.000
0.25	14.387	14.387	0.000
0.30	14.387	14.387	0.000

3-Phase fault at bus: **S002-SB-01E001**
[BUS A]
Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 111.742 kA Method C
Steady State = 45.283 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-SB-01E001 [BUS A]	Total	0.00	9.849	-49.370	5.0	50.343
Bus10	S002-SB-01E001 [BUS A]	8.71	8.218	-45.355	5.5	46.094
S002-SB-PSS01N01 [BUS A]	S002-SB-01E001 [BUS A]	11.28	0.682	-1.638	2.4	1.774
S002-MCC1-4 [BUS]	S002-SB-01E001 [BUS A]	3.17	0.264	-0.491	1.9	0.557
S002-6300-MK-1A	S002-SB-01E001 [BUS A]	105.00	0.473	-1.378	2.9	1.457
S002-SB-01E001 [A]_L	S002-SB-01E001 [BUS A]	105.00	0.213	-0.507	2.4	0.550

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	49.396	63.105	39.272
0.02	49.027	53.825	22.214
0.03	48.669	50.225	12.408
0.04	48.324	48.819	6.931
0.05	47.997	48.162	3.990
0.06	47.837	47.889	2.242
0.07	47.681	47.697	1.260
0.08	47.529	47.534	0.708
0.09	47.381	47.383	0.398
0.10	47.238	47.238	0.231
0.15	46.972	46.972	0.013
0.20	46.721	46.721	0.001
0.25	46.485	46.485	0.000
0.30	46.485	46.485	0.000

3-Phase fault at bus: **S002-SB-01E001**
[BUS B]
Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 108.032 kA Method C
Steady State = 44.571 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-SB-01E001 [BUS B]	Total	0.00	9.553	-47.766	5.0	48.711
Bus23	S002-SB-01E001 [BUS B]	10.28	8.192	-44.607	5.4	45.353
S002-SB-PSS01N01 [BUS B]	S002-SB-01E001 [BUS B]	6.37	0.386	-0.925	2.4	1.002
S002-MCC1-6 [BUS]	S002-SB-01E001 [BUS B]	9.18	0.506	-1.019	2.0	1.137
S002-6500-MP-100B	S002-SB-01E001 [BUS B]	105.00	0.221	-0.628	2.8	0.665
S002-SB-01E001 [B]_L	S002-SB-01E001 [BUS B]	105.00	0.247	-0.587	2.4	0.637

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	47.759	60.955	37.877
0.02	47.458	52.018	21.299
0.03	47.167	48.631	11.843
0.04	46.889	47.349	6.585
0.05	46.626	46.777	3.760
0.06	46.497	46.544	2.102
0.07	46.372	46.386	1.175
0.08	46.250	46.254	0.657
0.09	46.132	46.133	0.367
0.10	46.017	46.018	0.211
0.15	45.812	45.812	0.012
0.20	45.620	45.620	0.001
0.25	45.439	45.439	0.000
0.30	45.439	45.439	0.000

3-Phase fault at bus: **S002-SB-02N001**
[BUS A]
Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 62.751 kA Method C
Steady State = 28.142 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-SB-02N001 [BUS A]	Total	0.00	9.025	-29.921	3.3	31.253
Bus27	S002-SB-02N001 [BUS A]	3.22	7.934	-27.324	3.4	28.453
Area Lighting DB (A8)	S002-SB-02N001 [BUS A]	0.00	0.000	0.000	999.9	0.000
Lump22	S002-SB-02N001 [BUS A]	105.00	1.091	-2.597	2.4	2.817

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	30.878	35.427	17.367
0.02	30.599	31.369	6.907
0.03	30.328	30.451	2.731
0.04	30.068	30.088	1.080
0.05	29.821	29.824	0.434
0.06	29.701	29.702	0.172
0.07	29.585	29.585	0.068
0.08	29.472	29.472	0.027
0.09	29.363	29.363	0.011
0.10	29.256	29.256	0.004
0.15	29.048	29.048	0.000
0.20	28.852	28.852	0.000
0.25	28.668	28.668	0.000
0.30	28.668	28.668	0.000

3-Phase fault at bus: **S002-SB-02N001**

[BUS B]

Nominal kV = 0.380

Voltage c Factor = 1.05 (Maximum If)

Peak Value = 62.495 kA Method C

Steady State = 29.002 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-SB-02N001 [BUS B]	Total	0.00	8.842	-29.744	3.4	31.031
Bus24	S002-SB-02N001 [BUS B]	3.32	8.180	-28.168	3.4	29.332
Lump23	S002-SB-02N001 [BUS B]	105.00	0.662	-1.576	2.4	1.710

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	30.710	35.304	17.416
0.02	30.543	31.330	6.976
0.03	30.383	30.510	2.782
0.04	30.228	30.249	1.109
0.05	30.082	30.085	0.448
0.06	30.011	30.011	0.179
0.07	29.942	29.942	0.072
0.08	29.875	29.875	0.029
0.09	29.809	29.809	0.011
0.10	29.746	29.746	0.005
0.15	29.626	29.626	0.000
0.20	29.513	29.513	0.000
0.25	29.408	29.408	0.000
0.30	29.408	29.408	0.000

3-Phase fault at bus: **S002-SB-03E001**
[BUS A]
Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 78.524 kA Method C
Steady State = 30.531 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-SB-03E001 [BUS A]	Total	0.00	9.257	-36.167	3.9	37.333
Bus28	S002-SB-03E001 [BUS A]	13.62	6.736	-30.166	4.5	30.909
S002-DB03E001-20_LI	S002-SB-03E001 [BUS A]	105.00	0.167	-0.398	2.4	0.432
S002-SB-03E001 [A]_C	S002-SB-03E001 [BUS A]	105.00	1.531	-3.645	2.4	3.953
S002-SB-03E001 [A]_S	S002-SB-03E001 [BUS A]	105.00	0.823	-1.958	2.4	2.124

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Ide
0.01	36.454	44.030	24.693
0.02	35.814	37.756	11.954
0.03	35.193	35.650	5.688
0.04	34.597	34.702	2.707
0.05	34.029	34.055	1.338
0.06	33.756	33.762	0.642
0.07	33.490	33.491	0.308
0.08	33.231	33.231	0.147
0.09	32.979	32.979	0.071
0.10	32.735	32.735	0.035
0.15	32.260	32.260	0.001
0.20	31.812	31.812	0.000
0.25	31.391	31.391	0.000
0.30	31.391	31.391	0.000

3-Phase fault at bus: **S002-SB-03E001**
[BUS B]
Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 77.752 kA Method C
Steady State = 30.516 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-SB-03E001 [BUS B]	Total	0.00	9.082	-35.771	3.9	36.906
Bus33	S002-SB-03E001 [BUS B]	13.62	6.721	-30.152	4.5	30.892
S002-DB03N001-22_LSB	S002-SB-03E001 [BUS B]	105.00	0.704	-1.677	2.4	1.819
S002-SB-03E001 [B]_C	S002-SB-03E001 [BUS B]	105.00	1.657	-3.943	2.4	4.277
S002-SB-03E001 [B]_I	S002-SB-03E001 [BUS B]	105.00	0.000	0.000	2.4	0.000

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Ide
0.01	36.184	43.711	24.524
0.02	35.581	37.520	11.906
0.03	34.996	35.455	5.687
0.04	34.434	34.541	2.716
0.05	33.898	33.925	1.345
0.06	33.641	33.647	0.647
0.07	33.390	33.392	0.311
0.08	33.146	33.147	0.150
0.09	32.909	32.909	0.072
0.10	32.679	32.679	0.036
0.15	32.228	32.228	0.001
0.20	31.802	31.802	0.000
0.25	31.402	31.402	0.000
0.30	31.402	31.402	0.000

3-Phase fault at bus: **S002-SB-03N001**
[BUS A]
Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 105.577 kA Method C
Steady State = 40.650 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-SB-03N001 [BUS A]	Total	0.00	10.548	-47.359	4.5	48.520
Bus67	S002-SB-03N001 [BUS A]	9.36	7.696	-40.571	5.3	41.295
S002-SB-03N001 [A]_C	S002-SB-03N001 [BUS A]	105.00	2.852	-6.788	2.4	7.363

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	47.945	59.888	35.886
0.02	47.552	51.409	19.536
0.03	46.826	47.972	10.424
0.04	46.126	46.460	5.562
0.05	45.457	45.563	3.102
0.06	45.138	45.169	1.670
0.07	44.827	44.836	0.899
0.08	44.524	44.527	0.484
0.09	44.229	44.230	0.261
0.10	43.942	43.943	0.146
0.15	43.357	43.357	0.007
0.20	42.803	42.803	0.000
0.25	42.283	42.283	0.000
0.30	42.283	42.283	0.000

3-Phase fault at bus: **S002-SB-03N001**
[BUS B]
Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 100.372 kA Method C
Steady State = 40.892 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-SB-03N001 [BUS B]	Total	0.00	9.390	-44.724	4.8	45.700
Bus69	S002-SB-03N001 [BUS B]	9.42	7.749	-40.818	5.3	41.547
S002-SB-03N001 [B]_C	S002-SB-03N001 [BUS B]	105.00	1.641	-3.907	2.4	4.237

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	45.317	56.993	34.563
0.02	44.897	48.742	18.975
0.03	44.489	45.661	10.281
0.04	44.096	44.447	5.571
0.05	43.722	43.833	3.111
0.06	43.543	43.576	1.696
0.07	43.368	43.378	0.924
0.08	43.198	43.201	0.504
0.09	43.032	43.033	0.275
0.10	42.871	42.872	0.154
0.15	42.552	42.552	0.008
0.20	42.250	42.250	0.000
0.25	41.967	41.967	0.000
0.30	41.967	41.967	0.000

3-Phase fault at bus: **S002-SB-04E001**
[BUS A]
Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 28.229 kA Method C
Steady State = 13.985 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-SB-04E001 [BUS A]	Total	0.00	8.908	-14.406	1.6	16.938
Bus157	S002-SB-04E001 [BUS A]	0.00	0.000	0.000	999.9	0.000
S002-TR-04DE001_PRI	S002-SB-04E001 [BUS A]	91.56	7.855	-12.119	1.5	14.442
688-PM-01A	S002-SB-04E001 [BUS A]	105.00	0.137	-0.238	1.7	0.275
688-PM-01B	S002-SB-04E001 [BUS A]	105.00	0.137	-0.238	1.7	0.275
S002-688-FM-01-04	S002-SB-04E001 [BUS A]	105.00	0.468	-1.115	2.4	1.209
S002-688-FM-05-07	S002-SB-04E001 [BUS A]	105.00	0.079	-0.187	2.4	0.203
HVAC-001/002	S002-SB-04E001 [BUS B]	105.00	0.116	-0.255	2.2	0.280
HVAC-002	S002-SB-04E001 [BUS B]	105.00	0.116	-0.255	2.2	0.280
S002-SB-04E001 [BUS B]	S002-SB-04E001 [BUS A]	0.00	0.232	-0.509	2.2	0.560

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	16.074	16.476	3.618
0.02	15.843	15.853	0.570
0.03	15.620	15.621	0.088
0.04	15.408	15.408	0.014
0.05	15.209	15.209	0.002
0.06	15.110	15.110	0.000
0.07	15.015	15.015	0.000
0.08	14.922	14.922	0.000
0.09	14.833	14.833	0.000
0.10	14.746	14.746	0.000
0.15	14.597	14.597	0.000
0.20	14.456	14.456	0.000
0.25	14.325	14.325	0.000
0.30	14.325	14.325	0.000

3-Phase fault at bus: **S002-SB-04E001**
[BUS B]
Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 28.229 kA Method C
Steady State = 13.985 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-SB-04E001 [BUS B]	Total	0.00	8.908	-14.406	1.6	16.938
HVAC-001/002	S002-SB-04E001 [BUS B]	105.00	0.116	-0.255	2.2	0.280
HVAC-002	S002-SB-04E001 [BUS B]	105.00	0.116	-0.255	2.2	0.280
Bus157	S002-SB-04E001 [BUS A]	0.00	0.000	0.000	999.9	0.000
S002-TR-04DE001_PRI	S002-SB-04E001 [BUS A]	91.56	7.855	-12.119	1.5	14.442
688-PM-01A	S002-SB-04E001 [BUS A]	105.00	0.137	-0.238	1.7	0.275
688-PM-01B	S002-SB-04E001 [BUS A]	105.00	0.137	-0.238	1.7	0.275
S002-688-FM-01-04	S002-SB-04E001 [BUS A]	105.00	0.468	-1.115	2.4	1.209
S002-688-FM-05-07	S002-SB-04E001 [BUS A]	105.00	0.079	-0.187	2.4	0.203
S002-SB-04E001 [BUS A]	S002-SB-04E001 [BUS B]	0.00	8.677	-13.897	1.6	16.383

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	16.074	16.476	3.618
0.02	15.843	15.853	0.570
0.03	15.620	15.621	0.088
0.04	15.408	15.408	0.014
0.05	15.209	15.209	0.002
0.06	15.110	15.110	0.000
0.07	15.015	15.015	0.000
0.08	14.922	14.922	0.000
0.09	14.833	14.833	0.000
0.10	14.746	14.746	0.000
0.15	14.597	14.597	0.000
0.20	14.456	14.456	0.000
0.25	14.325	14.325	0.000
0.30	14.325	14.325	0.000

Project: UQ LT LPG Export - ST & MT
Location: Umm Qasr, Iraq
Contract: UQ83
Engineer: Hemanathan Gopalan
Filename: Umm Qasr-ST

ETAP
24.0.2C

Study Case: SC_Max

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Revision: Base
Config.: 1_NE

3-Phase fault at bus: **S002-SB-05N001A**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 115.240 kA Method C
Steady State = 43.205 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-SB-05N001A	Total	0.00	9.685	-50.447	5.2	51.369
S002-SB-1AC001 [BUS A]	S002-SB-05N001A	93.96	7.023	-43.411	6.2	43.975
TYPICAL MOTOR FEEDER 2 A	S002-SB-05N001A	105.00	0.339	-0.807	2.4	0.875
Bus9	TYPICAL MOTOR FEEDER 1 A	19.32	0.158	-0.411	2.6	0.440
Bus12	TYPICAL MOTOR FEEDER 1 A	19.05	0.157	-0.412	2.6	0.441
Bus53	TYPICAL MOTOR FEEDER 1 A	18.78	0.156	-0.413	2.6	0.442
Bus54	TYPICAL MOTOR FEEDER 1 A	18.78	0.156	-0.413	2.6	0.442
Bus85	TYPICAL MOTOR FEEDER 1 A	18.51	0.155	-0.414	2.7	0.443
Bus149	TYPICAL MOTOR FEEDER 1 A	18.24	0.155	-0.416	2.7	0.443
Bus151	TYPICAL MOTOR FEEDER 1 A	18.51	0.155	-0.414	2.7	0.443
Bus154	TYPICAL MOTOR FEEDER 1 A	18.24	0.155	-0.416	2.7	0.443
Bus161	TYPICAL MOTOR FEEDER 1 A	17.96	0.154	-0.417	2.7	0.444
Bus162	TYPICAL MOTOR FEEDER 1 A	18.24	0.155	-0.416	2.7	0.443
Bus164	TYPICAL MOTOR FEEDER 1 A	17.96	0.154	-0.417	2.7	0.444
Bus165	TYPICAL MOTOR FEEDER 1 A	17.69	0.153	-0.418	2.7	0.445
Bus170	TYPICAL MOTOR FEEDER 1 A	17.96	0.154	-0.417	2.7	0.444
Bus171	TYPICAL MOTOR FEEDER 1 A	17.69	0.153	-0.418	2.7	0.445
Bus172	TYPICAL MOTOR FEEDER 1 A	17.42	0.153	-0.419	2.7	0.446
TYPICAL MOTOR FEEDER 1 A	S002-SB-05N001A	0.00	2.323	-6.230	2.7	6.649

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	48.757	64.110	41.626

Project: UQ LT LPG Export - ST & MT
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Engineer: Hemanathan Gopalan
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3-Phase fault at bus: **S002-SB-05N001A**

Nominal kV	=	0.380	
Voltage c Factor	=	1.05	(Maximum If)
Peak Value	=	115.240	kA Method C
Steady State	=	43.205	kA rms

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

<u>TD (S)</u>	<u>Ib sym</u>	<u>Ib asym</u>	<u>Idc</u>
0.02	48.163	54.201	24.860
0.03	47.593	49.766	14.543
0.04	47.052	47.815	8.508
0.05	46.544	46.836	5.221
0.06	46.292	46.395	3.083
0.07	46.048	46.084	1.821
0.08	45.812	45.824	1.076
0.09	45.583	45.588	0.635
0.10	45.363	45.365	0.393
0.15	44.995	44.995	0.029
0.20	44.649	44.649	0.002
0.25	44.329	44.329	0.000
0.30	44.329	44.329	0.000

3-Phase fault at bus: S002-SB-05N001B

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 121.703 kA Method C
Steady State = 43.174 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-SB-05N001B	Total	0.00	10.984	-53.665	4.9	54.777
S002-SB-1AC001 [BUS B]	S002-SB-05N001B	93.86	7.008	-43.367	6.2	43.930
TYPICAL MOTOR FEEDER 2 B	S002-SB-05N001B	105.00	0.243	-0.579	2.4	0.628
POWER FEEDER 1/2	S002-SB-05N001C	105.00	1.444	-3.437	2.4	3.728
Bus188	POWER SOCKET FEEDER 1 B	0.00	0.000	0.000	999.9	0.000
Bus189	POWER SOCKET FEEDER 1 B	0.00	0.000	0.000	999.9	0.000
Bus190	POWER SOCKET FEEDER 1 B	0.00	0.000	0.000	999.9	0.000
Bus173	TYPICAL MOTOR FEEDER 1 B	16.58	0.150	-0.422	2.8	0.448
Bus174	TYPICAL MOTOR FEEDER 1 B	16.86	0.151	-0.421	2.8	0.447
Bus175	TYPICAL MOTOR FEEDER 1 B	17.14	0.152	-0.420	2.8	0.446
Bus176	TYPICAL MOTOR FEEDER 1 B	16.86	0.151	-0.421	2.8	0.447
Bus177	TYPICAL MOTOR FEEDER 1 B	17.14	0.152	-0.420	2.8	0.446
Bus178	TYPICAL MOTOR FEEDER 1 B	17.42	0.153	-0.419	2.7	0.446
Bus179	TYPICAL MOTOR FEEDER 1 B	17.14	0.152	-0.420	2.8	0.446
Bus180	TYPICAL MOTOR FEEDER 1 B	17.42	0.153	-0.419	2.7	0.446
Bus181	TYPICAL MOTOR FEEDER 1 B	17.69	0.153	-0.418	2.7	0.445
Bus182	TYPICAL MOTOR FEEDER 1 B	17.42	0.153	-0.419	2.7	0.446
Bus183	TYPICAL MOTOR FEEDER 1 B	17.69	0.153	-0.418	2.7	0.445
Bus184	TYPICAL MOTOR FEEDER 1 B	17.96	0.154	-0.417	2.7	0.444
Bus185	TYPICAL MOTOR FEEDER 1 B	17.69	0.153	-0.418	2.7	0.445
Bus186	TYPICAL MOTOR FEEDER 1 B	17.96	0.154	-0.417	2.7	0.444
Bus187	TYPICAL MOTOR FEEDER 1 B	18.24	0.155	-0.416	2.7	0.443
S002-SB-05N001B	POWER SOCKET FEEDER 1 B	0.00	0.000	0.000	9999.0	0.000
TYPICAL MOTOR FEEDER 1 B	S002-SB-05N001B	0.00	2.289	-6.281	2.7	6.685

(Cont.)

3-Phase fault at bus: S002-SB-05N001B

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 121.703 kA Method C
Steady State = 43.174 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-SB-05N001C	S002-SB-05N001B	0.00	1.444	-3.437	2.4	3.728

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	51.801	67.590	43.417
0.02	50.852	56.994	25.737
0.03	49.939	52.096	14.835
0.04	49.068	49.807	8.551
0.05	48.245	48.530	5.250
0.06	47.843	47.941	3.064
0.07	47.452	47.486	1.789
0.08	47.074	47.085	1.044
0.09	46.707	46.711	0.609
0.10	46.352	46.354	0.378
0.15	45.718	45.718	0.026
0.20	45.123	45.123	0.002
0.25	44.568	44.568	0.000
0.30	44.568	44.568	0.000

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Location: Umm Qasr, Iraq

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Contract: UQ83

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Engineer: Hemanathan Gopalan

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Filename: Umm Qasr-ST

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3-Phase fault at bus: S002-SB-05N001C

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 121.703 kA Method C
Steady State = 43.174 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-SB-05N001C	Total	0.00	10.984	-53.665	4.9	54.777
POWER FEEDER 1/2	S002-SB-05N001C	105.00	1.444	-3.437	2.4	3.728
S002-SB-1AC001 [BUS B]	S002-SB-05N001B	93.86	7.008	-43.367	6.2	43.930
TYPICAL MOTOR FEEDER 2 B	S002-SB-05N001B	105.00	0.243	-0.579	2.4	0.628
Bus188	POWER SOCKET FEEDER 1 B	0.00	0.000	0.000	999.9	0.000
Bus189	POWER SOCKET FEEDER 1 B	0.00	0.000	0.000	999.9	0.000
Bus190	POWER SOCKET FEEDER 1 B	0.00	0.000	0.000	999.9	0.000
Bus173	TYPICAL MOTOR FEEDER 1 B	16.58	0.150	-0.422	2.8	0.448
Bus174	TYPICAL MOTOR FEEDER 1 B	16.86	0.151	-0.421	2.8	0.447
Bus175	TYPICAL MOTOR FEEDER 1 B	17.14	0.152	-0.420	2.8	0.446
Bus176	TYPICAL MOTOR FEEDER 1 B	16.86	0.151	-0.421	2.8	0.447
Bus177	TYPICAL MOTOR FEEDER 1 B	17.14	0.152	-0.420	2.8	0.446
Bus178	TYPICAL MOTOR FEEDER 1 B	17.42	0.153	-0.419	2.7	0.446
Bus179	TYPICAL MOTOR FEEDER 1 B	17.14	0.152	-0.420	2.8	0.446
Bus180	TYPICAL MOTOR FEEDER 1 B	17.42	0.153	-0.419	2.7	0.446
Bus181	TYPICAL MOTOR FEEDER 1 B	17.69	0.153	-0.418	2.7	0.445
Bus182	TYPICAL MOTOR FEEDER 1 B	17.42	0.153	-0.419	2.7	0.446
Bus183	TYPICAL MOTOR FEEDER 1 B	17.69	0.153	-0.418	2.7	0.445
Bus184	TYPICAL MOTOR FEEDER 1 B	17.96	0.154	-0.417	2.7	0.444
Bus185	TYPICAL MOTOR FEEDER 1 B	17.69	0.153	-0.418	2.7	0.445
Bus186	TYPICAL MOTOR FEEDER 1 B	17.96	0.154	-0.417	2.7	0.444
Bus187	TYPICAL MOTOR FEEDER 1 B	18.24	0.155	-0.416	2.7	0.443
S002-SB-05N001B	POWER SOCKET FEEDER 1 B	0.00	0.000	0.000	9999.0	0.000
TYPICAL MOTOR FEEDER 1 B	S002-SB-05N001B	0.00	2.289	-6.281	2.7	6.685

Project:

UQ LT LPG Export - ST & MT

Location:

Umm Qasr, Iraq

Contract:

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Engineer:

Hemanathan Gopalan

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3-Phase fault at bus: S002-SB-05N001C

Nominal kV

=

0.380

Voltage c Factor

=

1.05

(Maximum If)

Peak Value

=

121.703

kA

Method C

Steady State

=

43.174

kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-SB-05N001B	S002-SB-05N001C	0.00	9.540	-50.228	5.3	51.126

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	51.801	67.590	43.417
0.02	50.852	56.994	25.737
0.03	49.939	52.096	14.835
0.04	49.068	49.807	8.551
0.05	48.245	48.530	5.250
0.06	47.843	47.941	3.064
0.07	47.452	47.486	1.789
0.08	47.074	47.085	1.044
0.09	46.707	46.711	0.609
0.10	46.352	46.354	0.378
0.15	45.718	45.718	0.026
0.20	45.123	45.123	0.002
0.25	44.568	44.568	0.000
0.30	44.568	44.568	0.000

3-Phase fault at bus: **S002-SB-PSS1A01**
[BUS-A]
Nominal kV = 132.000
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 45.690 kA Method C
Steady State = 18.060 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-SB-PSS1A01 [BUS-A]	Total	0.00	2.395	-18.943	7.9	19.094
U1 [Max]	132kV GRID	110.00	2.213	-17.924	8.1	18.060
TR-PSS1AC02_132kV BUS	Bus81	0.03	0.093	-0.523	5.6	0.531
TR-PSS1AC01_132kV BUS	Bus79	0.03	0.089	-0.497	5.6	0.505
Bus79	S002-SB-PSS1A01 [BUS-A]	0.00	0.089	-0.497	5.6	0.505
S002-SB-PSS1A01 [BUS-A]	Bus80	0.00	0.000	0.000	9999.0	0.000
Bus81	S002-SB-PSS1A01 [BUS-A]	0.00	0.093	-0.523	5.6	0.531
132kV GRID	Bus83	0.00	1.106	-8.962	8.1	9.030
Bus83	S002-SB-PSS1A01 [BUS-A]	0.00	1.106	-8.962	8.1	9.030
132kV GRID	Bus84	0.00	1.106	-8.962	8.1	9.030
Bus84	S002-SB-PSS1A01 [BUS-A]	0.00	1.106	-8.962	8.1	9.030
Bus87	S002-SB-PSS1A01 [BUS-A]	0.00	0.000	0.000	9999.0	0.000
Bus88	S002-SB-PSS1A01 [BUS-A]	0.00	0.000	0.000	9999.0	0.000

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	19.026	26.339	18.214
0.02	18.989	22.639	12.327
0.03	18.947	20.697	8.328
0.04	18.899	19.718	5.627
0.05	18.838	19.221	3.818
0.06	18.807	18.983	2.582
0.07	18.776	18.857	1.746
0.08	18.746	18.783	1.180
0.09	18.716	18.733	0.798
0.10	18.687	18.695	0.542
0.15	18.623	18.624	0.077
0.20	18.562	18.562	0.011
0.25	18.503	18.503	0.002

Project:

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Umm Qasr, Iraq

UQ83

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3-Phase fault at bus: S002-SB-PSS1A01

[BUS-A]

Nominal kV

=

132.000

Voltage c Factor

=

1.10

(Maximum If)

Peak Value

=

45.690

kA Method C

Steady State

=

18.060

kA rms

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.30	18.503	18.503	0.000

3-Phase fault at bus: **S002-SB-PSS01N01**
[BUS A]
Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 27.820 kA Method C
Steady State = 12.143 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-SB-PSS01N01 [BUS A]	Total	0.00	4.673	-13.561	2.9	14.343
S002-SB-01E001 [BUS A]	S002-SB-PSS01N01 [BUS A]	78.61	3.903	-11.728	3.0	12.361
A01	S002-SB-PSS01N01 [BUS A]	105.00	0.770	-1.832	2.4	1.987

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	14.034	15.667	6.965
0.02	13.838	14.047	2.417
0.03	13.648	13.673	0.834
0.04	13.465	13.468	0.288
0.05	13.292	13.292	0.101
0.06	13.208	13.208	0.035
0.07	13.126	13.126	0.012
0.08	13.047	13.047	0.004
0.09	12.970	12.970	0.001
0.10	12.895	12.895	0.001
0.15	12.751	12.751	0.000
0.20	12.615	12.615	0.000
0.25	12.487	12.487	0.000
0.30	12.487	12.487	0.000

3-Phase fault at bus: **S002-SB-PSS01N01**
[BUS B]
Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 26.014 kA Method C
Steady State = 12.090 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-SB-PSS01N01 [BUS B]	Total	0.00	4.301	-12.655	2.9	13.366
S002-SB-01E001 [BUS B]	S002-SB-PSS01N01 [BUS B]	78.24	3.888	-11.672	3.0	12.302
Lump4	S002-SB-PSS01N01 [BUS B]	105.00	0.413	-0.984	2.4	1.067

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	13.113	14.669	6.576
0.02	13.010	13.214	2.309
0.03	12.911	12.937	0.807
0.04	12.817	12.820	0.282
0.05	12.727	12.727	0.100
0.06	12.683	12.683	0.035
0.07	12.641	12.641	0.012
0.08	12.600	12.600	0.004
0.09	12.560	12.560	0.001
0.10	12.521	12.521	0.001
0.15	12.449	12.449	0.000
0.20	12.382	12.382	0.000
0.25	12.319	12.319	0.000
0.30	12.319	12.319	0.000

3-Phase fault at bus:

S002-TR-04DE001_PR

Nominal kV

=

3.300

Voltage c Factor

=

1.10

(Maximum If)

Peak Value

=

31.334

kA

Method C

Steady State

=

12.991

kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-TR-04DE001_PRI	Total	0.00	7.953	-15.857	2.0	17.740
S002-SB-1AD001 [BUS A]	S002-TR-04DE001_PRI	51.42	7.831	-15.608	2.0	17.462
S002-SB-04E001 [BUS A]	S002-TR-04DE001_PRI	15.28	0.121	-0.250	2.1	0.277

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	17.302	18.227	5.733
0.02	17.055	17.126	1.565
0.03	16.757	16.762	0.391
0.04	16.448	16.448	0.098
0.05	16.148	16.148	0.034
0.06	16.002	16.002	0.009
0.07	15.858	15.858	0.002
0.08	15.716	15.716	0.001
0.09	15.577	15.577	0.000
0.10	15.441	15.441	0.000
0.15	15.163	15.163	0.000
0.20	14.897	14.897	0.000
0.25	14.643	14.643	0.000
0.30	14.643	14.643	0.000

3-Phase fault at bus:

S002-TR-04DE002_PRI			
Nominal kV	=	3.300	
Voltage c Factor	=	1.10	(Maximum If)
Peak Value	=	29.297	kA Method C
Steady State	=	13.114	kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-TR-04DE002_PRI	Total	0.00	6.697	-14.716	2.2	16.168
S002-SB-1AD001 [BUS B]	S002-TR-04DE002_PRI	47.61	6.697	-14.716	2.2	16.168

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	15.886	16.961	5.944
0.02	15.712	15.811	1.761
0.03	15.516	15.524	0.489
0.04	15.323	15.324	0.136
0.05	15.134	15.134	0.047
0.06	15.041	15.041	0.014
0.07	14.949	14.949	0.004
0.08	14.858	14.858	0.001
0.09	14.768	14.768	0.000
0.10	14.680	14.680	0.000
0.15	14.500	14.500	0.000
0.20	14.326	14.326	0.000
0.25	14.158	14.158	0.000
0.30	14.158	14.158	0.000

Project: UQ LT LPG Export - ST & MT
Location: Umm Qasr, Iraq
Contract: UQ83
Engineer: Hemanathan Gopalan
Filename: Umm Qasr-ST

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3-Phase fault at bus: **S003-DB11E001-01**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 18.453 kA Method C
Steady State = 12.165 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S003-DB11E001-01	Total	0.00	10.696	-6.804	0.6	12.676
S003-SB-11E001 [BUS A]	S003-DB11E001-01	79.37	10.675	-6.753	0.6	12.631
Lump43	S003-DB11E001-01	105.00	0.021	-0.050	2.4	0.055

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	12.641	12.642	0.142
0.02	12.609	12.609	0.001
0.03	12.575	12.575	0.000
0.04	12.541	12.541	0.000
0.05	12.507	12.507	0.000
0.06	12.490	12.490	0.000
0.07	12.473	12.473	0.000
0.08	12.456	12.456	0.000
0.09	12.438	12.438	0.000
0.10	12.421	12.421	0.000
0.15	12.384	12.384	0.000
0.20	12.347	12.347	0.000
0.25	12.311	12.311	0.000
0.30	12.311	12.311	0.000

3-Phase fault at bus: S003-DB11E001-02

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 20.301 kA Method C
Steady State = 13.274 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S003-DB11E001-02	Total	0.00	11.405	-7.919	0.7	13.885
S003-SB-11E001 [BUS B]	S003-DB11E001-02	79.06	11.384	-7.869	0.7	13.839
Lump46	S003-DB11E001-02	105.00	0.021	-0.050	2.4	0.054

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	13.840	13.842	0.235
0.02	13.813	13.813	0.003
0.03	13.785	13.785	0.000
0.04	13.758	13.758	0.000
0.05	13.730	13.730	0.000
0.06	13.716	13.716	0.000
0.07	13.702	13.702	0.000
0.08	13.689	13.689	0.000
0.09	13.675	13.675	0.000
0.10	13.661	13.661	0.000
0.15	13.632	13.632	0.000
0.20	13.604	13.604	0.000
0.25	13.577	13.577	0.000
0.30	13.577	13.577	0.000

3-Phase fault at bus: **S003-DB-11N001-03**
[BUS A]
Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 22.610 kA Method C
Steady State = 15.159 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S003-DB-11N001-03 [BUS A]	Total	0.00	13.579	-7.642	0.6	15.582
S003-SB-11N001 [BUS A]	S003-DB-11N001-03 [BUS A]	87.90	13.483	-7.415	0.5	15.388
Lump49	S003-DB-11N001-03 [BUS A]	105.00	0.095	-0.227	2.4	0.246

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	15.503	15.503	0.120
0.02	15.485	15.485	0.001
0.03	15.468	15.468	0.000
0.04	15.452	15.452	0.000
0.05	15.437	15.437	0.000
0.06	15.430	15.430	0.000
0.07	15.422	15.422	0.000
0.08	15.415	15.415	0.000
0.09	15.409	15.409	0.000
0.10	15.402	15.402	0.000
0.15	15.391	15.391	0.000
0.20	15.381	15.381	0.000
0.25	15.372	15.372	0.000
0.30	15.372	15.372	0.000

3-Phase fault at bus: **S003-DB-11N001-03**
[BUS B]
Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 22.763 kA Method C
Steady State = 15.160 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S003-DB-11N001-03 [BUS B]	Total	0.00	13.741	-7.584	0.6	15.695
S003-SB-11N001 [BUS B]	S003-DB-11N001-03 [BUS B]	88.56	13.646	-7.357	0.5	15.503
Lump56	S003-DB-11N001-03 [BUS B]	105.00	0.095	-0.227	2.4	0.246

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	15.617	15.617	0.111
0.02	15.599	15.599	0.001
0.03	15.583	15.583	0.000
0.04	15.567	15.567	0.000
0.05	15.552	15.552	0.000
0.06	15.544	15.544	0.000
0.07	15.537	15.537	0.000
0.08	15.530	15.530	0.000
0.09	15.524	15.524	0.000
0.10	15.517	15.517	0.000
0.15	15.506	15.506	0.000
0.20	15.496	15.496	0.000
0.25	15.487	15.487	0.000
0.30	15.487	15.487	0.000

3-Phase fault at bus: **S003-DB11N001-04**
[BUS A]
Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 22.553 kA Method C
Steady State = 15.159 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S003-DB11N001-04 [BUS A]	Total	0.00	13.564	-7.599	0.6	15.548
S003-SB-11N001 [BUS A]	S003-DB11N001-04 [BUS A]	87.92	13.486	-7.414	0.5	15.390
Lump1	S003-DB11N001-04 [BUS A]	105.00	0.078	-0.185	2.4	0.201

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	15.480	15.480	0.111
0.02	15.465	15.465	0.001
0.03	15.452	15.452	0.000
0.04	15.439	15.439	0.000
0.05	15.427	15.427	0.000
0.06	15.421	15.421	0.000
0.07	15.415	15.415	0.000
0.08	15.409	15.409	0.000
0.09	15.404	15.404	0.000
0.10	15.399	15.399	0.000
0.15	15.390	15.390	0.000
0.20	15.382	15.382	0.000
0.25	15.375	15.375	0.000
0.30	15.375	15.375	0.000

3-Phase fault at bus: **S003-DB11N001-04**
[BUS B]
Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 22.732 kA Method C
Steady State = 15.160 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S003-DB11N001-04 [BUS B]	Total	0.00	13.733	-7.561	0.6	15.677
S003-SB-11N001 [BUS B]	S003-DB11N001-04 [BUS B]	88.57	13.647	-7.356	0.5	15.504
Lump3	S003-DB11N001-04 [BUS B]	105.00	0.086	-0.205	2.4	0.222

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	15.604	15.605	0.106
0.02	15.589	15.589	0.001
0.03	15.574	15.574	0.000
0.04	15.559	15.559	0.000
0.05	15.546	15.546	0.000
0.06	15.539	15.539	0.000
0.07	15.533	15.533	0.000
0.08	15.527	15.527	0.000
0.09	15.521	15.521	0.000
0.10	15.515	15.515	0.000
0.15	15.505	15.505	0.000
0.20	15.496	15.496	0.000
0.25	15.488	15.488	0.000
0.30	15.488	15.488	0.000

Project: UQ LT LPG Export - ST & MT
Location: Umm Qasr, Iraq
Contract: UQ83
Engineer: Hemanathan Gopalan
Filename: Umm Qasr-ST

ETAP
24.0.2C

Study Case: SC_Max

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SN: SL-KENTUAE
Revision: Base
Config.: 1_NE

3-Phase fault at bus: **S003-DB11N001-11**

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 14.954 kA Method C
Steady State = 10.211 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S003-DB11N001-11	Total	0.00	9.456	-4.217	0.4	10.354
S003-SB-11N001 [BUS B]	S003-DB11N001-11	94.64	9.456	-4.217	0.4	10.354

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	10.354	10.354	0.014
0.02	10.354	10.354	0.000
0.03	10.354	10.354	0.000
0.04	10.354	10.354	0.000
0.05	10.354	10.354	0.000
0.06	10.354	10.354	0.000
0.07	10.354	10.354	0.000
0.08	10.354	10.354	0.000
0.09	10.354	10.354	0.000
0.10	10.354	10.354	0.000
0.15	10.354	10.354	0.000
0.20	10.354	10.354	0.000
0.25	10.354	10.354	0.000
0.30	10.354	10.354	0.000

3-Phase fault at bus: S003-DB11N001-12

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 14.954 kA Method C
Steady State = 10.211 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S003-DB11N001-12	Total	0.00	9.456	-4.217	0.4	10.354
S003-SB-11N001 [BUS B]	S003-DB11N001-12	94.64	9.456	-4.217	0.4	10.354

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	10.354	10.354	0.014
0.02	10.354	10.354	0.000
0.03	10.354	10.354	0.000
0.04	10.354	10.354	0.000
0.05	10.354	10.354	0.000
0.06	10.354	10.354	0.000
0.07	10.354	10.354	0.000
0.08	10.354	10.354	0.000
0.09	10.354	10.354	0.000
0.10	10.354	10.354	0.000
0.15	10.354	10.354	0.000
0.20	10.354	10.354	0.000
0.25	10.354	10.354	0.000
0.30	10.354	10.354	0.000

3-Phase fault at bus: S003-MDP-B

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 50.507 kA Method C
Steady State = 24.784 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S003-MDP-B	Total	0.00	8.168	-24.485	3.0	25.811
S003-SB-11N001 [BUS A]	S003-MDP-B	58.51	8.168	-24.485	3.0	25.811

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	25.711	28.819	13.017
0.02	25.656	26.090	4.740
0.03	25.599	25.656	1.708
0.04	25.542	25.550	0.616
0.05	25.486	25.487	0.229
0.06	25.458	25.458	0.083
0.07	25.429	25.429	0.030
0.08	25.401	25.401	0.011
0.09	25.373	25.373	0.004
0.10	25.344	25.344	0.001
0.15	25.287	25.287	0.000
0.20	25.231	25.231	0.000
0.25	25.176	25.176	0.000
0.30	25.176	25.176	0.000

3-Phase fault at bus: S003-SB01E1

Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 24.686 kA Method C
Steady State = 11.249 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S003-SB01E1	Total	0.00	4.469	-12.160	2.7	12.955
S003-SB-11N001 [BUS B]	S003-SB01E1	84.73	3.905	-10.819	2.8	11.502
Lump51	S003-SB01E1	105.00	0.564	-1.341	2.4	1.455

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	12.649	13.930	5.836
0.02	12.508	12.648	1.880
0.03	12.371	12.386	0.602
0.04	12.240	12.242	0.193
0.05	12.116	12.116	0.063
0.06	12.056	12.056	0.020
0.07	11.998	11.998	0.006
0.08	11.941	11.941	0.002
0.09	11.885	11.885	0.001
0.10	11.832	11.832	0.000
0.15	11.731	11.731	0.000
0.20	11.637	11.637	0.000
0.25	11.548	11.548	0.000
0.30	11.548	11.548	0.000

3-Phase fault at bus: **S003-SB-11C001**
[BUS A]
Nominal kV = 11.000
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 17.836 kA Method C
Steady State = 9.587 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S003-SB-11C001 [BUS A]	Total	0.00	6.495	-8.890	1.4	11.009
S002-SB-1AC001 [BUS A]	S003-SB-11C001 [BUS A]	80.74	6.293	-8.199	1.3	10.336
Bus56	S003-SB-11C001 [BUS A]	0.02	0.051	-0.123	2.4	0.134
Bus19	S003-SB-11C001 [BUS A]	0.01	0.056	-0.136	2.5	0.147
Bus168	S003-SB-11C001 [BUS A]	0.04	0.080	-0.393	4.9	0.401
Bus66	S003-SB-11C001 [BUS A]	0.05	0.015	-0.038	2.6	0.040

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	10.903	11.087	2.012
0.02	10.842	10.848	0.355
0.03	10.780	10.780	0.054
0.04	10.714	10.714	0.008
0.05	10.650	10.650	0.002
0.06	10.618	10.618	0.000
0.07	10.587	10.587	0.000
0.08	10.557	10.557	0.000
0.09	10.527	10.527	0.000
0.10	10.497	10.497	0.000
0.15	10.461	10.461	0.000
0.20	10.407	10.407	0.000
0.25	10.327	10.327	0.000
0.30	10.327	10.327	0.000

3-Phase fault at bus: **S003-SB-11C001**
[BUS B]
Nominal kV = 11.000
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 18.251 kA Method C
Steady State = 9.553 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S003-SB-11C001 [BUS B]	Total	0.00	6.494	-9.107	1.4	11.186
S002-SB-1AC001 [BUS B]	S003-SB-11C001 [BUS B]	80.27	6.215	-8.183	1.3	10.276
Bus55	S003-SB-11C001 [BUS B]	0.02	0.080	-0.205	2.6	0.220
Bus20	S003-SB-11C001 [BUS B]	0.01	0.054	-0.149	2.8	0.159
Bus163	S003-SB-11C001 [BUS B]	0.07	0.126	-0.520	4.1	0.535
Bus78	S003-SB-11C001 [BUS B]	0.07	0.020	-0.050	2.5	0.053

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	11.065	11.279	2.187
0.02	10.996	11.004	0.412
0.03	10.923	10.923	0.066
0.04	10.852	10.852	0.011
0.05	10.784	10.784	0.003
0.06	10.751	10.751	0.000
0.07	10.718	10.718	0.000
0.08	10.686	10.686	0.000
0.09	10.655	10.655	0.000
0.10	10.625	10.625	0.000
0.15	10.564	10.564	0.000
0.20	10.507	10.507	0.000
0.25	10.452	10.452	0.000
0.30	10.452	10.452	0.000

3-Phase fault at bus: **S003-SB-11D001**
[BUS A]
Nominal kV = 3.300
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 16.986 kA Method C
Steady State = 6.380 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S003-SB-11D001 [BUS A]	Total	0.00	1.770	-7.728	4.4	7.928
Bus50	S003-SB-11D001 [BUS A]	1.51	1.477	-6.377	4.3	6.546
S003-7200-MP-03A	S003-SB-11D001 [BUS A]	110.00	0.108	-0.560	5.2	0.570
S003-7100-MK-2A	S003-SB-11D001 [BUS A]	110.00	0.056	-0.234	4.1	0.240
S003-7100-MK-1B	S003-SB-11D001 [BUS A]	110.00	0.064	-0.277	4.3	0.284
S003-7100-MK-1A	S003-SB-11D001 [BUS A]	110.00	0.065	-0.280	4.3	0.287

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	7.733	9.490	5.501
0.02	7.601	8.075	2.725
0.03	7.472	7.592	1.343
0.04	7.347	7.377	0.662
0.05	7.227	7.235	0.332
0.06	7.169	7.171	0.164
0.07	7.113	7.113	0.081
0.08	7.057	7.057	0.040
0.09	7.003	7.003	0.020
0.10	6.951	6.951	0.010
0.15	6.847	6.847	0.000
0.20	6.749	6.749	0.000
0.25	6.657	6.657	0.000
0.30	6.657	6.657	0.000

3-Phase fault at bus: **S003-SB-11D001**
[BUS B]
Nominal kV = 3.300
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 18.010 kA Method C
Steady State = 6.330 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S003-SB-11D001 [BUS B]	Total	0.00	1.964	-8.226	4.2	8.457
Bus49	S003-SB-11D001 [BUS B]	1.50	1.452	-6.336	4.4	6.500
Bus37	S003-SB-11D001 [BUS B]	1.34	0.235	-0.591	2.5	0.636
S003-7100-MK-1C	S003-SB-11D001 [BUS B]	110.00	0.064	-0.276	4.3	0.284
S003-7100-MK-2B	S003-SB-11D001 [BUS B]	110.00	0.056	-0.234	4.1	0.240
S003-7100-MK-2C	S003-SB-11D001 [BUS B]	110.00	0.056	-0.234	4.2	0.241
S003-7200-MP-02B	S003-SB-11D001 [BUS B]	110.00	0.101	-0.555	5.5	0.564

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	8.280	10.103	5.788
0.02	8.130	8.620	2.865
0.03	7.956	8.079	1.403
0.04	7.787	7.817	0.686
0.05	7.625	7.632	0.346
0.06	7.546	7.548	0.170
0.07	7.470	7.470	0.084
0.08	7.395	7.395	0.041
0.09	7.322	7.322	0.020
0.10	7.251	7.251	0.010
0.15	7.107	7.107	0.000
0.20	6.970	6.970	0.000
0.25	6.840	6.840	0.000
0.30	6.840	6.840	0.000

3-Phase fault at bus: **S003-SB-11E001**
[BUS A]
Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 69.427 kA Method C
Steady State = 31.428 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S003-SB-11E001 [BUS A]	Total	0.00	12.013	-33.963	2.8	36.025
Bus166	S003-SB-11E001 [BUS A]	15.09	10.432	-30.201	2.9	31.951
S003-DB11E001-01	S003-SB-11E001 [BUS A]	0.34	0.021	-0.050	2.4	0.055
Lump42	S003-SB-11E001 [BUS A]	105.00	1.560	-3.713	2.4	4.027

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	35.627	39.515	17.093
0.02	35.216	35.704	5.882
0.03	34.818	34.875	1.998
0.04	34.434	34.440	0.679
0.05	34.067	34.068	0.239
0.06	33.892	33.892	0.082
0.07	33.721	33.721	0.028
0.08	33.554	33.554	0.010
0.09	33.392	33.392	0.003
0.10	33.234	33.234	0.001
0.15	32.922	32.922	0.000
0.20	32.627	32.627	0.000
0.25	32.350	32.350	0.000
0.30	32.350	32.350	0.000

3-Phase fault at bus: **S003-SB-11E001**
[BUS B]
Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 75.988 kA Method C
Steady State = 31.612 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S003-SB-11E001 [BUS B]	Total	0.00	9.434	-35.368	3.7	36.605
Bus167	S003-SB-11E001 [BUS B]	15.20	7.857	-31.224	4.0	32.197
S003-DB11E001-02	S003-SB-11E001 [BUS B]	0.31	0.021	-0.050	2.4	0.054
S003-7500-K-1A	S003-SB-11E001 [BUS B]	105.00	0.372	-1.275	3.4	1.328
Lump48	S003-SB-11E001 [BUS B]	105.00	1.185	-2.820	2.4	3.058

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Ide
0.01	35.865	42.639	23.060
0.02	35.443	36.991	10.589
0.03	35.032	35.358	4.789
0.04	34.636	34.704	2.166
0.05	34.259	34.274	1.020
0.06	34.076	34.079	0.465
0.07	33.898	33.898	0.212
0.08	33.724	33.724	0.097
0.09	33.555	33.555	0.044
0.10	33.390	33.390	0.021
0.15	33.075	33.075	0.000
0.20	32.776	32.776	0.000
0.25	32.496	32.496	0.000
0.30	32.496	32.496	0.000

3-Phase fault at bus: **S003-SB-11N001**
[BUS A]
Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 119.995 kA Method C
Steady State = 53.139 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S003-SB-11N001 [BUS A]	Total	0.00	15.337	-56.072	3.7	58.132
S003-DB-11N001-03 [BUS A]	S003-SB-11N001 [BUS A]	1.40	0.097	-0.224	2.3	0.244
S003-DB11N001-04 [BUS A]	S003-SB-11N001 [BUS A]	1.14	0.079	-0.183	2.3	0.200
S003-MDP-B	S003-SB-11N001 [BUS A]	0.00	0.000	0.000	999.9	0.000
Bus56	S003-SB-11N001 [BUS A]	81.20	13.956	-52.854	3.8	54.665
S003-7400-MP-4A	S003-SB-11N001 [BUS A]	105.00	0.118	-0.221	1.9	0.251
Lump2	S003-SB-11N001 [BUS A]	105.00	1.088	-2.589	2.4	2.808

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	57.466	67.809	35.997
0.02	57.129	59.439	16.409
0.03	56.803	57.274	7.331
0.04	56.490	56.585	3.275
0.05	56.193	56.214	1.549
0.06	56.049	56.053	0.700
0.07	55.909	55.910	0.316
0.08	55.773	55.773	0.143
0.09	55.640	55.640	0.065
0.10	55.512	55.512	0.031
0.15	55.270	55.270	0.001
0.20	55.041	55.041	0.000
0.25	54.827	54.827	0.000
0.30	54.827	54.827	0.000

3-Phase fault at bus: **S003-SB-11N001**
[BUS B]
Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 125.140 kA Method C
Steady State = 53.128 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S003-SB-11N001 [BUS B]	Total	0.00	16.092	-58.485	3.6	60.658
S003-SB01E1	S003-SB-11N001 [BUS B]	9.73	0.508	-1.219	2.4	1.320
S003-DB-11N001-03 [BUS B]	S003-SB-11N001 [BUS B]	1.40	0.097	-0.224	2.3	0.244
S003-DB11N001-04 [BUS B]	S003-SB-11N001 [BUS B]	1.26	0.087	-0.202	2.3	0.220
S003-DB11N001-11	S003-SB-11N001 [BUS B]	0.00	0.000	0.000	999.9	0.000
S003-DB11N001-12	S003-SB-11N001 [BUS B]	0.00	0.000	0.000	999.9	0.000
Bus55	S003-SB-11N001 [BUS B]	81.33	13.821	-52.978	3.8	54.752
S003-7400-P-21B	S003-SB-11N001 [BUS B]	105.00	0.339	-0.911	2.7	0.972
Lump52	S003-SB-11N001 [BUS B]	105.00	1.239	-2.950	2.4	3.199

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	59.557	70.411	37.560
0.02	59.002	61.456	17.194
0.03	58.464	58.969	7.698
0.04	57.948	58.050	3.447
0.05	57.456	57.480	1.641
0.06	57.218	57.223	0.744
0.07	56.986	56.987	0.337
0.08	56.761	56.761	0.153
0.09	56.541	56.541	0.069
0.10	56.329	56.329	0.034
0.15	55.925	55.925	0.001
0.20	55.545	55.545	0.000
0.25	55.188	55.188	0.000
0.30	55.188	55.188	0.000

3-Phase fault at bus: **TR-PSS1AC01_11kV**
BUS
Nominal kV = 11.000
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 77.939 kA Method C
Steady State = 23.148 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
TR-PSS1AC01_11kV BUS	Total	0.00	2.588	-30.493	11.8	30.602
S002-SB-1AC001 [BUS A]	TR-PSS1AC01_11kV BUS	0.10	1.630	-7.291	4.5	7.471
TR-PSS1AC01_132kV BUS	TR-PSS1AC01_11kV BUS	96.12	0.958	-23.202	24.2	23.222

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	29.909	45.917	34.840
0.02	29.546	42.130	30.033
0.03	29.138	38.405	25.018
0.04	28.692	35.463	20.841
0.05	28.143	34.207	19.444
0.06	27.864	32.418	16.569
0.07	27.590	30.993	14.119
0.08	27.322	29.854	12.031
0.09	27.060	28.937	10.252
0.10	26.803	28.619	10.032
0.15	26.264	26.704	4.830
0.20	25.749	25.853	2.326
0.25	25.260	25.285	1.120
0.30	25.260	25.266	0.539

3-Phase fault at bus:

TR-PSS1AC01_132k

Nominal kV

Voltage c Factor

Peak Value

Steady State

=

=

=

=

132.000

1.10

45.184

17.919

(Maximum If)

kA Method C

kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
TR-PSS1AC01_132kV BUS	Total	0.00	2.428	-18.789	7.7	18.945
Bus79	TR-PSS1AC01_132kV BUS	0.93	2.339	-18.292	7.8	18.441
TR-PSS1AC01_11kV BUS	TR-PSS1AC01_132kV BUS	25.08	0.089	-0.497	5.6	0.505

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	18.878	26.025	17.914
0.02	18.842	22.347	12.017
0.03	18.800	20.450	8.048
0.04	18.752	19.511	5.390
0.05	18.692	19.040	3.624
0.06	18.661	18.819	2.429
0.07	18.631	18.702	1.628
0.08	18.601	18.633	1.091
0.09	18.571	18.585	0.731
0.10	18.542	18.548	0.492
0.15	18.479	18.479	0.067
0.20	18.418	18.418	0.009
0.25	18.359	18.359	0.001
0.30	18.359	18.359	0.000

3-Phase fault at bus: **TR-PSS1AC02_11kV**
BUS
Nominal kV = 11.000
Voltage c Factor = 1.10 (Maximum If)
Peak Value = 76.963 kA Method C
Steady State = 22.950 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
TR-PSS1AC02_11kV BUS	Total	0.00	2.524	-30.077	11.9	30.183
S002-SB-1AC001 [BUS B]	TR-PSS1AC02_11kV BUS	0.13	1.557	-7.073	4.5	7.243
TR-PSS1AC02_132kV BUS	TR-PSS1AC02_11kV BUS	91.82	0.967	-23.004	23.8	23.024

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	29.501	45.343	34.434
0.02	29.139	41.546	29.615
0.03	28.737	37.872	24.667
0.04	28.271	34.949	20.547
0.05	27.711	33.615	19.028
0.06	27.428	31.849	16.188
0.07	27.150	30.443	13.773
0.08	26.877	29.320	11.718
0.09	26.609	28.415	9.969
0.10	26.347	28.068	9.677
0.15	25.793	26.202	4.608
0.20	25.263	25.359	2.194
0.25	24.760	24.782	1.045
0.30	24.760	24.765	0.497

3-Phase fault at bus:

TR-PSSIAC02_132k

Nominal kV

Voltage c Factor

Peak Value

Steady State

=

=

=

=

132.000

1.10

45.185

17.919

(Maximum If)

kA Method C

kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
TR-PSSIAC02_132kV BUS	Total	0.00	2.428	-18.790	7.7	18.946
Bus81	TR-PSSIAC02_132kV BUS	0.93	2.335	-18.267	7.8	18.415
TR-PSSIAC02_11kV BUS	TR-PSSIAC02_132kV BUS	25.41	0.093	-0.523	5.6	0.531

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	18.878	26.026	17.915
0.02	18.842	22.348	12.018
0.03	18.800	20.451	8.049
0.04	18.752	19.511	5.390
0.05	18.692	19.040	3.625
0.06	18.661	18.819	2.430
0.07	18.631	18.702	1.629
0.08	18.601	18.633	1.092
0.09	18.571	18.585	0.732
0.10	18.542	18.548	0.493
0.15	18.479	18.479	0.067
0.20	18.418	18.418	0.009
0.25	18.359	18.359	0.001
0.30	18.359	18.359	0.000

Project: UQ LT LPG Export - ST & MT
Location: Umm Qasr, Iraq
Contract: UQ83
Engineer: Hemanathan Gopalan
Filename: Umm Qasr-ST

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Revision: Base
Config.: 1_NE

3-Phase fault at bus: **TYPICAL MOTOR**
FEEDER 1 A
Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 115.240 kA Method C
Steady State = 43.205 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
TYPICAL MOTOR FEEDER 1 A	Total	0.00	9.685	-50.447	5.2	51.369
Bus9	TYPICAL MOTOR FEEDER 1 A	19.32	0.158	-0.411	2.6	0.440
Bus12	TYPICAL MOTOR FEEDER 1 A	19.05	0.157	-0.412	2.6	0.441
Bus53	TYPICAL MOTOR FEEDER 1 A	18.78	0.156	-0.413	2.6	0.442
Bus54	TYPICAL MOTOR FEEDER 1 A	18.78	0.156	-0.413	2.6	0.442
Bus85	TYPICAL MOTOR FEEDER 1 A	18.51	0.155	-0.414	2.7	0.443
Bus149	TYPICAL MOTOR FEEDER 1 A	18.24	0.155	-0.416	2.7	0.443
Bus151	TYPICAL MOTOR FEEDER 1 A	18.51	0.155	-0.414	2.7	0.443
Bus154	TYPICAL MOTOR FEEDER 1 A	18.24	0.155	-0.416	2.7	0.443
Bus161	TYPICAL MOTOR FEEDER 1 A	17.96	0.154	-0.417	2.7	0.444
Bus162	TYPICAL MOTOR FEEDER 1 A	18.24	0.155	-0.416	2.7	0.443
Bus164	TYPICAL MOTOR FEEDER 1 A	17.96	0.154	-0.417	2.7	0.444
Bus165	TYPICAL MOTOR FEEDER 1 A	17.69	0.153	-0.418	2.7	0.445
Bus170	TYPICAL MOTOR FEEDER 1 A	17.96	0.154	-0.417	2.7	0.444
Bus171	TYPICAL MOTOR FEEDER 1 A	17.69	0.153	-0.418	2.7	0.445
Bus172	TYPICAL MOTOR FEEDER 1 A	17.42	0.153	-0.419	2.7	0.446
S002-SB-1AC001 [BUS A]	S002-SB-05N001A	93.96	7.023	-43.411	6.2	43.975
TYPICAL MOTOR FEEDER 2 A	S002-SB-05N001A	105.00	0.339	-0.807	2.4	0.875
S002-SB-05N001A	TYPICAL MOTOR FEEDER 1 A	0.00	7.362	-44.217	6.0	44.826

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	48.757	64.110	41.626

Project: UQ LT LPG Export - ST & MT
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Engineer: Hemanathan Gopalan
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3-Phase fault at bus: **TYPICAL MOTOR**
FEEDER 1 A
Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 115.240 kA Method C
Steady State = 43.205 kA rms

<u>Breaking and DC Fault Current (kA)</u>			
Based on Total Bus Fault Current			
<u>TD (S)</u>	<u>Ib sym</u>	<u>Ib asym</u>	<u>Idc</u>
0.02	48.163	54.201	24.860
0.03	47.593	49.766	14.543
0.04	47.052	47.815	8.508
0.05	46.544	46.836	5.221
0.06	46.292	46.395	3.083
0.07	46.048	46.084	1.821
0.08	45.812	45.824	1.076
0.09	45.583	45.588	0.635
0.10	45.363	45.365	0.393
0.15	44.995	44.995	0.029
0.20	44.649	44.649	0.002
0.25	44.329	44.329	0.000
0.30	44.329	44.329	0.000

3-Phase fault at bus: **TYPICAL MOTOR**
FEEDER 1 B
Nominal kV = 0.380
Voltage c Factor = 1.05 (Maximum If)
Peak Value = 121.703 kA Method C
Steady State = 43.174 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
TYPICAL MOTOR FEEDER 1 B	Total	0.00	10.984	-53.665	4.9	54.777
Bus173	TYPICAL MOTOR FEEDER 1 B	16.58	0.150	-0.422	2.8	0.448
Bus174	TYPICAL MOTOR FEEDER 1 B	16.86	0.151	-0.421	2.8	0.447
Bus175	TYPICAL MOTOR FEEDER 1 B	17.14	0.152	-0.420	2.8	0.446
Bus176	TYPICAL MOTOR FEEDER 1 B	16.86	0.151	-0.421	2.8	0.447
Bus177	TYPICAL MOTOR FEEDER 1 B	17.14	0.152	-0.420	2.8	0.446
Bus178	TYPICAL MOTOR FEEDER 1 B	17.42	0.153	-0.419	2.7	0.446
Bus179	TYPICAL MOTOR FEEDER 1 B	17.14	0.152	-0.420	2.8	0.446
Bus180	TYPICAL MOTOR FEEDER 1 B	17.42	0.153	-0.419	2.7	0.446
Bus181	TYPICAL MOTOR FEEDER 1 B	17.69	0.153	-0.418	2.7	0.445
Bus182	TYPICAL MOTOR FEEDER 1 B	17.42	0.153	-0.419	2.7	0.446
Bus183	TYPICAL MOTOR FEEDER 1 B	17.69	0.153	-0.418	2.7	0.445
Bus184	TYPICAL MOTOR FEEDER 1 B	17.96	0.154	-0.417	2.7	0.444
Bus185	TYPICAL MOTOR FEEDER 1 B	17.69	0.153	-0.418	2.7	0.445
Bus186	TYPICAL MOTOR FEEDER 1 B	17.96	0.154	-0.417	2.7	0.444
Bus187	TYPICAL MOTOR FEEDER 1 B	18.24	0.155	-0.416	2.7	0.443
S002-SB-1AC001 [BUS B]	S002-SB-05N001B	93.86	7.008	-43.367	6.2	43.930
TYPICAL MOTOR FEEDER 2 B	S002-SB-05N001B	105.00	0.243	-0.579	2.4	0.628
POWER FEEDER 1/2	S002-SB-05N001C	105.00	1.444	-3.437	2.4	3.728
Bus188	POWER SOCKET FEEDER 1 B	0.00	0.000	0.000	999.9	0.000
Bus189	POWER SOCKET FEEDER 1 B	0.00	0.000	0.000	999.9	0.000
Bus190	POWER SOCKET FEEDER 1 B	0.00	0.000	0.000	999.9	0.000
S002-SB-05N001B	POWER SOCKET FEEDER 1 B	0.00	0.000	0.000	9999.0	0.000
S002-SB-05N001B	TYPICAL MOTOR FEEDER 1 B	0.00	8.695	-47.384	5.4	48.175

(Cont.)

3-Phase fault at bus: **TYPICAL MOTOR**

FEEDER 1 B

Nominal kV = 0.380

Voltage c Factor = 1.05 (Maximum If)

Peak Value = 121.703 kA Method C

Steady State = 43.174 kA rms

Contribution		Voltage & Initial Symmetrical Current (rms)				
From Bus ID	To Bus ID	% V From Bus	kA Real	kA Imaginary	X/R Ratio	kA Magnitude
S002-SB-05N001C	S002-SB-05N001B	0.00	1.444	-3.437	2.4	3.728

Breaking and DC Fault Current (kA)

Based on Total Bus Fault Current

TD (S)	Ib sym	Ib asym	Idc
0.01	51.801	67.590	43.417
0.02	50.852	56.994	25.737
0.03	49.939	52.096	14.835
0.04	49.068	49.807	8.551
0.05	48.245	48.530	5.250
0.06	47.843	47.941	3.064
0.07	47.452	47.486	1.789
0.08	47.074	47.085	1.044
0.09	46.707	46.711	0.609
0.10	46.352	46.354	0.378
0.15	45.718	45.718	0.026
0.20	45.123	45.123	0.002
0.25	44.568	44.568	0.000
0.30	44.568	44.568	0.000

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Location: Umm Qasr, Iraq

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Contract: UQ83

SN: SL-KENTUAE

Engineer: Hemanathan Gopalan

Study Case: SC_Max

Revision: Base

Filename: Umm Qasr-ST

Config.: 1_NE

Load Terminal Fault Short-Circuit Current

Bus ID	Load					Feeder Z (ohm)		I"k (kA)		
	ID	Type	kV	HP/kW	kVA	Heater & Cable R	Cable X	System	Load	Total
Area Lighting DB (A8)	Flood Light-1	Static Load	0.380	10.66	10.66	0.00000	0.00000	2.17	0.000	2.170
BUS-MTR1_ENCRYO	614-KM-01A	Ind. Motor	11.000	5400.00	6473.27	0.00678	0.01146	0.52	2.429	2.584
BUS-MTR2_ENCRYO	614-KM-01B	Ind. Motor	11.000	5400.00	6473.27	0.00678	0.01146	0.52	2.429	2.584
Bus2	Chiller-15	Ind. Motor	0.415	295.00	357.10	0.00000	0.00000	0.59	3.346	3.569
Bus4	Chiller-16	Ind. Motor	0.415	295.00	357.10	0.00000	0.00000	0.59	3.346	3.569
Bus9	614-FM-06AA	Ind. Motor	0.380	37.00	47.06	0.00000	0.00000	2.34	0.488	2.617
Bus12	614-FM-06AB	Ind. Motor	0.380	37.00	47.06	0.00000	0.00000	2.38	0.488	2.654
Bus14	Pump-15	Ind. Motor	0.415	18.00	23.06	0.00000	0.00000	4.27	0.108	4.327
Bus15	Pump-16	Ind. Motor	0.415	18.00	23.06	0.00000	0.00000	3.90	0.108	3.954
Bus47	S002-40-MP-04A	Ind. Motor	3.300	250.00	297.25	0.00000	0.00000	12.54	0.378	12.802
Bus53	614-FM-06AC	Ind. Motor	0.380	37.00	47.06	0.00000	0.00000	2.42	0.488	2.691
Bus54	614-FM-06BA	Ind. Motor	0.380	37.00	47.06	0.00000	0.00000	2.42	0.488	2.691
Bus57	Chiller-9	Ind. Motor	0.415	295.00	357.10	0.00000	0.00000	0.59	3.346	3.569
Bus59	Chiller-10	Ind. Motor	0.415	295.00	357.10	0.00000	0.00000	0.59	3.346	3.569
Bus60	Pump-9	Ind. Motor	0.415	18.00	23.06	0.00000	0.00000	5.99	0.108	6.042
Bus62	Pump-10	Ind. Motor	0.415	18.00	23.06	0.00000	0.00000	5.28	0.108	5.336
Bus68	Chiller-11	Ind. Motor	0.415	295.00	357.10	0.00000	0.00000	0.59	3.346	3.569
Bus70	Chiller-12	Ind. Motor	0.415	295.00	357.10	0.00000	0.00000	0.59	3.346	3.569
Bus72	Pump-11	Ind. Motor	0.415	18.00	23.06	0.00000	0.00000	7.60	0.108	7.663
Bus73	Pump-12	Ind. Motor	0.415	18.00	23.06	0.00000	0.00000	6.51	0.108	6.563
Bus74	Chiller-13	Ind. Motor	0.415	295.00	357.10	0.00000	0.00000	0.59	3.346	3.569
Bus75	Chiller-14	Ind. Motor	0.415	295.00	357.10	0.00000	0.00000	0.59	3.346	3.569
Bus76	Pump-13	Ind. Motor	0.415	18.00	23.06	0.00000	0.00000	5.99	0.108	6.042
Bus77	Pump-14	Ind. Motor	0.415	18.00	23.06	0.00000	0.00000	4.63	0.108	4.681
Bus85	614-FM-06BB	Ind. Motor	0.380	37.00	47.06	0.00000	0.00000	2.45	0.488	2.729
Bus89	S002-40-MP-5C	Ind. Motor	11.000	560.00	655.43	0.05331	0.04944	19.87	0.189	20.057
Bus90	S002-40-MP-05A	Ind. Motor	11.000	560.00	655.43	0.05421	0.05027	19.75	0.189	19.933
Bus91	S002-DB1AN001-01_Power DB	Static Load	0.380	46.67	46.67	0.00000	0.00000	3.88	0.000	3.882
Bus92	S002-6100-MK-1D	Ind. Motor	11.000	6900.00	7991.07	0.00745	0.01261	25.71	2.075	27.781
Bus93	Chiller-1	Ind. Motor	0.415	295.00	357.10	0.00000	0.00000	0.59	3.346	3.569
Bus94	Chiller-2	Ind. Motor	0.415	295.00	357.10	0.00000	0.00000	0.59	3.346	3.569
Bus95	Pump-1	Ind. Motor	0.415	18.00	23.06	0.00000	0.00000	8.57	0.114	8.636
Bus96	Pump-2	Ind. Motor	0.415	18.00	23.06	0.00000	0.00000	7.26	0.114	7.322
Bus97	Chiller-3	Ind. Motor	0.415	295.00	357.10	0.00000	0.00000	0.59	3.346	3.569
Bus98	Chiller-4	Ind. Motor	0.415	295.00	357.10	0.00000	0.00000	0.59	3.346	3.569
Bus99	Pump-3	Ind. Motor	0.415	18.00	23.06	0.00000	0.00000	6.30	0.114	6.361

Bus ID	Load					Feeder Z (ohm)		I"k (kA)		
	ID	Type	kV	HP/kW	kVA	Heater &		System	Load	Total
						Cable R	Cable X			
Bus100	Pump-4	Ind. Motor	0.415	18.00	23.06	0.00000	0.00000	4.97	0.114	5.031
Bus101	Chiller-5	Ind. Motor	0.415	295.00	357.10	0.00000	0.00000	0.59	3.346	3.569
Bus102	Chiller-6	Ind. Motor	0.415	295.00	357.10	0.00000	0.00000	0.59	3.346	3.569
Bus103	Pump-5	Ind. Motor	0.415	18.00	23.06	0.00000	0.00000	4.50	0.114	4.557
Bus104	Pump-6	Ind. Motor	0.415	18.00	23.06	0.00000	0.00000	6.31	0.114	6.364
Bus105	Chiller-7	Ind. Motor	0.415	295.00	357.10	0.00000	0.00000	0.59	3.346	3.569
Bus106	Chiller-8	Ind. Motor	0.415	295.00	357.10	0.00000	0.00000	0.59	3.346	3.569
Bus107	Pump-7	Ind. Motor	0.415	18.00	23.06	0.00000	0.00000	8.64	0.114	8.707
Bus108	Pump-8	Ind. Motor	0.415	18.00	23.06	0.00000	0.00000	6.89	0.114	6.945
Bus110	S002-6100-MP-10A	Ind. Motor	11.000	1050.00	1215.28	0.00000	0.00000	25.20	0.316	25.519
Bus113	S002-6100-MP-10C	Ind. Motor	11.000	1050.00	1255.98	0.00000	0.00000	24.81	0.326	25.134
Bus114	S002-6100-MK-1C	Ind. Motor	11.000	6900.00	7982.68	0.00723	0.01223	25.50	1.977	27.473
Bus115	S002-6100-MP-10B	Ind. Motor	11.000	1050.00	1255.98	0.00000	0.00000	24.85	0.326	25.179
Bus117	S002-40-MP-05B	Ind. Motor	11.000	560.00	655.43	0.05331	0.04944	19.69	0.189	19.880
Bus119	S002-6200-MP-02A	Ind. Motor	3.300	470.00	574.71	0.00000	0.00000	22.32	0.730	23.034
Bus120	S002-6100-MP-11A	Ind. Motor	3.300	950.00	1105.68	0.00000	0.00000	19.85	1.170	20.969
Bus121	S002-6100-MP-9C	Ind. Motor	3.300	950.00	1105.68	0.00000	0.00000	20.18	1.064	21.198
Bus122	S002-6100-MP-9A	Ind. Motor	3.300	950.00	1105.68	0.00000	0.00000	20.32	1.064	21.341
Bus123	S002-6100-MP-8D	Ind. Motor	3.300	200.00	248.62	0.00000	0.00000	5.31	0.335	5.487
Bus124	S002-6100-MP-8B	Ind. Motor	3.300	200.00	248.62	0.00000	0.00000	5.40	0.335	5.577
Bus125	S002-6100-MP-7C	Ind. Motor	3.300	187.00	232.46	0.00000	0.00000	3.51	0.201	3.602
Bus126	S002-6100-MP-7A	Ind. Motor	3.300	187.00	232.46	0.00000	0.00000	3.51	0.201	3.602
Bus127	S002-6100-MP-6D	Ind. Motor	3.300	187.00	232.46	0.00000	0.00000	4.37	0.201	4.468
Bus128	S002-6100-MP-6B	Ind. Motor	3.300	187.00	232.46	0.00000	0.00000	4.37	0.201	4.468
Bus129	S002-6100-MP-4D	Ind. Motor	3.300	187.00	232.46	0.00000	0.00000	7.19	0.201	7.300
Bus130	S002-6100-MP-4B	Ind. Motor	3.300	187.00	232.46	0.00000	0.00000	7.19	0.201	7.300
Bus131	S002-6100-MP-3C	Ind. Motor	3.300	187.00	232.46	0.00000	0.00000	5.14	0.201	5.240
Bus132	S002-6100-MP-3A	Ind. Motor	3.300	187.00	232.46	0.00000	0.00000	5.14	0.201	5.240
Bus133	S002-6100-MP-3B	Ind. Motor	3.300	187.00	232.46	0.00000	0.00000	5.10	0.201	5.203
Bus134	S002-6100-MP-3D	Ind. Motor	3.300	187.00	232.46	0.37910	0.05116	2.65	0.201	2.746
Bus135	S002-6100-MP-4A	Ind. Motor	3.300	187.00	232.46	0.26145	0.03528	3.78	0.201	3.880
Bus136	S002-6100-MP-4C	Ind. Motor	3.300	187.00	232.46	0.26145	0.03528	3.78	0.201	3.880
Bus137	S002-6100-MP-6A	Ind. Motor	3.300	187.00	232.46	0.45754	0.06174	2.23	0.201	2.318
Bus138	S002-6100-MP-6C	Ind. Motor	3.300	187.00	232.46	0.45754	0.06174	2.23	0.201	2.318
Bus139	S002-6100-MP-7B	Ind. Motor	3.300	187.00	232.46	0.00000	0.00000	3.49	0.201	3.590
Bus140	S002-6100-MP-7D	Ind. Motor	3.300	187.00	232.46	0.00000	0.00000	3.49	0.201	3.590
Bus141	S002-6100-MP-8A	Ind. Motor	3.300	200.00	248.62	0.00000	0.00000	5.35	0.335	5.536
Bus142	S002-6100-MP-8C	Ind. Motor	3.300	200.00	248.62	0.00000	0.00000	5.26	0.335	5.449
Bus143	S002-6100-MP-9B	Ind. Motor	3.300	950.00	1105.68	0.00000	0.00000	18.36	1.064	19.386

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Location:	Umm Qasr, Iraq	24.0.2C	Date:	03-08-2026
Contract:	UQ83		SN:	SL-KENTUAE
Engineer:	Hemanathan Gopalan	Study Case: SC_Max	Revision:	Base
Filename:	Umm Qasr-ST		Config.:	1_NE

Bus ID	Load					Feeder Z (ohm)		I"k (kA)		
	ID	Type	kV	HP/kW	kVA	Heater &		System	Load	Total
						Cable R	Cable X			
Bus144	S002-6100-MP-11B	Ind. Motor	3.300	950.00	1105.68	0.00000	0.00000	18.26	1.064	19.293
Bus145	S002-6100-MP-11C	Ind. Motor	3.300	950.00	1105.68	0.00000	0.00000	18.04	1.064	19.072
Bus146	S002-40-MP-04B	Ind. Motor	3.300	250.00	297.25	0.17648	0.02381	5.44	0.343	5.628
Bus149	614-FM-06BC	Ind. Motor	0.380	37.00	47.06	0.00000	0.00000	2.49	0.488	2.769
Bus151	614-FM-06CA	Ind. Motor	0.380	37.00	47.06	0.00000	0.00000	2.45	0.488	2.729
Bus152	S002-6600-MP-1B	Ind. Motor	0.380	15.00	18.14	0.00000	0.00000	2.02	0.174	2.138
Bus154	614-FM-06CB	Ind. Motor	0.380	37.00	47.06	0.00000	0.00000	2.49	0.488	2.769
Bus157	EOT CRANE	Static Load	0.380	36.88	36.88	0.00000	0.00000	5.33	0.000	5.326
Bus158	S002-6600-MP-1A	Ind. Motor	0.380	15.00	18.14	0.00000	0.00000	2.02	0.174	2.140
Bus159	S002-6600-MP-2A	Ind. Motor	0.380	18.50	21.76	0.00000	0.00000	1.31	0.208	1.431
Bus160	688-KM-01	Ind. Motor	3.300	1000.00	1137.58	0.00000	0.00000	19.91	2.174	21.963
Bus161	614-FM-06CC	Ind. Motor	0.380	37.00	47.06	0.00000	0.00000	2.54	0.488	2.810
Bus162	614-FM-06DA	Ind. Motor	0.380	37.00	47.06	0.00000	0.00000	2.49	0.488	2.769
Bus164	614-FM-06DB	Ind. Motor	0.380	37.00	47.06	0.00000	0.00000	2.54	0.488	2.810
Bus165	614-FM-06DC	Ind. Motor	0.380	37.00	47.06	0.00000	0.00000	2.58	0.488	2.853
Bus169	S002-6600-MP-2B	Ind. Motor	0.380	18.50	21.76	0.00000	0.00000	1.31	0.208	1.431
Bus170	614-FM-06EA	Ind. Motor	0.380	37.00	47.06	0.00000	0.00000	2.54	0.488	2.810
Bus171	614-FM-06EB	Ind. Motor	0.380	37.00	47.06	0.00000	0.00000	2.58	0.488	2.853
Bus172	614-FM-06EC	Ind. Motor	0.380	37.00	47.06	0.00000	0.00000	2.62	0.488	2.896
Bus173	614-FM-06KC	Ind. Motor	0.380	37.00	47.06	0.00000	0.00000	2.77	0.488	3.040
Bus174	614-FM-06KB	Ind. Motor	0.380	37.00	47.06	0.00000	0.00000	2.72	0.488	2.991
Bus175	614-FM-06KA	Ind. Motor	0.380	37.00	47.06	0.00000	0.00000	2.67	0.488	2.943
Bus176	614-FM-06JC	Ind. Motor	0.380	37.00	47.06	0.00000	0.00000	2.72	0.488	2.991
Bus177	614-FM-06JB	Ind. Motor	0.380	37.00	47.06	0.00000	0.00000	2.67	0.488	2.943
Bus178	614-FM-06JA	Ind. Motor	0.380	37.00	47.06	0.00000	0.00000	2.63	0.488	2.898
Bus179	614-FM-06HC	Ind. Motor	0.380	37.00	47.06	0.00000	0.00000	2.67	0.488	2.943
Bus180	614-FM-06HB	Ind. Motor	0.380	37.00	47.06	0.00000	0.00000	2.63	0.488	2.898
Bus181	614-FM-06HA	Ind. Motor	0.380	37.00	47.06	0.00000	0.00000	2.58	0.488	2.854
Bus182	614-FM-06GC	Ind. Motor	0.380	37.00	47.06	0.00000	0.00000	2.63	0.488	2.898
Bus183	614-FM-06GB	Ind. Motor	0.380	37.00	47.06	0.00000	0.00000	2.58	0.488	2.854
Bus184	614-FM-06GA	Ind. Motor	0.380	37.00	47.06	0.00000	0.00000	2.54	0.488	2.811
Bus185	614-FM-06FC	Ind. Motor	0.380	37.00	47.06	0.00000	0.00000	2.58	0.488	2.854
Bus186	614-FM-06FB	Ind. Motor	0.380	37.00	47.06	0.00000	0.00000	2.54	0.488	2.811
Bus187	614-FM-06FA	Ind. Motor	0.380	37.00	47.06	0.00000	0.00000	2.50	0.488	2.770
Bus188	614-W-01	Static Load	0.380	47.06	47.06	0.00000	0.00000	5.93	0.000	5.933
Bus189	614-W-02	Static Load	0.380	47.06	47.06	0.00000	0.00000	11.54	0.000	11.540
Bus190	614-W-03	Static Load	0.380	47.06	47.06	0.00000	0.00000	9.67	0.000	9.670
BUS_S002-CMP3	614-KM-11	Ind. Motor	11.000	3150.00	3706.10	0.00678	0.01146	27.66	1.070	28.729
S002-DBASP1-2-22	MOV_0.93	Ind. Motor	0.380	3.72	1.45	0.00000	0.00000	8.73	0.021	8.753

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ETAP

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Location: Umm Qasr, Iraq

24.0.2C

Date: 03-08-2026

Contract: UQ83

SN: SL-KENTUAE

Engineer: Hemanathan Gopalan

Study Case: SC_Max

Revision: Base

Filename: Umm Qasr-ST

Config.: 1_NE

Bus ID	Load					Feeder Z (ohm)		I"k (kA)		
	ID	Type	kV	HP/kW	kVA	Heater &		System	Load	Total
						Cable R	Cable X			
S002-DBASP1-2-22	MOV_0.55	Ind. Motor	0.380	1.10	0.86	0.00000	0.00000	8.74	0.012	8.753
S002-DBASP1-2-22	MOV_1.86	Ind. Motor	0.380	11.16	2.91	0.00000	0.00000	8.71	0.042	8.753
S002-DBASP1-2-22	MOV_2.59	Ind. Motor	0.380	10.36	4.05	0.00000	0.00000	8.70	0.058	8.753
S002-DBASP1-2-22	MOV_0.4	Ind. Motor	0.380	6.40	0.63	0.00000	0.00000	8.74	0.009	8.753
S002-DBASP1-2-22	MOV_0.3	Ind. Motor	0.380	0.90	0.47	0.00000	0.00000	8.75	0.007	8.753
S002-DBASP1-2-22	MOV_2.59 [F]	Ind. Motor	0.380	7.77	4.05	0.00000	0.00000	8.70	0.058	8.753
S002-MCC1-4 [BUS]	S002-MCC1-4_M	Ind. Motor	0.380	15.00	21.04	0.22672	0.01584	0.96	0.201	1.084
S002-MCC1-9 [BUS]	S002-6100-MP-1B	Ind. Motor	0.380	55.00	70.75	0.02616	0.00999	6.88	0.677	7.429
S002-SB-01E001 [BUS A]	S002-6300-MK-1A	Ind. Motor	0.380	132.00	163.29	0.01456	0.00680	11.98	1.563	13.232
S002-SB-01E001 [BUS B]	S002-6500-MP-100B	Ind. Motor	0.380	55.00	69.51	0.00000	0.00000	48.05	0.665	48.711
S002-SB-04E001 [BUS A]	688-PM-01A	Ind. Motor	0.380	22.00	28.69	0.00000	0.00000	16.66	0.275	16.938
S002-SB-04E001 [BUS A]	688-PM-01B	Ind. Motor	0.380	22.00	28.69	0.00000	0.00000	16.66	0.275	16.938
S002-SB-04E001 [BUS A]	688-KM-01-OH	Static Load	0.380	4.33	4.33	0.00000	0.00000	16.94	0.000	16.938
S002-SB-04E001 [BUS A]	DB04E001-01	Static Load	0.380	11.69	11.69	0.00000	0.00000	16.94	0.000	16.938
S002-SB-04E001 [BUS A]	688-W-01	Static Load	0.380	44.44	44.44	0.00000	0.00000	16.94	0.000	16.938
S002-SB-04E001 [BUS B]	HVAC-001/002	Ind. Motor	0.380	25.00	29.24	0.00000	0.00000	16.66	0.280	16.938
S002-SB-04E001 [BUS B]	S002-UP-04V002	Static Load	0.380	16.66	16.66	0.00000	0.00000	16.94	0.000	16.938
S002-SB-04E001 [BUS B]	HVAC-002	Ind. Motor	0.380	25.00	29.24	0.00000	0.00000	16.66	0.280	16.938
S002-SB-04E001 [BUS B]	S002-UP-04V001	Static Load	0.380	16.66	16.66	0.00000	0.00000	16.94	0.000	16.938
S003-SB-11D001 [BUS A]	S003-7200-MP-03A	Ind. Motor	3.300	500.00	601.31	0.06580	0.04036	6.10	0.579	6.669
S003-SB-11D001 [BUS A]	S003-7100-MK-2A	Ind. Motor	3.300	210.00	251.49	0.12553	0.02884	6.01	0.242	6.241
S003-SB-11D001 [BUS A]	S003-7100-MK-1B	Ind. Motor	3.300	250.00	297.22	0.11507	0.02643	6.12	0.286	6.390
S003-SB-11D001 [BUS A]	S003-7100-MK-1A	Ind. Motor	3.300	250.00	300.66	0.13076	0.03004	5.92	0.289	6.192
S003-SB-11D001 [BUS B]	S003-7100-MK-1C	Ind. Motor	3.300	250.00	297.22	0.12553	0.02884	6.27	0.286	6.538
S003-SB-11D001 [BUS B]	S003-7100-MK-2B	Ind. Motor	3.300	210.00	251.49	0.13076	0.03004	6.22	0.242	6.448
S003-SB-11D001 [BUS B]	S003-7100-MK-2C	Ind. Motor	3.300	210.00	251.49	0.11158	0.02563	6.50	0.242	6.725
S003-SB-11D001 [BUS B]	S003-7200-MP-02B	Ind. Motor	3.300	502.00	588.77	0.02119	0.01300	7.39	0.567	7.951
S003-SB-11E001 [BUS B]	S003-7500-K-1A	Ind. Motor	0.380	140.00	175.04	0.00733	0.00680	14.48	1.396	15.798
S003-SB-11N001 [BUS A]	S003-7400-MP-4A	Ind. Motor	0.380	22.00	27.16	0.05230	0.01066	4.17	0.260	4.340
S003-SB-11N001 [BUS B]	S003-7400-P-21B	Ind. Motor	0.380	90.00	108.89	0.02075	0.00985	8.98	1.042	9.780

* The terminal short-circuit current of the single-phase loads is calculated in the Panel/UPS/1Ph System Device Duty calculation.

Short-Circuit Summary Report

3-Phase Fault Currents

Bus		Device		Device Capacity (kA)				Short-Circuit Current (kA)					
				Making									
ID	kV	ID	Type	Peak	Ib sym	Ib asym	Idc	I"k	ip	Ib sym	Ib asym	Idc	Ik
S002-TR-CMP2_VFD~2	11.646	S002-TR-CMP2_VFD~2	Bus					4.314	11.695				4.136
S002-TR-CMP2_VFD~3	11.646	S002-TR-CMP2_VFD~3	Bus					4.314	11.695				4.136
S002-TR-CMP1_VFD~2	11.220	S002-TR-CMP1_VFD~2	Bus					4.164	11.288				3.991
S002-TR-CMP1_VFD~3	11.220	S002-TR-CMP1_VFD~3	Bus					4.164	11.288				3.991
Cable201~	0.462	Cable201~	Bus					48.132	87.082				46.074
	0.462	QS1_Chiller-8	SPST	143.000				48.132	87.082				
Cable200~	0.462	Cable200~	Bus					50.838	93.051				48.545
	0.462	QS1_Chiller-7	SPST	143.000				50.838	93.051				
Cable153~	0.446	Cable153~	Bus					41.999	74.562				40.591
	0.446	QS1_Chiller=9	SPST	143.000				41.999	74.562				
Cable192~	0.462	Cable192~	Bus					45.512	81.452				43.656
	0.462	QS1_Chiller-3	SPST	143.000				45.512	81.452				
Cable163~	0.446	Cable163~	Bus					49.131	89.953				47.206
	0.446	QS1_Chiller-11	SPST	143.000				49.131	89.953				
Cable197~	0.462	Cable197~	Bus					45.319	81.043				43.480
	0.462	QS1_Chiller-6	SPST	143.000				45.319	81.043				
Cable154~	0.446	Cable154~	Bus					40.361	71.177				39.061
	0.446	QS1_Chiller-10	SPST	143.000				40.361	71.177				
Cable164~	0.446	Cable164~	Bus					46.509	84.164				44.783
	0.446	QS1_Chiller-12	SPST	143.000				46.509	84.164				
Cable168~	0.446	Cable168~	Bus					43.771	78.284				42.242
	0.446	QS1_Chiller-13	SPST	143.000				43.771	78.284				
Cable169~	0.446	Cable169~	Bus					40.048	70.537				38.768
	0.446	QS1_Chiller-14	SPST	143.000				40.048	70.537				
Cable40~	0.446	Cable40~	Bus					37.432	65.252				36.314
	0.446	QS1_Chiller-15	SPST	143.000				37.432	65.252				
Cable41~	0.446	Cable41~	Bus					36.117	62.644				35.077
	0.446	QS1_Chiller-16	SPST	143.000				36.117	62.644				
Cable188~	0.462	Cable188~	Bus					51.584	94.727				49.203
	0.462	QS1_Chiller-1	SPST	143.000				51.584	94.727				
Cable189~	0.462	Cable189~	Bus					49.247	89.521				47.074
	0.462	QS1_Chiller-2	SPST	143.000				49.247	89.521				
Cable193~	0.462	Cable193~	Bus					41.961	74.042				40.381
	0.462	QS1_Chiller-4	SPST	143.000				41.961	74.042				
Cable196~	0.462	Cable196~	Bus					39.632	69.309				38.222

3-Phase Fault Currents

Bus		Device		Device Capacity (kA)				Short-Circuit Current (kA)					
				Making									
ID	kV	ID	Type	Peak	Ib sym	Ib asym	Idc	I"k	ip	Ib sym	Ib asym	Idc	Ik
Cable196~	0.462	QS1_Chiller-5	SPST	143.000				39.632	69.309				
132kV GRID	132.000	132kV GRID	Bus					19.094	45.690				18.060
	132.000	=E02-Q0	CB	100.000	40.000	43.629	17.421	19.094	45.690	18.829	19.132	3.395	
	132.000	=E06-Q0	CB	100.000	40.000	43.629	17.421	19.094	45.690	18.829	19.132	3.395	
200A MDB Agreko	0.415	200A MDB Agreko	Bus					49.236	84.661				46.822
	0.415	CB155	CB	220.000	100.000	101.269		49.236	84.661	48.655	48.656	0.263	
	0.415	QM	CB	40.000	36.000	36.276		49.236	84.661 *	48.685 *	48.687 *	0.506	
Area Lighting DB (A8)	0.380	Area Lighting DB (A8)	Bus					2.170	3.131				2.166
	0.380	CB79	CB	374.000	85.000	130.613		2.170	3.131	2.170	2.170	0.003	
BUS-MTR1_ENCRYO	11.000	BUS-MTR1_ENCRYO	Bus					2.574	6.047				0.520
BUS-MTR2_ENCRYO	11.000	BUS-MTR2_ENCRYO	Bus					2.574	6.047				0.520
Bus1	11.000	Bus1	Bus					30.246	76.701				22.883
Bus2	0.380	Bus2	Bus					3.569	7.485				0.592
Bus4	0.380	Bus4	Bus					3.569	7.485				0.592
Bus7	11.000	Bus7	Bus					22.287	44.357				18.184
Bus8	11.000	Bus8	Bus					21.301	41.748				17.536
Bus9	0.380	Bus9	Bus					2.617	4.021				2.337
Bus10	0.380	Bus10	Bus					54.377	123.006				49.213
Bus11	0.380	Bus11	Bus					14.126	32.860				2.287
Bus12	0.380	Bus12	Bus					2.654	4.071				2.373
Bus14	0.380	Bus14	Bus					4.327	6.241				4.266
Bus15	0.380	F1_Pump-15	SPST	143.000				4.327	6.241				
	0.380	Bus15	Bus					3.954	5.703				3.894
	0.380	F1_Pump-16	SPST	143.000				3.954	5.703				
Bus17	11.000	Bus17	Bus					21.548	42.146				17.662
Bus18	11.000	Bus18	Bus					21.334	41.828				17.536
Bus19	11.000	Bus19	Bus					10.919	17.669				9.518
Bus20	11.000	Bus20	Bus					11.102	18.093				9.491
Bus21	11.000	Bus21	Bus					29.803	75.642				22.670
Bus22	3.300	Bus22	Bus					28.769	70.335				17.848
Bus23	0.380	Bus23	Bus					53.492	121.337				49.213
Bus24	0.380	Bus24	Bus					31.983	64.579				29.935
Bus25	3.300	Bus25	Bus					25.040	62.606				18.093
Bus27	0.380	Bus27	Bus					32.142	64.701				29.020
Bus28	0.380	Bus28	Bus					41.682	89.771				34.953
Bus29	3.450	Bus29	Bus					23.976	55.331				17.642
Bus30	3.450	Bus30	Bus					27.442	61.728				17.420

3-Phase Fault Currents

Bus		Device		Device Capacity (kA)				Short-Circuit Current (kA)					
				Making									
ID	kV	ID	Type	Peak	Ib sym	Ib asym	Idc	I"k	ip	Ib sym	Ib asym	Idc	Ik
Bus31	3.450	Bus31	Bus					27.448	61.747				17.420
Bus32	3.450	Bus32	Bus					24.028	55.479				17.642
Bus33	0.380	Bus33	Bus					41.272	89.031				34.933
Bus35	0.380	Bus35	Bus					14.009	32.588				4.384
Bus36	0.400	Bus36	Bus					22.866	44.983				15.470
	0.400	CB210	CB	220.000	100.000	101.269		22.866	44.983	20.954	20.983	1.095	
	0.400	CB151	CB	220.000	100.000	101.269		22.866	44.983	20.954	20.983	1.095	
Bus37	3.300	Bus37	Bus					7.633	15.040				5.774
Bus38	0.400	Bus38	Bus					59.814	119.736				56.822
Bus40	0.400	Bus40	Bus					56.620	111.106				53.880
	0.400	Q9A	SPST	143.000				56.620	111.106				
Bus41	0.380	Bus41	Bus					25.904	60.258				5.846
Bus42	11.000	Bus42	Bus					29.077	71.229				22.284
Bus43	11.000	Bus43	Bus					29.141	70.431				22.296
Bus45	0.400	Bus45	Bus					56.620	111.106				53.880
	0.400	Q9B	SPST	143.000				56.620	111.106				
Bus46	11.000	Bus46	Bus					22.044	43.972				18.050
Bus47	3.300	Bus47	Bus					12.802	18.999				10.753
Bus48	0.400	Bus48	Bus					56.620	111.106				53.880
	0.400	Q9C	SPST	143.000				56.620	111.106				
Bus49	3.300	Bus49	Bus					8.538	18.211				6.414
Bus50	3.300	Bus50	Bus					8.014	17.197				6.465
Bus51	11.000	Bus51	Bus					23.490	44.311				18.915
	11.000	RMU TOFS	Fuse		63.000	83.839		23.490	44.311	23.147	25.460	10.604	
Bus52	0.400	Bus52	Bus					56.318	110.376				53.170
	0.400	Q10A	SPST	143.000				56.318	110.376				
Bus53	0.380	Bus53	Bus					2.691	4.122				2.410
Bus54	0.380	Bus54	Bus					2.691	4.122				2.410
Bus55	11.000	Bus55	Bus					11.095	18.078				9.485
Bus56	11.000	Bus56	Bus					10.876	17.586				9.485
Bus57	0.380	Bus57	Bus					3.569	7.485				0.592
Bus58	0.380	Bus58	Bus					5.146	11.972				0.935
Bus59	0.380	Bus59	Bus					3.569	7.485				0.592
Bus60	0.380	Bus60	Bus					6.042	8.715				5.970
Bus61	0.380	Bus61	Bus					12.777	29.723				2.287
Bus62	0.380	Bus62	Bus					5.336	7.697				5.269
Bus63	0.400	Bus63	Bus					56.318	110.376				53.170

Project: UQ LT LPG Export - ST & MT
Location: Umm Qasr, Iraq
Contract: UQ83
Engineer: Hemanathan Gopalan
Filename: Umm Qasr-ST

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3-Phase Fault Currents

Bus		Device		Device Capacity (kA)				Short-Circuit Current (kA)					
				Making				I"k	ip	Ib sym	Ib asym	Idc	Ik
ID	kV	ID	Type	Peak	Ib sym	Ib asym	Idc						
Bus63	0.400	Q10B	SPST	143.000				56.318	110.376				
Bus64	0.415	Bus64	Bus					58.451	114.565				55.164
	0.415	Q10C	SPST	143.000				58.451	114.565				
Bus65	0.400	Bus65	Bus					60.251	121.036				56.763
Bus66	11.000	Bus66	Bus					9.747	15.536				8.620
Bus67	0.380	Bus67	Bus					52.346	116.165				44.482
Bus68	0.380	Bus68	Bus					3.569	7.485				0.592
Bus69	0.380	Bus69	Bus					49.662	111.237				44.773
Bus70	0.380	Bus70	Bus					3.569	7.485				0.592
Bus71	0.380	Bus71	Bus					12.229	28.447				3.653
Bus72	0.380	Bus72	Bus					7.663	11.055				7.578
Bus73	0.380	Bus73	Bus					6.563	9.467				6.486
Bus74	0.380	Bus74	Bus					3.569	7.485				0.592
Bus75	0.380	Bus75	Bus					3.569	7.485				0.592
Bus76	0.380	Bus76	Bus					6.042	8.715				5.970
	0.380	F1_Pump-13	SPST	143.000				6.042	8.715				
Bus77	0.380	Bus77	Bus					4.681	6.752				4.617
	0.380	F1_Pump-14	SPST	143.000				4.681	6.752				
Bus78	11.000	Bus78	Bus					9.891	15.854				8.593
Bus79	132.000	Bus79	Bus					19.094	45.690				18.060
	132.000	=E03-Q0	CB	100.000	40.000	43.629	17.421	19.094	45.690	18.829	19.132	3.395	
Bus80	132.000	Bus80	Bus					19.094	45.690				18.060
Bus81	132.000	Bus81	Bus					19.094	45.690				18.060
	132.000	=E07-Q0	CB	100.000	40.000	43.629	17.421	19.094	45.690	18.829	19.132	3.395	
Bus82	0.415	Bus82	Bus					49.236	84.661				46.822
	0.415	QM	CB	40.000	36.000	36.276		49.236	84.661 *	48.685 *	48.687 *	0.506	
Bus83	132.000	Bus83	Bus					19.094	45.690				18.060
	132.000	=E02-Q0	CB	100.000	40.000	43.629	17.421	19.094	45.690	18.829	19.132	3.395	
Bus84	132.000	Bus84	Bus					19.094	45.690				18.060
	132.000	=E06-Q0	CB	100.000	40.000	43.629	17.421	19.094	45.690	18.829	19.132	3.395	
Bus85	0.380	Bus85	Bus					2.729	4.175				2.449
Bus87	132.000	Bus87	Bus					19.094	45.690				18.060
	132.000	=E01-Q0	CB	100.000	40.000	43.629	17.421	19.094	45.690	18.829	19.132	3.395	
Bus88	132.000	Bus88	Bus					19.094	45.690				18.060
	132.000	=E08-Q0	CB	100.000	40.000	43.629	17.421	19.094	45.690	18.829	19.132	3.395	
Bus89	11.000	Bus89	Bus					24.438	51.297				19.516
Bus90	11.000	Bus90	Bus					24.351	50.999				19.462

3-Phase Fault Currents

Bus		Device		Device Capacity (kA)				Short-Circuit Current (kA)					
				Making									
ID	kV	ID	Type	Peak	Ib sym	Ib asym	Idc	I"k	ip	Ib sym	Ib asym	Idc	Ik
Bus91	0.380	Bus91	Bus					3.882	5.601				3.829
Bus92	11.000	Bus92	Bus					29.099	71.816				22.133
Bus93	0.380	Bus93	Bus					3.569	7.485				0.592
Bus94	0.380	Bus94	Bus					3.569	7.485				0.592
Bus95	0.400	Bus95	Bus					8.636	12.458				8.531
Bus96	0.400	Bus96	Bus					7.322	10.562				7.232
Bus97	0.380	Bus97	Bus					3.569	7.485				0.592
Bus98	0.380	Bus98	Bus					3.569	7.485				0.592
Bus99	0.400	Bus99	Bus					6.361	9.176				6.281
Bus100	0.400	Bus100	Bus					5.031	7.257				4.961
Bus101	0.380	Bus101	Bus					3.569	7.485				0.592
Bus102	0.380	Bus102	Bus					3.569	7.485				0.592
Bus103	0.400	Bus103	Bus					4.557	6.573				4.490
Bus104	0.400	F1_Pump-5	SPST	143.000				4.557	6.573				
	0.400	Bus104	Bus					6.364	9.181				6.284
	0.400	F1_Pump-6	SPST	143.000				6.364	9.181				
Bus105	0.380	Bus105	Bus					3.569	7.485				0.592
Bus106	0.380	Bus106	Bus					3.569	7.485				0.592
Bus107	0.400	Bus107	Bus					8.707	12.560				8.602
	0.400	F1_Pump-7	SPST	143.000				8.707	12.560				
	0.400	Bus108	Bus					6.945	10.019				6.860
Bus108	0.400	F1_Pump-8	SPST	143.000				6.945	10.019				
Bus110	11.000	Bus110	Bus					25.519	55.131				20.151
Bus112	11.000	Bus112	Bus					21.561	42.178				17.662
Bus113	11.000	Bus113	Bus					25.134	54.292				19.930
Bus114	11.000	Bus114	Bus					28.732	71.120				21.961
Bus115	11.000	Bus115	Bus					25.179	54.458				19.958
Bus117	11.000	Bus117	Bus					24.166	50.847				19.361
Bus119	3.300	Bus119	Bus					23.034	42.865				15.504
Bus120	3.300	Bus120	Bus					20.969	37.092				14.538
Bus121	3.300	Bus121	Bus					21.198	37.654				14.665
Bus122	3.300	Bus122	Bus					21.341	38.039				14.728
Bus123	3.300	Bus123	Bus					5.487	7.961				5.127
Bus124	3.300	Bus124	Bus					5.577	8.091				5.207
Bus125	3.300	Bus125	Bus					3.602	5.211				3.449
Bus126	3.300	Bus126	Bus					3.602	5.211				3.449
Bus127	3.300	Bus127	Bus					4.468	6.461				4.263

3-Phase Fault Currents

Bus		Device		Device Capacity (kA)				Short-Circuit Current (kA)					
				Making									
ID	kV	ID	Type	Peak	Ib sym	Ib asym	Idc	I"k	ip	Ib sym	Ib asym	Idc	Ik
Bus128	3.300	Bus128	Bus					4.468	6.461				4.263
Bus129	3.300	Bus129	Bus					7.300	10.565				6.779
Bus130	3.300	Bus130	Bus					7.300	10.565				6.779
Bus131	3.300	Bus131	Bus					5.240	7.577				4.973
Bus132	3.300	Bus132	Bus					5.240	7.577				4.973
Bus133	3.300	Bus133	Bus					5.203	7.529				4.981
Bus134	3.300	Bus134	Bus					5.207	7.532				4.981
Bus135	3.300	Bus135	Bus					7.207	10.444				6.798
Bus136	3.300	Bus136	Bus					7.207	10.444				6.798
Bus137	3.300	Bus137	Bus					4.450	6.435				4.269
Bus138	3.300	Bus138	Bus					4.450	6.435				4.269
Bus139	3.300	Bus139	Bus					3.590	5.196				3.452
Bus140	3.300	Bus140	Bus					3.590	5.196				3.452
Bus141	3.300	Bus141	Bus					5.536	8.045				5.216
Bus142	3.300	Bus142	Bus					5.449	7.918				5.135
Bus143	3.300	Bus143	Bus					19.386	35.797				14.888
Bus144	3.300	Bus144	Bus					19.293	35.400				14.823
Bus145	3.300	Bus145	Bus					19.072	34.749				14.693
Bus146	3.300	Bus146	Bus					9.988	14.626				9.071
Bus149	0.380	Bus149	Bus					2.769	4.230				2.488
Bus151	0.380	Bus151	Bus					2.729	4.175				2.449
Bus152	0.380	Bus152	Bus					2.138	3.084				2.017
Bus154	0.380	Bus154	Bus					2.769	4.230				2.488
Bus157	0.380	Bus157	Bus					5.326	7.743				5.027
Bus158	0.380	Bus158	Bus					2.140	3.087				2.017
Bus159	0.380	Bus159	Bus					1.431	2.084				1.312
Bus160	3.450	Bus160	Bus					21.963	39.475				15.002
Bus161	0.380	Bus161	Bus					2.810	4.287				2.529
Bus162	0.380	Bus162	Bus					2.769	4.230				2.488
Bus163	11.000	Bus163	Bus					11.054	18.022				9.449
Bus164	0.380	Bus164	Bus					2.810	4.287				2.529
Bus165	0.380	Bus165	Bus					2.853	4.345				2.572
Bus166	0.380	Bus166	Bus					41.308	80.053				36.597
Bus167	0.380	Bus167	Bus					41.907	88.807				36.803
Bus168	11.000	Bus168	Bus					10.912	17.668				9.509
Bus169	0.380	Bus169	Bus					1.431	2.083				1.311
Bus170	0.380	Bus170	Bus					2.810	4.287				2.529

3-Phase Fault Currents

Bus		Device		Device Capacity (kA)				Short-Circuit Current (kA)					
				Making									
ID	kV	ID	Type	Peak	Ib sym	Ib asym	Idc	I"k	ip	Ib sym	Ib asym	Idc	Ik
Bus171	0.380	Bus171	Bus					2.853	4.345				2.572
Bus172	0.380	Bus172	Bus					2.896	4.406				2.616
Bus173	0.380	Bus173	Bus					3.040	4.603				2.756
Bus174	0.380	Bus174	Bus					2.991	4.535				2.708
Bus175	0.380	Bus175	Bus					2.943	4.470				2.661
Bus176	0.380	Bus176	Bus					2.991	4.535				2.708
Bus177	0.380	Bus177	Bus					2.943	4.470				2.661
Bus178	0.380	Bus178	Bus					2.898	4.407				2.615
Bus179	0.380	Bus179	Bus					2.943	4.470				2.661
Bus180	0.380	Bus180	Bus					2.898	4.407				2.615
Bus181	0.380	Bus181	Bus					2.854	4.346				2.572
Bus182	0.380	Bus182	Bus					2.898	4.407				2.615
Bus183	0.380	Bus183	Bus					2.854	4.346				2.572
Bus184	0.380	Bus184	Bus					2.811	4.287				2.529
Bus185	0.380	Bus185	Bus					2.854	4.346				2.572
Bus186	0.380	Bus186	Bus					2.811	4.287				2.529
Bus187	0.380	Bus187	Bus					2.770	4.230				2.488
Bus188	0.380	Bus188	Bus					5.933	8.674				5.834
Bus189	0.380	Bus189	Bus					11.540	17.075				11.128
Bus190	0.380	Bus190	Bus					9.670	14.242				9.389
Bus_LVDB-1	0.415	Bus_LVDB-1	Switchboard	143.000				58.906	115.763				55.556
	0.415	QA_LVDB-1	SPST	143.000				58.906	115.763				
	0.415	QB_LVDB-1	SPST	143.000				58.906	115.763				
	0.415	QC_LVDB-1	SPST	143.000				58.906	115.763				
	0.415	Q1_LVDB-4	CB	143.000	65.000	65.383		58.906	115.763	58.050	58.073	1.619	
	0.415	Q3_LVDB-1	CB	143.000	65.000	65.383		58.906	115.763	58.050	58.073	1.619	
	0.415	Q4_LVDB-1	CB	143.000	65.000	65.383		58.906	115.763	58.050	58.073	1.619	
Bus_LVDB-2	0.415	Q6_LVDB-1	CB	143.000	65.000	65.383		58.906	115.763	58.050	58.073	1.619	
	0.415	Bus_LVDB-2	Switchboard	143.000				59.107	116.230				56.203
	0.415	QA_LVDB-2	SPST	143.000				59.107	116.230				
	0.415	QB_LVDB-2	SPST	143.000				59.107	116.230				
	0.415	QC_LVDB-2	SPST	143.000				59.107	116.230				
	0.415	Q1_LVDB-2	CB	143.000	65.000	65.383		59.107	116.230	58.438	58.461	1.635	
	0.415	Q3_LVDB-2	CB	143.000	65.000	65.383		59.107	116.230	58.438	58.461	1.635	
Bus_MDB-1	0.415	Q4_LVDB-2	CB	143.000	65.000	65.383		59.107	116.230	58.438	58.461	1.635	
	0.415	Q6_LVDB-2	CB	143.000	65.000	65.383		59.107	116.230	58.438	58.461	1.635	
	0.415	Bus_MDB-1	MCC	143.000				51.108	96.178				48.514

Project: UQ LT LPG Export - ST & MT
Location: Umm Qasr, Iraq
Contract: UQ83
Engineer: Hemanathan Gopalan
Filename: Umm Qasr-ST

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3-Phase Fault Currents

Bus		Device		Device Capacity (kA)				Short-Circuit Current (kA)					
				Making									
ID	kV	ID	Type	Peak	Ib sym	Ib asym	Idc	I"k	ip	Ib sym	Ib asym	Idc	Ik
Bus_MDB-1	0.415	QM_MDB-1	SPST	143.000				51.108	96.178				
	0.415	Q1_MDB-1	CB	143.000	65.000	66.372		51.108	96.178	50.509	50.556	2.174	
	0.415	Q2_MDB-1	CB	143.000	65.000	66.372		51.108	96.178	50.509	50.556	2.174	
	0.415	Q5_MDB-1	CB	143.000	65.000	66.372		51.108	96.178	50.509	50.556	2.174	
	0.415	Q6_MDB-1	CB	143.000	65.000	66.372		51.108	96.178	50.509	50.556	2.174	
Bus_MDB-2	0.415	Bus_MDB-2	MCC	143.000				51.638	97.446				48.995
	0.415	QM_MDB-2	SPST	143.000				51.638	97.446				
	0.415	Q1_MDB-2	CB	143.000	65.000	66.372		51.638	97.446	51.029	51.080	2.280	
	0.415	Q2_MDB-2	CB	143.000	65.000	66.372		51.638	97.446	51.029	51.080	2.280	
	0.415	Q6_MDB-2	CB	143.000	65.000	66.372		51.638	97.446	51.029	51.080	2.280	
Bus_MDB-3	0.415	Q5_MDB--2.	CB	143.000	65.000	66.372		51.638	97.446	51.029	51.080	2.280	
	0.415	Bus_MDB-3	MCC	143.000				51.907	98.093				49.239
	0.415	QM_MDB-3	SPST	143.000				51.907	98.093				
	0.415	Q1_MDB-3	CB	143.000	65.000	66.372		51.907	98.093	51.293	51.346	2.336	
	0.415	Q2_MDB-3	CB	143.000	65.000	66.372		51.907	98.093	51.293	51.346	2.336	
Bus_MDB-4	0.415	Q5_MDB-3	CB	143.000	65.000	66.372		51.907	98.093	51.293	51.346	2.336	
	0.415	Q6_MDB-3	CB	143.000	65.000	66.372		51.907	98.093	51.293	51.346	2.336	
	0.415	Bus_MDB-4	MCC	143.000				54.156	103.596				51.278
	0.415	QM_MDB-4	SPST	143.000				54.156	103.596				
	0.415	Q1_MDB-4	CB	143.000	65.000	66.372		54.156	103.596	53.498	53.574	2.859	
Bus_MDB-5	0.415	Q2_MDB-4	CB	143.000	65.000	66.372		54.156	103.596	53.498	53.574	2.859	
	0.415	Q5_MDB-4	CB	143.000	65.000	66.372		54.156	103.596	53.498	53.574	2.859	
	0.415	Q6_MDB-4	CB	143.000	65.000	66.372		54.156	103.596	53.498	53.574	2.859	
	0.415	Bus_MDB-5	MCC	143.000				51.527	97.134				49.259
	0.415	QM_MDB-5	SPST	143.000				51.527	97.134				
Bus_MDB-6	0.415	Q1_MDB-5	CB	143.000	65.000	66.372		51.527	97.134	51.044	51.093	2.238	
	0.415	Q2_MDB-5	CB	143.000	65.000	66.372		51.527	97.134	51.044	51.093	2.238	
	0.415	Q5_MDB-5	CB	143.000	65.000	66.372		51.527	97.134	51.044	51.093	2.238	
	0.415	Q6_MDB-5	CB	143.000	65.000	66.372		51.527	97.134	51.044	51.093	2.238	
	0.400	Bus_MDB-6	MCC	143.000				49.665	93.624				47.478
Bus_MDB-7	0.400	QM_MDB-6	SPST	143.000				49.665	93.624				
	0.400	Q1_MDB-6	CB	143.000	65.000	66.372		49.665	93.624	49.200	49.248	2.157	
	0.400	Q2_MDB-6	CB	143.000	65.000	66.372		49.665	93.624	49.200	49.248	2.157	
	0.400	Q5_MDB-6	CB	143.000	65.000	66.372		49.665	93.624	49.200	49.248	2.157	
	0.400	Q6_MDB-6	CB	143.000	65.000	66.372		49.665	93.624	49.200	49.248	2.157	
Bus_MDB-7	0.400	Bus_MDB-7	MCC	143.000				49.665	93.623				47.478
	0.400	QM_MDB-7	SPST	143.000				49.665	93.623				

3-Phase Fault Currents

Bus		Device		Device Capacity (kA)				Short-Circuit Current (kA)					
				Making									
ID	kV	ID	Type	Peak	Ib sym	Ib asym	Idc	I"k	ip	Ib sym	Ib asym	Idc	Ik
Bus_MDB-7	0.400	Q1_MDB-7	CB	143.000	65.000	66.372		49.665	93.623	49.200	49.248	2.157	
	0.400	Q2_MDB-7	CB	143.000	65.000	66.372		49.665	93.623	49.200	49.248	2.157	
	0.400	Q5_MDB-7	CB	143.000	65.000	66.372		49.665	93.623	49.200	49.248	2.157	
	0.400	Q6_MDB-7	CB	143.000	65.000	66.372		49.665	93.623	49.200	49.248	2.157	
Bus_MDB-8	0.400	Bus_MDB-8	MCC	143.000				49.665	93.623				47.478
	0.400	QM_MDB-8	SPST	143.000				49.665	93.623				
	0.400	Q1_MDB-8	CB	143.000	65.000	66.372		49.665	93.623	49.201	49.248	2.157	
	0.400	Q2_MDB-8	CB	143.000	65.000	66.372		49.665	93.623	49.201	49.248	2.157	
	0.400	Q5_MDB-8	CB	143.000	65.000	66.372		49.665	93.623	49.201	49.248	2.157	
	0.400	Q6_MDB-8	CB	143.000	65.000	66.372		49.665	93.623	49.201	49.248	2.157	
BUS_S002-CMP3	11.000	BUS_S002-CMP3	Bus					30.108	76.691				22.866
	11.000	IAC001=R17	CB	125.000	50.000	54.165	20.830	30.108	76.691	27.487	32.440	17.229	
CAMP BUS	11.000	CAMP BUS	Bus					15.318	24.999				13.240
LV SWBD_TOFS	0.400	LV SWBD_TOFS	Switchgear	75.000				28.609	57.679				27.193
	0.400	CB224	CB	75.000	36.000	36.276		28.609	57.679	28.199	28.319	2.603	
	0.400	CB225	CB	75.000	36.000	36.276		28.609	57.679	28.199	28.319	2.603	
MV RMU-TOFS	11.000	MV RMU-TOFS	Switchgear	62.500				23.490	44.311				18.915
	11.000	CB220	CB	62.500	25.000	27.083	10.415	23.490	44.311	22.106	22.106	0.149	
	11.000	CB219	CB	62.500	25.000	27.083	10.415	23.490	44.311	22.106	22.106	0.149	
	11.000	RMU TOFS	Fuse		63.000	83.839		23.490	44.311	23.147	25.460	10.604	
	0.380	POWER SOCKET	Bus					54.777	121.703				43.174
FEEDER 1 B S002-ASP1-1 [BUS A]	0.380	FEEDER 1 B S002-ASP1-1 [BUS A]	Switchboard					27.514	59.348				22.802
	0.380	ASP1-1=IC1	CB	105.000	50.000	50.383		27.514	59.348	26.483	27.060	5.557	
S002-ASP1-1B [BUS B]	0.380	S002-ASP1-1B [BUS B]	Switchboard					24.334	53.465				23.132
	0.380	ASP1-1=IC2	CB	105.000	50.000	50.383		24.334	53.465	24.236	24.811	5.313	
S002-ASP1-2A	0.380	S002-ASP1-2A	Switchboard					29.216	63.168				23.846
	0.380	CB178	CB	220.000	100.000	101.269		29.216	63.168	27.848	28.244	4.715	
	0.380	ASP1-2A=IC1	CB	105.000	50.000	50.383		29.216	63.168	27.975	28.629	6.087	
S002-ASP1-2B	0.380	S002-ASP1-2B	Switchboard					27.037	59.249				23.967
	0.380	CB179	CB	220.000	100.000	101.269		27.037	59.249	25.816	26.263	4.829	
	0.380	ASP1-2A=IC2	CB	105.000	50.000	50.383		27.037	59.249	25.876	26.599	6.160	
S002-DBASP1-2-22	0.380	S002-DBASP1-2-22	Panelboard					8.753	13.490				7.672
	0.380	CB15	CB	32.000	16.000	16.043		8.753	13.490	7.895	7.895	0.003	
S002-MCC1-1 [BUS]	0.380	S002-MCC1-1 [BUS]	MCC	105.000				28.613	61.390				22.612
	0.380	MCC1-1=Q1	CB	105.000	50.000	50.383		28.613	61.390	27.057	27.629	5.594	
S002-MCC1-2 [BUS]	0.380	S002-MCC1-2 [BUS]	MCC	105.000				24.980	55.432				23.712
	0.380	MCC1-2=Q1	CB	105.000	50.000	50.383		24.980	55.432	24.876	25.582	5.968	

3-Phase Fault Currents

Bus		Device		Device Capacity (kA)				Short-Circuit Current (kA)					
				Making									
ID	kV	ID	Type	Peak	Ib sym	Ib asym	Idc	I"k	ip	Ib sym	Ib asym	Idc	Ik
S002-MCC1-3 [BUS]	0.380	S002-MCC1-3 [BUS]	MCC	105.000				29.912	64.429				23.593
	0.380	MCC1-3=Q1	CB	105.000	50.000	50.383		29.912	64.429	28.285	28.933	6.091	
S002-MCC1-4 [BUS]	0.380	S002-MCC1-4 [BUS]	MCC	105.000				15.441	22.717				14.674
	0.380	MCC1-4=Q1	CB	105.000	50.000	50.383		15.441	22.717	15.162	15.162	0.000	
S002-MCC1-5 [BUS]	0.380	S002-MCC1-5 [BUS]	MCC	105.000				27.625	59.007				21.747
	0.380	MCC1-5=Q1	CB	105.000	50.000	50.383		27.625	59.007	26.060	26.568	5.171	
S002-MCC1-6 [BUS]	0.380	S002-MCC1-6 [BUS]	MCC	105.000				12.168	18.018				11.062
	0.380	MCC1-6=Q1	CB	105.000	50.000	50.383		12.168	18.018	11.666	11.666	0.001	
S002-MCC1-7 [BUS]	0.380	S002-MCC1-7 [BUS]	MCC	105.000				27.398	58.482				21.574
	0.380	MCC1-7=Q1	CB	105.000	50.000	50.383		27.398	58.482	25.847	26.343	5.092	
S002-MCC1-8 [BUS]	0.380	S002-MCC1-8 [BUS]	MCC	105.000				26.611	56.856				23.361
	0.380	MCC1-8=Q1	CB	105.000	50.000	50.383		26.611	56.856	25.469	25.906	4.741	
	0.380	CB17	CB	110.000	50.000	54.209		26.611	56.856	25.757	28.852	13.001	
S002-MCC1-9 [BUS]	0.380	S002-MCC1-9 [BUS]	MCC	105.000				28.606	59.944				22.869
	0.380	CB241	CB	220.000	100.000	101.269		28.606	59.944	26.876	27.077	3.288	
	0.380	MCC1-9=Q1	CB	105.000	50.000	50.383		28.606	59.944	27.023	27.381	4.418	
	0.380	CB18	CB	220.000	100.000	101.269		28.606	59.944	26.876	27.077	3.288	
S002-PC1-1A	0.380	S002-PC1-1A	Switchgear	105.000				38.994	86.320				30.119
	0.380	PC1-1A-3A	CB	105.000	50.000	50.383		38.994	86.320	36.725	38.182	10.446	
	0.380	PC1-1A-2A	CB	105.000	50.000	50.383		38.994	86.320	36.725	38.182	10.446	
	0.380	PC1-1A-5A	CB	105.000	50.000	50.383		38.994	86.320	36.725	38.182	10.446	
	0.380	PC1-1A-4A	CB	105.000	50.000	50.383		38.994	86.320	36.725	38.182	10.446	
	0.380	PC1-1A-1A	CB	105.000	50.000	50.383		38.994	86.320	36.725	38.182	10.446	
	0.380	CB16	CB	220.000	100.000	101.269		38.994	86.320	36.493	37.440	8.368	
S002-PC1-1B	0.380	S002-PC1-1B	Switchgear	105.000				32.433	74.055				30.313
	0.380	PC1B-1-2B	CB	105.000	50.000	50.383		32.433	74.055	32.252	33.777	10.035	
	0.380	PC1-1A-5B	CB	105.000	50.000	50.383		32.433	74.055	32.252	33.777	10.035	
	0.380	PC1-1A-4B	CB	105.000	50.000	50.383		32.433	74.055	32.252	33.777	10.035	
	0.380	PC1-1B-1B	CB	105.000	50.000	50.383		32.433	74.055	32.252	33.777	10.035	
S002-PC1-2A	0.380	S002-PC1-2A	Switchgear	105.000				39.413	87.155				30.192
	0.380	PC1-2A-4A	CB	105.000	50.000	50.383		39.413	87.155	36.981	38.442	10.500	
	0.380	PC1-2A-3A	CB	105.000	50.000	50.383		39.413	87.155	36.981	38.442	10.500	
	0.380	PC1-2A-5A	CB	105.000	50.000	50.383		39.413	87.155	36.981	38.442	10.500	
	0.380	PC1-2A-2A	CB	105.000	50.000	50.383		39.413	87.155	36.981	38.442	10.500	
	0.380	PC1-2A-1A	CB	105.000	50.000	50.383		39.413	87.155	36.981	38.442	10.500	
S002-PC1-2B	0.380	S002-PC1-2B	Switchgear	105.000				34.961	78.774				30.387
	0.380	PC1-2A-3B	CB	105.000	50.000	50.383		34.961	78.774	33.420	34.975	10.311	

Project:	UQ LT LPG Export - ST & MT	ETAP	Page:	344
Location:	Umm Qasr, Iraq	24.0.2C	Date:	03-08-2026
Contract:	UQ83		SN:	SL-KENTUAE
Engineer:	Hemanathan Gopalan	Study Case: SC_Max	Revision:	Base
Filename:	Umm Qasr-ST		Config.:	1_NE

3-Phase Fault Currents

Bus		Device		Device Capacity (kA)				Short-Circuit Current (kA)					
				Making									
ID	kV	ID	Type	Peak	Ib sym	Ib asym	Idc	I"k	ip	Ib sym	Ib asym	Idc	Ik
S002-PC1-2B	0.380	PC1-2A-5B	CB	105.000	50.000	50.383		34.961	78.774	33.420	34.975	10.311	
	0.380	CB240	CB	220.000	100.000	101.269		34.961	78.774	33.331	34.365	8.366	
	0.380	PC1-2A-1B	CB	105.000	50.000	50.383		34.961	78.774	33.420	34.975	10.311	
S002-SB-1AC001 [BUS A]	11.000	S002-SB-1AC001 [BUS A]	Switchgear	125.000				30.544	77.725				23.082
	11.000	1AC001=L1 [I/C 1]	CB	125.000	50.000	54.165	20.830	30.544	77.725	27.939	33.069	17.691	
	11.000	1AC001=L12	CB	125.000	50.000	54.165	20.830	30.544	77.725	27.939	33.069	17.691	
	11.000	1AC001=L11	CB	125.000	50.000	54.165	20.830	30.544	77.725	27.939	33.069	17.691	
	11.000	1AC001=L10	CB	125.000	50.000	54.165	20.830	30.544	77.725	27.939	33.069	17.691	
	11.000	1AC001=L5	CB	125.000	50.000	54.165	20.830	30.544	77.725	27.939	33.069	17.691	
	11.000	1AC001=L4	CB	125.000	50.000	54.165	20.830	30.544	77.725	27.939	33.069	17.691	
	11.000	1AC001=L3	CB	125.000	50.000	54.165	20.830	30.544	77.725	27.939	33.069	17.691	
	11.000	1AC001=L2	CB	125.000	50.000	54.165	20.830	30.544	77.725	27.939	33.069	17.691	
	11.000	1AC001=L14	CB	125.000	50.000	54.165	20.830	30.544	77.725	27.939	33.069	17.691	
	11.000	1AC001=L6	CB	125.000	50.000	54.165	20.830	30.544	77.725	27.939	33.069	17.691	
	11.000	1AC001=L9	CB	125.000	50.000	54.165	20.830	30.544	77.725	27.939	33.069	17.691	
	11.000	1AC001=L7	CB	125.000	50.000	54.165	20.830	30.544	77.725	27.939	33.069	17.691	
	11.000	1AC001=L15	CB	125.000	50.000	54.165	20.830	30.544	77.725	27.939	33.069	17.691	
	11.000	1AC001=L13	CB	125.000	50.000	54.165	20.830	30.544	77.725	27.939	33.069	17.691	
	11.000	1AC001=L16	CB	125.000	50.000	54.165	20.830	30.544	77.725	27.939	33.069	17.691	
S002-SB-1AC001 [BUS B]	11.000	S002-SB-1AC001 [BUS B]	Switchgear	125.000				30.108	76.691				22.866
	11.000	1AC001=R2	CB	125.000	50.000	54.165	20.830	30.108	76.691	27.487	32.440	17.229	
	11.000	1AC001=R3	CB	125.000	50.000	54.165	20.830	30.108	76.691	27.487	32.440	17.229	
	11.000	1AC001=R4	CB	125.000	50.000	54.165	20.830	30.108	76.691	27.487	32.440	17.229	
	11.000	1AC001=R5	CB	125.000	50.000	54.165	20.830	30.108	76.691	27.487	32.440	17.229	
	11.000	1AC001=R6	CB	125.000	50.000	54.165	20.830	30.108	76.691	27.487	32.440	17.229	
	11.000	1AC001=R7	CB	125.000	50.000	54.165	20.830	30.108	76.691	27.487	32.440	17.229	
	11.000	1AC001=R8	CB	125.000	50.000	54.165	20.830	30.108	76.691	27.487	32.440	17.229	
	11.000	1AC001=R10	CB	125.000	50.000	54.165	20.830	30.108	76.691	27.487	32.440	17.229	
	11.000	1AC001=R11	CB	125.000	50.000	54.165	20.830	30.108	76.691	27.487	32.440	17.229	
	11.000	1AC001=R13	CB	125.000	50.000	54.165	20.830	30.108	76.691	27.487	32.440	17.229	
	11.000	1AC001=R1 [I/C 2]	CB	125.000	50.000	54.165	20.830	30.108	76.691	27.487	32.440	17.229	
	11.000	1AC001=R16	CB	125.000	50.000	54.165	20.830	30.108	76.691	27.487	32.440	17.229	
	11.000	1AC001=R12	CB	125.000	50.000	54.165	20.830	30.108	76.691	27.487	32.440	17.229	
	11.000	1AC001=R14	CB	125.000	50.000	54.165	20.830	30.108	76.691	27.487	32.440	17.229	
	11.000	1AC001=R15	CB	125.000	50.000	54.165	20.830	30.108	76.691	27.487	32.440	17.229	
	11.000	1AC001=R17	CB	125.000	50.000	54.165	20.830	30.108	76.691	27.487	32.440	17.229	

Project: UQ LT LPG Export - ST & MT
Location: Umm Qasr, Iraq
Contract: UQ83
Engineer: Hemanathan Gopalan
Filename: Umm Qasr-ST

ETAP
24.0.2C

Study Case: SC_Max

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Date: 03-08-2026
SN: SL-KENTUAE
Revision: Base
Config.: 1_NE

3-Phase Fault Currents

Bus		Device		Device Capacity (kA)				Short-Circuit Current (kA)					
				Making				I"k	ip	Ib sym	Ib asym	Idc	Ik
ID	kV	ID	Type	Peak	Ib sym	Ib asym	Idc						
S002-SB-1AD001 [BUS A]	3.300	S002-SB-1AD001 [BUS A]	Switchgear	125.000				28.555	69.410				17.554
	3.300	1AD001=L2 [I/C 1]	CB	125.000	50.000	54.165	20.830	28.555	69.410	23.499	26.156	11.488	
	3.300	1AD001=L4	CB	125.000	50.000	54.165	20.830	28.555	69.410	23.499	26.156	11.488	
	3.300	1AD001=L3	CB	125.000	50.000	54.165	20.830	28.555	69.410	23.499	26.156	11.488	
	3.300	1AD001=L5	Fuse		63.000	83.839		28.555	69.410	27.170	39.928	29.258	
	3.300	1AD001=L6	Fuse		63.000	83.839		28.555	69.410	27.170	39.928	29.258	
	3.300	1AD001=L7	Fuse		63.000	83.839		28.555	69.410	27.170	39.928	29.258	
	3.300	1AD001=L8	Fuse		63.000	83.839		28.555	69.410	27.170	39.928	29.258	
	3.300	1AD001=L11	Fuse		63.000	83.839		28.555	69.410	27.170	39.928	29.258	
	3.300	1AD001=L12	Fuse		63.000	83.839		28.555	69.410	27.170	39.928	29.258	
	3.300	1AD001=L13	Fuse		63.000	83.839		28.555	69.410	27.170	39.928	29.258	
	3.300	1AD001=L15	Fuse		63.000	83.839		28.555	69.410	27.170	39.928	29.258	
	3.300	1AD001=L18	Fuse		63.000	83.839		28.555	69.410	27.170	39.928	29.258	
	3.300	1AD001=L19	Fuse		63.000	83.839		28.555	69.410	27.170	39.928	29.258	
	3.300	1AD001=L20	Fuse		63.000	83.839		28.555	69.410	27.170	39.928	29.258	
	3.300	1AD001=L21	Fuse		63.000	83.839		28.555	69.410	27.170	39.928	29.258	
	3.300	1AD001=L14	Fuse		63.000	83.839		28.555	69.410	27.170	39.928	29.258	
	3.300	1AD001=L16	Fuse		63.000	83.839		28.555	69.410	27.170	39.928	29.258	
	3.300	1AD001=L17	Fuse		63.000	83.839		28.555	69.410	27.170	39.928	29.258	
	3.300	1AD001=L25	Fuse		63.000	83.839		28.555	69.410	27.170	39.928	29.258	
	3.300	1AD001=L26	CB	125.000	50.000	54.165	20.830	28.555	69.410	23.499	26.156	11.488	
S002-SB-1AD001 [BUS B]	3.300	S002-SB-1AD001 [BUS B]	Switchgear	125.000				24.744	61.464				17.790
	3.300	1AD001=R1 [I/C 2]	CB	125.000	50.000	54.165	20.830	24.744	61.464	21.614	24.283	11.067	
	3.300	1AD001=R2	CB	125.000	50.000	54.165	20.830	24.744	61.464	21.614	24.283	11.067	
	3.300	1AD001=R3	CB	125.000	50.000	54.165	20.830	24.744	61.464	21.614	24.283	11.067	
	3.300	1AD001=R4	Fuse		63.000	83.839		24.744	61.464	23.842	35.701	26.573	
	3.300	1AD001=R5	Fuse		63.000	83.839		24.744	61.464	23.842	35.701	26.573	
	3.300	1AD001=R6	Fuse		63.000	83.839		24.744	61.464	23.842	35.701	26.573	
	3.300	1AD001=R7	Fuse		63.000	83.839		24.744	61.464	23.842	35.701	26.573	
	3.300	1AD001=R10	Fuse		63.000	83.839		24.744	61.464	23.842	35.701	26.573	
	3.300	1AD001=R11	Fuse		63.000	83.839		24.744	61.464	23.842	35.701	26.573	
	3.300	1AD001=R12	Fuse		63.000	83.839		24.744	61.464	23.842	35.701	26.573	
	3.300	1AD001=R13	Fuse		63.000	83.839		24.744	61.464	23.842	35.701	26.573	
	3.300	1AD001=R14	Fuse		63.000	83.839		24.744	61.464	23.842	35.701	26.573	
	3.300	1AD001=R15	Fuse		63.000	83.839		24.744	61.464	23.842	35.701	26.573	
	3.300	1AD001=R16	Fuse		63.000	83.839		24.744	61.464	23.842	35.701	26.573	

Project: UQ LT LPG Export - ST & MT
Location: Umm Qasr, Iraq
Contract: UQ83
Engineer: Hemanathan Gopalan
Filename: Umm Qasr-ST

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Date: 03-08-2026
SN: SL-KENTUAE
Revision: Base
Config.: 1_NE

3-Phase Fault Currents

Bus		Device		Device Capacity (kA)				Short-Circuit Current (kA)					
				Making									
ID	kV	ID	Type	Peak	Ib sym	Ib asym	Idc	I"k	ip	Ib sym	Ib asym	Idc	Ik
S002-SB-1AD001 [BUS B]	3.300	1AD001=R17	Fuse		63.000	83.839		24.744	61.464	23.842	35.701	26.573	
	3.300	1AD001=R18	Fuse		63.000	83.839		24.744	61.464	23.842	35.701	26.573	
	3.300	1AD001=R19	Fuse		63.000	83.839		24.744	61.464	23.842	35.701	26.573	
	3.300	1AD001=R20	CB	125.000	50.000	54.165	20.830	24.744	61.464	21.614	24.283	11.067	
S002-SB-1AN001 [BUS A]	0.380	S002-SB-1AN001 [BUS A]	Switchboard	105.000				15.696	24.793				14.079
	0.380	CB217	CB	105.000	50.000	54.209		15.696	24.793	15.408	15.420	0.614	
	0.380	CB92	CB	110.000	50.000	54.209		15.696	24.793	15.408	15.420	0.614	
	0.380	CB144	CB	105.000	50.000	54.209		15.696	24.793	15.408	15.420	0.614	
	0.380	1AN001=IC1	CB	105.000	50.000	50.383		15.696	24.793	15.272	15.272	0.019	
S002-SB-1AN001 [BUS B]	0.380	S002-SB-1AN001 [BUS B]	Switchboard	105.000				14.728	23.677				14.116
	0.380	CB215	CB	220.000	100.000	101.269		14.728	23.677	14.519	14.519	0.012	
	0.380	1AN001=IC2	CB	105.000	50.000	50.383		14.728	23.677	14.529	14.529	0.028	
	0.380	CB6	CB	110.000	50.000	54.209		14.728	23.677	14.568	14.587	0.747	
S002-SB-01E001 [BUS A]	0.380	S002-SB-01E001 [BUS A]	Switchgear	220.000				50.343	111.742				45.283
	0.380	01E001+A04.RA01	CB	264.000	120.000	120.055		50.343	111.742	47.837	47.889	2.242	
	0.380	01E001+A06.RA01	CB	264.000	120.000	120.055		50.343	111.742	47.837	47.889	2.242	
	0.380	01E001+A07.PA01	CB	264.000	120.000	120.055		50.343	111.742	47.837	47.889	2.242	
	0.380	01E001+A03.AA01	CB	220.000	100.000	102.111		50.343	111.742	48.669	50.225	12.408	
S002-SB-01E001 [BUS B]	0.380	S002-SB-01E001 [BUS B]	Switchgear	220.000				48.711	108.032				44.571
	0.380	01E001+B02.AA01	CB	220.000	85.000	86.794		48.711	108.032	47.167	48.631	11.843	
	0.380	01E001+B07.RA01	CB	264.000	120.000	120.055		48.711	108.032	46.497	46.544	2.102	
	0.380	01E001+B06.PA01	CB	264.000	120.000	120.055		48.711	108.032	46.497	46.544	2.102	
	0.380	01E001+B04.KA01	CB	264.000	120.000	136.404		48.711	108.032	47.607	55.274	28.086	
S002-SB-02N001 [BUS A]	0.380	S002-SB-02N001 [BUS A]	Switchgear	125.000				31.253	62.751				28.142
	0.380	CB180	CB	220.000	100.000	101.269		31.253	62.751	30.223	30.282	1.884	
	0.380	CB80	CB	374.000	85.000	130.613		31.253	62.751	31.077	45.619	33.396	
	0.380	02N001+Q1	CB	187.000	85.000	86.794		31.253	62.751	30.328	30.451	2.731	
S002-SB-02N001 [BUS B]	0.380	S002-SB-02N001 [BUS B]	Switchgear	125.000				31.031	62.495				29.002
	0.380	02N001+Q2	CB	187.000	85.000	86.794		31.031	62.495	30.383	30.510	2.782	
	0.380	CB185	CB	220.000	100.000	101.269		31.031	62.495	30.320	30.381	1.926	
S002-SB-03E001 [BUS A]	0.380	S002-SB-03E001 [BUS A]	Switchgear	143.000				37.333	78.524				30.531
	0.380	03E001+A03.BA01	CB	143.000	65.000	66.372		37.333	78.524	35.193	35.650	5.688	
S002-SB-03E001 [BUS B]	0.380	S002-SB-03E001 [BUS B]	Switchgear	143.000				36.906	77.752				30.516
	0.380	03E001+B02.BA01	CB	143.000	65.000	66.372		36.906	77.752	34.996	35.455	5.687	

3-Phase Fault Currents

Bus		Device		Device Capacity (kA)				Short-Circuit Current (kA)					
				Making									
ID	kV	ID	Type	Peak	Ib sym	Ib asym	Idc	I"k	ip	Ib sym	Ib asym	Idc	Ik
S002-SB-03N001 [BUS A]	0.380	S002-SB-03N001 [BUS A]	Switchgear	220.000				48.520	105.577				40.650
	0.380	03N001+A03.BA01	CB	220.000	80.000	81.689		48.520	105.577	46.826	47.972	10.424	
S002-SB-03N001 [BUS B]	0.380	S002-SB-03N001 [BUS B]	Switchgear	220.000				45.700	100.372				40.892
	0.380	03N001+B03.BA01	CB	220.000	80.000	81.689		45.700	100.372	44.489	45.661	10.281	
S002-SB-04E001 [BUS A]	0.380	S002-SB-04E001 [BUS A]	MCC	52.500				16.938	28.229				13.985
	0.380	SB-04E001=CB1	CB	187.000	85.000	91.304		16.938	28.229	15.843	15.853	0.570	
	0.380	SB-04E001=CB3	CB	84.000	40.000	40.307		16.938	28.229	15.620	15.621	0.088	
S002-SB-04E001 [BUS B]	0.380	S002-SB-04E001 [BUS B]	MCC	52.500				16.938	28.229				13.985
	0.380	SB-04E001=CB3	CB	84.000	40.000	40.307		16.938	28.229	15.620	15.621	0.088	
S002-SB-05N001A	0.380	S002-SB-05N001A	Switchgear	220.000				51.369	115.240				43.205
S002-SB-05N001B	0.380	S002-SB-05N001B	Switchgear	220.000				54.777	121.703				43.174
S002-SB-05N001C	0.380	S002-SB-05N001C	Bus					54.777	121.703				43.174
S002-SB-PSS1A01 [BUS-A]	132.000	S002-SB-PSS1A01 [BUS-A]	Switchgear	100.000				19.094	45.690				18.060
S002-SB-PSS01N01 [BUS A]	0.380	S002-SB-PSS01N01 [BUS A]	Switchboard	105.000				14.343	27.820				12.143
	0.380	PSS1N01+Q1	CB	105.000	50.000	50.383		14.343	27.820	13.648	13.673	0.834	
S002-SB-PSS01N01 [BUS B]	0.380	S002-SB-PSS01N01 [BUS B]	Switchboard	105.000				13.366	26.014				12.090
	0.380	PSS1N01+Q2	CB	105.000	50.000	50.383		13.366	26.014	12.911	12.937	0.807	
S002-TR-04DE001_PRI	3.300	S002-TR-04DE001_PRI	Bus					17.740	31.334				12.991
S002-TR-04DE002_PRI	3.300	S002-TR-04DE002_PRI	Bus					16.168	29.297				13.114
S003-DB11E001-01	0.380	S003-DB11E001-01	Switchrack	52.500				12.676	18.453				12.165
	0.380	11N001-01 [I/C1]	CB	52.500	25.000	25.192		12.676	18.453	12.575	12.575	0.000	
S003-DB11E001-02	0.380	S003-DB11E001-02	Switchrack	52.500				13.885	20.301				13.274
	0.380	DB11E001-02 [I/C1]	CB	52.500	25.000	25.192		13.885	20.301	13.785	13.785	0.000	
S003-DB-11N001-03 [BUS A]	0.380	S003-DB-11N001-03 [BUS A]	Switchrack	52.500				15.582	22.610				15.159
	0.380	11N001-03 [I/C1]	CB	52.500	25.000	25.192		15.582	22.610	15.468	15.468	0.000	
S003-DB-11N001-03 [BUS B]	0.380	S003-DB-11N001-03 [BUS B]	Switchrack	52.500				15.695	22.763				15.160
	0.380	11N001-03 [I/C2]	CB	52.500	25.000	25.192		15.695	22.763	15.583	15.583	0.000	
S003-DB11N001-04 [BUS A]	0.380	S003-DB11N001-04 [BUS A]	Switchrack	52.500				15.548	22.553				15.159
	0.380	11N001-04 [I/C1]	CB	52.500	25.000	25.192		15.548	22.553	15.452	15.452	0.000	
S003-DB11N001-04 [BUS B]	0.380	S003-DB11N001-04 [BUS B]	Switchrack	52.500				15.677	22.732				15.160
	0.380	11N001-04 [I/C2]	CB	52.500	25.000	25.192		15.677	22.732	15.574	15.574	0.000	
S003-DB11N001-11	0.380	S003-DB11N001-11	Bus					10.354	14.954				10.211
	0.380	CB85	CB	187.000	85.000	86.794		10.354	14.954	10.354	10.354	0.000	

3-Phase Fault Currents

Bus		Device		Device Capacity (kA)				Short-Circuit Current (kA)					
				Making									
ID	kV	ID	Type	Peak	Ib sym	Ib asym	Idc	I"k	ip	Ib sym	Ib asym	Idc	Ik
S003-DB11N001-12	0.380	S003-DB11N001-12	Bus					10.354	14.954				10.211
	0.380	CB74	CB	187.000	85.000	86.794		10.354	14.954	10.354	10.354	0.000	
S003-MDP-B	0.380	S003-MDP-B	Switchgear	73.500				25.811	50.507				24.784
	0.380	MDP-B [I/C1]	CB	220.000	100.000	101.269		25.811	50.507	25.576	25.602	1.136	
S003-SB01E1	0.380	S003-SB01E1	Switchrack	52.500				12.955	24.686				11.249
	0.380	SB01E1 [I/C1]	CB	52.500	25.000	25.192		12.955	24.686	12.371	12.386	0.602	
S003-SB-11C001 [BUS A]	11.000	S003-SB-11C001 [BUS A]	Switchgear	100.000				11.009	17.836				9.587
	11.000	11C001=L2 [I/C 1]	CB	100.000	40.000	42.689	14.911	11.009	17.836	10.618	10.618	0.000	
	11.000	11C001=L4	CB	100.000	40.000	42.689	14.911	11.009	17.836	10.618	10.618	0.000	
	11.000	11C001=L3	CB	100.000	40.000	42.689	14.911	11.009	17.836	10.618	10.618	0.000	
	11.000	11C001=L5	CB	100.000	40.000	42.689	14.911	11.009	17.836	10.618	10.618	0.000	
	11.000	11C001=L8	CB	100.000	40.000	42.689	14.911	11.009	17.836	10.618	10.618	0.000	
	11.000	11C001=L6	CB	100.000	40.000	42.689	14.911	11.009	17.836	10.618	10.618	0.000	
	11.000	11C001=L7	CB	100.000	40.000	42.689	14.911	11.009	17.836	10.618	10.618	0.000	
S003-SB-11C001 [BUS B]	11.000	S003-SB-11C001 [BUS B]	Switchgear	100.000				11.186	18.251				9.553
	11.000	11C001=R1 [I/C 2]	CB	100.000	40.000	42.689	14.911	11.186	18.251	10.751	10.751	0.000	
	11.000	11C001=R2	CB	100.000	40.000	42.689	14.911	11.186	18.251	10.751	10.751	0.000	
	11.000	11C001=R3	CB	100.000	40.000	42.689	14.911	11.186	18.251	10.751	10.751	0.000	
	11.000	11C001=R4	CB	100.000	40.000	42.689	14.911	11.186	18.251	10.751	10.751	0.000	
	11.000	11C001=R6	CB	100.000	40.000	42.689	14.911	11.186	18.251	10.751	10.751	0.000	
	11.000	11C001=R5	CB	100.000	40.000	42.689	14.911	11.186	18.251	10.751	10.751	0.000	
S003-SB-11D001 [BUS A]	3.300	S003-SB-11D001 [BUS A]	Switchgear	100.000				7.928	16.986				6.380
	3.300	11D001=L2 [I/C 1]	CB	100.000	40.000	43.332	16.664	7.928	16.986	7.198	7.202	0.233	
	3.300	11D001=L7	CB	100.000	40.000	43.332	16.664	7.928	16.986	7.198	7.202	0.233	
	3.300	11D001=L6	Fuse		63.000	83.839		7.928	16.986	7.733	9.490	5.501	
	3.300	11D001=L5	Fuse		63.000	83.839		7.928	16.986	7.733	9.490	5.501	
	3.300	11D001=L4	Fuse		63.000	83.839		7.928	16.986	7.733	9.490	5.501	
S003-SB-11D001 [BUS B]	3.300	S003-SB-11D001 [BUS B]	Switchgear	100.000				8.457	18.010				6.330
	3.300	11D001=R1 [I/C 2]	CB	100.000	40.000	43.332	16.664	8.457	18.010	7.585	7.589	0.243	
	3.300	11D001=R2	Fuse		63.000	83.839		8.457	18.010	8.280	10.103	5.788	
	3.300	11D001=R3	Fuse		63.000	83.839		8.457	18.010	8.280	10.103	5.788	
	3.300	11D001=R4	Fuse		63.000	83.839		8.457	18.010	8.280	10.103	5.788	
	3.300	11D001=R5	CB	100.000	40.000	43.332	16.664	8.457	18.010	7.585	7.589	0.243	
S003-SB-11E001 [BUS A]	0.380	S003-SB-11E001 [BUS A]	Switchgear	143.000				36.025	69.427				31.428
	0.380	11E001+A02 [I/C1]	CB	187.000	85.000	86.794		36.025	69.427	34.818	34.875	1.998	

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Location: Umm Qasr, Iraq

24.0.2C

Date: 03-08-2026

Contract: UQ83

SN: SL-KENTUAE

Engineer: Hemanathan Gopalan

Study Case: SC_Max

Revision: Base

Filename: Umm Qasr-ST

Config.: 1_NE

3-Phase Fault Currents

Bus		Device		Device Capacity (kA)				Short-Circuit Current (kA)					
				Making									
ID	kV	ID	Type	Peak	Ib sym	Ib asym	Idc	I"k	ip	Ib sym	Ib asym	Idc	Ik
S003-SB-11E001 [BUS A]	0.380	CB204	CB	220.000	100.000	101.269		36.025	69.427	34.662	34.686	1.298	
	0.380	CB153	CB	220.000	100.000	101.269		36.025	69.427	34.662	34.686	1.298	
S003-SB-11E001 [BUS B]	0.380	S003-SB-11E001 [BUS B]	Switchgear	143.000				36.605	75.988				31.612
	0.380	11E001+B02 [I/C2]	CB	187.000	85.000	86.794		36.605	75.988	35.032	35.358	4.789	
	0.380	11E001+A05	CB	154.000	70.000	79.569		36.605	75.988	35.653	38.833	15.391	
	0.380	CB237	CB	220.000	100.000	100.045		36.605	75.988	34.076	34.079	0.465	
	0.380	CB205	CB	220.000	100.000	101.269		36.605	75.988	34.871	35.045	3.487	
S003-SB-11N001 [BUS A]	0.380	S003-SB-11N001 [BUS A]	Switchgear	220.000				58.132	119.995				53.139
	0.380	11N001+A02 [I/C1]	CB	220.000	85.000	86.794		58.132	119.995	56.803	57.274	7.331	
	0.380	CB146	CB	220.000	100.000	101.269		58.132	119.995	56.676	56.924	5.311	
	0.380	CB208	CB	220.000	100.000	101.269		58.132	119.995	56.676	56.924	5.311	
S003-SB-11N001 [BUS B]	0.380	S003-SB-11N001 [BUS B]	Switchgear	220.000				60.658	125.140				53.128
	0.380	CB172	CB	440.000	200.000	200.000		60.658	125.140	55.188	55.188	0.000	
	0.380	11N001+B02 [I/C2]	CB	220.000	100.000	101.269		60.658	125.140	58.255	58.522	5.582	
	0.380	CB190	CB	220.000	100.000	101.269		60.658	125.140	58.255	58.522	5.582	
	0.380	CB203	CB	220.000	100.000	101.269		60.658	125.140	58.255	58.522	5.582	
	0.380	CB84	CB	187.000	85.000	86.794		60.658	125.140	58.464	58.969	7.698	
	0.380	CB75	CB	187.000	85.000	86.794		60.658	125.140	58.464	58.969	7.698	
TR-PSS1AC01_11kV BUS	11.000	TR-PSS1AC01_11kV BUS	Bus					30.602	77.939				23.148
TR-PSS1AC01_132kV BUS	132.000	TR-PSS1AC01_132kV BUS	Bus					18.945	45.184				17.919
TR-PSS1AC02_11kV BUS	11.000	TR-PSS1AC02_11kV BUS	Bus					30.183	76.963				22.950
TR-PSS1AC02_132kV BUS	132.000	TR-PSS1AC02_132kV BUS	Bus					18.946	45.185				17.919
TYPICAL MOTOR FEEDER 1 A	0.380	TYPICAL MOTOR FEEDER 1 A	Bus					51.369	115.240				43.205
TYPICAL MOTOR FEEDER 1 B	0.380	TYPICAL MOTOR FEEDER 1 B	Bus					54.777	121.703				43.174

ip is calculated using method C

Ib does not include decay of non-terminal faulted induction motors

Ik is the maximum steady state fault current

Idc is based on X/R from Method C and Ib as specified above

LV CB duty determined based on service rating.

Total through current is used for device duty.

* Indicates a device with calculated duty exceeding the device capability.

Indicates a device with calculated duty exceeding the device marginal limit. (95 % times device capability)

Short-Circuit Summary Report

Bus ID	Device ID	Device Capacity			3-Phase Short-Circuit Duty Results		
		Ithr (kA)	Tkr (sec.)	Rated Thermal Energy (MJ)	Ith (kA)	Tk (sec.)	Thermal Energy (MJ)
132kV GRID	=E02-Q0	40.000	1.00	1600.00	19.352	1.00	374.49
132kV GRID	=E06-Q0	40.000	1.00	1600.00	19.352	1.00	374.49
200A MDB Agreko	CB155	100.000	1.00	10000.00	49.396	1.00	2439.95
200A MDB Agreko	QM	65.000	1.00	4225.00	49.396	1.00	2439.95
Area Lighting DB (A8)	CB79	5.000	1.00	25.00	2.173	1.00	4.72
Bus36	CB210	100.000	1.00	10000.00	20.130	1.00	405.23
Bus36	CB151	100.000	1.00	10000.00	20.130	1.00	405.23
Bus79	=E03-Q0	40.000	1.00	1600.00	19.352	1.00	374.49
Bus81	=E07-Q0	40.000	1.00	1600.00	19.352	1.00	374.49
Bus82	QM	65.000	1.00	4225.00	49.396	1.00	2439.95
Bus83	=E02-Q0	40.000	1.00	1600.00	19.352	1.00	374.49
Bus84	=E06-Q0	40.000	1.00	1600.00	19.352	1.00	374.49
Bus87	=E01-Q0	40.000	1.00	1600.00	19.352	1.00	374.49
Bus88	=E08-Q0	40.000	1.00	1600.00	19.352	1.00	374.49
Bus_LVDB-1	Q1_LVDB-4	65.000	1.00	4225.00	56.886	1.00	3236.00
Bus_LVDB-1	Q3_LVDB-1	65.000	1.00	4225.00	56.886	1.00	3236.00
Bus_LVDB-1	Q4_LVDB-1	65.000	1.00	4225.00	56.886	1.00	3236.00
Bus_LVDB-1	Q6_LVDB-1	65.000	1.00	4225.00	56.886	1.00	3236.00
Bus_LVDB-2	Q1_LVDB-2	65.000	1.00	4225.00	59.421	1.00	3530.84
Bus_LVDB-2	Q3_LVDB-2	65.000	1.00	4225.00	59.421	1.00	3530.84
Bus_LVDB-2	Q4_LVDB-2	65.000	1.00	4225.00	59.421	1.00	3530.84
Bus_LVDB-2	Q6_LVDB-2	65.000	1.00	4225.00	59.421	1.00	3530.84
Bus_MDB-1	Q1_MDB-1	65.000	1.00	4225.00	51.339	1.00	2635.67
Bus_MDB-1	Q2_MDB-1	65.000	1.00	4225.00	51.339	1.00	2635.67
Bus_MDB-1	Q5_MDB-1	65.000	1.00	4225.00	51.339	1.00	2635.67
Bus_MDB-1	Q6_MDB-1	65.000	1.00	4225.00	51.339	1.00	2635.67
Bus_MDB-2	Q1_MDB-2	65.000	1.00	4225.00	51.873	1.00	2690.84
Bus_MDB-2	Q2_MDB-2	65.000	1.00	4225.00	51.873	1.00	2690.84
Bus_MDB-2	Q6_MDB-2	65.000	1.00	4225.00	51.873	1.00	2690.84
Bus_MDB-2	Q5_MDB--2.	65.000	1.00	4225.00	51.873	1.00	2690.84
Bus_MDB-3	Q1_MDB-3	65.000	1.00	4225.00	52.144	1.00	2719.05
Bus_MDB-3	Q2_MDB-3	65.000	1.00	4225.00	52.144	1.00	2719.05
Bus_MDB-3	Q5_MDB-3	65.000	1.00	4225.00	52.144	1.00	2719.05
Bus_MDB-3	Q6_MDB-3	65.000	1.00	4225.00	52.144	1.00	2719.05
Bus_MDB-4	Q1_MDB-4	65.000	1.00	4225.00	54.415	1.00	2961.05
Bus_MDB-4	Q2_MDB-4	65.000	1.00	4225.00	54.415	1.00	2961.05

Bus ID	Device ID	Device Capacity			3-Phase Short-Circuit Duty Results		
		Ithr (kA)	Tkr (sec.)	Rated Thermal Energy (MJ)	Ith (kA)	Tk (sec.)	Thermal Energy (MJ)
Bus_MDB-4	Q5_MDB-4	65.000	1.00	4225.00	54.415	1.00	2961.05
Bus_MDB-4	Q6_MDB-4	65.000	1.00	4225.00	54.415	1.00	2961.05
Bus_MDB-5	Q1_MDB-5	65.000	1.00	4225.00	51.761	1.00	2679.21
Bus_MDB-5	Q2_MDB-5	65.000	1.00	4225.00	51.761	1.00	2679.21
Bus_MDB-5	Q5_MDB-5	65.000	1.00	4225.00	51.761	1.00	2679.21
Bus_MDB-5	Q6_MDB-5	65.000	1.00	4225.00	51.761	1.00	2679.21
Bus_MDB-6	Q1_MDB-6	65.000	1.00	4225.00	49.890	1.00	2489.03
Bus_MDB-6	Q2_MDB-6	65.000	1.00	4225.00	49.890	1.00	2489.03
Bus_MDB-6	Q5_MDB-6	65.000	1.00	4225.00	49.890	1.00	2489.03
Bus_MDB-6	Q6_MDB-6	65.000	1.00	4225.00	49.890	1.00	2489.03
Bus_MDB-7	Q1_MDB-7	65.000	1.00	4225.00	49.890	1.00	2489.03
Bus_MDB-7	Q2_MDB-7	65.000	1.00	4225.00	49.890	1.00	2489.03
Bus_MDB-7	Q5_MDB-7	65.000	1.00	4225.00	49.890	1.00	2489.03
Bus_MDB-7	Q6_MDB-7	65.000	1.00	4225.00	49.890	1.00	2489.03
Bus_MDB-8	Q1_MDB-8	65.000	1.00	4225.00	49.890	1.00	2489.02
Bus_MDB-8	Q2_MDB-8	65.000	1.00	4225.00	49.890	1.00	2489.02
Bus_MDB-8	Q5_MDB-8	65.000	1.00	4225.00	49.890	1.00	2489.02
Bus_MDB-8	Q6_MDB-8	65.000	1.00	4225.00	49.890	1.00	2489.02
BUS_S002-CMP3	IAC001=R17	50.000	1.00	2500.00	28.009	1.00	784.52
LV SWBD_TOFS	CB224	36.000	1.00	1296.00	28.776	1.00	828.05
LV SWBD_TOFS	CB225	36.000	1.00	1296.00	28.776	1.00	828.05
MV RMU-TOFS	CB220	25.000	1.00	625.00	21.751	1.00	473.11
MV RMU-TOFS	CB219	25.000	1.00	625.00	21.751	1.00	473.11
S002-ASP1-1 [BUS A]	ASP1-1=IC1	50.000	1.00	2500.00	25.773	1.00	664.25
S002-ASP1-1B [BUS B]	ASP1-1=IC2	50.000	1.00	2500.00	24.539	1.00	602.16
S002-ASP1-2A	CB178	100.000	1.00	10000.00	27.257	1.00	742.96
S002-ASP1-2A	ASP1-2A=IC1	50.000	1.00	2500.00	27.257	1.00	742.96
S002-ASP1-2B	CB179	100.000	1.00	10000.00	25.794	1.00	665.32
S002-ASP1-2B	ASP1-2A=IC2	50.000	1.00	2500.00	25.794	1.00	665.32
S002-DBASP1-2-22	CB15	16.000	1.00	256.00	8.269	1.00	68.37
S002-MCC1-1 [BUS]	MCC1-1=Q1	50.000	0.50	1250.00	27.414	0.50	375.76
S002-MCC1-2 [BUS]	MCC1-2=Q1	50.000	0.50	1250.00	25.419	0.50	323.07
S002-MCC1-3 [BUS]	MCC1-3=Q1	50.000	0.50	1250.00	28.656	0.50	410.60
S002-MCC1-4 [BUS]	MCC1-4=Q1	50.000	0.50	1250.00	15.489	0.50	119.96
S002-MCC1-5 [BUS]	MCC1-5=Q1	50.000	0.50	1250.00	26.440	0.50	349.53
S002-MCC1-6 [BUS]	MCC1-6=Q1	50.000	0.50	1250.00	11.792	0.50	69.53
S002-MCC1-7 [BUS]	MCC1-7=Q1	50.000	0.50	1250.00	26.222	0.50	343.81
S002-MCC1-8 [BUS]	MCC1-8=Q1	50.000	0.50	1250.00	25.958	0.50	336.91
S002-MCC1-8 [BUS]	CB17	50.000	1.00	2500.00	25.299	1.00	640.04

Bus ID	Device ID	Device Capacity			3-Phase Short-Circuit Duty Results		
		Ithr (kA)	Tkr (sec.)	Rated Thermal Energy (MJ)	Ith (kA)	Tk (sec.)	Thermal Energy (MJ)
S002-MCC1-9 [BUS]	CB241	100.000	1.00	10000.00	26.506	1.00	702.57
S002-MCC1-9 [BUS]	MCC1-9=Q1	50.000	0.50	1250.00	27.421	0.50	375.95
S002-MCC1-9 [BUS]	CB18	100.000	1.00	10000.00	26.506	1.00	702.57
S002-PC1-1A	PC1-1A-3A	50.000	1.00	2500.00	35.865	1.00	1286.33
S002-PC1-1A	PC1-1A-2A	50.000	1.00	2500.00	35.865	1.00	1286.33
S002-PC1-1A	PC1-1A-5A	50.000	0.50	1250.00	37.299	0.50	695.60
S002-PC1-1A	PC1-1A-4A	50.000	0.50	1250.00	37.299	0.50	695.60
S002-PC1-1A	PC1-1A-1A	50.000	1.00	2500.00	35.865	1.00	1286.33
S002-PC1-1A	CB16	100.000	1.00	10000.00	35.865	1.00	1286.33
S002-PC1-1B	PC1B-1-2B	50.000	0.50	1250.00	32.141	0.50	516.53
S002-PC1-1B	PC1-1A-5B	50.000	0.50	1250.00	32.141	0.50	516.53
S002-PC1-1B	PC1-1A-4B	50.000	0.50	1250.00	32.141	0.50	516.53
S002-PC1-1B	PC1-1B-1B	50.000	1.00	2500.00	31.418	1.00	987.07
S002-PC1-2A	PC1-2A-4A	50.000	1.00	2500.00	36.163	1.00	1307.79
S002-PC1-2A	PC1-2A-3A	50.000	1.00	2500.00	36.163	1.00	1307.79
S002-PC1-2A	PC1-2A-5A	50.000	0.50	1250.00	37.637	0.50	708.28
S002-PC1-2A	PC1-2A-2A	50.000	1.00	2500.00	36.163	1.00	1307.79
S002-PC1-2A	PC1-2A-1A	50.000	1.00	2500.00	36.163	1.00	1307.79
S002-PC1-2B	PC1-2A-3B	50.000	1.00	2500.00	33.231	1.00	1104.29
S002-PC1-2B	PC1-2A-5B	50.000	0.50	1250.00	34.200	0.50	584.83
S002-PC1-2B	CB240	100.000	1.00	10000.00	33.231	1.00	1104.29
S002-PC1-2B	PC1-2A-1B	50.000	1.00	2500.00	33.231	1.00	1104.29
S002-SB-1AC001 [BUS A]	1AC001=L1 [I/C 1]	50.000	1.00	2500.00	28.368	1.00	804.75
S002-SB-1AC001 [BUS A]	1AC001=L12	50.000	1.00	2500.00	28.368	1.00	804.75
S002-SB-1AC001 [BUS A]	1AC001=L11	50.000	1.00	2500.00	28.368	1.00	804.75
S002-SB-1AC001 [BUS A]	1AC001=L10	50.000	1.00	2500.00	28.368	1.00	804.75
S002-SB-1AC001 [BUS A]	1AC001=L5	50.000	1.00	2500.00	28.368	1.00	804.75
S002-SB-1AC001 [BUS A]	1AC001=L4	50.000	1.00	2500.00	28.368	1.00	804.75
S002-SB-1AC001 [BUS A]	1AC001=L3	50.000	1.00	2500.00	28.368	1.00	804.75
S002-SB-1AC001 [BUS A]	1AC001=L2	50.000	1.00	2500.00	28.368	1.00	804.75
S002-SB-1AC001 [BUS A]	1AC001=L14	50.000	1.00	2500.00	28.368	1.00	804.75
S002-SB-1AC001 [BUS A]	1AC001=L6	50.000	1.00	2500.00	28.368	1.00	804.75
S002-SB-1AC001 [BUS A]	1AC001=L9	50.000	1.00	2500.00	28.368	1.00	804.75
S002-SB-1AC001 [BUS A]	1AC001=L7	50.000	1.00	2500.00	28.368	1.00	804.75
S002-SB-1AC001 [BUS A]	1AC001=L15	50.000	1.00	2500.00	28.368	1.00	804.75
S002-SB-1AC001 [BUS A]	1AC001=L13	50.000	1.00	2500.00	28.368	1.00	804.75
S002-SB-1AC001 [BUS A]	1AC001=L16	50.000	1.00	2500.00	28.368	1.00	804.75
S002-SB-1AC001 [BUS B]	1AC001=R2	50.000	1.00	2500.00	28.009	1.00	784.52
S002-SB-1AC001 [BUS B]	1AC001=R3	50.000	1.00	2500.00	28.009	1.00	784.52

Bus ID	Device ID	Device Capacity			3-Phase Short-Circuit Duty Results		
		Ithr (kA)	Tkr (sec.)	Rated Thermal Energy (MJ)	Ith (kA)	Tk (sec.)	Thermal Energy (MJ)
S002-SB-1AC001 [BUS B]	1AC001=R4	50.000	1.00	2500.00	28.009	1.00	784.52
S002-SB-1AC001 [BUS B]	1AC001=R5	50.000	1.00	2500.00	28.009	1.00	784.52
S002-SB-1AC001 [BUS B]	1AC001=R6	50.000	1.00	2500.00	28.009	1.00	784.52
S002-SB-1AC001 [BUS B]	1AC001=R7	50.000	1.00	2500.00	28.009	1.00	784.52
S002-SB-1AC001 [BUS B]	1AC001=R8	50.000	1.00	2500.00	28.009	1.00	784.52
S002-SB-1AC001 [BUS B]	1AC001=R10	50.000	1.00	2500.00	28.009	1.00	784.52
S002-SB-1AC001 [BUS B]	1AC001=R11	50.000	1.00	2500.00	28.009	1.00	784.52
S002-SB-1AC001 [BUS B]	1AC001=R13	50.000	1.00	2500.00	28.009	1.00	784.52
S002-SB-1AC001 [BUS B]	1AC001=R1 [I/C 2]	50.000	1.00	2500.00	28.009	1.00	784.52
S002-SB-1AC001 [BUS B]	1AC001=R16	50.000	1.00	2500.00	28.009	1.00	784.52
S002-SB-1AC001 [BUS B]	1AC001=R12	50.000	1.00	2500.00	28.009	1.00	784.52
S002-SB-1AC001 [BUS B]	1AC001=R14	50.000	1.00	2500.00	28.009	1.00	784.52
S002-SB-1AC001 [BUS B]	1AC001=R15	50.000	1.00	2500.00	28.009	1.00	784.52
S002-SB-1AC001 [BUS B]	1AC001=R17	50.000	1.00	2500.00	28.009	1.00	784.52
S002-SB-1AD001 [BUS A]	1AD001=L2 [I/C 1]	50.000	1.00	2500.00	24.697	1.00	609.92
S002-SB-1AD001 [BUS A]	1AD001=L4	50.000	1.00	2500.00	24.697	1.00	609.92
S002-SB-1AD001 [BUS A]	1AD001=L3	50.000	1.00	2500.00	24.697	1.00	609.92
S002-SB-1AD001 [BUS A]	1AD001=L26	50.000	1.00	2500.00	24.697	1.00	609.92
S002-SB-1AD001 [BUS B]	1AD001=R1 [I/C 2]	50.000	1.00	2500.00	22.539	1.00	508.01
S002-SB-1AD001 [BUS B]	1AD001=R2	50.000	1.00	2500.00	22.539	1.00	508.01
S002-SB-1AD001 [BUS B]	1AD001=R3	50.000	1.00	2500.00	22.539	1.00	508.01
S002-SB-1AD001 [BUS B]	1AD001=R20	50.000	1.00	2500.00	22.539	1.00	508.01
S002-SB-1AN001 [BUS A]	CB92	50.000	1.00	2500.00	14.921	1.00	222.62
S002-SB-1AN001 [BUS A]	1AN001=IC1	50.000	1.00	2500.00	14.921	1.00	222.62
S002-SB-1AN001 [BUS B]	CB215	100.000	1.00	10000.00	14.765	1.00	218.01
S002-SB-1AN001 [BUS B]	1AN001=IC2	50.000	1.00	2500.00	14.765	1.00	218.01
S002-SB-1AN001 [BUS B]	CB6	50.000	1.00	2500.00	14.765	1.00	218.01
S002-SB-01E001 [BUS A]	01E001+A04.RA01	120.000	1.00	14400.00	48.235	1.00	2326.59
S002-SB-01E001 [BUS A]	01E001+A06.RA01	120.000	1.00	14400.00	48.235	1.00	2326.59
S002-SB-01E001 [BUS A]	01E001+A07.PA01	120.000	1.00	14400.00	48.235	1.00	2326.59
S002-SB-01E001 [BUS A]	01E001+A03.AA01	100.000	1.00	10000.00	48.235	1.00	2326.59
S002-SB-01E001 [BUS B]	01E001+B02.AA01	100.000	1.00	10000.00	46.870	1.00	2196.78
S002-SB-01E001 [BUS B]	01E001+B07.RA01	120.000	1.00	14400.00	46.870	1.00	2196.78
S002-SB-01E001 [BUS B]	01E001+B06.PA01	120.000	1.00	14400.00	46.870	1.00	2196.78
S002-SB-01E001 [BUS B]	01E001+B04.KA01	120.000	1.00	14400.00	46.870	1.00	2196.78
S002-SB-02N001 [BUS A]	CB180	100.000	1.00	10000.00	29.850	1.00	891.03
S002-SB-02N001 [BUS A]	CB80	50.000	1.00	2500.00	29.850	1.00	891.03
S002-SB-02N001 [BUS A]	02N001+Q1	75.000	1.00	5625.00	29.850	1.00	891.03
S002-SB-02N001 [BUS B]	02N001+Q2	75.000	1.00	5625.00	29.917	1.00	895.01

Bus ID	Device ID	Device Capacity			3-Phase Short-Circuit Duty Results		
		Ithr (kA)	Tkr (sec.)	Rated Thermal Energy (MJ)	Ith (kA)	Tk (sec.)	Thermal Energy (MJ)
S002-SB-02N001 [BUS B]	CB185	100.000	1.00	10000.00	29.917	1.00	895.01
S002-SB-03E001 [BUS A]	03E001+A03.BA01	65.000	1.00	4225.00	34.813	1.00	1211.96
S002-SB-03E001 [BUS B]	03E001+B02.BA01	65.000	1.00	4225.00	34.520	1.00	1191.64
S002-SB-03N001 [BUS A]	03N001+A03.BA01	80.000	1.00	6400.00	45.599	1.00	2079.29
S002-SB-03N001 [BUS B]	03N001+B03.BA01	80.000	1.00	6400.00	43.707	1.00	1910.33
S002-SB-04E001 [BUS A]	SB-04E001=CB1	20.000	1.00	400.00	15.762	1.00	248.45
S002-SB-04E001 [BUS A]	SB-04E001=CB3	40.000	1.00	1600.00	15.762	1.00	248.45
S002-SB-04E001 [BUS B]	SB-04E001=CB3	40.000	1.00	1600.00	15.762	1.00	248.45
S002-SB-PSS01N01 [BUS A]	PSS1N01+Q1	50.000	1.00	2500.00	13.471	1.00	181.47
S002-SB-PSS01N01 [BUS B]	PSS1N01+Q2	50.000	1.00	2500.00	12.772	1.00	163.13
S003-DB11E001-01	11N001-01 [I/C1]	25.000	1.00	625.00	12.694	1.00	161.14
S003-DB11E001-02	DB11E001-02 [I/C1]	25.000	1.00	625.00	13.905	1.00	193.36
S003-DB-11N001-03 [BUS A]	11N001-03 [I/C1]	25.000	1.00	625.00	15.603	1.00	243.46
S003-DB-11N001-03 [BUS B]	11N001-03 [I/C2]	25.000	1.00	625.00	15.717	1.00	247.01
S003-DB11N001-04 [BUS A]	11N001-04 [I/C1]	25.000	1.00	625.00	15.569	1.00	242.39
S003-DB11N001-04 [BUS B]	11N001-04 [I/C2]	25.000	1.00	625.00	15.698	1.00	246.44
S003-DB11N001-11	CB85	100.000	1.00	10000.00	10.367	1.00	107.48
S003-DB11N001-12	CB74	100.000	1.00	10000.00	10.367	1.00	107.48
S003-MDP-B	MDP-B [I/C1]	100.000	1.00	10000.00	25.945	1.00	673.16
S003-SB01E1	SB01E1 [I/C1]	25.000	1.00	625.00	12.245	1.00	149.93
S003-SB-11C001 [BUS A]	11C001=L2 [I/C 1]	40.000	3.00	4800.00	10.082	3.00	304.93
S003-SB-11C001 [BUS A]	11C001=L4	40.000	3.00	4800.00	10.082	3.00	304.93
S003-SB-11C001 [BUS A]	11C001=L3	40.000	3.00	4800.00	10.082	3.00	304.93
S003-SB-11C001 [BUS A]	11C001=L5	40.000	3.00	4800.00	10.082	3.00	304.93
S003-SB-11C001 [BUS A]	11C001=L8	40.000	3.00	4800.00	10.082	3.00	304.93
S003-SB-11C001 [BUS A]	11C001=L6	40.000	3.00	4800.00	10.082	3.00	304.93
S003-SB-11C001 [BUS A]	11C001=L7	40.000	3.00	4800.00	10.082	3.00	304.93
S003-SB-11C001 [BUS B]	11C001=R1 [I/C 2]	40.000	3.00	4800.00	10.150	3.00	309.05
S003-SB-11C001 [BUS B]	11C001=R2	40.000	3.00	4800.00	10.150	3.00	309.05
S003-SB-11C001 [BUS B]	11C001=R3	40.000	3.00	4800.00	10.150	3.00	309.05
S003-SB-11C001 [BUS B]	11C001=R4	40.000	3.00	4800.00	10.150	3.00	309.05
S003-SB-11C001 [BUS B]	11C001=R6	40.000	3.00	4800.00	10.150	3.00	309.05
S003-SB-11C001 [BUS B]	11C001=R5	40.000	3.00	4800.00	10.150	3.00	309.05
S003-SB-11D001 [BUS A]	11D001=L2 [I/C 1]	40.000	1.00	1600.00	7.365	1.00	54.24
S003-SB-11D001 [BUS A]	11D001=L7	40.000	1.00	1600.00	7.365	1.00	54.24
S003-SB-11D001 [BUS B]	11D001=R1 [I/C 2]	40.000	1.00	1600.00	7.695	1.00	59.22
S003-SB-11D001 [BUS B]	11D001=R5	40.000	1.00	1600.00	7.695	1.00	59.22
S003-SB-11E001 [BUS A]	11E001+A02 [I/C1]	100.000	1.00	10000.00	34.099	1.00	1162.72
S003-SB-11E001 [BUS A]	CB204	100.000	1.00	10000.00	34.099	1.00	1162.72

Bus ID	Device ID	Device Capacity			3-Phase Short-Circuit Duty Results		
		Ithr (kA)	Tkr (sec.)	Rated Thermal Energy (MJ)	Ith (kA)	Tk (sec.)	Thermal Energy (MJ)
S003-SB-11E001 [BUS A]	CB153	100.000	1.00	10000.00	34.099	1.00	1162.72
S003-SB-11E001 [BUS B]	11E001+B02 [I/C2]	100.000	1.00	10000.00	34.620	1.00	1198.53
S003-SB-11E001 [BUS B]	11E001+A05	70.000	1.00	4900.00	34.620	1.00	1198.53
S003-SB-11E001 [BUS B]	CB237	100.000	1.00	10000.00	34.620	1.00	1198.53
S003-SB-11E001 [BUS B]	CB205	100.000	1.00	10000.00	34.620	1.00	1198.53
S003-SB-11N001 [BUS A]	11N001+A02 [I/C1]	85.000	1.00	7225.00	55.774	1.00	3110.79
S003-SB-11N001 [BUS A]	CB146	100.000	1.00	10000.00	55.774	1.00	3110.79
S003-SB-11N001 [BUS A]	CB208	100.000	1.00	10000.00	55.774	1.00	3110.79
S003-SB-11N001 [BUS B]	CB172	100.000	1.00	10000.00	57.568	1.00	3314.11
S003-SB-11N001 [BUS B]	11N001+B02 [I/C2]	100.000	1.00	10000.00	57.568	1.00	3314.11
S003-SB-11N001 [BUS B]	CB190	100.000	1.00	10000.00	57.568	1.00	3314.11
S003-SB-11N001 [BUS B]	CB203	100.000	1.00	10000.00	57.568	1.00	3314.11
S003-SB-11N001 [BUS B]	CB84	100.000	1.00	10000.00	57.568	1.00	3314.11
S003-SB-11N001 [BUS B]	CB75	100.000	1.00	10000.00	57.568	1.00	3314.11

Ithr = Rated short-time withstand current (Icw for low voltage circuit breaker)
Tkr = Rated short-time
Ith = Thermal equivalent short-time current
Tk = User-defined short-time

* Indicates a device with calculated duty exceeding the device capability.

Indicates a device with calculated duty exceeding the device marginal limit. (95 % times device capability)